

Morphology & Syntax

(509)

Virtual University of Pakistan

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WHAT IS MORPHOLOGY?

Topic- 001: Introduction

In this book, we will study some basic notions in morphology.

- What does morphology study?
- What are the goals of morphological analysis?
- What is its place in general theory of grammar?

The term morphology coined early in the 19th century in a biological context (the study of the form and structure of organisms) is generally attributed to the German poet, novelist, playwright, and philosopher Johann Wolfgang von Goethe (1749– 1832). Its etymology is Greek: morph- means ‘shape, form’, and morphology is the study of form or forms. In linguistics morphology refers to the mental system involved in word formation or to the branch of linguistics that deals with words, their internal structure, and how they are formed.

According to Fabregas & Scalise (2012), morphology is the part of linguistics that studies grammatical properties of words and how words relate to each other in a language. The chief task of morphology is to study how words such as the following are related to each other:

- (1) a. deep
b. deepen
- (2) a. dog
b. dogs

(1b) and (2b) simply seem more complex than (1a) and (2a) as on the formal side they contain additional (-en) and (-s) respectively, and on the semantic side, they have more complex meaning that presuppose the meaning of (1a) and (2a) such as ; (1b) and (2b) ‘become deep or cause something to become deep’ and ‘more dogs’. The (1b) and (2b) are BASE from which words in (1b) and (2b) have been built. The central question in morphology is ‘How words are related to each other?’ Are there some aspects of morphological research which have something to do with morphology?

Topic- 002: Morphemes

A major way in which morphologists investigate words, their internal structure, and how they are formed is through the identification and study of morphemes, often defined as the smallest linguistic pieces with a grammatical function. This definition is not meant to include all morphemes, but it is the usual one and a good starting point. A morpheme may consist of a word, such as hand, or a meaningful piece of a word, such as the -ed of looked, that cannot be divided into smaller meaningful parts. Another way in which morphemes have been defined is as a pairing between sound and meaning.

Some morphemes have no concrete form or no continuous form, as we will see, and some do not have meanings in the conventional sense of the term. You may also run across the term morph. The term ‘morph’ is sometimes used to refer specifically to the phonological realization of a morpheme. For example, the English past tense morpheme that we spell -ed has various morphs. It is realized as [t] after the voiceless [p] of jump (cf. jumped), as [d] after the voiced [l] of repel (cf. repelled), and as [ed] after the voiceless [t] of root or the voiced [d] of wed (cf. rooted and wedded). We can also call these morphs allomorphs or variants. The appearance of one morph over another in this case is determined by voicing and the place of articulation of the final consonant of the verb stem.

Now consider the word reconsideration. We can break it into three morphemes: re-, consider, and -ation. Consider is called the stem. A stem is a base unit to which another morphological piece is attached. The stem can be simple, made up of only one part, or complex, itself made up of more than one piece. Here it is best to consider ‘consider’ a simple stem. Although it consists historically of more than one part, most present-day speakers would treat it as an unanalyzable form. We could also call consider the root. A root is like a stem in constituting the core of the word to which other pieces attach, but the term refers only to morphologically simple units. For example, disagree is the stem of disagreement, because it is the base to which -ment attaches, but agree is the root. Taking disagree now, agree is both the stem to which dis- attaches and the root of the entire word.

Returning now to reconsideration, re- and -ation are both affixes, which means that they are attached to the stem. Affixes like re- that go before the stem are prefixes, and those like -ation that go after are suffixes. Why have we not broken -ation down further into two pieces, -ate and -ion, which function independently elsewhere. In this particular word they do not do so (cf. *reconsiderate), and hence we treat -ation as a single morpheme. It is important to take seriously the idea that the grammatical function of a morpheme, which may include its meaning, must be constant. Consider the English words lovely and quickly. They both end with the suffix -ly. But is it the same in both words? No – when we add -ly to the adjective quick, we create an adverb that is often synonymous with “rapidly”: The students quickly assimilated the concept. When we add -ly to the noun love, we create an adjective: What a lovely day! What on the surface appears to be a single morpheme turns out to be two. One attaches to adjectives and creates adverbs; the other attaches to nouns and creates adjectives.

There are two other sorts of affixes that you will encounter, infixes and circumfixes. Both are classic challenges to the notion of morpheme. Infixes are segmental strings that do not attach to the front or back of a word, but rather somewhere in the middle. The Tagalog infix -um- is illustrated below (McCarthy and Prince 1993: 101–5; French 1988). It creates an agent from a verb stem and appears before the first vowel of the word:

(1) Root -um /sulat/ /s-um-ulat/ ‘one who wrote’ /gradwet/ /gr-um-adwet/ ‘one who graduated’

The existence of infixes challenges the traditional notion of a morpheme as an indivisible unit. We want to call the stem *sulat* ‘write’ a morpheme, and yet the infix -um- breaks it up. This seems to be a property of -um- rather than *sulat*. Our definition of morphemes as the smallest linguistic pieces with a grammatical function survives this challenge. Circumfixes are affixes that come in two parts. One attaches to the front of the word and the other to the back. Circumfixes are controversial because it is possible to analyze them as consisting of a prefix and a suffix that apply to a stem simultaneously. One example is Indonesian *ke* ... -*an*. It applies to the stem *besar* ‘big’ to form a noun *ke-besar-an* meaning

‘bigness, greatness’ (MacDonald 1976: 63; Beard 1998: 62). Like infixes, the existence of circumfixes challenges the traditional notion of morpheme (but not the definition used here) because they involve discontinuity.

Topic- 003: Morphology in Action

We would like to explore the idea of morphology more deeply by examining some data. These are examples of **morphology in action** – morphological facts of everyday life. Let’s see how novel words come into existence and how word play takes place.

If you had been walking down the street in Ithaca, New York, several years ago, you might have looked up and seen a sign for the music store “Rebop,” a name that owes its inspiration to the jazz term rebop. Rebop was originally one of the many nonsense expressions that jazz musicians threw into their vocal improvisations, starting in the early 1920s. In the 1940s, rebop became interchangeable with bebop, a term of similar origin, as the term for the rhythmically and harmonically eccentric music played by young black musicians. By the 1950s the name of this musical style was quite firmly established as simply bop. Today, the original use of rebop is known only to experts, so that most people who pass by the store will be likely to interpret the word as composed of the word bop and the prefix re-, which means approximately ‘again’. This prefix can attach only to verbs, so we must interpret bop as a verb here. Rebop must therefore mean ‘bop again’, if it means anything at all. And this music store, appropriately, specialized in selling used CDs. There’s something going on here with English morphology. Rebop is not a perfectly well-formed English word. The verb bop means something like ‘bounce’, but the prefix re- normally attaches only to a verb whose meaning denotes an accomplishment. The verb rebop therefore makes little sense.

Consider now the following phrases, taken from a Toni Braxton song: Unbreak my heart, uncry these tears. We have never seen anyone unbreak something, and you certainly can’t uncry tears, but every English speaker can understand these words. We all know what it means to unbreak somebody’s heart or to wish that one’s heart were unbroken. If we asked somebody, “unbreak my heart,” we would be asking them to reverse the process of having our heart broken. We can visualize “uncry these tears,” too – think of a film running backwards. We can understand these words because we know the meaning of un-, which, when attached to a verb, reverses or undoes an action. The fact that these particular actions, breaking a heart and crying tears, cannot be reversed only adds poignancy to the song.

All human beings have this capacity for generating and understanding novel words. Sometimes someone creates an entirely new word, as J. R. R. Tolkien did when he coined the now-familiar term hobbit. But more often than not, we build new words from pre-existing pieces, as with unbreak and uncry, or as with hobbitish and hobbit-like, built by adding suffixes to the stem hobbit. We could easily go on to create more words on these patterns. Novel words are all around us.

These are everyday morphological facts, the kind you run across every day as a literate speaker of English. What all these words – rebop, unbreak, uncry, hobbit, hobbitish, and hobbit-like, – have in common is their newness. When we saw or heard them for the first time, they leapt out at us. It is interesting that novel words do this to us, because novel sentences generally do not. When you hear a new sentence, you generally don’t realize that it is the first time that you’ve heard it, and you don’t say to yourself, “What a

remarkable sentence,” Many people have made the observation before that morphology differs from syntax in this way.

Topic- 004: Abstract Morphological Facts

Let’s move to some more abstract morphological facts. These are the kind of morphological facts that you don’t notice every day. They are so embedded in the language that you don’t even think about them. They are more common than the ones we have just looked at, but deeper and more complex. If you speak English and are concerned about your health, you might say:

(2) I eat one melon a day.

Let’s imagine that we are even more concerned about our health than you are. We don’t just eat one melon a day, rather:

(3) We eat two melons a day.

It is a fact about standard American or British English that we cannot say:

(4) *We eat two melon a day.

However, if we were speaking Indonesian or Japanese, we would say the equivalent of two melon (three melon, four melon, etc.) because these languages don’t use morphological plurals in sentences like this.

Indonesian: Saiga makan dua buah semangka (se) tiap hari

I eat two fruit melon every day

‘I eat two melons every day.’

Japanese: mainichi futatsu-no meron-o tabemasu

every.day two- gen melon-obj eat.imp

‘I eat two melons every day.’

The morphological grammar of English tells us that we have to put an **-s** on melon whenever we are talking about more than one. This fact of English is so transparent that native speakers don’t notice it. If we happen to be speakers of a language without obligatory plural marking, however, we will notice and may have trouble with it. We have now observed something about English morphology. If a word is plural, it takes the suffix **-s**. Living creatures don’t eat only melons, however: The evil giant at the top of the beanstalk eats two melons, three fish, and four children a day.

Everyone agrees that fish is plural, even though there is no plural marker. Children is also plural, but it has a very unusual plural suffix, **-ren**, plus an internal change. We do not always mark plural words with an **s-like** thing; there are other ways in which we can mark plurals. Native speakers of English know this, and they do not need to think about it before making a plural. Consider the following:

Today they claim that they will fix the clock tower by Friday, but yesterday they claimed that it would take at least a month.

In this example, we use two different forms of the verb claim. One is present tense, and the other is past. Again, this is not true for all languages. If we were speaking Vietnamese, for example, we wouldn't make any distinction between *claim* and *claimed* – we wouldn't mark the verb at all. If we were speaking Chinese, we would not distinguish between *claim* and *claimed* in a sentence like this, because the adverb *zuótian* 'yesterday' is sufficient to indicate past tense:

If we were to leave out *zuótian*-n 'yesterday', we would need to use the particle **le** after the verb to show that the action took place in the past. Whether or not a speaker must indicate past tense in Chinese depends on context. Notice what happens in English when we use some other verbs besides claim:

Today they say ... but yesterday they said ... tell us, told us, know, knew, etc.

That these verbs and others do not add **-t, -d, or ed** to make their past tense is an elementary fact about English morphology. The next observation about English morphology has to do with pronouns. Here is an exchange between an American mother, who has just watched a billiard ball break through a window, and her 6-year-old boy, who is standing inside:

- (1) Who just threw a pool ball through the basement window?
- (2) Not me.

In this context, a 6-year-old wouldn't respond Not I, though if he were to answer with a sentence, the response would be I didn't, not Me didn't. Without formally knowing anything at all about subjects and objects, English-speaking 6-year-olds (and children even younger) master the pronoun system of the spoken language.

In some other languages, marking for dual is even more pervasive. This is the case in Ancient Greek, as shown by the following examples:

	Ancient Greek	English
Singular	stratiô:tes	One horse
Dual	stratiô:ta	Two horse
Plural three or more than	stratiô:tai	Three or more horses

While English does not have special affixes to mark the dual, it keeps track of the distinction through words like *all* and *both*. There are even languages in the world like Manam (Papua New Guinea: Gregersen 1976) and Larike (Central Maluku, Indonesia: Laidig and Laidig 1990) that distinguish not only singular, dual, and plural, but also trial.

Topic- 005: Background and Beliefs

Our purpose here is a general introduction to morphology and morphological analysis from the point of view of a morphologist. The purpose is not to advocate any particular theory or to give the truth (whatever that is), but rather to get you, the students, to where you can look for it by yourself. Still, it is

inevitable that some of our remarks will be colored by our own beliefs and background. We would therefore like to present some of our foundational beliefs about linguistics and linguistic methodology. First, we believe that languages differ from one another. You might be thinking, “Of course they do!” But we mean this in a very special way. Some linguists are always looking for ways that languages are similar, and at times, we do that, too. But we believe that if you focus only on the similarities between languages, you miss out on all of the exciting ways in which they differ. What’s more, you may find parallels and similarities where none really exists. We try to approach linguistic analysis with as open a mind as possible, and to do this, it is first necessary to appreciate the uniqueness and diversity of the world’s languages. Our second foundational belief is that languages, which we can write with a small *l*, are different from Language, with a capital *L*. There are thousands of individual languages in the world. But we may also speak of language in general to mean the general phenomenon of Language that encompasses all individual languages. This Language is related to Noam Chomsky’s notion of Universal Grammar, which posits that all languages are alike in basic ways. Each of these two uses of the word language is equally important to linguistics. Individual languages have features that are not characteristic of Language in general. For example, one feature of English is that its regular way of forming plural nouns is to add /z/. We would never claim, however, that this is universally true, or that it is a property of Language.

To tie this belief in with the preceding one, we strongly believe that morphological theory and morphological analysis must be grounded in morphological description. If we want to appreciate what morphology really is, it’s best to have some idea of what the morphology of individual languages is like. At the same time, we must have a reasonably well-thought-out general theory of the morphology of Language, so that we can compare our descriptions of individual languages within a wider context. In short, linguists need to pay equal attention to both small-*l* language and capital-*L* Language. Our next belief is that morphology is a distinct component of languages or grammars. If you are not already familiar with some of the controversy surrounding morphology, this needs an explanation.

The fact that some languages, such as Vietnamese, do not have morphologically complex words has led some people to conclude that morphology should not be a separate branch of linguistics. The reasoning is that linguistics is generally understood to deal with properties of Language. If there are languages that don’t have morphology, then morphology is not a property of all languages and of Language, and morphological phenomena should be treated in syntax or phonology. We disagree. It has been shown elsewhere (e.g., Aronoff 1994) that some aspects of morphology cannot be attributed to syntax or phonology, or anything else.

One piece of evidence that morphology is separate from syntax, phonology, and other branches of linguistics is that words in some languages are grouped into largely arbitrary classes that determine their forms in different environments. Latin nouns fall into five distinct classes, called declensions, which have little or nothing to do with syntax or phonology and cannot be explained by either. They are purely morphological in their significance. The uniquely morphological nature of these classes is truly brought home by the fact that Latin nouns also fall into syntactic agreement classes (usually called genders), and the two systems cross-cut one another: two nouns may belong to the same gender but to different declensions and vice versa. Their mere existence in many languages shows that morphology must be given some independent status in linguistics.

Morphology, probably more than any other component of language, interacts with all the rest, but it still has properties of its own. We also believe that morphologies are systems. This is a very old observation. Because of it, it is impossible to talk about isolated facts in a language – everything holds together. This belief together with the second one, above, is the reason why we'll be looking carefully at the morphology of a particular languages, throughout this book.

So far, we have given you our beliefs about the nature of language and morphology. We also have some that pertain to methodology. So far, we have given you our beliefs about the nature of language and morphology. We also have some that pertain to methodology. The first is that we should take an attitude of skeptical realism. Albert Einstein said that a physicist must be both a realist and a nominalist, a realist in the sense that you must believe what you ultimately find will be real, but a nominalist in the sense that you must never believe you've found what you're looking for. Martin Joos made a similar statement about linguistics. On the one hand, you should always believe that what you are looking for is God's truth, but on the other, you should consider all that you have found so far as hocus-pocus. We believe strongly in the value of having a linguistic theory, but we believe equally strongly that you should never trust it completely.

Our other methodological belief can be summed up as a motto: anything goes. This methodological belief is associated with the Against Method of Paul Feyerabend, a twentieth-century philosopher who felt that if we insisted on a single rule of scientific methodology, one that would not inhibit progress, it would be "Anything goes." We take a noholds-barred approach to linguistics. We'll use any tool or method that will tell us how language works. This attitude stems in part from our skepticism about particular theories. People who are wedded to individual theories tend to believe in using tools that are rooted in that theory. Our tools are not theory-based in that way. If a tool does the job, we are happy to use it, whether it is a traditional linguistic tool (e.g., native speaker consultants, dictionaries, written grammars), an experimental tool (e.g., imaging technology), or a statistical tool.

Topic- 006: Introduction to Morphological Analysis

Two basic approaches: analysis and synthesis

There are two complementary approaches to morphology, analytic and synthetic. The linguist needs both. The analytic approach has to do with breaking words down, and it is usually associated with American structuralist linguistics of the first half of the twentieth century. There is a good reason for this. These linguists were often dealing with languages that they had never encountered before, and there were no written grammars of these languages to guide them. It was therefore crucial that they should have very explicit methods of linguistic analysis. No matter what language we're looking at, we need analytic methods that are independent of the structures we are examining; preconceived notions might interfere with an objective, scientific analysis. This is especially true when dealing with unfamiliar languages. The second approach to morphology is more often associated with theory than with methodology, perhaps unfairly. This is the synthetic approach. It basically says, "I have a lot of little pieces here. How do I put them together?" This question presupposes that you already know what the pieces are. Analysis must in some way precede synthesis. Say that you've broken a clock and taken it apart, and now you have to put

all the little pieces back together. There's a catch: you don't know how. You could always go by trial and error. But the most efficient way would be to have some theory of how the clock goes together.

But the primary way in which morphologists determine the pieces they are dealing with is by examination of language data. They pull words apart carefully, taking great care to note where each piece came from to begin with. We have described analysis and synthesis in terms of the morphologist studying language, but the two notions are equally applicable to speakers of a language. Speakers apply morphological analysis when they read or hear a complex word they have never encountered before. In order to understand it, they pull it apart and ask themselves whether they recognize any of the pieces. Speakers use synthesis whenever they create new forms from pre-existing pieces.

Analytic principles

Before we try out some problems, we would like to give you some basic analytic principles used in morphology. They are taken from Eugene Nada's (1949; revised edition 1965) textbook Morphology. The first principle is given below:

Principle 1: Forms with the same meaning and the same sound shape in all their occurrences are instances of the same morpheme. Step one in morphological analysis is to look for elements that have the same form and the same meaning. This is the basic type-token problem. Let's say that we have a bunch of coins. Each is a token, a form. If we look at them carefully, we see that three of them look very much the same (they are all nickels), and two of them are identical (they both say 1997). These two coins are tokens of exactly the same type: they have identical forms and identical values. We may further say that the three coins are all tokens of a larger type that includes all nickels, not just those minted in 1997. But five pennies, though they have the same value as a nickel, do not together comprise the same type as the nickel, because, although identical in value to the nickel, they are different in form.

One thing that makes a morpheme a morpheme is that it recurs, and thus speakers are able to identify it and give it a meaning. This isn't the whole story, as Principle 2 tells us:

Principle 2: Forms with the same meaning but different sound shapes may be instances of the same morpheme if their distributions do not overlap. When two or more instances of a given morpheme occur with different shapes, we call them allomorphs. Allomorphs will be introduced in details. The regular plural marker in English has several allomorphs – voiceless alveolar fricative /s/, voiced alveolar fricative /z/, etc. As in the example, the distributions of these forms do not overlap, and they all have the same meaning. We can infer that they are instances of the same morpheme.

Principle 3: Not all morphemes are segmental. Normally, when we think of morphemes, we think of forms that can be pronounced in some sense, e.g., chicken, the, un-, -ize. But some morphemes can't be pronounced on their own. They are dependent on other morphemes for their realization. In English, for example, vowel alternations may serve to differentiate basic and past forms of the verb. We refer to these alternations as ablaut. Look at the following:

- run ran,
- speak spoke,
- eat ate, etc.

We know that there is a past tense marker distinguishing the words in the second column from those in the first. But what is it? It is not the /æ/ of ran or the /o/ of spoke but rather the difference between these vowels and the vowels of the basic verb, which is not segmental at all. We must look at both the present and past tense forms of these verbs, because it is the contrast between them that is important. Another type of non-segmental morpheme in English is shown in the following:

- breath (N) breathe (V)
- cloth (N) clothe (V)
- house (N) house (V)

In each pair, the noun ends in a voiceless fricative ([θ, s]), while the verb ends in a voiced fricative ([ð, z]). Assuming that the noun is basic, we say that the morpheme that marks the verbs consists of the phonological feature [+voice].

Principle 4: A morpheme may have zero as one of its allomorphs provided it has a non-zero allomorph. Fish generally displays no special marking in the plural: one fish, ten fish-Ø. We can say that it has a zero plural, and that this zero plural is an allomorph of the usual plural [z], because other words in the language, like frogs, have non-zero plurals. This is an analytic procedure, not a theoretical point. We cannot posit a zero unless it contrasts with some non-zero variant. In Japanese, where sakana means both ‘fish (sg)’ and ‘fish (pl)’, we cannot posit a zero plural (*sakana-Ø) because nowhere in the language does -ØPL contrast with a non-zero allomorph.

Lesson 02

WORDS AND LEXEMES

Topic- 007: What is a Word?

A single word can have multiple uses and interpretations. Occasionally a headline-writer underestimates this fact and ends up writing amusing headlines when no humor is intended:

- MINERS REFUSE TO WORK AFTER DEATH
- EYE DROPS OFF SHELF
- LOCAL HIGH SCHOOL DROPOUTS CUT IN HALF

Death in the first headline can be understood as ‘the act or fact of dying’ or as ‘the death of a specific person’ – the intended meaning. Words like noun, verb, adjective, and adverb refer to what linguists call lexical category. They are labels that tell us how a word is generally used in a sentence. A noun can be the subject of a sentence, but not so a verb. In many cases, identical-sounding or identical-looking words can belong to multiple categories, and that is what is going on in the sentences mentioned above. Lexical category is basic information about a word, but there is much more that, as linguists, we want to say. For example, Death ‘act or fact of dying’ and death ‘the dying of a specific individual’ are different.

What is a Word?

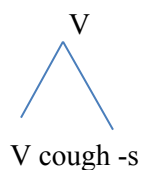
There are various ways to define a word, but no definition is entirely satisfactory. Scholars have acknowledged this fact over and over again. Here we present some of the reasons why what seems like a relatively simple task proves to be so problematic.

Defining words syntactically

One way that people have attempted to define words is to call them the smallest unit of syntax. This seems reasonable: sentences are built by combining words according to particular patterns. But even this simple definition runs into problems. Take a sentence like the following:

- Harry coughs every time he steps outside.

Everyone would agree that Harry, every, and outside are words, and that -s is not. But at the same time, some people (though not all) would argue that -s is indeed a unit of syntax and that it occupies a particular position in a syntactic tree. The following diagram illustrates how we might break cough off from -s syntactically:



Calling words the minimal units of syntax raises the question, “What is syntax?” If we think of syntax as the component of the human grammar that governs the ordering of items, then -s should be a word. After all, it is subject to ordering principles. It must follow cough; we don’t say s-cough. If we respond by saying that syntax governs the ordering of not just any item, but only words, then we are back where we started. What is a word? Another characteristic of words is that they are the smallest unit of language that can stand alone:

A: When are you going to the store?

B: Tomorrow.

A: What did the emperor wear to the procession?

B: Nothing!

We recognize the ability of words to stand alone by saying that they are free forms. Units that are incapable of standing alone, such as affixes, are correspondingly called bound forms. This characteristic of words also runs into problems. Certain forms that native speakers would identify as words are not capable of standing alone and therefore do not meet this definition:

A: Whose book is this?

B: *My.

My is a word, as we would all agree. But it generally does not stand alone. The reasons why *my* cannot stand on its own have more to do with syntax than with morphology: it is a determiner, and it generally appears alongside a noun. Speakers would use *mine* in this context instead. Nevertheless, this example shows that a potential diagnostic for wordhood – can it stand alone? – is not universally reliable. Once in a while we even get a supposedly bound form appearing on its own. In the musical *Camelot*, Queen Guenevere sings the following lines:

It’s May, it’s May,

the month of “yes, you may”

The time for every frivolous whim,

proper or im ...

When all the world is brimming with fun,

wholesome or un

The prefix *im-* is used on its own to rhyme with *whim*, and *un-* is used to rhyme with *fun*. We are dealing with a creative word play here. Both *im-* and *un-* are stressed here, which means that in some sense, the songwriter has turned them into words. We are not proposing otherwise. We present this example to help demonstrate that words are difficult to define, and that traditional notions such as bound and free are not always reliable.

Defining words phonologically

Words tend to be important units phonologically as well as syntactically. For example, the word is typically the domain of stress assignment. In French, stress always falls on the last syllable of a word. In Cairene Arabic, stress falls on one of the three final syllables, depending on syllable weight. In Polish, main stress falls on the penultimate (next-to-last) or antepenultimate (third-to-last) syllable (Hayes 1995: 67–8). Even this generalization is not absolute. Clitics (from Classical Greek *klinein* ‘to lean’) are grammatical words that are unable to stand on their own phonologically and must instead ‘lean’ on an adjacent word – be incorporated into its prosodic structure. This means that clitics often have an effect on the position of word stress. In Modern Greek, for example, stress is always on one of the last three syllables of a word. When a genitive clitic such as *mas* ‘our’ follows or leans on a word that is stressed on the third-to-last syllable, stress readjustment occurs (Nespor and Vogel 1986):

a. o ánthropos ‘the person’

b. o ánthropòs mas ‘our person’

This example demonstrates that the word-plus-clitic sequence functions as a single word as far as stress assignment is concerned in Modern Greek.

Grammatical words

Despite the elusiveness of a definition of word, speakers both literate and illiterate have clear awareness about what is and what isn’t a word. Children readily learn to break utterances up into words when learning to write. Some written languages, such as Chinese, represent words with symbols called logograms. For now, we are simply going to assume that we know a word when we see one. The term grammatical word or morphosyntactic word is virtually synonymous with word but is generally used to refer specifically to different forms of a single word that occur depending on the syntactic context. You would be justified in thinking, for example, that rabbit and rabbits are tokens of the same word. But they absolutely must be considered to be different grammatical words. The first occurs in contexts appropriate for a singular noun, and the second in contexts appropriate for a plural noun. Even though forms *like* and, *into*, and *lovely* have only one form, they are also considered grammatical words.

Topic- 008: Types of Words

Words are generally thought of having space before or after in terms of written words. This way they are easily identifiable. But words may be defined in different ways from different perspectives, with each perspective picking out a somewhat different object. Linguists distinguish phonological words, grammatical words and lexemes. We are most concerned with grammatical words and lexemes, but we begin with phonological words, just to make sure that we have made the proper distinctions.

Phonological words

A phonological word can be defined as *a string of sounds that behaves as a unit for certain kinds of phonological processes*, especially stress or accent. In English, every phonological word has a main stress. Elements that are written as separate words but do not have their own stress are therefore not phonological words in English. Consider again the sentence: *The hot dogs ran for the lake*. Think now in

terms of word stress. The sentence has seven words, but only four words have stresses, there is no stress on *the* or *for*.

Prepositions like *for* sometimes have stress, but as often as not are also included in the stress domain of the following word. We therefore say that the string *for the lake*, which we write as three separate words, is a single phonological word. Items like *the* and *for*, which are phonologically dependent on adjacent words, are called clitics (see Zwicky and Pullum 1983; Zwicky 1985). Syntactically, clitics are like distinct words, but they cannot stand alone phonologically and need to be incorporated into the prosodic structure of an adjacent word, the host. Proclitics precede their host and enclitics follow it. Well-known examples of clitics are the contracted form of the English auxiliary verb *be* ('m, 's, and 're), as in *Mary's here* or *We're in this together*. These auxiliaries are unable to stand on their own.

Content words vs. function words

Words are also distinguished between content words and function words. Content words “have meaning in that they refer to objects, events, and abstract concepts; they are marked as being characteristic of particular social, ethnic, and regional dialects and of particular contexts; and convey information about the feelings and attitudes of language users.” Function words also have meaning, but in a different way.

Most nouns, verbs, adjectives, and adverbs are content words. Function words are often best defined by their function. Examples of function words are determiners, pronouns, conjunctions, and certain verbs – those with little or no meaning such as *be*, *should*, or *must*. One generalization we can make is that while content words are an open class and it is possible to coin new ones, function words are a closed class. A person cannot easily invent a new preposition or conjunction.

Content words	Function words
Nouns: baby, bargain, Jane Verbs: publicize, hurtle, sleep Adjectives: peaceful, quick, bright Adverbs: readily, carefully	Pronouns: I, him, our Auxiliary Verbs: am, was, should Demonstratives: this, those Adverbs: very, not Prepositions: in, by

Look at following two examples - a with both content and function words and b only with content words.

- a) If we took the function words out of speech, it would be hard to figure out what was going on:
- b) took function words speech hard figure going on

Lack of function words create comprehension problem.

Lexemes

We need one more distinction, which will allow us to tell grammatical words apart from lexemes. Let's say we run across the word *dog* and notice that we have at least three tokens of the word:

- a. dog1: [noun], a canine
- b. dog2: [noun], a hooked or U-shaped device used for gripping heavy objects
- c. dog3: [verb], to follow closely and persistently

These three words sound alike: they are tokens of the same phonological word. Semantically, the meanings of (a) and (c) are related, and that of (b) is not. Despite the semantic relatedness of (a) and (c), however, all three tokens can be said to have distinct meanings. When we want to distinguish among phonologically similar forms on the basis of their differing meanings, we call each a separate lexeme. A lexeme is a word with a specific sound and a specific meaning. Its shape may vary depending on syntactic context. Thus we have *dog* and *dogs*, distinct grammatical word forms of the same lexeme *dog*¹. We use the term *paradigm* to refer to the set of all the inflected forms that a lexeme assumes. A lexeme is a theoretical construct. It is not a sound form (e.g., *dog*), but rather a sign or set of signs, with sound form, syntax, and meaning all bound together.

Topic- 009: Empirical Tests for Wordhood

It is difficult to come up with a definition that tells us whether something is a word. There are empirical tests that can tell us whether something isn't a word.

Fixed order of elements

The first empirical test has to do with the fixed order of elements within a word. Take a morphologically complex word like *unbreakable*. We can't say *breakableun* or *unablebreak*. The same doesn't hold for sentences. To use examples from the *Mad Hatter* and the *March Hare* in Lewis Carroll's *Alice in Wonderland*, "I see what I eat" is just as grammatical as "I eat what I see," and "I like what I get" and "I get what I like" are equally acceptable. When we change the order of words in a sentence, we generally also change the meaning. When we change the order of morphemes in a word, we generally end up with something ungrammatical.

However, in English we cannot change the order of words in a sentence and still have a grammatical result. We don't say **get like I what I*, for instance. But there are languages such as Latin where you can order the words of particular sentences any possible way and still have a grammatical result. In contrast, there are no languages in which you can arrange morphemes any such way.

Non-separability and integrity

Non-separability and integrity are two more diagnostics for wordhood. Words differ from larger units, such as phrases, in that they cannot be broken up by the insertion of segmental or phrasal material. This characteristic of words is called non-separability. Likewise, syntactic processes cannot apply to pieces of words. This is integrity. Adjectives and adverbs, for example, modify words, not morphemes. Words and phrases are often displaced to the beginning of a sentence or questioned, but not morphemes:

- a. Now, that one I like. Which one do you like?
- b. **Possible, it's im-.* ** Which school did you see bus?* (i.e., Which school's school bus did you see?)

Non-separability and integrity diagnostics tell us that compounds like *doghouse*, *greenhouse*, and *school bus* consist of a single word, rather than a pair of words. Let's begin with *doghouse*. We know this is a single word because we can't put anything inside it or modify the internal components in any way. We can't distinguish between a *doghouse* and a **dogshouse*, where a *doghouse* is a house for one dog and a

dogshouse for two or more. The same restrictions hold for greenhouse. If we break up the components in any way (a) or try to modify only a part (b), the meaning ‘warm glassed-in structure for growing plants’ gets lost:

a. a green and blue house a greener house

b. a very greenhouse *a very greenhouse

Modifiers must modify the whole compound, not just a part as components are non-separable.

Stress

Now it is established that hot dog (the edible kind) is a compound. If the hot dog you are eating is hotter than mine, you wouldn’t say that you were eating a *hotter dog or a *very hot dog. Given this fact, compare the pronunciation of the sequence hot dogs in the following two sentences:

1. We ate two *hot* dogs each.
2. The *hot dogs* ran for the lake.

In the first sentence, the main stress in *hot dog* is on *hot*. In a neutral pronunciation of the second, both hot and dog are stressed. It is also possible to emphasize dogs by stressing it more heavily: The hot *dógs* ran for the lake – hot dogs, not hot children, or canines, not frankfurters. These two examples suggest that stress can also be used as a diagnostic for whether or not a sequence of words is a compound – compounds are normally stressed on their first component in English, while phrases are normally stressed on their last element. [Exercise 1]

Topic- 010: Inflection vs. Derivation

If we understand the difference between words and lexemes, we can understand the distinction made by morphologists between inflection and derivation. Inflection involves the formation of grammatical forms – past, present, future; singular, plural; masculine, feminine, neuter; and so on – of a single lexeme. The use of these grammatical forms is dependent upon the sentence structure. The forms *is*, *are*, and *being* are examples of inflected forms of the lexeme *be*, which happens to be highly irregular not only in English, but in many other languages as well.

English language regular verb lexemes have a lexical stem, which is the bare form with no affixes (e.g., play) and three more inflected forms, one each with the suffixes *-s*, *-ed*, and *-ing* (plays, played, and playing). Noun lexemes in English have a singular and plural form. Adjectives, adverbs, prepositions, and other parts of speech typically have only one form in English. As you can tell from the example of *play* given above, one way of inflection can be realized is through affixes. Further examples of affixal realization of inflection can be found in the following.

Examples of words + inflectional morphemes

- Nouns: wombat + s ox + en
- Verbs: brainwash + es dig + s escape + d rain + ing

Derivation involves the creation of one lexeme from another, such as selector or selection from select. Compounding is a special type of derivation, since it involves the creation of one lexeme from two or

more other lexemes. Derivation generally results in a change in lexical meaning or the lexical category of a particular word, while inflection does not; and the application or non-application of inflectional morphology generally depends on the syntactic context (e.g., what is the subject of the verb? is the noun singular or plural?), while the application of derivational morphology does not.

Examples of words + derivational affixes	
Nouns to nouns:	New York + ese fish + ery Boston + ian auto + biography vice + president
Verbs to verbs:	un + tie re + surface pre + register under + estimate
Adjectives to adjectives:	gray + ish a + moral sub + human il + legible
Nouns to adjectives:	hawk + ish poison + ous soul + ful iron + like
Verbs to nouns:	acquitt + al digg + er
Adjectives to adverbs:	sad + ly efficient + ly

We often will come across the terms word formation and lexeme formation, both referring to derivation, in the morphological literature. We avoid the term word formation, since it is used by some linguists to refer to both inflection and derivation or to morphology in general.

Topic- 011: Two Approaches to Morphology: Item-and-Arrangement, Item-and-Process

Hockett (1954) distinguishes between two basic approaches to morphology, which he calls item-and-arrangement and item-and-process. Both are associated with American structuralist linguistics, codified by Bloomfield (1933), but continue to be important today. Item-and arrangement and item-and-process represent two distinct points of view.

Item-and-arrangement proceeds from a picture of each language as a set of elements and the patterns in which those elements occur. *The item-and-process picture gives no independent status to the items, which arise instead through the construction of the patterns.*

Item-and-arrangement grew out of the structuralists' preoccupation with word analysis, and in particular, with techniques for breaking words down into their component morphemes, which are the items. Morphology is then seen as the arrangement of these morphemes into a particular order or structure. For example, books result from the concatenation (Occurrence one after another as if a sequence) of the two morphemes book and -s.

Item-and-process, as its name suggests, is an approach to morphology in which complex words result from the operation of processes on simpler words. Working in an item-and-process model, we might say that books results when the lexeme book undergoes the function ‘make plural’.

In regular cases, this function will add the segment /-z/ (cf. photos, lions), which is realized as /-s/ after most voiceless segments (cf. giraffes), and as /e z/ after sibilants and affricates (cf. roses).

Affixation in the item-and-process and item-and-arrangement models

Item-and-arrangement and item-and-process are almost equivalent to one another mathematically. Everything you can express in item-and-arrangement can be expressed in item-and-process, and almost anything you can express in item-and-process can be expressed in item-and-arrangement.

Let’s look at a simple English example of a lexical function within the item-and-process model, which we happen to be most comfortable with. The following function creates agent nouns from verbs:

X[V (er)]N

Think[V er]N,

Runn[V er]N,

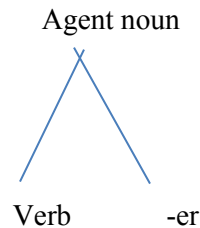
Fli[V er]N,

Hunt[V er]N

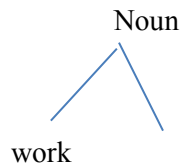
We generally think of lexeme-formation functions as having a phonological, a syntactic, and a semantic component. Phonologically, the function in examples takes a pre-existing string of segments and adds the suffix /ə/. Syntactically, it produces a noun from a verb. Semantically, it produces an agent of the verb. In the function in examples above, the phonology, syntax, and semantics are additive. When we derive an agent noun from a verb via the suffix /ə/, we add phonological information in the form of an additional segment, syntactic information in terms of the lexical category noun, and semantic information (the fact that the new noun represents an agent).

Furthermore, the old information is preserved, not lost: *worker* includes both the form and the meaning of the verb work, to which the suffix /ə/ has been added, as well as the fact that *work* is a verb, not a member of some other lexical category. The phonology, syntax, and semantics of most derivational functions are **additive** in this sense.

Additive functions like this one are easily recast in the item-and arrangement model. To express the above example in item-and-arrangement, we also need to break agent nouns into two parts. Then we put them back together. One way we can show this is through a tree structure.



Worker, then, is represented as follows:



This tree illustrates how the affix -er attaches to the stem work to form the agent noun worker.

Non-affixal phenomena and the item-and-process model

Many languages have morphological phenomena that do not involve affixes. These can be represented within the item-and-process model but pose problems for some morphological frameworks based on item and-arrangement, as we noted above.

English noun–verb pairs

An example of a non-affixal morphological phenomenon is illustrated in below. English possesses a number of noun–verb pairs that are distinguished phonologically by the location of stress. Verbs are stressed on the final syllable, and nouns on the first syllable. Examples can be found in Old English and in other related Germanic languages, suggesting that the difference existed already in proto-Germanic times, over two thousand years ago:

Verb	Noun
• Overflow	óverflow
• conduct	cónduct
• insert	insert
• réject	réject
• convíct	cónvict

It is possible to represent the shift in stress in these examples as an item, but the forms in these examples are more amenable to a processual analysis, where the appropriate function would be something like ‘shift stress’.

Topic- 012: The Lexicon

A discussion of words and lexemes would not be complete without a brief introduction to the lexicon. The word lexicon is from Greek *lexikós* ‘pertaining to words’ and often designates a book containing a list of words in a language along with their definitions.

Linguists use the term in particular to refer to the mental dictionary. Within linguistics, lexicon has taken on multiple definitions. This complicates matters, particularly because linguists do not always specify which definition they assume.

There are two widely accepted views of the lexicon. According to one, the lexicon is a list of the indivisible morphological units, or morphemes, in a language. This definition comes from Baudouin de Courtenay (b. 1845 in Radzymin, Poland; d. 1929 in Warsaw).

The second view of the lexicon, due more or less to Bloomfield (1933), is a list of irregular or arbitrary forms. Because they are irregular or arbitrary, they must be memorized. For example, a speaker of French must learn that the sound sequence [arbr] refers to a tree, and a speaker of English must learn that the word *slide* refers to a small square object that we put in a slide projector to project an image onto a screen or wall. It would be an error to assume that the first definition is equivalent to the second and that the list of irregular forms is a list of morphemes, which is to say a list of indivisible units. If we spoke a perfect language, this would be true. Every irregular form in the language would be indivisible.

But where natural language is concerned, this position is too extreme. A great deal of evidence suggests that even morphologically complex forms are present in a speaker’s lexicon. One morphologically complex word that must be considered to be listed in the lexicon is *representative*. If it were enough to say that *re-*, *present*, and *-ative* are stored in the lexicon, we would expect the meaning of *representative* to be a function of its parts, which it is not. A *representative* is always a person who represents something, but in the United States, the word most commonly refers to an elected member of a specific state or federal legislative body. Senators may represent us, and thus they are representatives, but a United States Senator is not a Representative.

Digging deeper, we find that most words ending in *-ative* are adjectives. *Representative* can be an adjective, but in this specific sense it is a noun, again an idiosyncrasy that must be listed in the lexicon along with the special meaning it has come to have in the context of United States government.

In this and many other cases, we are tempted to think that the meaning of a complex word is the sum of the meaning of its parts, because the difference between the meaning that we expect a word to have on the basis of the meanings of its parts and the meaning that it actually has is quite subtle. Still, there is no way out. If we know the meaning of the word *representative*, it must be the case that we store the whole word and its very specific meaning.

The abstract list or place where it is stored is the lexicon. The lexicon contains more than words. Affixes, such as English *re-*, can be assumed to be in the lexicon. Speakers know and understand such affixes and readily attach them to new stems. Some affixed inflected forms, like *says*, must also be in a lexicon. We know this because *says* is an exception to the general rule “Add /-z/ to the basic stem of a verb to form the third person singular present.” Say *say* and *says* out loud: *say* [sej] has a tense vowel, but *says* [sez] has a lax one. A speaker of English must memorize the third person singular of *say* because it does not follow

the normal rules of English. In other words, he or she stores it in the lexicon. Is says a simple form or a complex form? We can safely assume that it is complex because it has two parts, a stem and a recognizable third person ending /-z/. This tells us that complex forms may be stored in the lexicon.

Now consider a famous complex word, *antidisestablishmentarianism*, in which we easily recognize the pieces *anti-*, *dis-*, *establish*, *-ment*, *-ary*, *-an*, and *-ism*. However, these pieces together tell us very little about the meaning of the word, ‘opposition to denying special state recognition of a particular religion’ (it was used in Irish political discussion in the mid-nineteenth century). If you are a speaker of English who happens to know and use this word, then it must be stored in your lexicon, because its meaning cannot straightforwardly be determined from the meaning of its parts.

We have established the need to list inflected forms and complex lexemes like *representative* in the lexicon. We need to list some compounds, too. Some people might argue that you don’t need to list *doghouse* in the lexicon. However, there is no doubt that the compound *cathouse* is listed there, because its meaning, ‘brothel’, is not predictable from its form. Fixed phrases (with respect to ..., *butterflies in my stomach*), phrasal verbs (*run up a bill*, *catch up with Tom*), names of people and places (*Audrey Hepburn*, *Gulf of Mexico*), and proverbs (*Don’t count your chickens before they’re hatched*) all need to be memorized, too, and are arguably in the lexicon.

Psycholinguistic studies have even shown that perfectly regular complex forms, if frequent enough, are listed in the lexicon.

In sum, the most accurate conception of the lexicon is as a list of forms that you know. The lexicon is in some sense equivalent to your linguistic memory. It cannot simply be a list of indivisible morphological elements. Instead, it contains irregular forms, forms that are in some way unpredictable. Some are indivisible or unanalyzable morphologically, and others are not. We close this section by mentioning a third use of lexicon among linguists. Some linguists equate the lexicon with the morphological component of the grammar as a whole. We believe that the mental dictionary should be considered separately from the internal mechanisms involved in the formation and analysis of words.

Lesson 03

MORPHOLOGY AND PHONOLOGY**Topic- 013: Morphology and Phonology**

Let's now talk about some of the many interactions that take place between morphology and phonology. These interactions and the grammar that describes them are often called morphophonology or morphophonemics. We begin by looking at phonological processes such as assimilation and the effect they have on the shapes of morphemes. We then consider limitations on the phonological shape of morphological entities such as words and stems. From there we move on to two general types of affixes that are distinguished, in part, by phonological criteria.

Their phonological behavior reveals details about their underlying structure and the point at which they attach to their bases. We conclude with a look at secret languages in which morphology and phonology interact to disguise the shapes of words. Our learners have the rudimentary knowledge of linguistics that comes from taking the introductory course like Introduction to Linguistics. To understand the discussion in this chapter, you need to know three terms. They are onset, nucleus, and coda. The onset of a syllable is made up of the first consonant or consonants. The nucleus is the syllable's core, usually a simple vowel or a diphthong. The coda is made up of the consonant or consonants that follow the nucleus. All syllables must have nuclei. Onsets and codas are optional, though some languages require the former.

Topic- 014: Allomorphs

Kujamaat Jóola, the stem /baj-/ has two possible shapes, [baj-] and [b e j-], with [b e j-] occurring in the presence of a morpheme with an underlying tense vowel, and [baj-] elsewhere. In English the plural marker comes in several shapes, among them [s] as in lips, [z] as in balls, [e z] as in roses, [n] as in oxen, and null as in sheep.

Alternations based purely on phonological context, as with the Kujamaat Jóola facts. In the English data, only the first three allomorphs of the plural suffix depend on phonological context. The last two, as in oxen and sheep, are lexical, and we will not be concerned with them here. Like the English plural suffix -s, the English past tense suffix has three forms: [d], [t], and [ed]:

[d] blamed [blejmd],

[t] jumped [dʒʌmpt],

[ed] aided [eIdId]

The distribution of the three allomorphs is predictable. We can formulate the distribution of the allomorphs in even simpler terms: the English past tense suffix is /d/. Where we find [t], the /d/ has assimilated to the preceding segment in voicing. [e d], the [e] has been added by an automatic phonological rule of epenthesis (i.e., insertion of a phonological segment or segments) that is triggered by the fact that the final segment in the verb and the suffix itself agree in both place and continuancy

/d/ the basic form or the basic allomorph of the English past tense suffix. It is not always easy, or even possible, to determine the basic form of a morpheme. To do so, we must decide which form of a morpheme best accounts for the full range of data.

The forms on the left are in the nominative case, used for subjects, and those on the right are in the genitive case.

(2) aithiops ‘Ethiopian’ aithiopos ‘of an Ethiopian’ phleps ‘vein’ phlebos ‘of a vein’

Epenthesis

The insertion of a sound or letter within a word, e.g. the b in thimble. Similarly if you say film as "FIL-um," As morphologists, we are used to dealing with cases like this. Unlike syntax, which tends to be very regular, morphology is full of irregularities and exceptions.

In Kujamaat Jóola, the stem /baj-/ has two possible shapes, [baj-] and [b e j-], but their distributions don’t overlap. [b e j-] occurs in the presence of a morpheme with an underlyingly tense vowel, but [baj-] does not. This non-overlapping distribution allows us to conclude that the two forms are instances of the same morpheme. When two or more instances of a given morpheme occur with different shapes, we call them allomorphs.

Allomorphs

The term ‘allomorph’ was introduced, in two contexts. In Kujamaat Jóola, we noted that the stem /baj-/ has two possible shapes, [baj-] and [b e j-], with [b e j-] occurring in the presence of a morpheme with an underlying tense vowel, and [baj-] elsewhere. (Lax vowels, remember, are short. Tense vowels are also called long vowels; this name is slightly misleading because, in RP English at least, the tense vowels have variable length; they can be much longer than the lax vowels, but under certain conditions they become clipped, or shortened to roughly lax vowel length. Tense vowels, then, have more variable length than lax vowels. Tense vowels can be unchecked; that is, they can occur at the end of a *word*.)

In English, the plural marker comes in several shapes, among them [s] as in lips, [z] as in balls, [e z] as in roses, [n] as in oxen, and null as in sheep. Here we are most interested in alternations based purely on phonological context, as with the Kujamaat Jóola facts. In the English data, only the first three allomorphs of the plural suffix depend on phonological context. The last two, as in oxen and sheep, are lexical, and we will not be concerned with them here. Like the English plural suffix -s, the English past tense suffix has three forms: [d], [t], and [e d]:

[d]

- (1) blamed [blejmd],
- (2) triggered [trigerd],
- (3) realized [rije lajzd],
- (4) sighed [sajd],
- (5) rubbed [rʌbd]

[t]

- (6) jumped [dʒʌmpt],
- (7) licked [lɪkt],
- (8) hushed [hʌʃt],
- (9) laughed [læft],
- (10) itched [ɪft]

[e d]

- (11) aided [eided],

- (12) knighted [naited]

The distribution of the three allomorphs is predictable and parallel to the distribution of the three allomorphs of the English plural suffix.

[d] is found on verbs ending in a vowel or a voiced consonant, with the exception of alveolar stop /d/.

[t] is found on verbs that end with a voiceless consonant, with the exception of alveolar stop /t/. In verbs that end with /t/ or /d/ (generally realized in speech as [ɾ]), we find [e d].

We can formulate the distribution of the allomorphs in even simpler terms: the English past tense suffix is /d/. Where we find [t], the /d/ has assimilated to the preceding segment in voicing. Where we find

[e d], the [e] has been added by an automatic phonological rule of epenthesis (i.e., insertion of a phonological segment or segments) that is triggered by the fact that the final segment in the verb and the suffix itself agree in both place and continuancy. We can call /d/ the basic form or the basic allomorph of the English past tense suffix. It is not always easy, or even possible, to determine the basic form of a morpheme. To do so, we must decide which form of a morpheme best accounts for the full range of data. Consider the two words from Classical Greek given below. The forms on the left are in the nominative case, used for subjects, and those on the right are in the genitive case.

- (2) aithiops ‘Ethiopian’ aithiops ‘of an Ethiopian’ phleps ‘vein’ phlebos ‘of a vein’

On the basis of these two forms, we can hypothesize that the nominative in Greek is formed by the addition of -s and that the genitive is formed by the addition of -os. What then are the basic forms of the stems meaning ‘Ethiopian’ and ‘vein’? With respect to the first, the answer seems simple: comparing the nominative and genitive forms, we isolate aithiop-. Finding the basic stem meaning ‘vein’ is more difficult. It might be phlep-, the stem of the nominative, or phleb-, the stem of the genitive. Theoretically, it could also be neither.

Epenthesis (means the addition of one or more sounds to a word,) is another common source of allomorphy for example the insertion of a sound or letter within a word, e.g. the b in thimble. Similarly if you say film as "FIL-um," with two syllables, you've committed epenthesis. It isn't a punishable offense- in fact, it's not an offense at all. It is simply a natural way to break up an awkward cluster of consonants. It's easier for some people to say film as two syllables instead of one, just as it's easier for some to insert a "b" sound into cummerbund, pronouncing that word as "CUM-ber-bund." . Frequently languages epenthesize consonants in contexts where a final vowel in one word would otherwise come up against an initial vowel in the following word, particularly when the two have a strong syntactic bond. We call this

unwanted vowel–vowel contact **hiatus**. French is a language that doesn't tolerate hiatus well, and it has various ways of eliminating it in certain syntactic contexts. The plural article *les* [le] 'the' is pronounced [lez] when it is followed by a vowel-initial word

As morphologists, we are used to dealing with cases like this. Unlike syntax, which tends to be very regular, morphology is full of irregularities and exceptions.

Topic- 015: Prosodic Morphology

Prosodic morphology deals with the interaction of morphology and prosodic structure. Prosodic structure is particularly concerned with the timing units of languages, for example, the word and syllable, and vowel length. From this general category we present three phenomena: phonotactic constraints, root-and-pattern morphology, and reduplication.

Phonotactic constraints

Phonotactics (from Ancient Greek *phōnē* "voice, sound" and *tacticós* "having to do with arranging") is a branch of phonology that deals with restrictions in a language on the permissible combinations of phonemes. Phonotactics defines permissible syllable structure, consonant clusters and vowel sequences by means of phonotactic constraints. Phonotactic constraints limit the possible phonological shapes of stems and words. Phonotactic constraints are often, but not always, connected with syllable structure. At their most basic, phonotactic constraints determine the minimum length of content words in particular languages. For example, in Mohawk, each content word contains at least two syllables (Michelson 1988, cited by Hayes 1995: 47).

Other languages require that content words consist of at least a heavy syllable, where heavy means that the syllable contains a long vowel, diphthong, or a vowel and a weight bearing (moraic) consonant.² Many languages do not have minimal word constraints, including Romanian, Hungarian, and Icelandic (Hayes 1995: 88–9). Does English have a minimal word constraint? An analysis of nicknames suggests that it does (see McCarthy and Prince 1998: 287–8). Here is a list of some English first names and their corresponding short forms, or nicknames:

(9) English nicknames a. Alexander → Alex Caroline → Carrie Katherine → Cathy, Kitty b. Josephine → Jo Louisa, Louis → Lou Susan, Suzanne → Sue Tyler → Ty c. Beverly → Bev Christopher → Chris Robert → Rob, Bob Stephanie → Steph

Even if we were to keep going, we would find that any nickname we can think of falls into one of these three sets. They are either polysyllabic (9a) (in this case, bisyllabic) names or monosyllabic names (9b–c) that have either a diphthong, a coda, or both. No native English nicknames consist of a single, light syllable. From this data, it seems that English falls in with the set of languages that have a minimal word constraint. Even the shortest of names, nicknames, must consist minimally of a heavy syllable or two light syllables. And it is difficult, if not impossible, to find an English content word that contradicts this conclusion. We also find that languages have restrictions on the possible shapes of roots. Nida (1965: 66) reports that in the Mayan languages, roots are predominantly of the shape CVC and in Bantu they are generally CVCV. In Hebrew, Arabic, and other Semitic languages, roots generally consist of three consonants: CCC.

Root-and-pattern morphology

We ended the last section by saying that in Semitic languages such as Hebrew and Arabic, roots generally consist of three consonants. To form words, vowels are superimposed on this consonantal pattern. We call this type of morphology root-and-pattern.

Hebrew root with sample forms

The following forms share a triconsonantal root, M-L-K. In some, the root consonant /k/ is realized as the fricative [x]. M-L-K: melex ‘king’ malkah ‘queen’ malax ‘he reigned’ yimlox ‘he reigns, he will reign’ m alxut ‘royalty, royal power, reign, kingdom’ mamlaxah ‘kingdom, sovereignty, dominion’

In root-and-pattern morphology, the root consonants in a given inflectional or derivational paradigm combine with vowels and sometimes consonants in a fixed pattern. It is possible to think of the consonantal root being superimposed on a template. Thus, in the box above, malax ‘he reigned’ consists of a root m-l-x and a template _a_a_. The consonants fit into the empty slots of the template. Yimlox ‘he reigns’ can be thought of as the root m-l-x and a template yi_o_. (We use y here to represent IPA [j] in keeping with traditional Hebrew transliteration.)

The root-and-pattern morphology of Hebrew and Arabic is reflected in their writing systems, which use the primary symbols to represent consonants and diacritics to represent vowels. While by definition there are no exceptions to the minimal word constraints presented in the previous section, we do sometimes find exceptions to generalizations about possible root shapes.

In the Semitic languages, we do find roots consisting of two consonants (CC) and four (CCCC). But these are less numerous than triconsonantal roots.

Reduplication

In reduplication, a continuous substring from either the beginning or the end of a word is copied. Languages may use reduplication for inflection or derivation. Plural reduplication in Ilokano, an Austronesian language of the Philippines, is illustrated below (McCarthy and Prince 1998: 285). The reduplicated portion of the word is in italics:

(10) kaldĩŋ ‘goat’ kal-kaldĩŋ ‘goats’

púsa ‘cat’ pus-púsa ‘cats’

kláse ‘class’ klas-kláse ‘classes’

jyánitor ‘janitor’ jyan-jyánitor ‘janitors’ ró?ot

‘litter’ ro:-ró?ot ‘litter (pl)’

trák ‘truck’ tra:-trák ‘trucks’

The inclusion of borrowings in the data set above shows that reduplication is a productive means of forming plurals in Ilokano. Reduplication bears an important similarity to root-and-pattern morphology.

The reduplicant typically must follow a certain pattern or adhere to some other prosodic requirement. In Ilokano plurals, as you can see, the reduplicant is always of the shape C(C)VC or C(C) VV. [Exercise 7]

Topic- 016: Primary and Secondary Affixes

Over the years, our knowledge of morphological structure has been enhanced by work in phonology. We learn a lot by observing the phonological processes that take place or do not take place within particular sets of morphologically complex words. One distinction that has come out of work that pairs morphology and phonology is between primary and secondary affixes, also known as level 1 and level 2 affixes or class 1 and class 2 affixes. In English, this distinction is intimately connected with language history.

Primary affixes in English are often of Latin-Romance origin, while secondary affixes are often of native Germanic origin. (English is a Germanic language.) However, we de-emphasize this fact in the following discussion because etymology can only take us so far in morphological analysis. The primary–secondary distinction is a living process, regardless of its history, and in English, as in other languages of the world, it cannot be explained away as etymological residue. Below are some examples from Kiparsky (1983) of words bearing -(i)an, a primary affix (1), and ones bearing -ism, a secondary affix (2):

1.

- a. Mendel → Mendelian
- b. Mongol → Mongolian
- c. Parkinson → Parkinsonian
- d. Shakespeare → Shakespearian
- e. grammar → grammarian

2.

- a. Mendel → Mendelism
- b. Mongol → Mongolism
- c. Parkinson → Parkinsonism
- d. national → nationalism
- e. capital → capitalism

If you read the words in (1) and (2) to yourself, you hear an important difference. Words with -(i)an have a stress shift. The stress in Mendel is on the first syllable, while in Mendelian, it is on the second. Likewise, Parkinson is stressed on the first syllable, but Parkinsonian is stressed on the third. We can generalize by saying that in all words with -(i)an, stress is on the syllable that immediately precedes the suffix. Stems suffixed with -ism, on the other hand, are stressed on the same syllable as their unaffixed counterparts. Nationalism is stressed on the first syllable, just like national. This is the first and most famous difference between primary and secondary affixes in English.

Primary affixes cause a stress shift, while secondary affixes do not. If primary and secondary affixes both occur in the same word, we can make a second prediction. The primary affix will occur closer to the stem than the secondary affix. Therefore, Parkinsonianism is a possible word, but *Parkinsonismian is not. Now consider the words reparable and repairable. Both have repair as their stem, but it is slightly disguised in

the first. Semantically, both mean ‘capable of being repaired’, but only the second would be used to describe a broken appliance. Repairable has the additional sense of ‘liable to be paid back or recovered’ as with repairable damages. These words show that the suffix -able in English is actually two suffixes. One is primary, as in repairable, and the other is secondary, as in repairable.

Traditional usage among morphologists is to use the symbol ‘+’ to mark the juncture between a stem and a primary affix and to use ‘#’ to mark the juncture between a stem and a secondary affix. We use these symbols to differentiate between the two types of -able here.

Other word pairs that show the opposition between +able and #able are prefer+able [prɛ ˈf(e)r e bl] vs. prefer#able [pr e f ˈ e ˌr e bl] and compar+able [ka ˈmp(e)r e bl] vs. compar#able [kæmpæ ˈr e bl]. Phonologically, the stem in the first is stressed differently in combination with +able than it is in isolation: compare the pronunciation of prefer+able and prefer. This is typical of primary affixes.

In addition, the semantics of the forms containing the primary affix +able are less direct, or less compositional, than the semantics of the forms containing the secondary affix #able. If two models are not comparable (contains the primary affix +able), they are unlike. If they are not compárable (contains the secondary affix #able), it is not possible, in a literal sense, to compare them. The semantics of forms containing the secondary affix #able are so predictable that they are often not even listed in dictionaries. Looking at other +able vs. #able pairs, we discover other phonological characteristics of primary affixation. Consider the following words:

(3)

+able	#able
a. defensible	defendable
b. perceptible	perceivable
c. divisible	dividable

Forms with +able in (3) exhibit allomorphy in the stem. They use a form that is recognizable from nouns, namely defense, perception, and division, instead of the citation form of the lexeme: defend, perceive, and divide. The fact that +able can be spelled +ible is unimportant here. Further examples of stem allomorphs occurring with the primary affix +able are given in the leftmost column of (4):

(4) +able base #able base

- a. cultivable cultiv cultivatable cultivate
- b. educable educ educatable educate
- c. irrigable irrig irrigatable irrigate
- d. navigable navig navigatable navigate
- e. demonstrable demonstr demonstratable demonstrate

The second pair of columns in (14) shows that the secondary affix #able differs from +able in attaching to the citation form of the lexeme: cultivate, educate, irrigate, navigate, and demonstrate. Another difference between primary +able and secondary #able is that in words of the form X#able, X must be a transitive verb. In (4), cultivate, educate, irrigate, navigate, and demonstrate are all transitive. +able, on the other hand, is sometimes found on stems that are not transitive verbs, such as poss- (from Latin posse ‘to be able’), as in possible, or ris- (from Latin ride -re ‘to laugh’), as in risible. We have said that we would de-emphasize etymology in our discussion of primary and secondary affixes, but here it becomes important. Many words with primary affixes – including possible and risible – were borrowed directly into English from French or Latin. They are fossilized; speakers have learned to use them, not created them on their own. However, even such fossilized examples as possible and risible are significant, because they illustrate another point: speakers are able to isolate both primary and secondary affixes when presented with words that contain them. Possible may be stored in the lexicon as a whole, but it is nevertheless analyzable. A person, such as a child or a second language learner, encountering possible for the first time will recognize it as an adjective because it ends in -ible.

In talking about the distinction between primary and secondary affixes, we have focused on suffixes. Prefixes can be primary or secondary as well. An example of a primary prefix is in+, and an example of a secondary one is un#. Both mean ‘not’. Phonologically, in+ has allomorphs. It surfaces as ir-, im-, in-, and il- in words like irreplaceable, immortal, inoperable, and illegal. Un# does not have any allomorphs. Un- is of Germanic origin, with cognates in earlier stages of English, in contrast to in-, which came to English from Latin, through French. In+, like +able, attaches to allomorphic stems:

(5) in+ un# a. irregulable unregulatable b. inviolable unviolatable c. imperceptible unperceivable d. indivisible undividable

The stems it attaches to are stressed differently than the lexical stem of the corresponding lexeme:

(6) in+ un# a. irréparable unrepáirable (cf. repáir) b. irrévocable unrevókable (cf. revóke)

In+ may attach to non-lexical stems, while un# does not. (In+ attaches to lexical stems as well, for example, impalpable, impossible.)

(7) a. inept *unept, *ept d. inert *unert, *ert

Our objective was not to give a comprehensive account of the topic, but rather an overview, and thus we have left many issues untouched. Under the heading “lexical phonology,” phonologists have used the distinction between the two types of affixes to explore the possibility that different phonological rules apply at different levels of a morphological derivation. But as we have seen, primary and secondary affixes have semantic consequences as well. In particular, the semantics of primary affixes is less likely to be fully compositional. **[Exercises 8 and 9]**

Primary and secondary affixes in English

Primary	Secondary
Often of Latin-Romance origin	Often of Germanic origin
Cause a stress shift	Do not cause a stress shift
Usually occur closer to the stem than	Usually occur outside of primary affixes
The semantics of the derived form tend to be less compositional	The semantics of the derived form tend to be compositional
May attach to a non-basic allomorph	Attach to a stem's basic allomorph of the stem
The affix itself may have allomorphs	The affix itself does not have allomorphs
May attach to non-lexical stems	Attach to lexical stems (i.e., words from lexical categories N, V, Adj/Adv)

Topic- 017: Linguistic Exaptation, Levelling and Analogy

Rudes (1980) and Lass (1990) have both raised the question of what to do with “linguistic left-overs” (Rudes’s term) or “linguistic junk” (Lass’s term). In both cases, it has to do with morphemes that lose their semantic content or morphosyntactic function as a result of language change and are left as contentless, functionless strings of phonemes floating around in the system. Rudes’s and Lass’s investigations on this question cover a variety of cases. They show that languages are in general intolerant of useless elements, and speakers reanalyze them as having a new role. Lass calls this process linguistic exaptation, extending a term originally coined by Stephen Jay Gould and Elizabeth Vrba in the context of evolutionary biology to the study of language change. In evolutionary biology, exaptation occurs when a structure or feature that developed for a certain function through natural selection takes on a new and different function. Carstairs-McCarthy (1994) and Cameron-Faulkner and Carstairs McCarthy (2000), in their work on inflectional classes and gender, and stem alternation, respectively, suggest that linguistic exaptation is pervasive. Linguistic exaptation is therefore a natural consequence of a core psycholinguistic mechanism that makes it easier for speakers to master complex inflectional systems or to learn the meanings of new vocabulary items, and we expect to find it playing an important role in the evolution of inflectional systems cross linguistically. One language family in which linguistic exaptation has occurred is Germanic (Lass 1990: 83–7). There, Indo-European vowel alternations within verbal paradigms came to encode the present/past distinction (e.g., English write, wrote). They had originally been used to encode aspectual distinctions. The case that we focus on here, however, is the one discussed by Rudes: the development of the verbal suffix *-esc* in Romance. Our story begins with Latin, where the suffix *-sc* attached to sequences of verb stem plus theme vowel to form the inchoative aspect (which has the general meaning ‘to begin to’). Compare the Latin verb form *paleo* - ‘I am pale’ with its inchoative counterpart *pallesco* - ‘I begin to pale’. Similarly, *amo* - in Latin means ‘I love’, while *amasco* - means ‘I am beginning to love’, and *flo -re -re* is ‘to flower’, while *flo -rescere* is ‘to begin to flower’ (examples from Rudes 1980). English doesn’t have a productive inchoative aspect, but we do have pairs like *white* and *whiten* ‘become pale; begin to be pale’.

What happened was (iii). The suffix represented by *-esc* and *-isc* became productive once again and is a distinctive feature of the verbal morphology of certain Romance languages. Why? Although the classical conception of morpheme is a pairing between sound and meaning, we have defined it as the smallest grammatically significant unit in a word. What makes the development of *-esc* in the Romance languages distinctive is that its function was phonological, rather than syntactic or semantic. It is meaningless. We

can exemplify this by looking at the paradigm of a verb that contains this morpheme. As seen below, the suffix occurs in the first, second, and third person singular and third person plural of the present indicative of the verbs that have it (so-called fourth conjugation verbs) in Romanian, illustrated here by a citi ‘to read’ (Rudes 1980: 333). (It also occurs in these persons in the present subjunctive and imperative.) But the core meaning of verbs does not change depending on the subject:

In some cases, the augment appears in parallel forms in both languages, but in other cases, it doesn’t. Maltese has borrowed a morpheme and the rule that governs its distribution, but not the verb forms themselves.

Leveling & Analogy

The tendency for languages to prefer regular paradigms over irregular ones sometimes leads to leveling, the elimination of sound alternations that do not signal important differences in meaning. A classic example of leveling comes from Latin. In Prehistoric Latin, the stems of words like colo -s ‘color’ and hono -s ‘worth’ ended in -s throughout the nominal paradigm:⁵ (19)

- Prehistoric Latin
- Nominative col-os
- Genitive *col-os-es
- Dative *col-os-ei
- Accusative *col-os-em
- Ablative *col-os-i

Through regular sound change, intervocalic /s/ became /r/, a process called rhotacism:

- Old Latin
- Nominative col-os
- Genitive col-or-is
- Dative col-or-ei/-e
- Accusative col-or-em
- Ablative col-or-e

At this point, the paradigm was characterized by two stems, one ending in /s/ (the nominative form) and one ending in /r/ (the oblique forms). Eventually the final /s/ of the nominative form was replaced by /r/ in order to conform with the stem of the oblique forms. Note also that the /o/ preceding the /r/ of the nominative form shortened:

- Classical Latin
- Nominative color
- Genitive col-or-is
- Dative col-or-i
- Accusative col-or-em
- Ablative col-or-e

Hock gives the example of English plurals. It is because of four-part analogy that the plural of cow is cows, replacing the earlier form kine. The new plural cows generalizes the plural formation familiar from other words, such as stone, stones. Here, as in most instances of analogy, the pattern that serves as the basis of the analogy (in this case the regular plural suffix -s) is more productive.

stone: stone-s cow : X = cow-s

Sometimes the older form that existed before analogical leveling remains as a relic, used for special meanings. The old plural of brother is brethren. These days, it is used only to refer to fellow-members of a church or social organization, not to brothers in the literal sense of the term. The title of a book about the United States Supreme Court, *The Brethren*, uses this form to emphasize the special nature of the relationship among the members of this most elite of groups. Leveling and analogy are powerful forces in the development of languages over time. They are driven by a seemingly innate preference in speakers for phonological and morphological similarity between members of a paradigm or a class of words.

Topic- 018: Morpho-Phonology and Secret Languages

We now turn to two secret languages that are permutations of existing languages. Secret languages are found around the world and have been attested in English, French, Spanish, Dutch, Thai, Cuna (Sherzer 1970), and Haitian Creole, to name only a few. They can be seen as examples of creative language use, and thus they should be considered external to the mental grammar. What, then, is the place of these languages in a book on morphology? In the examples that we present, speakers go from the existing language to the secret language through the regular application of phonological rules in what might be considered a morphological derivation. Secret languages also exploit notions that are independently motivated in phonology and morphology, notably the syllable and onset. One secret language you may already be familiar with is Pig Latin. In one variation, words that start with vowels are suffixed with way [wej]. Words that begin with a consonant or consonant cluster shift the entire onset sequence to the end and are suffixed with ay [ej]:

- igpay atinlay ‘Pig Latin’
- eefray ormfoy ‘free form’
- inflectionway ‘inflection’

When speakers manipulate words in this fashion, they make use of their subconscious knowledge of linguistic entities such as onset and nucleus. Language games are therefore an instructive union of phonology and morphology. Another secret language is *verlan* [vɛʁlɑ̃], which is based on French. The word *verlan* is derived by reversing the syllables of *l’envers* [lɑ̃vɛʁ] ‘the other way around’. *Verlan* works best with words of two syllables, because in these the two syllables can simply be reversed: (26) Standard French *Verlan*

- a. pourri [pʁi] ripou [ʁipu] ‘rotten’ (generally refers to corrupt police)
- b. branché [bʁɑ̃ʃe] chébran [ʃebʁɑ̃] ‘plugged in, informed’
- c. pétard [petɑʁ] tarpé [tɑʁpe] ‘cannabis joint’
- d. bagnole [bɑnɔl] gnolba [ɲolba] ‘car’

Monosyllables are *verlanized* differently depending on whether they are open or closed. In open monosyllables, such as *pue* [py] ‘stinks’, the order of consonant and vowel is reversed, yielding forms like [yp]. Closed monosyllables, such as *femme* [fam] ‘woman’, are treated as if they end in a schwa (Bullock 1996: 185) and therefore as if they are underlyingly bisyllabic. The syllables are reversed, and

the final vowel is dropped: [fam e] → [fa.m e] → [m e .fa] → [mœf]. Note that the schwa is realized as [œ]; this is because schwa cannot be in stressed position in French.

All of verlan is based on standard French, and speakers of verlan invariably speak French, if not natively, then under more natural conditions. According to Bullock, the ‘rules’ of verlan are artificial compared to those of the standard – and natural – language. Returning to Pig Latin, in the version presented here, words that start with a consonant cluster postpose the entire cluster, then add the suffix -ay. But another version of the secret language postposes only the first letter (e.g., losetcay for closet). Any rule that operates on letters rather than on phonological entities such as onsets is unnatural. Languages are first and foremost oral, and orthographies are systems imposed on them by people. Because of characteristics like these, secret languages must be looked upon as somewhat artificial and should not on their own be used to draw conclusions about the workings of natural languages.

Lesson 04

DERIVATION AND THE LEXICON

Topic- 019: The Saussurean Sign

Ferdinand de Saussure (1857–1913), one of the first modern linguists, believed that language was a system of signs. He defined a linguistic sign as an arbitrary pairing between what he called the signifiant ‘signifier’, a particular sequence of sounds, and the signifié ‘signified’, the concept that is denoted by the sound sequence. These three terms, sign, signifier, and signified, are still standard in linguistics. Saussure (1969) distinguished between motivated and unmotivated signs. A sign is motivated to the extent that by inspection you can get clues as to what it means. A walk signal at a crosswalk is an example of a motivated sign, because the stylized image of a person walking indicates whether you should or should not cross the street. A stop sign is partially motivated. Signs can lose their motivation: consider the name of the basketball team, the Los Angeles Lakers. Motivation is not all-or-nothing, and signs can be partially motivated. The name of the Ithaca Lakers of central New York

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Topic- 020: Motivation and Compositionality

We can relate motivation to the logical notion of composition or compositionality. We say that something is logically compositional if it is defined entirely in terms of its parts. For example, the word doghouse is compositional, at least in its original sense, because its meaning is derivable from its two components, dog and house. Because it is purely compositional, we could also call doghouse a motivated sign (although if we break it down, the sound shapes of dog and house are purely arbitrary and unmotivated). But in the expression in the doghouse ‘in a state of disfavor or repudiation’, the word is not compositional, since there is nothing in particular about doghouses that suggests that an occupant of one should be in disfavor. Another example of a compositional form is the Kujamaat Jóola dubitive-incompletive, which expresses an action that was never completed or whose existence is in doubt. For emphasis, the dubitiveincompletive suffix (inc) is doubled (1b). It can even be repeated several times (Thomas and Sapir 1967: 350, note 2):

- a. ni- ŋar -ε:n
- b. 1sg.sub- take -inc
- ‘I was taking’
- c. ni- ŋar -ε:n -ε:n

- d. 1sg.sub- take -inc -inc
- e. 'I was taking (emphasis on incompletive aspect of action)'

Expressing emphasis through doubling should be considered compositional. The following set of English words is particularly significant in light of partial motivation and compositionality:

behead 'to remove someone's head'

befriend 'to make yourself a friend to someone'

besiege 'to lay siege to'

bewitch 'to place under one's power as if by magic'

Our first response to this list is to say that these are all morphologically complex words formed with the prefix be-. But are they motivated signs? In each case, the prefix be- has a different effect on the meaning of the stem (head, friend, siege, witch). In the word behead, it indicates 'deprive of' (as in the obsolete verbs beland and belimb), but it means nothing of the sort in the other forms. This suggests that these forms are not completely motivated. The meanings of the stems in (2) are never lost, simply transformed. The forms are therefore partially motivated. The question of whether or not the forms in are compositional is more difficult and depends in large part on our approach to linguistic analysis. It is undoubtedly the case that formation of words with the prefix be- is no longer productive in English. Some linguists will nevertheless want to analyze forms like behead and bewitch as having two parts. Our approach, however, will be to say that these forms are stored whole in the lexicon – they are memorized. We would say the same for forms like crayfish, raspberry, boysenberry, and cranberry.¹ While the isolatability of the stems fish and berry make it possible to isolate cray-, rasp-, boysen-, and cran- as affixes, doing so is largely an academic exercise; we will assume so, at least, until the opposite can be shown by experimental means. The preceding data lead into a major issue to be addressed in this chapter: the relationship between derivation and the lexicon. Recall that although some linguists consider the lexicon to be equivalent to the morphology, our definition of lexicon is a list of forms that speakers of a language know or memorize. One question we need to ask is whether lexemes formed by productive derivational processes are ever stored in the lexicon.

Compounding

One derivational process we have already discussed is compounding. Here are some basic examples:

(3) English compounds tool + bar amusement + park puppy + love coffee + house

To give a more extreme example, if someone asks us what Violet does for a living, we might respond:

She's a high voltage electricity grid systems supervisor.

There is evidence that high voltage electricity grid systems supervisor is a single noun. First, its distribution matches that of any other noun, so we can insert it into phrases like [a good N] or [N for hire]. Second, high voltage electricity grid systems supervisor behaves as a single unit for the purposes of wh-movement. Question-answer pairs that break it up are at the very least awkward. In chapter 2 we related this characteristic of words to the notion of lexical integrity: (5)

- a. Q: Which electricity grid systems supervisor did you see?
- b. A: ?The high voltage one.
- c. b. Q: Which systems supervisor did you see?
- d. A: *The high voltage electricity grid one. c.
- e. Q: Which supervisor did you see?

A: *The high voltage electricity grid systems one.

Contrast these with syntactic strings of modifier plus noun which are easily broken up, as shown in (6):

(6) Q: Which supervisor did you see? A: The tall one.

Continuing with the notion of lexical integrity, we can ask whether it is possible to describe part of the string high voltage electricity grid systems supervisor with a modifier. When we try, the result is very awkward (7):

? A very high voltage electricity grid systems supervisor

The most natural interpretation of (7) is the figurative one whereby very high voltage is used as an adjectival phrase modifying a smaller compound [electricity grid] giving the intermediate form very high voltage electricity grid, which in turn modifies [systems supervisor], giving the entire form in (7).

Finally, we can point to the structure of high voltage electricity grid systems supervisor as evidence that it is a single noun formed by compounding. Words in English are generally head-final, meaning that the lexical category of the form as a whole matches that of its final constituent. A dogsled is a kind of sled, not a kind of dog; the Red River Valley is a valley, not a color or a body of water; and affixed words like pollution take on the lexical category of the suffix (in this case, noun) rather than that of the stem (pollute, a verb).² As speakers of English we know this, and without ever having heard high voltage electricity grid systems supervisor before, we know it designates a type of supervisor. Phrases, in contrast to words, are less likely to be head-final. The head of [NP John's walking into work without a tie] is walking, not tie, and the head of [NP the house on the hill] is house, not hill.

Having established that high voltage electricity grid systems supervisor is a single word of the category N, we can ask ourselves, "Is this noun in my lexicon?" Probably not. This compound is formed by a very productive process and there is nothing irregular involved in it. It is absolutely compositional and fully motivated. Words like this that are used but not stored are called **nonce** forms or **hapax legomena** (hapax legomenon in the singular). Nonce means 'a particular occasion', and hapax legomena is a Greek term meaning 'said once' that is used to refer to words that occur only once in the recorded corpus of a given language. These are words that somebody made up, used, and then threw away. The existence of nonce forms is one type of evidence that speakers create words on the fly as they speak. In discussing compounds, linguists sometimes use the terms endocentric and exocentric. These terms are related to the notions of motivation and compositionality presented earlier. An endocentric compound is one that has a head. The head expresses the core meaning of the compound, and it belongs to the same lexical category as the compound as a whole. For example, goldfish is an endocentric compound. It has a head, fish, which determines both the meaning and the lexical category – noun – of the compound as a whole. Compounds whose lexical category or meaning are not determinable from the head are exocentric. Figurehead is just such a compound, because it is not a type of head. Whether a compound is endocentric or exocentric is

sometimes a matter of opinion. Fabb (1998: 67) gives the example of greenhouse, which is endocentric if you think of it as a type of house, but exocentric if you do not.

Endocentric compounds

- jackknife
- board game
- bluebird
- high chair
- sailboat

Exocentric compounds

- funny farm
- azybones
- loony bin
- scarecrow
- pickpocket

Topic- 021: Zero-derivation

Derived words are not necessarily found in the lexicon comes from first language acquisition.

English-speaking adults vocabularies of 20,000 to 50,000 words, Children ranging from about 50–600 words at age 2 to about 14,000 at age 6. Children frequently coin new words (Clark 1995: 393, 399–401). One way children do this is to use zero-derivation, or conversion, a productive derivational process in English. Zero-derivation changes the lexical category of a word without changing its phonological shape. A word formed by zero-derivation or any other productive derivational process becomes lexicalized: The English verbs: chair, leaf, ship, table, and weather. Mail from the French is another example: *I'm going to mailbox this parcel.

Affixation

The next type of derivational process we consider here is affixation. We have already looked at a number of affixes, the types of stems they attach to, and the words they produce. The affixation may involve prefixes, suffixes, infixes, and perhaps circumfixes. We can characterize the stem an affix attaches to as bound or free and as belonging to a particular lexical category. Take the following words formed via prefixation with re-:

a. reignite d. reanalyze

b. reboot e. recertify

Re- attaches to stems belonging to the lexical category Verb. It produces words that are also verbs. All of the stems to which it attaches in the preceding examples are free. Ignite, boot, read, analyze, certify, The same cannot be said for all stems that seem to bear the prefix re-, based on their meaning.

a. repeat

b. resuscitate

c. recognize

Zero-derivation

Further evidence that derived words are not necessarily found in the lexicon comes from first language acquisition. While English-speaking adults typically have production vocabularies of 20,000 to 50,000 words, children's vocabularies are much smaller, ranging from about 50–600 words at age 2 to about 14,000 at age 6. To make up for this, children frequently coin new words (Clark 1995: 393, 399–401). One way children do this is to use zero-derivation, or conversion, a productive derivational process in English. Zero-derivation changes the lexical category of a word without changing its phonological shape. The following are all examples of novel verbs formed by 2- to 5-year-olds by zero-derivation. These examples are taken from Clark (1995: 402); the children's ages are given in the format years; months:

- a. SC (2;4, as his mother prepared to brush his hair): Don't hair me.
- b. JA (2;6, seated in a rocking chair): Rocker me, mommy.
- c. SC (2;7, hitting baby sister with toy broom): I broomed her.
- d. SC (2;9, playing with toy lawnmower): I'm lawning.
- e. DM (3;0, pretending to be Superman): I'm supermanning.
- f. FR (3;3, of a doll that disappeared): I guess she magicked.
- g. KA (4;0, pretending to be a doctor fixing a broken arm): We're gonna cast that.
- h. RT (4;0): Is Anna going to babysitter me?
- i. CE (4;11): We already decorationed our tree.
- j. KA (5;0): Will you chocolate my milk?

It may happen over time that a word formed by zero-derivation or any other productive derivational process becomes lexicalized. So it is with the English verbs chair, leaf, ship, table, and weather. Another example of a verb that was originally derived via zero-derivation but is now listed in the lexicon is mail. In this case, we know that the noun came first because it was borrowed from the French male (Modern French malle) 'bag, trunk', referring to the receptacle in which letters were carried. Evidence that the verb is now stored in the lexicon comes from its frequency, as well as from the fact that its existence blocks the coining of potential but non-occurring derived forms, such as *mailbox 'to put in a mailbox in order to send to someone' (e.g., *I'm going to mailbox this parcel).

Affixation

The next type of derivational process we consider here is affixation. We have already looked at a number of affixes, the types of stems they attach to, and the words they produce. The affixation may involve prefixes, suffixes, infixes, and perhaps circumfixes. Since our primary focus in this section is on English,

we deal only with the first two types, although as you can see in the following, English does have a productive infixing process that incorporates swearwords.

Prefixes:

un + do

cyber + dieter³

hemi + sphere

Suffixes:

rough + age

arachno + phobia

cut + ie

Infixes: abso + effing + lutely

Circumfix: Kujamaat Jóola u - ... al (first person plural inclusive subject):

ub c ɲ ɛ:n-ɛ:n c rut-al c 'We had not yet sent him'

We can characterize the stem an affix attaches to as bound or free and as belonging to a particular lexical category. Take the following words formed via prefixation with re-:

a. reignite d. reanalyze

b. reboot e. recertify

c. reread f. rebuild

Re- attaches to stems belonging to the lexical category Verb. It produces words that are also verbs. All of the stems to which it attaches in the preceding examples are free. Ignite, boot, read, analyze, certify, and build may all stand alone; they do not need to bear a prefix or suffix. The same cannot be said for all stems that seem to bear the prefix re-, based on their meaning. The following words are all formed with re-, which imparts a sense of repetition or doing again. Yet the stems are unable to stand on their own. They are bound.

a. repeat

b. resuscitate

c. recognize

Sometimes an affix may attach to stems of more than one category. One such affix is -ish. It may attach to nouns, as in prudish, bookish, girlish, or childish, adjectives, as in oldish or smallish, or verbs, as in ticklish. In each case, the resulting form is an adjective. So far we have mentioned stem type (bound vs.

free) and lexical category as two factors that can constrain affixation. Another type of restriction we find is on the combination of affixes. Aronoff and Fuhrhop (2002) observe that many English speakers cannot say *dressinglest, though both -ing and -less are secondary suffixes, and the word has a logical meaning: *a dressinglest salad. The perceived ungrammaticality of dressinglest by those speakers can be related to the fact that the combination *-ingless in English is vanishingly rare (a notable exception is meaningless). In German, Aronoff and Fuhrhop observe another type of restriction on affixation. The following examples show how affixation can be recursive. In the following examples we have a verbal stem 'teach', which can be made into a noun with the addition of -er. This noun, in turn, can be made into an adjective, which can then be turned into a complex noun, Lehrerhaftigkeit 'teacherlikeness'.

a. lehr(en) 'teach' (verb) → b. Lehrer 'teacher' (person term) → c. lehrhaft 'like a teacher' (adjective) → d. Lehrerhaftigkeit 'teacherlikeness' Significantly, Lehrerhaftigkeit cannot undergo further derivational affixation. In fact, no noun in -igkeit can. Aronoff and Fuhrhop call it a closing suffix. The existence of closing suffixes in German means that affixation in that language can never be as recursive as German compounding, which can go on and on, just like in English.

Topic- 022: Other Derivational Processes: Blending acronym

So far we have looked at compounding, zero-derivation, and affixation. What are other examples of derivational processes?

Blending

Blending is a common derivational process in English. Blends, also called portmanteau words, are formed by combining parts of more than one word. Speakers may be aware that a word has been formed via blending, as with spork (< spoon + fork), or the history of the word may be obscured, as with motel (< motor + hotel). Further examples of blends are given below:

- a. smog < smoke + fog
- b. chunnel < channel + tunnel
- c. chortle < chuckle + snort
- d. bit < binary + digit

Neither blending nor any of the other derivational processes we have been talking about is limited to English, of course. While blends are rare or absent in many Indo-European languages, they are common in Hebrew, examples of which are given below (Bat-El 1996), and Japanese:

- a. prígurt 'fruit yogurt' < prí 'fruit' + yógurt 'yogurt'
- b. kadurégel 'football' < kadúr 'ball' + régel 'foot'
- c. maškár 'cold drink' < mašké 'drink' + kár 'cold'
- d. kalcéfet 'easy-to-make' < kál 'easy, light' + kacéfet 'ice cream' 'whipped cream'
- e. ramzór 'traffic light' < ramáz 'to hint' + ʔor 'light'

f. pomelít ‘hybrid of pomelo < poméla ‘pomelo’ + and grapefruit’ ?eškolít ‘grapefruit’

g. šmanmúx ‘dumpy’ < šmanmán ‘plump’ + namúx ‘short’

Blending is an example of creative language use. It generally does not adhere to strict constraints, as does affixation. For example, we noted above that bit comes from ‘binary digit’. Nothing in the grammar would have prevented speakers from coining the word *binit from the same parts. The word-formation processes discussed below, including acronym formation, clipping, folk etymology, and backformation, are treated by linguists separately from productive derivational (lexeme formation) phenomena.

Acronyms

Acronyms are formed by taking the initial letters of a string of words and combining them to form a new one. In linguistics, speech is generally considered primary and writing secondary, and because acronym formation is dependent on orthography rather than pronunciation, many would consider it to be external to the general phenomenon of lexeme formation or at the very least a special case of lexeme formation that we would expect to find only among literate users of a language. If you question our statement that acronym formation is orthographically based, just say the NATO aloud, followed by North Atlantic Treaty Organization. The pronunciation of the A and O reflects the name of the letters, not the initial sounds of the words. Speakers may be aware or unaware that a particular form originated as an acronym. How many acronyms on the left would you have been able to break down into their parts on the right?

a. scuba self-contained underwater breathing apparatus

b. radar radio detecting and ranging

c. NATO North Atlantic Treaty Organization

d. AIDS Acquired Immune-Deficiency Syndrome

e. FBI Federal Bureau of Investigation

FBI differs from the other acronyms in in being pronounced letter by letter, rather than as a word *[fbi], *[fbaj]. While it is still an acronym, some people prefer to call it an abbreviation or alphabetism for this reason. Like FBI, PDG is pronounced letter by letter. A characteristic of acronyms in some languages is that they can serve as the base for further morphological operations, particularly affixation, as in the following French examples:

a. cégétiste someone from the CGT (Confédération générale du travail)

Topic- 023: Clipping & Formation

Clipping is the creation of a new word by truncation of an existing one. Many nicknames are formed via this process:

Rob (< Robert) Trish (< Patricia) Sue (< Susan)

Other nouns may also be formed via clipping, as with, ad (< advertisement), co-ed (< co-educational), typo (< typographical error), or fan (< fanatic).

Folk etymology

Some words are formed via folk etymology. This happens when speakers reinterpret a form – typically a borrowing from another language – on the basis of words or morphemes that already exist in the language. Examples of folk etymologies are given below:

cockroach < Spanish cucaracha woodchuck < a pre-existing Algonquian form (e.g., Cree wuchak) witch
hazel < wych [= weak] hazel

Folk etymology is a historical process of reanalysis. We include it here because it is a source of words, but it is not a true derivational process.

Backformation

Backformation is the creation of a word by removing what appears to be an affix. Like folk etymology, it is an example of historical reanalysis and not a productive derivational process. It is responsible for the occurrence of words such as surveil, from surveillance, or liposuct, from liposuction. Some very familiar words were originally derived by backformation. Peddle and edit postdate peddler and editor, although the uninitiated might guess that it was the other way around. Historically, cherry comes from Old Norman French cherise, minus the s, which English speakers mistook for the plural affix. Similarly, pea comes from pease, which speakers also reanalyzed as a plural. (You may know the rhyme “Pease porridge hot.”) In order to recognize whether a form has resulted from backformation or clipping, it is often necessary to know its history.

The words in column B were derived from the forms in column A. Identify the word-formation processes responsible for each.

A	B
a. garage, sale garage sale	: a. compounding,
b. refuse + nik refusenik	b. affixation (suffixation)
c. promenade prom	, c. clipping,
d. baby-sitter baby-sit	d. backformation
e. blanket (n) blanket (v)	e. zero-derivation or conversion
f. cheese, hot dog cheese dog	f. blending
g. asparagus sparrow grass	g. folk etymology
h. under + estimate underestimate	h. affixation (prefixation)
i. photo opportunity photo op	i. clipping
j. chocolate, alcoholic chocoholic	j. blending
k. mothers against drunk driving MADD	k. acronym formation

Topic- 024: Derivation and Structure

Having considered the issue of lexical storage of derived forms, let’s go back to the notion of derivation and structure. We can schematize derivation as follows:

Input → Output Lexeme X Lexeme VY

If we can have lexeme X as an input and lexeme Y as an output, then it should also be possible to take lexeme Y as an input to a second function:

Input → Output Lexeme V Lexeme Z

This is precisely what we do when we form words like unfriendly:

Function 1: add -ly friend → friendly

Function 2: add un friendly → unfriendly

We can even go on to form unfriendliness from unfriendly via a function that adds -ness. In each case, the output of one derivation serves as the input to the next. To determine the order of functions leading to a form, it helps to consider other words that contain the same parts. Consider the example of unfriendly. Un- attaches to nouns only in exceptional cases (for example, uncola, a word once used in an ad campaign for a particular soda). However, it regularly attaches to adjectives. We use this fact in determining that the function 'Add -ly', which forms adjectives, must come before the function 'Add un-'.

Let's go back to the compound we came across earlier: high voltage electricity grid systems supervisor. This compound clearly has an internal structure. [High voltage] is a compound, as are [electricity grid] and [systems supervisor]. [[High voltage] [electricity grid]] is also a compound, and in turn, [[[high voltage] [electricity grid]] [systems supervisor]]. We have taken English words, made compounds of them, and then used those compound words to yield further compound words. The output of the first compounding function serves as the input to the second and third compounding functions. The same occurs with any kind of affix, and this gives derivational morphology a great deal of power. We can think of derivation as always being binary in a sense. We take a form and apply a function to it. We then take the output of that function and perform another function on it. We can keep on going, getting bigger and bigger things, simply by adding one thing at a time. (When we get to inflection, we will see that this is not always the case.)

The fact that speakers of many languages can add phonological material to either end of a word sometimes leads to complex structures. Take the two English words below:

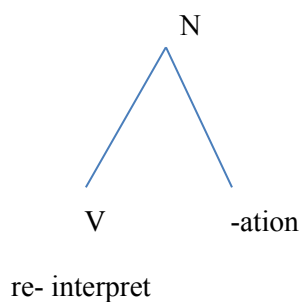
a. reinterpretation b. poststructuralist

These words have the following structures:

a. [[re- [interpret]V]V -ation]N

b. [post- [[[structur]N -al]A -ist]A]A

(a) tells us that reinterpretation is the act of reinterpreting, from reinterpret, not re- the act of interpreting. We start out with a verb, interpret, form a new verb via the prefix re-, and finally form a noun by adding the suffix -ation. In the case of poststructuralist, we start out with the noun structure, make an adjective via the adjectival suffix -al, create a new adjective by adding the suffix -ist, and a further one by adding the prefix post-. Poststructuralist (structuralist, too) can in turn be made into a noun by zero-derivation. The bracketing structures in (a) are convenient, in part because they are so compact. But the structure of morphologically complex words is made most clear when we use tree diagrams, like the following:



This diagram clearly shows that re- and the verb interpret form a unit, a verb, which attaches to the noun-forming suffix -ation. In order to draw a tree diagram, it is first necessary to break a word down into its components and to fully understand how they fit together.

The following argument demonstrates that even identical strings may have distinct structures. Consider the structure of the two words pseudonaturalistic and supernaturalistic:



In both cases we start out with the adjective natural, which we purposely have not broken down into nature and -al, although we could have. In (a) we first make a new adjective, naturalistic, which we then modify with the prefix pseudo-, yielding a word with the meaning ‘falsely naturalistic’. In (27b), however, we take the adjective natural and add the prefix super- to it, giving supernatural ‘pertaining to an existence outside the natural world’. It is to this form that we add the suffix -istic. English morphology is such that we could form a different supernaturalistic, this time with the same structure as pseudonaturalistic in (27a). This supernaturalistic would mean ‘really naturalistic’. Combining prefixation and suffixation leads to other potentially ambiguous forms in English. Three famous examples are given in below:

a. undressed b. unpacked c. unzipped

The ambiguity of the forms in these examples is due to the fact that the prefix un- has at least two distinct roles in English, depending on what it attaches to. When prefixed to a verb, un- is a so-called reversative with the basic meaning ‘undo the action of the verb’. If you unpack a suitcase, you return the suitcase to the state it was in before the packing action took place. If you untie a package, you return it to the state it

was in prior to being tied. When attached to adjectives, including participial adjectives like wounded or stressed, un- means ‘not’. If a soldier leaves the battlefield unwounded, it is not the case that he was first wounded and then unwounded, because it is impossible to unwind a person (we say instead that we cure them). The soldier in question is ‘not wounded’. This second un- is the one we see in forms like unafraid, uncertain, and un-American. Our analysis of an example like unzipped (c) depends on our interpretation of its prefix un-. One possibility is that its structure is as follows:

[[un-zip(p)] ed]

Here the prefixation of the reversative un- yields the meaning ‘cause to be zipped no longer’. The suffix -ed is then added to create the past tense or the past participle. The second possibility is that unzipped has the structure given below:

[un- [zipped]]

This form has the meaning ‘not zipped’, or, in the case of a computer file, ‘having never been stored on a zip disk’. The crucial semantic distinction between (29) and (30) is that only (29) requires that a zipping action have taken place at some past point. Structurally, [[un-zip(p)] ed] and [un- [zipped]] differ in the ordering of the affixation processes. Morphological structure depends not only on the elements you use, but on the order in which the elements have been applied. Let’s look at some other words. Is unwashed ambiguous? It is not; we cannot unwash something. Unwashed can only mean ‘not washed’.

Derivation forms complex lexemes, which may or may not be stored in the speaker’s mental lexicon. These complex lexemes, unlike simple signs like dog, have internal morphological structure. They may also serve as the basis of further derivation, leading to yet more complex lexemes. Derived lexemes that are not perfectly compositional must be retained in the mental lexicon. As a result, the lexicon does not consist solely of simple signs. Many of its members may be partially motivated complex words.

Lesson 05

WORDS & IT FORMS; INFLECTIONS

Topic- 025: Words and Grammar

I introduced the idea that some complex words have meanings that are so predictable that they do not have to be listed in a dictionary. Such words illustrate the fact that a word need not be a lexical item (while, conversely, idioms illustrate the fact that a lexical item need not be a word). However, I did not discuss the different varieties of no lexical-item words. In this chapter we will focus on one variety: words that do not have to be listed because they are merely grammatically conditioned variants of a word that is more basic, in some sense –and which itself may or may not be listed, depending on whether its meaning is predictable or not.

By way of illustrating the notions ‘more basic’ and ‘grammatically conditioned variant’, let us consider the words *performs*, *performed* and *performance* in (1)–(3):

- (1) This pianist performs in the local hall every week.
- (2) Mary told us that this pianist performed in the local hall.
- (3) The performance last week was particularly impressive.

All these words contain a suffix: *perform-s*, *perform-ed*, and *perform-ance*. However, the suffixes *-s* and *-ed* are dependent on the grammatical context in a way that the suffix *-ance* is not.

In (1), the reason why the verb *perform* has an *-s* suffix is that the subject of the verb (the noun phrase denoting the person doing the performing) is singular (*this pianist*), not plural (*these pianists*).

It is easy for a native speaker to check that (4) and (5) ‘feel wrong’:

- (4) *This pianist perform in the local hall every week.
- (5) *These pianists performs in the local hall every week.

Examples (4) and (5) are unacceptable because they violate a grammatical rule of English concerning ‘agreement’ between a verb and its subject: the *-s* suffix on the verb is obligatory when the subject is a singular noun phrase (that is, one for which *he*, *she* or *it* could be substituted), and forbidden when the subject is a plural noun phrase (one for which *they* could be substituted). The *-s* on the verb in (1) does not make any independent contribution to the meaning of the sentence, one might say; it simply reflects the fact that the subject of the sentence is singular rather than plural.

In (2), the aspect of the grammatical context that is relevant to the suffix *-ed* on *performed* is the fact that the verb *told* is in the past tense (that is, it refers to a past event, namely an earlier conversation with Mary). Mary’s actual words in this earlier conversation were probably ‘This pianist performs ...’, not ‘This pianist performed ...’. Why then is the word *performs* replaced by *performed* in the report of her words at (2)? The answer is that English grammar incorporates a rule about what is called ‘sequence of tenses’: if a verb of saying or thinking is in the past tense (as *told* is here), then a verb in any sentence

reported as having been said or thought is likely to be shifted backwards in tense, so to speak: *performs* is replaced by *performed*, *performed* in turn is replaced by *had performed*, and *will perform* is replaced by *would perform*. Again, the *-ed* on *performed* does not make any independent contribution to the meaning of the sentence –for example, it does not (as one might expect) indicate that the series of concerts has ceased since the conversation with Mary took place.

Instead, it is merely a grammatical consequence of the fact that the verb of saying is in the past tense (*told*) rather than the present (*tells*).

In (3), on the other hand, there is no grammatical factor that requires the presence of *-ance* on *performance*. The most one can say is that, in the context where *performance* occurs, one expects to find a noun rather than a verb such as *perform*, as illustrated by the unacceptability of (6):

(6) *The perform last week was particularly impressive.

However, there is nothing in this context that forces us to choose the noun *performance* in particular, or even another noun with the suffix *-ance*.

Any noun (or at least any noun with an appropriate meaning) will do, as in:

(7) The performer last week was particularly impressive.

(8) The concert last week was particularly impressive.

We can describe the difference between *performance* on the one hand and *performs* and *performed* on the other by saying that the latter pair are grammatically conditioned variant forms of the verb *perform*, whereas *performance* is not a variant form of the verb, but rather a noun derived from it. We have encountered here another important distinction: between derivational morphology, and so called inflectional morphology or inflection, which deals with the inflected forms of words, that is the kind of variation that words exhibit on the basis of their grammatical context. In

Sections 4.2–5 we will look in more detail at inflection in English, while

Sections 4.6 and 4.7 are concerned with kinds of inflection that require lexical listing because of unpredictability not of meaning but of shape. It is necessary first, however, to introduce some terms that are more precise than the ordinary term ‘word’, which I have relied on heavily up to now. I have called *performs* and *performed* ‘grammatically conditioned variants’ or ‘inflected forms’ of ‘the verb *perform*’. But if one compares (1) with (9), alongside the unacceptable examples (4) and (5), one can see that *perform* itself deserves to be called a grammatically conditioned variant too:

(9) These pianists perform in the local hall every week.

The fact that the verb appears with no suffix in (9), where the subject *these pianists* is plural, is just as much a matter of grammar as the fact that the verb appears with *-s* in (1), where the subject is singular. But it is awkward and confusing to describe *perform* in (9) as a form of itself! We need a new term for the more abstract kind of word of which the word forms *performs*, *performed* and *perform* are all inflectional variants. Let us call this more abstract kind of word a *lexeme*. Let us also introduce the convention that, where the distinction is important, words as lexemes are written in small capitals, while words as inflected

forms continue to be represented in italics. We can now say that *performs*, *performed* and *perform* are all inflected forms of the lexeme / *perform*, and we can describe the grammatical function of *performed* by calling it the past tense form of the verb /*perform*. Equally, *told* in (2) is the past tense form of the verb / *perform*, and *pianists* in (9) is the plural form of the lexeme /*pianist*.

The most straightforward way to define the term word form is to tie it so closely to pronunciation that pronunciation is its sole criterion: two word forms are the same if and only if they are pronounced the same, or are homophonous. (Let us not be sidetracked by the fact that two words can be pronounced the same but spelled differently in English, and vice versa; in most domains of linguistic research, spoken language is more important than written.) It follows that the same word form can belong to two quite different lexemes, as does *rows* in (10) and (11):

(10) There were four rows of seats.

(11) One person rows the boat.

In (10), *rows* is the plural of the noun /*row* meaning ‘line of people or things’, while in (11) it is one of the present tense forms of the verb /*row* meaning ‘propel with oars’ (more precisely, it is the form used with subjects that can be replaced by *he*, *she* or *it* : so-called ‘third person singular’ subjects).

Topic- 026: Regular and Irregular Inflections

The words that have to be listed in a dictionary and words that do not. One does not have to list *performs* and *performed* alongside *perform*, or *pianists* alongside *pianist*, because they are merely grammatically conditioned variants of one basic word – of one lexeme, in fact. But it is not correct to say that dictionaries never have anything to say about inflectional morphology.

This is because there are two reasons why a word form such as *pianists* does not have to be listed, and these reasons are independent. The first is that, once we know that an English word is a noun denoting a kind of thing that can be counted, then we can be confident that it will have a plural form with no idiosyncrasies of meaning: it will mean simply ‘more than one X’, whatever X may be. The second reason is that, unless otherwise specified, we can be confident that the plural form of any countable noun will be formed by adding to the singular form the suffix *-s* (or rather, the appropriate allomorph of this suffix); in other words, suffixing

-s is the regular method of forming plurals.

That qualification ‘unless otherwise specified’ is crucial, however. Any native speaker of English, after a moment’s thought, should be able to think of at least two or three nouns that form their plural in some other way than by adding *-s*: for example, *child* has the plural form *children*, *tooth* has the plural *teeth*, and *man* has the plural *men*. The complete list of such nouns in English is not long, but it includes some that are extremely common. What this means for the dictionary entries for *child*, *tooth*, *man* and the others is that, although nothing has to be said either about the fact that these nouns possess a plural form or about what it means, something does have to be said about how the plural is formed.

Thus, for example, a dictionary entry for *tooth* will look like this: *tooth noun (plural teeth)*. One of a set of hard white structures set in the jaw and used for biting and chewing.

Such nouns, in short, are irregular in their plural formation, and irregularity is a kind of idiosyncrasy that dictionaries need to acknowledge by indications such as ‘(*plural teeth*)’ here. One can easily visualise a variety of English with no irregularity, but this would be unlike any variety actually in use. Readers of George Orwell’s novel *Nineteen Eighty-Four* will recall that, in the politically purged variety of English called Newspeak, Orwell envisages the eradication of morphological irregularity along with opportunities for ‘thought crime’, so that the plural of *man* in Newspeak is not *men* but *mans*. In reality as opposed to fiction, this sort of regularisation is a well-known feature of the speech of young children and of non-native learners. The very fact that regularization takes place confirms that there is something about the irregular forms that requires them to be specially learned.

For English nouns, there is no difficulty in determining which is the regular method for forming the plural. However, the very fact that there is more than one method raises a potentially tricky question about morphemes and their allomorphs. The allomorphs of a morpheme may be distributed in a fashion that requires reference to individual lexical items, and also that allomorphs may differ from each other phonologically in idiosyncratic ways (as *-duce* differs from *-duct-*, and *-sume* from *-sump-*). If all this does not inhibit us from recognising them as allomorphs of one morpheme, what about the different plural suffixes exhibited by nouns such as *pianists*, *oxen*, *formulae* and *cacti* (these last three corresponding to the singular forms *ox*, *formula* and *cactus*)? Can we not classify *-s*, *-en*, *-ae* and *-i* as all allomorphs of a single ‘plural’ morpheme? Should we not also recognise a further allomorph that we might call ‘vowel change’, to accommodate *men* and *teeth*, which lack a suffix? Admittedly, these allomorphs are quite unlike each other in pronunciation – but if allomorphs are allowed to differ somewhat, why cannot we allow them to differ considerably? At what point, if any, does phonological divergence become too great?

This is a difficult question. Discussing it adequately would take us beyond the bounds of an elementary textbook on English word structure.

When reading any text in which the term ‘morpheme’ is used, it is important understand how the author is using it: whether in a more concrete sense, oriented towards pronunciation (in terms of which *-s*, *-en*, *-ae* and *-i* represent different morphemes), or a more abstract sense, oriented towards meaning or grammatical function (in terms of which *-s*, *-en*, *-ae* and *-i* are all allomorphs of one morpheme). A good way to avoid any confusion is to use terms such as ‘root’, ‘suffix’ and ‘prefix’, wherever possible, rather than ‘morpheme’. This is because, although there may be disagreement about whether to treat these plural suffixes as allomorphs of one morpheme, everyone agrees that they are distinct suffixes.

This question about suffixes with the same grammatical function has a bearing also on allomorphy affecting roots, such as *wife* and *wive-*. The phonological similarity between *wife* and *wive-*, and the fact that parallel alternations can be found (e.g. *knife* and *knive-*, *path* and *path-*, *house* and *house-*, in all of which a voiceless fricative consonant in the singular alternates with its voiced counterpart in the plural) are solid grounds for calling them allomorphs of one morpheme.

It is clear that *wife* and *wives* as expressing the singular and plural respectively of one lexeme. But does it follow that all the word forms of a lexeme must always share the same root morpheme? Does it ever happen that two word forms that behave grammatically like forms of one lexeme look so dissimilar that they seem to have no root morpheme in common (at least if ‘morpheme’ is given its more concrete sense)?

The answer is yes, but seldom at least in English. Consider the lexeme *go*. Because it is a verb, we expect it to have a past tense form, and this expectation is not disappointed. Surprisingly, however, what functions as the past tense form, namely *went*, is phonologically quite dissimilar to the verb's other forms *go*, *goes*, *going* and *gone*. Should we say, then, that *go* and *went* are allomorphs of one morpheme? Most linguists would say no; rather, they would treat this as showing that one lexeme may be represented by two (or more) quite distinct root morphemes (not allomorphs). The term given to this phenomenon is suppletion; *go* and *went* are said to be distinct roots (and hence distinct morphemes), standing in a suppletive relationship as representatives, in different grammatical contexts, of one lexeme. This view of suppletion, as a relationship between roots rather than between allomorphs, is consistent with the 'concrete' view of allomorphy outlined just now in relation to the plural suffixes.

From the point of view of allomorph, it may seem that *go* and *went*- stand in just the same relationship as the plural suffixes *-s*, *-en*, *-ae* and *-i*; hence, if the term 'suppletion' is used for the former relationship, it should be used for the latter too. In fact, however, 'suppletion' is generally applied only to roots, not to affixes. This is because suppletion is generally seen as a relationship between forms of the same lexeme, whereas allomorph need not be.

The discussion so far in this chapter has been rather general. In the remaining sections I will put flesh on the bones by discussing in more detail how inflection works in English, i.e. what grammatical words are associated with inflected lexemes, how these grammatical words are regularly expressed, and what kinds of irregularity they may display.

Because the role of inflectional morphology in English is much smaller than in languages such as German or Russian (although greater than in Chinese), what needs to be said about each word class is relatively circumscribed.

Topic- 027: Forms of Nouns

Most countable nouns in English have two word forms: a singular and a plural. Inflectionally, for any noun lexeme X, there are just two grammatical words, 'singular of X' and 'plural of X', contrasting in number.

Thus, to the lexeme *cat* there corresponds a singular form *cat*, consisting of just one morpheme, and a plural form *cats*, consisting of a root *cat* and the suffix *-s*. The suffix *-s* is the regular suffix for forming plurals. Irregular suffixes expressing plurality include *-i*, *-ae* and *-a* (as in *cacti*, *formulae*, *phenomena*) found with some relatively learned words borrowed from Latin or Greek; the suffix *-(r)en* that shows up only in *oxen*, *children* and *brethren*; and a very few others such as the Hebrew *-im* in *cherubim* and *kibbutzim*.

There are also some countable nouns that express their plural with no suffix at all like the (teeth, men) where there is a change in the vowel of the root – or, more precisely, an allomorph of the root with a different vowel from the singular. However, there are also some whose plurals display not even a vowel change: for example, *sheep*, *fish*, *deer*, *trout*. An obvious question, therefore, is: if the plural and singular forms of these nouns are the same, how can we tell whether they are singular or plural? The answer is: according to the syntactic context.

Consider the following examples:

(1) A deer was visible through the trees.

(2) Two deer were visible through the trees.

In (1) we can tell that deer is singular (more strictly, it represents the grammatical word ‘singular of the lexeme *deer*’ because it is accompanied by the indefinite article *a*, which only ever accompanies singular nouns (e.g. *a cat*, not **a cats*), and because the form of *deer* found in (1), agreeing in singular number with the subject *a deer*, is *was*, not *were*. In (2), for parallel reasons, we can tell that deer is plural: the numeral *two* accompanies only plural nouns (*two cats*, not **two cat*), and the form of *deer* in (2) is the plural *were*.

The class of nouns which are unchanged in the plural (sometimes called ‘zero-plural’ nouns, if they are analyzed as carrying a ‘zero suffix’) could conceivably be just as random as the class of those with vowel change (*tooth*, *man*, etc.). But in fact there seems to be a common semantic factor among the zero-plurals: they all denote animals, birds or fish that are either domesticated or hunted, usually for food. It is true that the relationship is not hard and-fast: there are plenty of domesticated and game animals which have regular -s plurals. Nevertheless, the correlation is sufficiently close to justify regarding zero-plurals as in some degree regular, obeying a minority pattern of plural formation that competes with the dominant pattern of -s-suffixation.

It may be noted that only some nouns have plural forms, namely nouns that refer to entities that are countable. That is why the forms *cats* and *pianists* exist, but not **astonishments* or **rices* – except perhaps in contexts where they can be interpreted as denoting countable entities, such as astonishing events or varieties of rice. But does that mean that all nouns referring to countable entities have both singular and plural forms? Not quite. There are a few nouns such as *scissors* and *pants* which exist only in an -s-plural form, and which appear only in plural syntactic contexts, even though they denote single countable entities, as is shown by the contrast between (3) and (4):

A WORD AND ITS FORMS: INFLECTION

(3) a. Those scissors belong in the top drawer.

b. Your pants have a hole in the seat.

(4) a. **That scissors belongs in the top drawer.*

b. **Your pants has a hole in the seat.*

This idiosyncratic lack of a morphological singular form (except in compounds such as *scissor factory*) creates a problem in contexts where the syntax seems to require such a form, as when the noun is preceded by the indefinite article *a* or *an*. We can say neither **a scissor* nor **a scissors*, and likewise neither **a pant* nor **a pants*. However, for these lexemes, there is a conventional circumlocution or periphrastic form: *pair of pants* and *pair of scissors* (as in *That pair of scissors belongs in the top drawer*).

If inflection is a matter of grammatically conditioned variation, as is thought generally, it is easy to agree that (say) the contrast between *performs* in (*This pianist performs ...*) and *perform* in (*These pianists*

perform ...) is inflectional, because it is a contrast imposed by the grammatical context (whether the subject noun phrase is singular or plural). But what about the noun phrases themselves? The choice between singular and plural there is determined not by grammar but by meaning, one may think – by what the speaker wants to say. If so, does this contrast really deserve to be called grammatically conditioned?

Despite the freedom to choose between, say, this pianist and these pianists as subjects of (9), there is still a sense in which English grammar affects the choice between singular and plural. It does so in the sense that it imposes the choice. In talking about a series of weekly piano concerts, we are free to be vague about the number of pianists who perform –except that we are forced by English grammar to be precise about whether there is one (that pianist) or more than one (these pianists). Likewise, if I see a cat or some cats in the garden, I cannot report what I have seen without making it clear whether there was just one cat, as in (16) or more than one cat, as in (7). A formulation that is deliberately vague on that issue, such as (18), is unacceptable:

(5) I saw a cat in the garden.

(6) I saw (some) cats in the garden.

(7) *I saw cat in the garden.

The best we can do to express the intended content of (8) is use a circumlocution like one or more cats or at least one cat. In this respect, English grammar contrasts with that of, for example, Chinese, where the singular–plural contrast is not expressed morphologically in nouns or verbs, and indeed is scarcely grammatically relevant at all. That does not mean that one cannot distinguish between one object and several when talking Chinese; it is just that the distinction is not imposed by Chinese grammar, which permits ambivalence about plurality. The singular–plural distinction is the only grammatical distinction that is expressed morphologically in English nouns.

Topic- 028: Forms of Pronouns & Determiners

In morphology we are mainly concerned with the behaviour of words which belong to **open classes**, namely nouns, adjectives, verbs and adverbs. These classes are so called because their membership can be added to, and indeed is added to constantly as new words come into use.

By contrast, one does not expect in English to encounter a new pronoun (a word such as *I* or *she* or *us*) or a new preposition (a word such as *in* or *at* or *without*). However, determiners deserve a mention here because some of them, like nouns, display a singular–plural contrast, and pronouns combine a singular–plural contrast with contrast unique to them, between subject and non-subject forms.

We have already encountered the distinction between *this* and *these*, as in *this pianist* and *these pianists*. These are the singular and plural forms of the determiner lexeme *this*. Other determiners include *the*, *a* (*an*) and *some*, but only one other determiner exhibits a singular–plural contrast: *that*, with singular and plural forms *that* and *those*. The determiners *that* and *this* demonstrate that number contrasts can have a grammatical effect inside noun phrase as well as between subject noun phrases and their accompanying verbs.

In many languages, the distinction that English expresses by word order in *John loves Mary* and *Mary loves John* is expressed by inflectional means on the words corresponding to *Mary* and *John*. In English, the same technique is used for one small closed class of lexemes, namely personal pronouns. If one replaces *John* and *Mary* with the appropriate pronouns in these two examples, the outcome is as in following (1 & 2):

(1) He loves her.

(2) She loves him.

He and *him* are sometimes said to contrast in **case**, *he* belonging to the **nominative case** and *him* belonging to the **accusative case**. This kind of inflection has only a marginal role in English, being limited to pronouns; but, if we treat (say) *he* as a lexeme, we must recognise it as having two forms: *he* and *him*. It is striking that the relationship between nominative and accusative forms is consistently suppletive, as in *I/me*, *she/her*, *we/us*, and *they/them*, except that for you the two forms are identical (*you*). This is consistent with the fact that pronouns are very common, and suppletion affects only very common words such as *go*.

If *he* and *him* are forms of the lexeme *he*, and *we* and *us* are forms of *we* (and so on), what are we to say about corresponding words with a possessive meaning: *his* and *our*, as well as *my*, *her*, *your* and *their*?

Syntactically and semantically, these words fulfil just the same role as noun phrases with the apostrophe-s discussed in the previous section: *his bicycle* means ‘the bicycle belonging to him’ just as *that man’s bicycle* means ‘the bicycle belonging to that man’. One possibility is to say that these are pronoun forms belonging to a third case, the genitive or possessive, which stand in for apostrophe-s forms in noun phrases that consistently of a personal pronoun. Another is to classify these words as determiners, because they perform a determiner-like role and cannot be combined with other determiners (we cannot say **the my hat* any more than we can say **the that hat*). But these are issues of syntax rather than morphology. For present purposes, we need merely note how *his*, *our* and the rest behave, while leaving their exact grammatical classification undecided.

Topic- 029: Forms of Verbs

We have already discussed some forms of English verbs in Sections 4.1 and 4.2, such as *performs*, *performed* and *perform*. In English, a verb lexeme has at most five distinct forms, as illustrated here with *give*:

(28) Give

a. third person singular present tense: *gives*

e.g. *Mary gives a lecture every year.*

b. past tense: *gave*

e.g. *Mary gave a lecture last week.*

c. progressive participle: *giving*

e.g. *Mary is giving a lecture today.*

d. perfect or passive participle: *given*

e.g. *Mary has given a lecture today.*

The lecture is always given by Mary.

e. basic form (used everywhere else): *give*

e.g. *Mary may give a lecture.*

Mary wants to give a lecture.

Mary and John give a lecture every year.

The contrast between present at (28a) and past at (28b) is a contrast of **tense**. The other dimensions of contrast manifested in (28a) are **person** (third person versus the rest) and **number** (singular versus plural, just as for nouns and pronouns). However, because only one word form (*gives*) exhibits these contrasts, they play a much smaller inflectional role in modern English verbs than in Old English verbs.

For the form labelled ‘perfect or passive participle’, two examples are given, because perfect and passive contexts can be distinguished clearly; however, it is a peculiarity of English verb morphology that the corresponding forms are always the same. Another way of putting this is that, for any verb V, the grammatical words ‘perfect participle of V’ and ‘passive participle of V’ are expressed by the same word form.

I said that a verb lexeme has at most five forms. In fact, most verbs have only four forms, because the past tense and the perfect (or passive) participle forms are the same. This is true for all regular verbs (those that form the past tense with the suffix *-ed*), such as perform (which I used for illustration in Section 4.1):

(29) perform

a. third person singular present tense: *performs*

b. past tense: *performed*

c. progressive participle: *performing*

d. perfect or passive participle: *performed*

e. basic form (used everywhere else): *perform*

When two grammatical words that are distinct for some lexemes are systematically identical for others, as here, these forms are said to be syncretised, or to exhibit **syncretism**. The same syncretism also occurs with some irregular verbs, such as Dig and sting (past = perfect participle *dug, stung*) and all those that use the suffix *-t*, such as bend, feel, and think (*bent, felt, taught*). In all, 150 or so verbs are irregular in

that they do not use the *-ed* suffix. I will not list them all here, however, because the study of these irregularities belongs to grammar rather than to word formation.

Other verbs or verb-like words whose behaviour belongs to grammar rather than word-formation are the auxiliaries, such as *be* and *have*, and modals, such as *can*, *must*, *may*. But they deserve mention here because their various forms distinguish an unusually small or large range of grammatical words. Instead of the usual verbal maximum of five forms, modals distinguish only two (e.g. *can*, *could*) or even just one (e.g. *must*), while *be* distinguishes eight (*am*, *is*, *are*, *was*, *were*, *being*, *been*, *be*).

Topic- 030: Forms of Adjective

Many English adjectives exhibit three forms, for example *green* here:

(30) Grass is green.

(31) The grass is greener now than in winter.

(32) The grass is greenest in early summer.

The grammatical words that *green*, *greener* and *greenest* express are the positive, comparative and superlative of green, contrasting on the dimension of **comparison**. Other adjectives with similar forms are:

(33) *Positive Comparative Superlative* happy happier happiest long longer longest pure purer purest untidy untidier untidiest good better best

All these exhibit a regular pattern of suffixation with *-er* and *-est*, except for *better* and *best*, which are suppletive.

The justification for saying that comparative and superlative forms of adjectives belong to inflectional rather than to derivational morphology is that there are some grammatical contexts in which comparative or superlative adjectives are unavoidable, anything else (even if semantically appropriate) being ill-formed:

a. This field is greener than that one.

b. *This field is green than that one.

c. *This field is fertile than that one.

a. The greenest fields of all are here.

b. *The green fields of all are here.

c. *The superior fields of all are here.

On the basis of our experience with plurals of countable nouns and past tense forms of verbs, then, you will probably expect that every adjective lexeme should possess a comparative and a superlative form (or,

at any rate, every adjective denoting a property that can be present to a greater or lesser degree). However, it is striking that many adjectives lack these forms:

(36) *Curiouser and curiouser!

(37) *This field is fertilier than that one.

(38) *The fertilest fields of all are here.

(You may recognise (36) from Lewis Carroll's *Alice's Adventures in Wonderland* as something that Alice scolds herself for saying.) But it is not that the content of (36)–(38) is inexpressible in English; rather, instead of the suffixes *-er* and *-est*, we use periphrastic forms with *more* or *most*:

(39) More and more curious!

(40) This field is more fertile than that one.

(41) The most fertile fields of all are here.

Broadly speaking, the suffixes *-er* and *-est* appear on adjectives whose basic form has one syllable, or two provided that the second syllable ends in a vowel (e.g. *tidy*, *yellow*), while longer adjectives usually require the periphrasis.

Lesson 06

WORDS & ITS RELATIVES: DERIVATION I**Topic- 031: Relationship between Lexemes**

We discussed the words *perform*, *performs*, and *performed* and *performance*. It was said that *perform*, *performs* and *performed* were grammatically conditioned variants of one lexeme *perform*, but *performance* was not one of these variants. The reason was that, whereas there are grammatical factors that determine the choice between *perform*, *performs* and *performed* (in appropriate contexts), there is no grammatical factor that requires specifically the presence of *-ance* on *performance*. To put it another way: there are contexts where, if any verb appears, it must carry the third person singular suffix *-s*, but there are no contexts where, if a noun appears, it must carry the suffix *-ance*. The suffix *-ance* is not one of the small class of suffixes (so-called ‘inflectional’ suffixes) whose use is tightly determined by grammar. What sort of suffix is it, then? A short answer is that, not being inflectional, it must be **derivational**, since the term ‘derivation’ is used for all aspects of word-structure involving affixation that is not inflectional. The purpose of this chapter is to put flesh on the bones of this purely negative definition, showing something of how derivation works in English.

Since *performance* is not a variant of the lexeme *perform*, it must belong to some other lexeme, which may itself have more than one form. What lexeme could this be? This question is easy to answer when we notice that, alongside *performance*, there is a plural form *performances*. Just as *cat* and *cats* are the two forms (singular and plural) of the lexeme *cat*, it makes sense to regard *performance* and *performances* as the two forms of a lexeme *performance*. This tells us something about the relationship between *perform* and *performance*: it is a relationship not between word forms but rather between lexemes. (Strictly, then, in terms of our typographical convention, we should call it a relationship between *perform* and *performance*.) Thus derivational morphology is concerned with one kind of relationship between lexemes. There are many ways in which lexemes can be related. We are not concerned here with relationships solely of meaning (such as the synonymy of AUBERGINE and EGGPLANT) or of sound (such as the homonymy of ROW ‘line of people or things’ and ROW ‘propel with oars’).

Rather, we are concerned mainly with relationships involving affixation, and the grammatical and semantic tasks that such affixation can perform.

As will be seen, both the affixes and their tasks are quite diverse. An encyclopedic coverage of all the English derivational processes would be impossible.

I will introduce the term *base* for the partially complete word form to which an affix is attached so as to create either an inflected word form or a new lexeme. (Equivalently, the base for an affixation process is what remains if the affix is removed.) Some bases are roots, whether bound (e.g. *wive-*, the base for *wives*) or free (e.g. *cat*, the base for *cats*). Others, however, already contain a root and one or more affixes, such as *helpful* in its capacity as the base for *helpfulness*.

Topic- 032: Word Classes & Conversion

To understand how adjectives can be derived from nouns, nouns from verbs, and so on. It is important therefore that terms for word classes such as ‘adjective’, ‘noun’ and ‘verb’ should be properly understood.

The word class to which a lexeme belongs is mainly determined by its meaning. That belief is however incorrect.

Verbs are ‘doing words’, while nouns are ‘thing words’ and adjectives ‘describing words’. The trouble with these meaning-based definitions is that, if one takes them seriously, they require us to lump together lexemes whose grammatical behaviour is quite different, and distinguish between ones whose grammatical behaviour is similar.

Consider again the lexeme *perform*, which looks like a prototypical ‘doing word’, denoting something that actors and musicians do. The lexeme *performance* denotes the same activity, surely. Does that mean that *perform* and *performance* belong to the same word class? That can hardly be right, since they occur in such different syntactic contexts, and since their inflectional behaviour is so different: *performance* has the two forms *performance* (singular) and *performances* (plural), while *perform* has the four forms *performs*, *performed*, *performing* and *perform*. In fact, as we have seen, *performance* is a noun and *perform* is a verb. This classification can be made, solely on the basis of their syntactic and inflectional behaviour, with no appeal to meaning – and indeed meaning may be positively misleading, since a *performance* is not obviously a ‘thing’.

Compare now the lexemes *perform* and *resemble*. Is the latter a ‘doing word’ too? That seems scarcely appropriate. *Resembling*, one may think, hardly counts as an activity. To say that (for example) my great-uncle William resembles a giraffe is not to report some action of his, but rather to describe him. Should we then lump *resemble* in with other supposed ‘describing words’ – adjectives such as *tall* and *interesting*?

Again, this meaning-motivated conclusion falls foul of syntactic and inflectional evidence. These adjectives have comparative and superlative forms (*taller*, *tallest*) or phrasal substitutes for them (*more interesting*, *most interesting*); on the other hand, *resemble* has a set of forms (*resembles*, *resembled*, *resembling* and *resemble*) exactly parallel to the forms of *perform*, and used in broadly parallel syntactic contexts. So to identify verbs as ‘doing words’ risks misleading us into neglect of the syntactic and inflectional parallels that justify classifying not only *perform* but also *resemble* as a verb.

Does that mean, then, that a lexeme cannot have both noun forms (singular and plural) and verb forms (past, third person singular present, and so on)? If part of identifying a lexeme is identifying what word class it belongs to, then that must be true – but trivially so, because it amounts to decreeing that a root that can carry verbal suffixes such as *-ed* and *-ing* as well as the noun plural suffix *-s* must belong to two lexemes, not one. The more interesting question, then, is: do such roots exist? The answer is certainly yes. For example, *hope* and *fear* have both noun forms (*her hope/fear for the future*) and verb forms (*she hoped/feared that it would rain*).

Other similarly ambivalent words are *wish*, *desire*, *father* (a verb in *He fathered seven children*), and *cook*. Does this mean that the concept ‘word class’, as I have used it, is too vague or inconsistent to be useful?

The answer is no, for two reasons. The first involves the proportion of our noun–verb vocabulary that is ambivalent in this way. Although numerous, it is still heavily outnumbered by the proportion that is either purely noun-like in its grammatical behaviour (e.g. *door*, *sister*, *desk*, *joy*) or purely verb-like (e.g. *hear*, *speak*, *write*, *believe*). Admittedly, one can imagine a language in which a far higher proportion of the

vocabulary is ambivalent in the way we are discussing, and in respect of such a language one might well argue that many or most lexemes did not belong to identifiable word classes. Such claims have in fact been made in relation to some languages in the Austronesian family, which contains (for example) Malay, Tagalog, and the languages of Polynesia, as well as some native languages of western Canada and the US Pacific coast. Even there, however, it seems generally necessary to distinguish nominal (i.e. 'nouny') and verbal syntactic structure, despite the fact that the class of lexemes that can occur in each type of structure is almost the same.

A second kind of reason has to do with English in particular. Let us compare hope and fear as verbs with other verbs that can be followed by that-clauses, as in (1):

- (1) a. She stated that it would rain.
- b. She knew that it would rain.
- c. She denied that it would rain.
- d. She admitted that it would rain.
- e. She acknowledged that it would rain.

For all of these sentences we can identify a nominal counterpart, that is a counterpart of the form her ... that it would rain:

- (2) a. her statement that it would rain
- b. her knowledge that it would rain
- c. her denial that it would rain
- d. her admission that it would rain
- e. her acknowledgement that it would rain

What is striking about the nouns in (2) is that they all involve a suffix added to the basic form of the verb in (1) (possibly with some other phonological change, as in knowledge and admission). There are few verb–noun pairs that one can use in the contexts of (1) and (2) such that the basic and suffixed forms are the other way round, the noun supplying the base and the verb being derived from it by means of a suffix. In morphological terms, therefore, it makes sense to say that the verbal construction in (1) is basic, the nominal construction in (2) being derived from it.

But this has implications for hope and fear as well. If we look only at (3) and (4), we have no basis for deciding whether these lexemes are basically nominal or basically verbal:

- (3) a. She hoped that it would rain.
- b. She feared that it would rain.
- (4) a. her hope that it would rain

b. her fear that it would rain

However, as soon as we notice that (3) and (4) are parallel to (1) and (2) respectively, we have a ground for concluding that hope and fear are basically verbal. The nominal contexts of (4) are parallel to those of (2), where the nouns are clearly derived from verbs; so it makes sense to say that the nouns hope and fear in (4) are derived from verbs too, even though they carry no affix.

The notion that derivation can occur without any overt change in shape may at first seem strange. Some linguists have accordingly decided that hope and fear, as nouns, are really ‘zero-derived’, carrying a phonologically empty and therefore unpronounceable ‘zero suffix’: Hope - $\emptyset$, fear - $\emptyset$. Others have preferred to say that one of the processes available in derivational morphology is conversion, whereby a lexeme belonging to one class can simply be ‘converted’ to another, without any overt change in shape. We do not need to decide here which is the better style of analysis, it may be referred as a phenomenon called ‘conversion’. Either way, these ambivalent words present the problem of determining which word class the basic form belongs to. Sometimes, as with hope and fear, a decisive argument involving parallels with affixed lexemes can be found.

Sometimes, despite the risks already mentioned of relying on meaning as a criterion, the basic meaning seems clearly appropriate to one word class rather than another; for example, few would deny that, even though father can function as a verb, it is the noun (as in my father) that is more basic. In respect of cook, working out the direction of conversion is left as an exercise at the end of the.

Topic- 033: Adverb Derived from Adjectives

In Chapter 2, I invited readers to think about the adjective dioecious, meaning ‘having male and female flowers on different plants’. Certainly, dioecious must be listed in any reasonably complete dictionary of English. I argued, however, that the corresponding adverb dioeciously would not have to be listed, because both its existence and its meaning can be taken for granted once the existence of dioecious is acknowledged.

This neatly illustrates the distinction between lexemes and lexical items: dioeciously is a distinct lexeme from dioecious, since it belongs to a different word class, but it is not a distinct lexical item.

This also illustrates a widespread though not universal characteristic of derivational processes: unlike inflection, they can change the word class of the bases to which they apply.

Some introductory treatments of English grammar talk as if not just many but all adverbs end in -ly. If that were true, it would be an unusual word class, all of its members being derived. In fact, simple or monomorphemic adverbs, though few in number, include some very common words (often, seldom never, soon), and some other adverbs are morphologically complex without containing -ly (nowhere, everywhere, today, yesterday,). Also, there are common adverbs that are formed by conversion: fast (as in The car was driven fast) and hard (as in They worked hard), derived from the adjective fast (as in a fast car) and hard (as in hard work).

Topic- 034: Noun Derived from Nouns and Other Word Classes

Not all derivational processes change word class. English has derivational processes that yield nouns with meanings such as ‘small X’, ‘female X’, ‘inhabitant of X’, ‘state of being an X’ and ‘devotee of or expert on X’. Here are some examples – though by no means a complete list, either of the affixes or of their possible meanings:

(5) ‘small X’: -let, -ette, -ie

e.g. droplet, booklet, cigarette, doggie

(6) ‘female X’: -ess, -ine

e.g. waitress, princess, heroine

(7) ‘inhabitant of X’: -er, -(i)an

e.g. Londoner, New Yorker, Texan, Glaswegian

(8) ‘state of being an X’: -ship, -hood

kingship, ladyship, motherhood, priesthood

(9) ‘devotee of or expert on X’: -ist, -ian

e.g. contortionist, Marxist, logician, historian

If you think about these, you should come to agree that all or nearly all of them must count as lexical items. Many of them have unpredictable meanings (a cigarette is not merely a small cigar, and a booklet is not merely a small book; brotherhood means not ‘the state of being a brother’ but rather ‘secret or semi-secret society’). Also, the very existence of some of these words seems arbitrary. Why is there a word actress (albeit less used now than formerly), but there has never been a word ‘writress’ to designate a woman writer? (I use quotation marks here to identify non-existent but plausible lexemes.) Why do we have droplet but not ‘grainlet’ or ‘lumpet’? It is merely an accident that some of these words have come into general use while others have not, so those that do exist must be lexically listed. This ‘gappiness’ also helps to confirm (should confirmation be needed) that these affixes are derivational rather than inflectional, even though they do not change word class.

Nouns derived from members of other word classes

Nouns derived from adjectives and from verbs are extremely numerous, and it should be easy for you to think of many other examples on the lines of those given here. Here are some suffixes used to derive nouns from adjectives:

(10) -ity, e.g. purity, equality, ferocity, sensitivity

(11) -ness, e.g. goodness, tallness, fierceness, sensitiveness

(12) -ism, e.g. radicalism, conservatism

All these three suffixes mean basically ‘property of being X’, where X is the base adjective. Of the three, -ness is the most widely applicable, and the great majority of nouns formed with it are not lexical items. For example, once one has learned dioecious, one can be confident of both the existence and the meaning of dioecious--ness. Even so, at least one noun in -ness is lexicalized: highness, which means not ‘property of being high’ (for which we use height), but rather ‘royal personage’, as in Her Royal Highness.

Some of these nouns are formed from bases other than the free form of the corresponding adjective, e.g. ferocity from feroc- (not ferocious), conservatism from conservat- (not conservative). The ferocity pattern is fairly general for adjectives in -ious (compare rapacity, capacity alongside rapacious and capacious) but not absolutely general (for example, to delicious and specious there correspond dioeciouness and speciousness, not ‘delicity or specity’). This gappiness is a reason for counting all nouns in -ity as lexical items, and its implications will be discussed further in Chapter 8.

Even more numerous are suffixes for deriving nouns from verbs. Here are just a few:

(13) -ance, -ence, e.g. performance, ignorance, reference, convergence

(14) -ment, e.g. announcement, commitment, development, engagement

(15) -ing, e.g. painting, singing, building, ignoring

(16) -(a)tion, e.g. denunciation, commission, organization, confusion

(17) -al, e.g. refusal, arrival, referral, committal

(18) -er, e.g. painter, singer, organiser, grinder

The suffixes in (13)–(17) all have much the same function (they form abstract nouns meaning ‘activity or result of Xing’), but they are certainly not freely interchangeable: for example, we have performance but no ‘performant or ‘performation’, and we have commitment, committal and commission but no ‘committance’. It is true that some verbs allow a choice of suffixes (e.g. commit), but the nouns thus formed are not synonyms: one can commit a crime, commit an accused person for trial, or commit oneself to a task, but, of the three nouns, only commission corresponds to the first meaning, only committal to the second, and only commitment to the third.

The suffix -er in (18) is the one most generally used for forming nouns denoting a person performing the action of the corresponding verb (agent nouns). But it is not the only agent suffix (typist and informant use other suffixes), and this is not its only function; for example, digger is more likely to denote a piece of machinery than a person, and we have already encountered -er in Section 5.4 with the meaning ‘inhabitant of’ (e.g. Londener).

This is an appropriate place to recall that, although affixation is by far the most common way in which lexemes are derived in English, it is not the only way. Some non-affixal ways of deriving abstract nouns (other than conversion) are:

(19) change in the position of the stress, e.g. nouns permit, transfer alongside verbs permit and transfer

(20) change in the final consonant, e.g. nouns belief, proof defence alongside verbs believe prove, defend

(21) change in a vowel, e.g. nouns *song*, *seat* alongside verbs *sing*, *sit*. By contrast with some languages, however, the derivational use that

English makes of vowel change is minimal. Languages that exploit it much more consistently are members of the Semitic family, such as Arabic and Hebrew.

Topic- 035: Adjectives Derived from Adjectives and Other Word Classes

In adjective prefixes predominate. The only suffix of note is *-ish*, meaning ‘somewhat X’, as in *greenish*, *smallish* and *remotish* ‘rather remote’. By contrast, the prefix *un-* meaning ‘not’ is extremely widespread: for example, *unhappy*, *unsure*, *unreliable* *undiscovered*. Because it is so common, most dictionaries do not attempt to list all *un-* adjectives.

Another negative prefix is *in-*, with allomorphs indicated by the variant spellings *il-*, *ir-* and *im-*, as *intangible*, *illegal*, *irresponsible* and *impossible*. For the present, it is worth noting the existence of pairs of more or less synonymous adjectives, one of which is negated with *un-* and the other with *in-* or one of its allomorphs: *eatable/uneatable* *edible/inedible* *readable/unreadable* *legible/illegible* *lawful/unlawful* *legal/illegal* *touchable/untouchable* *tangible/intangible*

Such examples confirm that the use of *in-* is lexically restricted. As the negative counterpart of *edible*, *unedible* sounds possible, especially if the speaker has limited education and has not encountered, or has momentarily forgotten, the form *inedible*. However, ‘*uneatable*’ as the counterpart of *eatable* is not a form that any English speaker would spontaneously use.

Adjectives derived from members of other word classes

Some of the processes that derive adjectives from verbs straddle the divide between derivation and inflection. We met the suffixes *-ed*, *-en* and *-ing*, and vowel change, in passive and progressive participle forms of verbs.

However, such forms in the following can also be adjectives:

- a. a not very *interesting* book
- b. The party-goers sounded very *drunk*.
- c. The car seemed more *damaged* than the lamp-post.

The modifier *very* and the comparative construction (*more ... than*) show that *interesting*, *drunk* and *damaged* are adjectives here, not forms of the verb lexemes *interest*, *drink* and *damage* (Notice that *very* cannot modify verbs, so one cannot say *That book very interested me.) As for *drunk*, its status as belonging to a distinct lexeme here is confirmed by its special meaning (‘intoxicated through drinking alcohol’), not predictable from the meaning of the verb *drink* (‘swallow liquid’).

Further suffixes that commonly form adjectives from verbs, with their basic meanings, are:

(24) *-able* ‘able to be Xed’: *breakable*, *readable*, *reliable*, *watchable*

(25) -ent, -ant ‘tending to X’: repellent, expectant, conversant

(26) -ive ‘tending to X’: repulsive, explosive, speculative

Expectations derived from these basic meanings can, as usual in derivation, be overridden; for example, conversant does not mean ‘tending to converse’. We have already encountered -able above, where the variant, or allomorph, -ible is also illustrated. What is striking about the -ible words in (22) is that their bases, although they have clearly identifiable verbal meanings such as ‘eat’, ‘read’ and ‘touch’, are bound rather than free. Some of these bound verb roots appear in a number of derived lexemes, such as the aud- root that occurs in (audible, audience, audible and auditory. Suffixes that form adjectives from nouns are more numerous. Here are some:

(27) -ful, e.g. joyful, hopeful, helpful, meaningful

(28) -less, e.g. joyless, hopeless, helpless, meaningless

(29) -al, e.g. original, normal, personal, national

(30) -ish, e.g. boyish, loutish, waspish, selfish

Generally seen, adjectives in -ful and -less tend to come in pairs, although the correspondence is not exact: we have slothful, slothless but ‘penniless but not pennifull. This confirms again that, even when the meaning of a potential word may be easily guessable (a ‘slothless’ person would be hardworking, and a ‘pennifull’ person would be well off), the existence of the word is not guaranteed.

Topic- 036: Verbs Derived from Verbs and Other Word Classes

This section is unusual in that all the affixes that I will mention in it are prefixes. Most prominent are re- and the negative or ‘reversive’ prefixes un-, de- and dis-, as in the following examples:

(31) paint, enter repaint, re-enter

(32) tie, tangle untie, untangle

(33) compose, sensitise decompose, desensitise

(34) entangle, believe disentangle, disbelieve

The prefix re- has already figured in our discussion in Chapter 2 of the relationship between morphemes and meaning. Semantically, the examples in (31)–(34) are mostly straightforward, although those with de- are less so: to decompose is not to undo the creative work of a musical composer!

Also worth mentioning here is the relationship between the verbs in the left and right columns in (35):

(35) Intransitive Transitive

Lie (past lay) lay (past laid)

rise (past rose) raise (past raised)

fall (past fell)

fell (past felled)

sit (past sat)

set (past set)

Transitive verbs (or verbs used transitively) are ones with an ‘object’ noun phrase, usually indicating the thing or person that is the goal of the action of the verb, as the book is the object of laid in (36a). Intransitive verbs, such as lay in (36b), lack such an object.

(36) a. Jill laid the book on the table.

b. The book lay on the table.

The transitive verbs in (35) are all causative, that is they mean ‘cause to X’, where X stands for the meaning of the corresponding intransitive.

Causative–in causative verb-pairs are common in English, but they nearly all involve conversion, as in (37), rather than either affixation or the kind of vowel change seen in (35):

(37) a. Jill boiled the water.

b. The water boiled.

The examples in (35) represent a residue of a vowel-change pattern that was more widespread at an earlier stage of the language. More will be said about such historical developments in Chapter 9.

Verbs derived from members of other word classes

Verbs derived from nouns and from adjectives are numerous. Some affixes for deriving verbs from nouns are:

(38) de-, e.g. debug, deforest, delouse

(39) -ise, e.g. organise, patronise, terrorise

(40) -(i)fy, e.g. beautify, gentrify, petrify

There are also some common verbs that are derived by replacing the final voiceless consonant of a noun with a voiced one, perhaps with some vowel change too (parallel to the relationship between belief and believe, although there it was the verb that seemed more basic):

(41) Nouns

Verbs

Bath

bathe

Breath

breathe

house [...s]

house [...z]

wreath

wreathe

A meaning for de- at (38) is clearly identifiable, namely ‘remove X from’ (compare its function in deriving verbs from verbs, e.g. desensitise). However, neither -ise nor -ify has a clear-cut meaning apart from its verb-forming function (organise does not share any obvious element of meaning with organ, for example). The suffixes -ise and -ify can derive verbs from adjectival bases too, as in nationalise, tenderise, intensify, purify. Hence, when the roots to which they are attached are bound (e.g. cauterise, sanitise, petrify, satisfy, magnify), it is often impossible to decide whether these roots are fundamentally nominal or adjectival. The suffix -ate shows the same sort of ambivalence. Words such as generate, rotate, replicate, and locate clearly contain a root and a suffix, because the same roots crop up elsewhere (e.g. in general, rotor, replica, local). However, because most of the bases to which -ate is attached are bound roots, it does not clearly favour either adjectival or nominal bases.

It will be evident by now that suffixes play a larger role than prefixes in English derivational morphology. But there is still one prefix to be mentioned: en- (with its allomorph em-), which forms verbs meaning ‘cause to become X’ or ‘cause to possess or enter X’ from a few adjectives and nouns: enfeeble enslave, empower, enrage, enthrone, entomb.

With the adjectives bold and live as bases, the prefix en- is combined with a suffix -en: embolden enliven. This suffix usually occurs without the prefix, however, and does so quite widely (e.g. tighten, loosen, stiffen, weaken, widen, redden, deepen toughen). These verbs have either an intransitive meaning, ‘become X’, or a transitive one, ‘cause to become X’. The adjectives that can constitute bases for such verbs share an unusual characteristic, however, which becomes evident when we consider some verbs in -en that are imaginable, yet do not occur: *greenen, narrowen, strongen, tallen. It turns out that the adjectives that can be bases for deriving -en verbs are all monosyllabic and all end in plosives (the sounds usually spelled p, b, t, d, (c)k and g in English) or fricatives (including the sounds usually spelled s, th, f and v). What is wrong with *greenen and the other unsuccessful candidates is that their bases end in a sound other than a plosive or a fricative – although with strong we get round this restriction (so to speak) by adding -en instead to the corresponding noun, strength(which ends in a fricative sound), so as to yield strengthen.

Can we then say that all adjectives ending in a plosive or a fricative, or at least a systematically identifiable subset of these adjectives, can be the base for a verb in -en? That is a question about productivity. However, the starting-point for an answer is to look for adjectives which end in plosives or fricatives but for which there is no corresponding verb in -en.

Lesson 07

WORDS & ITS RELATIVES: DERIVATION II**Topic- 037: Compounds versus Phrases**

It is now established that words are formed from other words, mainly by means of affixes. Now we look at compounds i.e. words formed by combining roots, and the much smaller category of phrasal words that is items that have the internal structure of phrases but function syntactically as words. Some types of compound are much commoner than others.

There are also some styles of writing (for example, newspaper headlines) in which compounds are especially frequent. But first we must deal with an issue that has not arisen so far, because until now all the complex words that we have looked at have contained at least one bound morpheme. Roots in English are mostly free rather than bound. How can we tell, then, whether a pair of such roots constitutes a compound word or a phrase that is a unit of sentence structure rather than a complex word?

A definite answer is not always possible, but there are enough clear cases to show that the distinction between compounds and phrases is valid. Consider the expressions a green house, with its literal meaning, and a greenhouse, meaning a glass structure (not usually green in colour!) Where delicate plants are reared. There is a difference in sound corresponding to the difference in meaning: in the first expression the main stress is on house, while in the second the main stress is on green. This pattern of semantic contrast between expressions stressed in different places is quite common, as in the following examples:

- | | |
|---------------------------------------|---------------------------------|
| (1) black bóard | blackboard |
| ‘board that is black’ | ‘board for writing on’ |
| (2) silk wórm | silkworm |
| ‘worm made of silk (e.g. a soft toy)’ | ‘caterpillar that spins silk’ |
| (3) hair nét | hairnet |
| ‘net made of hair’ | ‘net for covering hair’ |
| (4) white hóuse | (the) White House |
| ‘house that is white’ | ‘residence of the US President’ |
| (5) toy fáctory | tóy factory |
| ‘factory that is a toy | ‘factory where toys are made’ |
| (e.g. in a model city)’ | |

The items on the left in (1)–(5), like green hóuse, are phrases, because it is characteristic of phrases in English to be stressed on the last word, unless some contrast is being stated or implied (e.g. They live in a white house, not a yellow one! ’). The items on the right, stressed on the first element like gréenhouse, are

generally classified as compounds – though this stress pattern applies consistently only to compound nouns, not to compounds in other word classes.

Apart from stress, a second criterion traditionally used for distinguishing compounds from phrases is semantic: a compound tends to have a meaning that is more or less idiosyncratic or unpredictable. This is true of most of the compounds in (1)–(5). This criterion must be treated with caution, however, because, being semantically unpredictable does not correlate exactly with being a word. All the same, it is true that words are more likely to be lexical items than phrases are, so treating semantic idiosyncrasy as an indicator of compound status will not often be misleading. All the compounds in (1)–(5) are nouns, and compound nouns are indeed the commonest type of compound in English.

Topic- 038: Compound Verbs and Adjectives

Verbs formed by compounding are much less usual than verbs derived by affixation. Nevertheless, a variety of types exist which may be distinguished according to their structure:

- (1) verb–verb (VV): stir-fry, freeze-dry
- (2) noun–verb (NV): hand-wash, air-condition, steam-clean
- (3) adjective–verb (AV): dry-clean, whitewash
- (4) preposition–verb (PV): underestimate, outrun, overcook

Only the PV type is really common, however, and some compounds with under-, over- and out- do not need to be classed as lexical items. For example, out- can create a transitive verb meaning ‘outdo in Xing’ from any verb denoting a competitive or potentially competitive activity (e.g. outsail, outsing, outswim), while new words with over- can also be created freely (e.g. overpolish, overcriticise, over bleach).

All these compounds have a verb as the rightmost element, and also that, with most of them, the activity denoted by the compound as whole is a variety of the activity denoted by that rightmost element. Let us call these compounds right-headed, the rightmost element being the head. Most English compounds are right-headed, but not all.

On the analogy of (1-4) here are some examples of right-headed compound adjectives:

- (5) noun–adjective (NA): sky-high, coal-black, oil-rich
- (6) adjective–adjective (AA): grey-green, squeaky-clean, red-hot
- (7) preposition–adjective (PA): under full, overactive

As with verbs, it is the type with the preposition over as its first element that seems most productive, in that new adjectives of this type, with the meaning ‘too X’, are readily acceptable: for example, overindignant, oversmooth.

In overactive at (12), the head of the compound is the adjective active derived from the verb act. In structure, therefore, this adjective is not a mere string of morphemes (over + act + -ive), but rather a nested structure: [over[act-ive]]. Adjectives with a VA structure, corresponding to the VV verbs at

(2), would resemble a hypothetical ‘float-light’ ‘light enough to float’ or ‘sing-happy’ ‘happy enough to sing’. One actual example is fail-safe ‘designed to return to a safe condition if it fails or goes wrong’. However, other such compounds scarcely exist, even though it is easy enough to find plausible meanings for them. This reflects the relative reluctance of verbs to participate in compounding generally in English.

All the compounds in (5)–(7) are right-headed. There are also a few compound adjectives that are not right-headed.

Topic- 039: Compound Nouns and Headed and Headless Compounds

It is with nouns that compounding really comes into its own as a word forming process in English. That is not surprising. Cultural and technical change produces more novel artefacts than novel activities or novel properties. These changes therefore generate new vocabulary needs that are more readily answered by new nouns than by new verbs or adjectives. Examples can be found with each of the other main word classes supplying the left-hand element:

(13) verb–noun (VN): swearword, drophammer, playtime

(14) noun–noun (NN): hairnet, mosquito net, butterfly net, hair restorer

(15) adjective–noun (AN): blackboard, greenstone, faintheart

(16) preposition–noun (PN): in-group, outpost, overcoat

All of these have the main stress on the left – a characteristic important for distinguishing compound nouns from noun phrases. (The fact that hair restorer, butterfly net and mosquito net are spelled with a space does not affect the fact that, from the grammatical point of view, they each constitute one complex word.) Most of these are also right-headed.

Consider the four examples at (14). Does each one have a precise interpretation that is clearly the most natural, on the basis of the meanings of their two components? For hair restorer, the answer is surely yes: it most naturally denotes a substance for restoring hair growth. On the other hand, for hairnet, butterfly net and mosquito net the answer is less clear.

What tells us that a hairnet is for keeping one’s hair in place, while a butterfly net is for catching butterflies and a mosquito net is for keeping mosquitoes away? This information does not reside in the meaning of net, nor in the meanings of hair, butterfly and mosquito. The most that one can conclude from these individual meanings is that each is a net that has something to do with hair, butterflies and mosquitoes respectively.

Arriving at the precise meanings of these compounds depends on our knowledge of the world (that some people collect butterflies, and that mosquitoes can carry disease) rather than on purely linguistic knowledge.

The difference in precision with which we can interpret hair restorer on the one hand and hairnet etc. on the other hinges on the fact that restorer in hair restorer is derived from a verb (restore). Verbs, unlike most nouns and adjectives, impose expectations and requirements on the noun phrases that accompany them in the sentence. For example, with the verb sleep we expect to find one noun phrase as subject; with

eat we expect to find also a noun phrase as object; and with give we expect to find, or at least to be able to identify from the context, a third ‘indirect object’ noun phrase denoting the recipient of the gift. These expected or required nominal concomitants to a verb are called its arguments.

For present purposes, what matters is that, when the head of a NN compound is derived from a verb, as restorer is, the most natural way to interpret the whole compound is quite precise: the first element expresses the object argument of the verb (that is, the person or thing that undergoes the action). For example, an X-restorer, whatever X is, something or someone that restores X. Here are some more compounds whose second element is derived from a verb:

(17) sign-writer, slum clearance, crime prevention, wish-fulfilment

For all of these, the most natural interpretation is clear. To interpret any of them some other way – for example, to interpret crime prevention as meaning not ‘prevention of crime’ but ‘use of crime for preventive purposes’ – seems contrived and unnatural.

It is time to introduce some terminology, for convenience. Let us call a NN compound like hairnet or mosquito net, in which the right-hand noun is not derived from a verb and whose interpretation is therefore not precisely predictable on a purely linguistic basis, a primary or root compound. Let us call a NN compound like hair-restorer or slum clearance, in which the first element is interpreted as the object of the verb contained within the second, a secondary or verbal compound. (Yet another term sometime used is synthetic compound.) Paradoxically, then, although verbs are relatively rare as elements in compounds in English (the swearword pattern is unusual), verbal compounds, in the sense just defined, are common.

Secondary compounds are certainly right-headed, in that (for example) crime prevention denotes a kind of prevention and wish-fulfilment denotes a kind of fulfilment. In this respect they are like most Compounds and most compounds generally – but not all.

Headed and headless compounds

The AN compounds given at (15) included faint heart alongside blackboard and greenstone. However, whereas a greenstone is a kind of stone and a blackboard is a kind of board, a faint heart is not a kind of heart but a kind of person – someone who has a faint heart, metaphorically. So, although heart is a noun, it is not appropriate to call heart the head of the compound.

Rather, faint heart is headless, in the sense that its status as a noun is not determined by either of its two components. Similar headless AN compounds are loudmouth and redshank (a kind of bird that has red legs), and headless NN compounds are stickleback (a kind of fish with spines on its back) and sabre tooth.

A few VN-type compound nouns resemble secondary compounds in that the noun at the right is interpreted as the object of the verb:

(18) Pickpocket, killjoy, cutpurse

These too are headless, in that a pickpocket is not a kind of pocket, for example. An implication of these analyses is as follows: if the fact that heart and pocket are nouns is really irrelevant to the fact that faint heart and pickpocket are nouns too, we should expect there to be some headless nouns in which the

second element is not a noun at all – and likewise, perhaps, headless adjectives in which the second element is not an adjective. Both expectations turn out to be correct. Some nouns consist of a verb and a preposition or adverb:

(19) take-off, sell-out, wrap-up, sit-in nouns are sometimes formed from verbs by conversion that is with no affix. The nouns at (19) can be seen as a special case of this, where the base is a verb plus another word (sometimes constituting a lexical item), as illustrated in (20):

- (20) a. The plane took off at noon.
 b. The chairman wrapped the meeting up.
 c. The students sat in during the discussion.

As for headless adjectives, there are quite a number consisting of a preposition and a noun:

(21) Overland, in-house, with-profits, offshore, downmarket, upscale, underweight, over-budget

The adjectival status of these compounds can often be confirmed by their appropriateness in comparative contexts and with the modifier *very*:

- (22) a. They live in a very downmarket neighborhood.
 b. This year's expenditure is even more over-budget than last year's.

The fact that the word class of these headless compounds is not determined by any element inside them (that they have no internal 'Centre', one might say) has led some grammarians to call them *exocentric* – that is, having a 'centre' outside themselves, figuratively speaking. According to this approach, headed compounds would be regarded as having an internal 'centre'; and, sure enough, they are sometimes called *endocentric*.

Topic- 040: Blends and Acronyms

In all the examples that we have examined so far, the whole of each component root (or base) is reproduced in the compound. Sporadically, however, we encounter a kind of compound where at least one component is reproduced only partially. These are known as *blends*. A straightforward example is *smog*, blended from *smoke* and *fog*; a more elaborate one is *chortle* (first used by Lewis Carroll in *Through the Looking Glass*), blended from *chuckle* and *snort*.

Examples of partial blends, where only one component is truncated, are *talkathon* (from *talk* plus *marathon*) and *cheeseburger* (from *cheese* plus *hamburger*). The ready acceptance of *cheeseburger* and similar blends such as *beefburger* and *vegeburger* may have been encouraged by a feeling that *hamburger* is a compound whose first element is *ham* – scarcely appropriate semantically, since the meat in a *hamburger* (originally a kind of meat pattie from *Hamburg*) is *beef*.

An acronym is a word or name formed as an abbreviation from the initial components in a phrase or a word, usually individual letters (as in *NATO* or *laser*) and sometimes syllables (as in *Benelux*).

Their primary purpose is to save time and space – reducing phrases to a collection of letters far easier and quicker to say and understand – but they have also turned into something of a fashion statement among the younger generation.

Acronym are also pluralized like a noun phrase. Often a writer will add an 's' following an apostrophe, as in "PC's". However, some writers would prefer "DVDs" and "URLs".

Possessive plurals that also include apostrophes for mere pluralization and periods appear especially complex: for example, *the C.D.'s' labels* (the labels of the compact discs). In some instances, however, an apostrophe may increase clarity: for example, if the final letter of an abbreviation is *S*, as in *SOS's* (although abbreviations ending with *S* can also take *-es*, e.g. *SOSes*), or when pluralizing an abbreviation that has periods.

The most extreme kind of truncation that a component of a blend can undergo is reduction to just one sound (or letter), usually the first. Blends made up of initial letters are known as acronyms, of which well-known examples are NATO (for North Atlantic Treaty Organisation), ANZAC (for Australian and New Zealand Army Corps), RAM (for random access memory), SCSI (pronounced scuzzy, from small computer systems interface), and AIDS (from acquired immune deficiency syndrome). Intermediate between an acronym and a blend is sonar (from sound navigation and ranging).

The use of capital letters in the spelling of some of these words reflects the fact that speakers are aware of their acronym status. It does not follow that any string of capital letters represents an acronym. If the conventional way of reading the string is by pronouncing the name of each letter in turn, as with USA and RP (standing for the 'Received Pronunciation' of British English), then it is not an acronym but an abbreviation.

It is clear from these examples that blending and acronym are in active use for the creation of new vocabulary. However, they differ from derivational affixation and normal compounding in being more or less self-conscious, and are concentrated in areas where the demand for new noun vocabulary is greatest, such as (currently) information technology.

Whether you realize it or not, you probably use acronyms on a daily basis. Acronyms are simply abbreviations created from the first letter of each word in a phrase or series of words. The word ATM, for example, is an acronym that stands for automated teller machine.

An acronym (sometimes called an initialism) is simply a word formed by taking letters (usually the first) of each word in a phrase to form an abbreviation. Acronyms are thus a subset of abbreviations. (Notice that they may be pronounced as words themselves or spelt out depending on the case.)

The main importance of abbreviation is to provide a quick, easy and convenient way of recognizing or indicating something. Any abbreviations for any group of words should not be in a manner it makes hindrances to person. Today, in this technological advancement time, there is plenty of full form, abbreviations, acronyms especially when emergence of social media.

Topic- 041: Compounds Containing Bound Combining Forms

Most of the compounds that we have looked at so far involve roots that are free forms. But the vocabulary of English, especially in scientific and technical areas, includes a huge repertoire of compounds that are made up of bound roots, known as combining forms. Here are just a few:

(23) anthropology, sociology, cardiogram, electrocardiogram, retrograde, retrospect, plantigrade

For most of these, the meaning of the whole is clearly determinable from that of the parts: for example, anthrop(o)- ‘human’ plus -(o)logy ‘science or study’ yields a word that means ‘science or study of human beings’, and planti- ‘sole (of foot)’ and -grade ‘walking’ yields a word meaning ‘walking on the soles of the feet’. This semantic predictability is crucial to the coining of new technical terms using these elements.

Apart from containing bound roots, anthropology differs in two other ways from most compound nouns. Firstly, it has a central linking vowel-o- that cannot conclusively be assigned to either root. In this respect it resembles many combining-form compounds. Secondly, although it is a noun, its stress is not on the first element – unless the linking -o- belongs there. In this respect it resembles e.g. monogamy, philosophy and aristocracy. We encountered bound roots that could function as the base for derivational affixation, such as aud- in audible, audition etc.

Not surprisingly, some combining forms can function in this way too (in other words, the dividing line between combining forms and other bound roots is not sharp): for example, soci- and electr(o)- from (23) also occur, indeed much more commonly, in social and electric.

Given that combining forms, and the compounds that contain them, are so untypical of compounds in general, it is natural to ask how English has come to acquire them. In fact, they come mostly from Greek or Latin, through deliberate borrowings to supply new needs for technical vocabulary that arose partly from the revival of learning in Western Europe in the fifteenth and sixteenth centuries known as the Renaissance, and partly from the industrial revolution of the eighteenth century and its scientific spin-offs.

Topic- 042: Phrasal Words

In some of the compounds that we have looked at so far, relationships are expressed that are the same as ones expressed in syntax: for example, the verb–object relationship between hair and restore in hair restorer. On the other hand, the way in which the verb–object relationship is expressed in this compound is quite different from how it is expressed in syntax, in that the two words appear in the opposite order: we say this substance restores hair, not *This substance hair-restores. There is a clear difference between compound word structure and sentence structure here. But there are also complex items that function as words, yet whose internal structure is that of a clause or phrase rather than of a compound. There is no standard term for these items, so we will introduce the term phrasal words.

An example of a phrasal word is the noun jack-in-the-box. Structurally this has the appearance of a noun phrase in which the head noun, jack, is modified by a prepositional phrase, in the box, exactly parallel to the phrases people in the street or (a) book on the shelf. However, it forms its plural by suffixing -s not to the head noun (as in books on the shelf) but to the whole expression: not ‘jacks-in-the-box’ but jack-in-the-boxes, as in they jumped up and down like jack-in-the-boxes. Though structurally a phrase, then, it behaves as a word. Contrast this with another item which is at least as characteristic in meaning and

which has a superficially similar structure: brother-in-law. A crucial difference is that brother-in-law forms its plural by affixing -s not to the whole expression but to the head noun: brothers-in-law. Despite its hyphens, therefore, brother-in-law is not a word at all but a phrase.

Can phrases other than noun phrases constitute phrasal words? The answer is yes. Adjectival examples are dyed-in-the-wool (as in a dyed-in the-wool Republican) or couldn't-care-less (as in a couldn't-care-less attitude).

Syntactically, dyed-in-the-wool looks like an adjective phrase consisting of an adjective (dyed 'artificially coloured') modified by a prepositional phrase, just like suitable for the party or devoted to his children. However, such a phrase cannot entirely precede the noun it modifies (we say a man devoted to his children or suitable music for the party, not *a devoted to his children man or *suitable for the party music); therefore the behaviour of dyed-in-the-wool is that of a word rather than a phrase. As for couldn't-careless, its structure is that of a verb phrase, but again its behaviour is that of an adjective (e.g. Your attitude is even more couldn't-care-less than hers!).

This seems an appropriate point to mention a small and rather old fashioned class of lexical items exemplified by governor general, attorney general, court martial and lord lieutenant. How do they form their plural: like attorney generals, or like attorneys general? If you prefer the former, then these items may seem at first like further phrasal words – except for the fact that they differ from normal English noun phrases in having an adjective following the noun rather than preceding it. It seems better, therefore, to treat them as examples of something that we have not so far encountered: endocentric words which, untypically, have their head on the left rather than on the right. On the other hand, if you prefer the latter sort of plural (attorneys general), they seem more akin to brother(s)-in-law: not words but lexicalised phrases. If, finally, neither kind of plural sounds quite right to you, that is not surprising, because however these items are analysed, their structure is unusual.

Lesson 08

MORPHOLOGICAL STRUCTURES**Topic- 043: Introduction**

The meaning of a word is not obtained just by adding the individual meaning of each morpheme. The way they are combined provides aspects of the word's semantics in a compositional way. This is what is called a structure. Consider the undecidable

Three units

Prefix Un-, a verbal roots, decid(e), a suffix, -able

Let us derive the meaning step by step

Undecidable roughly means: 'cannot be decided'

The prefix un- has a negative also seen in adjective like unaware (not aware), unfriendly(not friendly), etc. this prefix is responsible for the negative meaning contained in the word, 'cannot be decided' the suffix – able has more complex meaning: it form adjectives from verbs, and its meaning is at the same time passive and modal, and can be glossed as ' that can or deserve to be PARTICIPLE', where participle is the one of the verb with which –able combines. We see this in enjoyable (can be enjoyed) ,and readable (that can be read). This suffix is responsible for another part of the meaning of the word. Once we have decomposed the meaning of the word and associated it with each one of its units, we can establish the correspondence in the following:

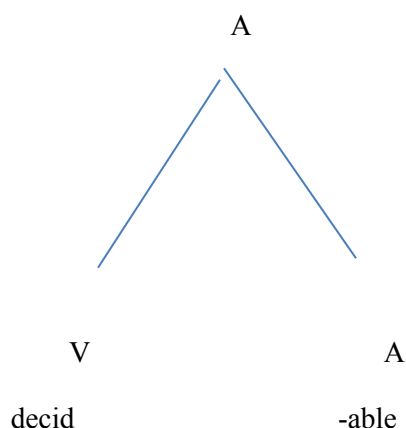
Un-	decid(e)	-able
not	that can be PARTICIPLE	

The question that arises at this point is why the meaning of the word is 'cannot be decided' and not 'can be not decided'. In the first case we refer to an entity that, in any case and not matter how hard one tries, cannot be decided; in the second case, we refer to something that can be decided, but which does not need to be. The way in which a speaker uses the word undecidable is the first and never the second. How can we account for this?

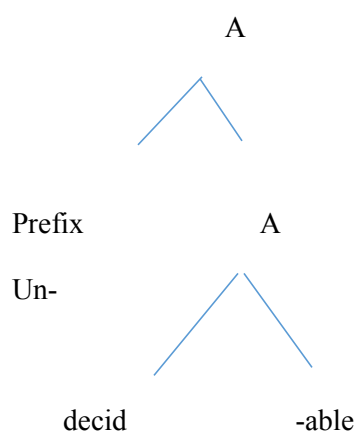
It is at this point that the notion of structure comes into play. The structure determines the morpheme with which another morpheme combines, and therefore how the meaning of each of them relates to the others. Identifying the structure of undecidable involves giving an account of how the three units in (1) combine with each other, and then of how their meanings affect each other. If we simply state that the word has three units, as we did in (1), we do not predict that only one of these two meanings is associated with the word. We need to represent also how the units are organized inside the word, that is, the structure of the word.

The difference between the real meaning of the word, 'cannot be decided', and the unattested meaning, 'can be not decided', has to do with the position that the negative morpheme occupies in the structure: in the first case it modifies that suffix- able, and in the second it relates to the verb. We need to propose a

structure where this property is clearly accounted for. In a first step, we combine the suffix-able and the verb decide.



This gives us the meaning ‘can be decided’. One of the units is more important than the other from the point of view of the grammatical category. In this case, this unit is the suffix-able, which combines with a verb, but gives as a result an adjective. The structure represents a possible word in English, the adjective decidable, formed from the verb decide. This complex word, in a second step, combines with the negative prefix un-, as in (3).



In (3), the prefix un- combines with an adjective, decidable, not with the verbal root decid(e). This accounts for the fact that, in the meaning of the whole word (‘cannot be decided’), the negation affects the combination of verb and –able, and not just the verb decid(e). The structure in (3) accounts for the fact that the meaning ‘can be not decided’ does not correspond to the word undecidable. This meaning would need a structure where the negative prefix attaches to the verb ([un[decide]] and then the whole is turned into an adjective ([un[decide]]able)), which is not theoretically impossible, but is not the way in which the word has been stored in the English lexicon.

There is a second reason why un- must combine with the structure built by the suffix -able and not with the verbal root directly. If we make a list of the English words that contain the prefix un-, we will see that the vast majority of them are adjectives (unaware, unclear, untrue, etc.) This tells us that, when the prefix

has a negative meaning, it generally combines with adjectives. The prefix un- can combine with verbs, but in these cases it has a reversative meaning, that is, the meaning of reversing an action that had already been performed, as in undo, untie, unzip, unlock or unfold, which do not mean ‘not doing’, ‘not zipping’, etc. The reversative meaning is not the one used in this word, as we expect if the prefix combines with something that is now an adjective. Thus the structure in (3) not only gives an account of the meaning of the word undecidable, but also motivates why the prefix that appears inside it has the meaning that it has.

Let’s now define what is considered a structure in morphology.

The morphological structure is the organisation of the units that form a word making it possible to account for its semantic and formal properties.

As for the formal properties, the structure shows several things:

- A) The grammatical properties of the whole word. The whole structure is an adjective. This means that the whole word is an adjective.
- B) The unit that is responsible for the fact that the word belongs to that particular category. The word is adjective because its last segment is –able. Units like these which determine the category of the word are called HEADS
- C) The combination of the units that produce possible words, and what their meaning is in each step. The structure in (3) explain the prefix un- has a negative, and not reversative meaning in the word undecidable.

Topic- 044: Properties of Morphological Structures

Any structure is different from a linear series of combined. In the latter, all relationships between units are identical, while in the former there is always a unit which stands in a privileged position inside the structure. This unit is the head, which imposes its properties on the whole word. In the word un-decid(e)-able, the head is the suffix –able, as it transmits a number of properties to the whole word:

- a) The grammatical category. The word undecidable is an adjective, and the affix –able is responsible for this categorization. Words ending in –able are normality adjectives.
- b) The semantics. The word undecidable denotes a quality, the quality of being unable to be firmly established or refuted. This meaning contribution is associated with affix –able, as all words ending in –able (or -ible) denote the quality of being able to undergo a particular process (demonstrable, deducible, inexpressible...)

The head inside a morphological structure is identified because of the properties of the whole word are the same properties that the morpheme itself contain.

The properties by the head are crucial in determining the behaviour of the word. The grammatical category is crucial when the word is used in syntactic context, as it determines what kind of other words it can combine with, and the meaning of the word determines its semantic behaviour.

A third class which is generally referred to as ‘formal’ or morphological’, in this sense its members do not affect the syntactic distribution of the word or its semantic behaviour. But specify some morphological properties of the word, such as its gender, its conjugation class or whether its inflection is regular or

irregular. Consider the English verb undersell consist of two units under- and –sell the unit –sell is the head. It imposes its grammatical category and meaning. This unit is also responsible for irregular inflection.

However, in syntactic analysis a head is the only unit in a structure which can stand alone. In the phrase

‘The big boy’ the head is ‘boy’ and whose absence will make an ill-formed sentence.

*The big failed the examination.

The head in morphology is frequently a bound morpheme which cannot stand alone without having been combined with other morpheme. As in government –ment cannot stand alone. Thus syntactic criterion is not followed by morphology.

Topic- 045: A-Morphous Morphology

The existence of morphological structure has certain controversies. A number of proposals claim that words do not have an internal structure. Let’s look at this influential theory that argues for this view.

A-Morphous Morphology

This theory proposed by Anderson (1992) argues that words do not have internal structure. Anderson gives a number of arguments against that analysis that proposes internal structure in words. A structure is defined as a set of units organized in a particular way. A prerequisite for having a structure is identifying a set of units that can be organized this commits a theory that argues for word-internal structure.

Anderson’s first argument against this view of word-internal structure is that the relevant units, morphemes do not exist on the basis of the problems that the classical notion of morpheme encounter.

Anderson’s view of morphology is REALIZATION, a system where there are no one-to-one correspondences between the meaning and the form of the word. In a system like this the meaning of a complex word cannot be established by identifying smaller units which contribute their meaning to the whole as in cranberry morpheme having two units’ cran- and –berry. The meaning cannot be explained by combining the meaning of the two.

A-morphous morphology proposes that word meaning cannot be reconstructed in that way because units we have segmented are not real. The word cannot be divided into smaller units, and, by implication it is impossible to identify any internal structure.

A-morphous provides a second argument to show that show the intuitions of a speaker about meaning-form correlation are not reliable. The argument has to do with PHONOAESTHETICS, that is, sequences of sounds which evoke a particular meaning. Anderson notices the words such as slide, slip, slither, slink and slippery share, in the intuition of most speakers, a part of their meaning and phonological sequence, /sl/, but it would not be justified to propose that sl- is a morpheme in English.

Topic- 046: Exocentricity

In these cases, it has been proposed that the word lacks a head and, therefore, it is dubious whether there is an underlying structure in that object.

Exocentricity has been studied especially in the domain of compounding, so we will concentrate on these cases here. The two cases presented in (21) have been claimed to illustrate exocentricity in compounding, though for different reasons.

(21) a. to grandstand

b. Pick pocket

The case in (21a) is exocentric because the unit that seems to impose its grammatical category (verb) on the whole object does not impose the rest of its properties on the word. The participle of grandstand is not grandstood, which is surprising if the head of construction is the verb stand, whose participle is stood. This case is to be contrasted with understand, derived from stand by the prefix under-.

The compound in (21b) is exocentric in the sense that none of its elements can explain the semantics of the word: a pick pockets is a kind of thief, and thus a kind of human being, not a kind of pocket or a kind of pick. Also, the word is a noun, but the unit is also a noun inside and compound (pocket) is an unlikely candidate for the head of the structure, as it does not seem to be responsible for introducing pick-rather the contrary, as the gloss would be ‘someone who picks the pockets of other people’, not ‘a pocket used to pick’ or ‘something similar. Williams (1981) proposed that these objects are created through exocentric rules, that is transform the properties of the object, but do not add any new unit that introduces these properties. However, accepting exocentric rules amounts in practice to accepting that some words do not have a complete morphological structure.

As we can see, Exocentricity is used in practice to label all the situations where the properties that an object has are not the immediate reflection of the properties of one, and only one, of its units. For this reason, it seems appropriate to classify the different kinds of exocentricity depending on the kind of property that shows the mismatch. We will discuss, in turn, category features, semantic features, and morphological features such as whether a verb is irregular or regular, or to which gender it belongs.

Topic- 047: Bracketing Paradoxes

A crucial property of a structure is that both the formal and the semantic properties of the word need to be represented in the structure. For this reason, the situations where the semantic and the formal properties of an object do not coincide are problematic for the idea that words have internal structure. These situations are known as BRACKETING PARADOXES (Williams 1981, Beard 1991). There are three kinds of bracketing paradox.

One kind of bracketing paradox is the situation in which the formal properties of one of the units requires it to combine with a particular base, but the meaning of the word suggests that it combines with a unit smaller than the base. Take the word international. The formal properties of the word require the prefix inter- to combine with the adjective national, as the word **internation* does not exist in English. This structure is represented in (24a). However, this structure gives the wrong semantics for the word, ‘relation between at least two national things.’ The word means more precisely ‘among nations’. Thus, as far as the meaning of the word goes, the prefix does not combine with the adjective national, but with the noun nation from which national is formed (as in the structure in (24b)). However, (24b) is impossible on formal grounds, because the prefix cannot combine directly with the noun nation, as **internation* does not

exist, and this structure contains inside it this impossible word. Thus, the two possible structures make wrong predictions at some level.

(24) a. [inter [[nation]N al]A]A

b. [[inter[nation]N]N al]A

Another kind of paradox comes when a unit combines with a word but semantically modifies not only the word, but a bigger structure in which the word is included. Consider the expression *Vulgar Latinist*. This does not refer to Latinist who happens to be *Vulgar*, but to someone who studies *Vulgar Latin*. This requires the suffix *-ist* to combine with whole phrase *Vulgar Latin*, and not only with the noun *Latin*, because that would give us the wrong semantics. However it is generally assumed that one of the properties of affixes is that they cannot combine with phrases, but only with morphological objects (§ 7.2.1)

Last kind of paradox takes into account phonological factors. In the case *unhappy*, there is a paradox in the sense that the meaning of the adjective (more *unhappy*) require the suffix *-er* to combine with the trisyllabic adjective *unhappy*. It is known that generally trisyllabic adjective take the adverb *more* in the comparative. Avoiding this phonological infraction implies proposing that *-er* combines with *happy* and then *un-* is combines with *happier*. This gives the wrong semantics as it mean; not more happy' which the speaker does not interpret. There is no analysis for bracketing in the general case which covers the three paradoxes

Topic- 048: Paresynthesis

The process of forming words by both compounding and the addition of derivational suffixes (Ex.: *big-hearted* from *big heart* + *-ed*, not *big* + *hearted*). The phenomenon known as *Parasynthesis* constitutes a problem for morphological structures because it casts doubt on one of its crucial properties, namely binary branching. Remember that non-binary branching was rejected before on the grounds that when more than two units are combined at a time the relationship which is established between them is not unambiguously stated.

Parasynthesis is the situation where two different affixes - normally a prefix and a suffix - seem to be added simultaneously to the same base. One example of this is the Spanish verb *engrandecer* 'to enlarge', from the adjective *grande* 'big.' Neither the set formed by the base and the suffix (a hypothetical verb **grandecer*) nor that which includes the prefix and the base (a hypothetical adjective **engrande*) constitutes a word in Spanish. In consequence, if it is assumed that only existing words can form the basis for further derivation, one is forced to conclude that the verb *engrandecer* has been formed by adding two units at the same time, the prefix *en-* and the suffix *-ece*, to the base *grande*, as represented in (26a), where three units, and not two, have been put together. The same structure, with a zero suffix, has also been proposed in English for words such as *enlarge* (27b); in this way, the analysis avoids saying that the prefix *en-* is responsible for the category change, since in English prefixes generally do not have this ability.

It is crucial in *Parasynthesis* that the combination with only one of the two affixes does not produce an existent word in the language. In other words, English *discontented* is not a case of *Parasynthesis*, because

the root and the prefix form an existent word (discontent); equally, the root and the suffix form one (contented).

Different proposals have been made to avoid the ternary structure in (26). One of the most widely accepted is to consider the prefix and the suffix to be a discontinuous morpheme, a circumfix (§1.2.1). In this way, the structure would be binary branching, as only two units, [en . . . ece] and [grande], are combined. A phonological property of the first unit would force it to appear around the base and not only to one of its sides. A variation of this analysis, framed in the DM approach, is to consider these discontinuous affixes as single units in the structure which are forced, by a process called fission, to materialize as two different morphemes.

The analysis of Parasyntesis as a circumfix, however, has an empirical problem. Notice that the circumfix that should be proposed in the case of *engrandecer* has a second part which is identical to a Spanish suffix (27) and a first part that, again, is identical to a Spanish prefix (28). The fact that these units exist independently of each other suggests that it is not accurate to propose that a circumfix is responsible for the verb *engrandecer*.

Another analysis of the parasyntetic formations that is compatible with the binary branching hypothesis can be found in Scalise (1983), where it is proposed that first the suffix is added, transforming the base (a noun or an adjective, as *grande*) into verb (*grandecer*, which would be a possible but non-existent word in Spanish), and only then is the prefix added (*engrandecer*). The weak side of this analysis is that has to postulate an intermediate phase with a possible but non-existent word hypothesis, *grandecer*, a possibility that some linguists accept but others deny.

Lesson 09**DERIVATION AND SEMANTICS****Topic- 049: Introduction**

The Utne Reader quoted the following Bucharest sign in its March–April 1996 issue: The lift is being fixed for the next day.

During that time we regret that you will be unbearable.

Unbearable is a perfectly good English word derived from the verb bear ‘carry’, or is it? Why is this sign so funny? First of all, unbearable already exists in English with the meaning ‘difficult to tolerate’, and this is the meaning we first think of when we read the sign. But this isn’t the whole story. If we replace unbearable with a similar but unambiguous word, like transportable, the sentence is no longer humorous, but it still sounds less than native. It’s because words like unbearable, untransportable, and uneatable describe inherent qualities of people or things, qualities that don’t change simply because an elevator is out of order. To take another example, the fact that someone is allergic to chocolate doesn’t make the chocolate cupcakes in the window uneatable.

In this simple sign, there is a complex interaction between affixation and semantics going on, and that is the sort of thing that we explore in this chapter. We begin by introducing a fundamental problem in lexical semantics, the study of word meaning: the meanings of individual lexemes can be highly diverse. We then examine in some detail the semantics of derived lexemes to see what generalizations we can draw.

Topic- 050: The Polysemy Problem

The most fundamental aspect of a word’s meaning is that it refers to some entity or relation (real or imaginary) in the world. We can refer to this entity or relation as the word’s semantic type. The word reptile refers to all individuals in the world that are reptiles. Verbs like respect or love refer to relationships between individuals. Formal approaches to grammar have provided us with terminology that allows us to make even more fine-tuned distinctions between words. We can differentiate bear from teddy bear by saying that the first is animate but the second is not, or foliage from leaf and literature from book on the basis of the mass/count distinction.

Verbs are given labels such as ergative, accusative, transitive, or intransitive. The main problem of lexical semantics is that the meanings of individual lexemes are highly diverse. We call this the problem of polysemy. As an example, take the verb lose. Lose has different meanings in the following sentences: They lost their passports; Jake lost his job; Sarah lost her husband to cancer; I lost my temper; we both lost ten pounds.

But all of the meanings of lose reflected here are related – they are all instances of the same lexeme. Because lose has more than one related meaning, we say that it is polysemous. There are many types of polysemy. In the next several examples, we present some general types of polysemy, focusing on nouns. Wholesome nouns are inherently mass or count nouns, others can be either, as with watermelon:

(1) a. I love watermelon. (mass)

b. I sold three watermelons. (count)

Besides the mass/count alternation, there are several other well-known alternations we find in noun meaning. A few are illustrated below:

(2) Figure–ground reversal

a. Hugh broke the window.

b. The kids climbed through the window.

(3) Container–contained alternation

a. A hot glass put under cold water will shatter.

b. Franny downed the glass in two seconds flat.

(4) Place–people alternation

a. The president and his family live in the White House.

b. The White House announced yesterday that the peace talks will continue.

The sentences in (2–4) contain pairs of words with very different – even contradictory – interpretations. Yet they represent single lexemes. In (2a) window refers to a solid barrier, but in (2b) an aperture; glass (3a–b) can refer to the container or to the liquid inside. Examples like these show that the same phonetic string can convey different, but related meanings depending on the linguistic and pragmatic context. This introduction to polysemy leads into the discussion which focuses on the semantics of derived lexemes.

Topic- 051: The Semantics of Derived Lexemes

When somebody makes up a word, they are inventing it for use under a particular circumstance. Sometimes the circumstance can be very peculiar. Take the sentence in (6):

(6) Joe was Houdini'd and died.

In order to understand this sentence, you have to know something about Houdini – a famous escape artist – and how he died. He died following a series of punches to his stomach (a fan was testing the strength of his abdominal muscles, which Houdini prided himself on).

So when we say “Joe was Houdini'd and died,” we mean he was punched in the stomach and died in the way that Houdini did. This is a dramatic example of how you might need to know pragmatic factors in order to understand a particular lexeme. We say more about examples like (6) below.

The second factor that can affect a word's meaning is its history. We might think of every lexeme not just as a word and its meaning, but as the word and every time it has ever been used: every time we hear the word, we revise its lexical entry in some way. That this indeed goes on is particularly evident from first-language acquisition research.

Children in earlier stages of language acquisition may **under extend** a word by using it to refer to only a subset of its actual referents, or **overextend** a word by using it to refer to objects or individuals that are typically covered by the word, as well as to others that are “perceptually similar” (Clark 1993: 33). For example, a child might under extend the word *dog* by using it to refer to more typical examples of the species, but not to varieties like Chihuahua or Pekingese (Kay and Anglin 1982), or overextend *tree* by using it to refer to potted plants, trees, and even balsam fir wreaths.

Such under- and overextensions are generally short-lived, which indicates that children revise lexical entries as they are exposed to more and more tokens of a word. It is not unreasonable to think that the meaning of a word is a compilation of every single use of that word that you have ever heard or said. Every word has a history. It has your own personal history – how you have heard the word. It has the history of the word as it has been used by other people. Over time, the meanings of words can become more complex and diverse, making the task of the morphologist looking for semantic patterns of word formation more complicated than it would be if the semantics of word formation were purely compositional (as the semantics of syntactic constructions are often considered to be). A syntactic construction may have pragmatics to deal with, but it doesn’t have history.

One question you might want to ask is what kinds of meanings arise via lexeme-formation rules. Are derived forms like lexemes, with potentially very complicated meanings? Or are they like syntactic collocations, with simple meanings?

Topic- 052: The Semantics of Affixation

Let’s take the English suffix *-ism*. This affix has some very highly lexicalized meanings, one of which is ‘doctrinal system of principles’. We find this meaning in words like the following, and many others having to do with religion, philosophy, science, politics, or the arts:

(7) Catholicism	Platonism	romanticism	McCarthyism
Judaism	Marxism	realism	socialism
Buddhism	idealism	surrealism	fascism

The suffix *-ism* has an even more specific and lexicalized meaning:

‘a peculiarity of speech’. We talk about *colloquialisms*, *spoonerisms*, and *Obamaisms* (*Obamaism* can either be a system of beliefs or a peculiarity of speech). So *-ism* is an example of a suffix with two very highly lexicalized meanings, both of which might be considered to be more characteristic of words than of affixes.

The German suffix *-ei* is like *-ism* in having at least two very highly lexicalized meanings. The first, illustrated by the words in (8), attaches to a noun and makes another noun meaning ‘the place in which X works’. The second, seen in (9), attaches to a verb stem and creates a noun referring to the ‘act of doing X’:

(8) a. Bäcker ‘baker’ → Bäckerei ‘bakery’

b. Drucker 'printer' → Druckerei 'printing office, print shop'

c. Sattler 'saddler' → Sattlerei 'saddlery, saddler's workshop'

d. Tischler 'joiner' → Tischlerei 'joinery'

(9) a. plaudern 'chat (v)' → Plauderei 'chat (n)'

b. zittern 'tremble' → Zitterei 'trembling'

c. prügeln 'clobber' → Prügelei 'brawl, fight'

d. quengeln 'whine' → Quengelei 'whining'

English has a cognate suffix of French origin, *-(e)ry*, as in *bakery, tannery, winery, bribery, flattery, foolery*.

Topic- 053: The Semantics of Zero Derivation

We now address a type of word formation which is much more abstract: zero-derivation. Zero-derivation results in lexemes whose interpretation is context-dependent. The data on zero-derived verbs discussed here come from Clark and Clark (1979), but the analysis is that of Aronoff (1980). The peculiarity of zero-derived verbs is that they often have a wide range of meanings. To give you just one example, the verb to sand denotes two very different actions. The most common meaning is 'to rub with sandpaper'. The second meaning is 'to spread or cover with sand', as is done in winter to make roads less slippery.

While sand is well established as a verb, zero-derivation is a productive derivational process in English (cf. 4.2.2), as shown by the following nonce forms presented by Clark and Clark. All of these sentences are actual quotations:

(10) a. Ruth Buzzi house guested with Bill Dodge (Herb Caen, SF Chronicle)

b. He wristed the ball over the net (tennis commentator)

c. When you're starting to Sunday School members, then I think you're going too far (a Californian legislator)

d. Will you cigarette me? (Mae West)

e. We all Wayned and Cagneyed (NY Times magazine)

Clark and Clark classify noun-to-verb derivations into various categories. For each category, they give numerous examples, of which we have given only a few:

(11) Location (N is at a place) blanket, saddle, roof (put something at N) kennel, ground, cellar

Duration (spend the duration of N) summer, holiday, vacation, weekend

Agent (N acts) jockey, referee, umpire, pilot

Goal (make into N) fool, orphan, baby, cripple, pile, loop, powder

Instrument (use N) ship, nail, glue, shampoo, fork

Miscellaneous lunch, hay, whale, dog

The descriptions given in (11) for each of the categories are slightly vague. One of the meanings given for the Location category is ‘put something at N’. To be more precise, verbs like *kennel*, *ground* (e.g., a teenager), or *cellar* involve keeping, not simply putting. Likewise, the true description of the Instrument category is much trickier than ‘use N’ because often you don’t use the noun – you use something else. For example, while shipping originally took place via ship, today we ship things by truck or air. We might redefine this category as ‘do what you do with N’. Even this, however, needs to be interpreted fairly broadly.

The miscellaneous category includes some interesting words. We use the verb *whale* to mean ‘catch whales’ or *fish* to mean ‘catch fish’ but are hard pressed to come up with many other verbs of this type. We don’t use a verb *deer* to mean ‘hunt deer’ or *butterfly* for ‘catch butterflies’.

The verb *dog* patterns with words like *clown* in having the meaning ‘act like a dog’. *Cub*, *foal*, and *pup* all mean to ‘give birth to’ these animals.

The zero-derived verbs with the most extensive semantic possibilities are probably those that are derived from personal names. To understand them, you need to know something about the person and often about a particular event. To understand the sentence in (6), repeated in

(12), you have to know the circumstances of Houdini’s death:

(12) Joe was Houdini’d and died.

To understand the sentences in (13) (from Clark and Clark 1979: 784), you have to know other things about Houdini, namely that he was famous for sensational and seemingly impossible escapes (13a) and that he made a lifetime crusade of showing phony mediums and spiritualists to be frauds:

(13) a. My sister Houdini’d her way out of the locked closet.

b. I would love to Houdini those ESP experiments.

The semantic obscurity of verbs derived from personal names results in speakers forgetting very rapidly that there even was a zero-derivation.

While any English speaker can see the connection between *nurse* (noun) and *nurse* (verb) or *bottle* (noun) and *bottle* (verb), most aren’t aware that *boycott* and *lynch* are of the same ilk.² Robert Lees, in his classic book on English nominals (1960), derived the meanings of denominal verbs from sentences containing them. The verb *summer* would be derived from the phrase *spend the summer* and the verb *kennel* from *keep in a kennel*. March and (1969) has a similar strategy: he associates a denominal verb with a sentence containing the noun from which it is derived. Hale and Keyser (1993) also relate the lexical semantics of verbs to a syntactic structure.

Given the sometimes idiosyncratic array of meanings that zero-derived verbs may have, what kind of semantics can we write for the noun-to-verb rule? We can potentially go two ways. We can both be very

inclusive, or specific, and formulate precise rules that generate all the cases. Or we can do the opposite and write a very general – what we call a sparse – rule. A sparse rule says very little but, as you will see, yields the right answer.

Let's review all of the examples of zero-derived verbs that we have given here and ask what is going on, in general. Each verb has something to do with the noun. Because it is a verb, it has the meaning of some action or activity, and on basic Gricean principles of cooperation (Grice 1975), we know that that action or activity is connected to the noun. Grice tells us that when people speak to one another, they have to assume that they are being cooperative. We give a more specific answer below, but first we need to examine a fact about evaluative adjectives like *good*, *bad*, or *wonderful*.

The meanings of evaluative adjectives may seem to be infinite. *Wonderful* means something very different in the sentences *Moby-Dick is a wonderful book* and *George is a wonderful nurse*. For one thing, the first sentence is true only if *Moby-Dick* is both a book and wonderful. The second sentence can be true even if George is not a nurse but a stockbroker. This is because *wonderful* can be construed as describing either the noun *nurse* or the activity associated with it. George is a *wonderful nurse* as long as he takes care of sick people (such as his kids) quite wonderfully. However, we do not need to say that the meanings of evaluative adjectives are infinite or even that they are polysemous. Instead, following Katz (1964), we can say that evaluative adjectives modify “that component of the meaning of a noun which has to do with the particular respect in which evaluations are made, within the language, of things in the extension of the noun” (p. 751).

(The noun's **extension** is the set of entities it picks out in the world.) We can refer to this as the **evaluative domain** of the word. The evaluative domain of simple lexemes like *nurse*, *knife*, or *dog* is the same as the evaluative domain of zero-derived verbs like *pilot* or *shampoo*. If Mary is good at piloting, it means that she is a good pilot. If this is good shampoo, it is good for what you do with shampoo. This tells us that the mechanisms by which speakers assign meanings to evaluative adjectives and to zero-derived verbs on the basis of context are likely the same.

To restate this analysis, the wide array of meanings of zero-derived verbs results from two properties. The first is that the rule by which they are formed is very simple, specifying only that we take a noun and form a verb. The second is that conversational convention dictates only that the verb have something to do with the noun. We can reasonably pare this analysis down even further and say that the proper analysis of zero-derived verbs is that they are simply verbs. The fact that they denote an activity connected with the noun is derivable on purely conversational grounds. While other derivational formations don't have as dramatic an array of potential meanings as zero-derived verbs, we still find variety.

Topic- 054: English Agent Nouns in -er

We next give you an example of another derivation that works in the same way as zero-derived verbs but is a little simpler. Marchand gives examples of English agent nouns in -er. He points out that they fall into four basic categories, which can be broken down even further into two separate sets.

The four basic categories are listed below:	
Persons:	baker, dancer, gambler, driver
Animals:	pointer, retriever, warbler, trotter
Material objects:	blotter, eraser, fertilizer, shutter
Immaterial objects:	reminder, clincher, thriller, eye-opener
These nouns can be further divided into their habitual and non-habitual uses.	
Habitually:	He is a gambler. we usually mean that he gambles regularly.
non-habitually:	All ticket-holders may enter

There are two possible analyses of agent nouns, both of which are reasonable. We won't try to choose between them here. One is the strategy to assign the derivation a sparse semantic rule. So an agent noun is 'someone or something connected with what the base denotes', or alternatively, 'somebody or something whose function or characteristic is to perform a particular act'. For now we assume the latter. It permits the categories person, animal, material object, and immaterial object, as well as a habitual or non-habitual interpretation. The other method which linguists might use to account for possible meanings of X-er agent nouns involves prototypes, also called archetypes. The idea is that not all members of a given category are equal.

Reasonably the prototypical agentive is a person who habitually performs a particular type of action. So the prototypical agentive is a word like baker, dancer, gambler, or driver, in the habitual sense. These reflect the core meaning of this particular formation. Other forms, like retriever, blotter, or clincher, involve relaxation of the core meaning. We use them to distinguish one specimen from other members of its class.

A pointer

A retriever

When we claim that persons are the prototypical members of the category of agentive nouns, we can also argue that some other members of the category – material objects like screwdrivers, for example – aren't agents at all. Instead, they are instruments, because they don't have will. In short, the second method of analyzing agentives is to establish a central case, the prototype, and to work out from that to get the others.

Lesson 10

INFLECTION

Topic- 055: Introduction

What is the plural of *euro*, the name of the European currency? The answer seems obvious: *euros* – one euro, two euros, voilà! After all, *-s* is the productive plural marker in English and is normally used with new coinages, which is just what we do with words like *modem* or *byte* (*modems*, *bytes*). But spend a little time in Ireland, the only largely English-speaking country in the euro zone, and you will soon discover that a pint of Guinness in the local pub sets you back four euro and not four euros. Why?

The answer lies in the multilingual nature of the euro zone. The name of the currency itself bespeaks such multilingualism: its devisers had to come up with a name that could be both written and pronounced in all of the languages in the zone. Indeed, though it is pronounced differently in each of the languages ([ojro] in German, [oero] in French, and [ɛvro] in Greek, for example), it is pronounceable in all of them. But the plural also must be not only pronounced but also written in each of the languages.

Look at a 10-euro bill and you will see the clever solution to this multilingual problem that was found. First, unlike the US 10-dollar bill, where *ten dollars* is written out in full, or the bilingual Canadian bill, which says *dix-ten dollars*, the word for ‘ten’ is written in numerals: *10*. The reason is simple: *10* can be written identically while still being pronounced differently in each of the languages of the zone, and still mean exactly the same thing. The second problem is that the plural form of *euro*, which follows the numeral, must also be written uniformly on the bill and, though several major European languages besides English use the plural marker *-s* (French, German, and Spanish do), not all would tolerate it (Italian, for example, never uses an *-s* plural).

The bill reads simply *10 EURO/EYPW* and the spoken languages have all followed the same practice of not using any plural marker either, making the plural of euro *euro* in Ireland. The linguistic point of this story is that even when several languages share the same word, as they do with *euro*, the inflectional systems of the languages differ, and this difference normally affects the forms of the words. Inflection varies from one language to another more than any other systematic aspect of language. This chapter will deal with inflectional systems, showing how they vary across languages widely but still remain quite relentlessly systematic within themselves.

Topic- 0056: What is Inflection?

We encountered inflection here in more detail. The word itself comes from traditional Latin grammar. Its root *flect-*, which we see in the English word *flex*, means ‘bend’. We give this etymology to evoke the image of a speaker “bending,” or, perhaps more clearly, altering the shape of a word so it will fit in a particular position within a sentence. Every sentence is a syntactic frame with positions for a series of words. In order to fill one of those positions, you take a lexeme from the lexicon and bend it to fit. In this way, inflectional morphology is determined by syntax.

What kinds of things do lexemes express through inflection? In general we speak of inflection expressing **morphosyntactic** information, syntactic information that is expressed morphologically. This includes the abstract syntactic categories of, tense, aspect number, and case. Specific values for these categories, such

as past, imperfective, plural, or genitive, are generally referred to as **morphosyntactic features** or **morphosyntactic properties**, the latter a term from Matthews (1991). Inflection is the realization of morphosyntactic features through morphological means. What those means are will be addressed later in this chapter. In order to fully understand inflection, we must situate it in the grammar. Since we are claiming that the syntax provides the morphology with morphosyntactic features, the job of the morphology must be to get from there to the actual phonological realization:



This diagram portrays the relationship between the syntax, morphology, and phonology as derivational, but it is equally possible to model a non-derivational, parallel relationship. Either way, a diagram like (1) is bound to be deceptively simple. We are still left asking precisely how words become inflected. We now turn to an exploration of the answers to that question.

The answers to that question is exponence. Exponence refers to the realization of morphosyntactic features via inflection. In seas, the morpheme[z] is the exponent of the morphosyntactic feature plural, In sailed, [d] is the exponent of past tense or past participle. In both cases there is a one-to-one relationship between form and meanings, since one morpheme realizes one morphosyntactic feature, a situation that Matthews calls simple exponence. Lastly, inflection for case, number, and gender in many Indo-European languages involves cumulative exponence. like Latin -o: in am-o: 'I love',

Context-free and context sensitive inflection.

Context-free inflection when there is a simple directional mapping between a morphosyntactic feature and a particular phonological string. as /-in/, refers to context-free inflection: All present participles in English bear the same suffix. Inflection for past tense in English is context sensitive in the sense that the feature [past] is realized as many things depending on the lexeme it attaches to, with /-d/ suffixation being the default case.

Inherent vs. assigned inflection

Nouns and pronouns are marked as having a particular gender in the speaker's mental lexicon. For them, gender is inherent. For any other lexical category that reflects the gender of nouns and pronouns, such as adjective and verb, gender cannot be inherent. It must be assigned.

Government vs. concord

Another word for concord is agreement. Concord or agreement occurs when one element in a sentence takes on the morphosyntactic features of another element. One example of concord is noun–adjective

agreement in the Romance languages or German. Adjectives take on the number and gender of the noun they modify. Government is more or less what it sounds like: one word dictates the form of another. Case assignment by verbs is usually thought of in this way. When a noun is required to appear in objective case, for example, it cannot be said that it agrees with (reflects the case of) the verb. This is because verbs don't have case. The same holds for prepositions.

Inflectional categories

While most languages have morphological inflection of some sort, the actual inflectional categories can differ quite widely across languages. In this section, we briefly survey both the most common categories and some of the ways languages may differ. It is convenient to make a first broad cut into nominal and verbal categories, though the nominal categories often appear on adjectives and verbs through concord. The most common nominal categories are number, gender and case.

Topic- 057: Inflection vs. Derivation

Within a lexeme-based theory of morphology, the difference between derivation and inflection boils down to this: derivation gives you new lexemes, and inflection gives you the forms of a lexeme that are determined by syntactic environment. But what exactly does this mean, and is there really a need for such a distinction? The process goes deeper into the relation between morphology and syntax.

Differences between inflection and derivation

The first question we can ask about the distinction between inflection and derivation is whether there is any formal basis for distinguishing the two types of morphology. Can we tell them apart because they do different things to words?

Distinction between inflectional and derivational morphemes

- Inflection does not change the core lexical meaning or the lexical category of the word to which it applies. Derivation does the former and may do the latter.
- Inflection is the realization of morphosyntactic features, i.e., those that are relevant to the syntax, such as case and number. Derivation is not.
- Inflectional morphology is more productive than derivational morphology.
- Derivational morphology tends to occur closer to the root or stem than inflectional morphology.
- Derived lexemes are more likely to be stored in the lexicon than inflected forms.

Despite the generalizations made above, the morphological form that inflection and derivation may take is very similar. Cross-linguistically, both can be expressed through prefixal, suffixal, or non-segmental

means. The difference between inflection and derivation is therefore not so much a difference in form as a difference in function – what they do and what they tell us.

Topic- 058: Inventory of Inflectional Morphology Types: I

What are the ways in which speakers inflect, or “bend,” lexemes to make them fit into a certain syntactic slot? We present a list of inflectional morphology types here. Although it is not comprehensive, it will give you a broad understanding of how inflection may be expressed.

Affixation and stem alternations

Since our focus in the non-Kujamaat-Jóola portions of this book has been on English, we have had ample opportunity to look at the expression of inflection through affixation. In English, this is the most common means by which inflectional categories are expressed. For that reason we do not address affixation on its own here. Instead we present it with stem alternations, another means of expressing inflection in the world’s languages.

We wish to emphasize that just as affixation does not necessarily occur with stem alternations, stem alternations do not necessarily occur with affixation. We present them together merely for organizational purposes. Latin is one language where in many cases, affixation interacts obligatorily with stem alternations in the expression of inflectional categories. The perfective stem *rēx-* can be contrasted with the present stem *reg-*.

Apophony

In some cases affixation is supplemented or replaced by apophony, or vowel changes within a root, as shown below for English (14) and the Bernese dialect of Swiss German (15): (14) a. sing, sang, sung b. drive, drove, driven

(15) a. *suuffe* [su:f e] ‘drink (inf)’ *gsoffe* [°gs c f: e] ‘drunk (past part)’

b. *schwimme* [ʃvɪm: e] ‘swim (inf)’ *gschwomme* [°qʃvʊm: e] ‘swum’

c. *pfyffe* [pfɪ:f e] ‘whistle (inf)’ *pfiffe* [pfɪf: e] ‘whistled (past part)’

Other terms for apophony are internal change and, particularly when referring to English and other Germanic languages, ablaut. All three of these terms are sometimes applied to the vowel changes that apply to roots in Semitic languages, a topic that we address in the next section under the heading root-and-pattern morphology.

In the context of Germanic linguistics, ablaut is often reserved for apophony in verb paradigms, as in (14) and (15). A second term, umlaut, is used to describe the apophony found in singular–plural noun pairs like *goose*~*geese* or *foot*~*feet*. Umlaut is a phonologically conditioned alternation in which a vowel assimilates in part to a succeeding vowel. The term is used even when the succeeding vowel has been lost. For instance, the umlaut seen in the plural forms of the noun pairs *goose*, *geese* and *foot*, *feet* resulted from vowel harmony with a high vowel in the plural suffix, which has since disappeared. *ts* and addition of a past tense indicator [t]).

Topic- 059: Inventory of Inflectional Morphology Types: II

In the Semitic language family, inflection often involves internal variations in vocalic and syllabic patterns, while the consonantal frame stays fairly stable. We call this root-and-pattern morphology. Root-and-pattern is illustrated in (16) for the expression of number in certain Arabic nouns (examples from McCarthy and Prince 1990: 212, 217). The inclusion of the loanword ‘bank’ is to show that this particular way of forming the plural (referred to as the broken plural) is productive. What all of the plural forms in (16) have in common is that they begin with the syllable pattern CVCVV+:

(16) Root	Singular	Plural	Gloss
Jndb	jundub	janaadib	‘locust’
sltˤn	sult-aan	salaatˤiin	‘sultan’
ʕnb	ʕinab	ʕanaab	‘grape’
nfs	nafs	nufuus	‘soul’
bnk	bank	bunuuk	‘bank’

Root-and-pattern morphology is yet another way of “bending” a form to fit a particular syntactic context.

Reduplication

Another means for expressing inflection that we find in certain languages is reduplication. Like root-and-pattern, reduplication has already been discussed, but from a phonological perspective. We present examples of the Indonesian plural here. It is formed via full reduplication (examples are from Sneddon 1996: 16).

(17)

kuda-kuda ‘horses’

rumah-rumah ‘houses’

singkatan-singkatan ‘abbreviations’

perubahan-perubahan ‘changes’

Indonesian plural reduplication is not obligatory. Speakers of Indonesian have the option of using the unduplicated form to refer to either singular or plural. So kuda not only means ‘horse’, but also ‘horses’; rumah can refer to one house or more than one; and so on. The reduplicated plural is most likely to be used when the number of the noun is not clear from the context, as in the examples below (Sneddon 1996: 17):7 (18)

- a. Rumah-nya dekat pohon-pohon mangga itu
house-3sg.poss near tree-redup mango that
‘His house is near those mango trees’

- b. Pada pinggang-nya terikat bumbung-bumbung
 loc waist-3sg.poss pass.tie water.container-redupkosongempty
 'At his waist are tied empty bamboo water containers'

Without reduplication of *pohon*, (18a) would be ambiguous between 'His house is near that mango tree' and 'His house is near those mango trees'. Likewise, if *bumbung* were not reduplicated in (18b), the sentence could have the interpretation, 'At his waist is tied an empty bamboo water container', as well as the one given above. Such ambiguity is not characteristic of plural reduplication cross-linguistically.

It is also the case that reduplication in Indonesian has functions beside plural marking. As Sneddon (1996: 16) also notes, the meaning of a reduplicated form may be "different but nevertheless related to the meaning of the single base." Among his examples are *gula* 'sugar' and *gula-gula* 'sweets', *laki* 'husband' and *laki-laki* 'man', *mata* 'eye' and *matamata* 'spy', and *langit* 'sky' and *langit-langit* 'ceiling'. Children acquiring languages without plural reduplication sometimes spontaneously use reduplication to express the plural. An English speaking child may say *shoe* for one shoe, but *shoe shoe* for two.

Suppletion

The last type of inflection presented here is suppletion. Suppletion is said to take place when the syntax requires a form of a lexeme that is not morphologically predictable. In English, the paradigm for the verb *be* is characterized by suppletion. *Am*, *are*, *is*, *was*, *were*, and *be* have completely different phonological shapes, and they are not predictable on the basis of the paradigms of other English verbs. We also find suppletion with pronouns.

Compare *me* and *me* or *her* and *her*. Suppletion is most likely to be found in the paradigms of high-frequency words, as seen in the following box.

Examples of suppletion

French *aller* 'to go' / *vais* 'go (1sg)'

être 'to be' / *suis* 'am (1sg)'

Spanish *ir* 'to go' / *fue* 'went (1sg)'

ser 'to be' / *fue* 'was (1sg)'

Finnish *hyvä* 'good (nom. sg)' / *parempi* 'better', *paras* 'best'

Greek [ɛnas] 'a, one (m.nom.sg)' / [mja] 'a, one (f.nom.sg)'

Swedish *ett* 'one', *två* 'two' / *första* 'first', *andra* 'second'

We can look to historical linguistics for an explanation for how suppletive forms arise. The paradigm of the verb 'to go' in French, for example, comes from three different Latin sources. The infinitive, *aller*, and the first person and second person plural forms in the present, *allons* 'we go' and *allez* 'you (pl) go', come from Latin *ambulāre* 'to walk, to walk along'. The stem of future and conditional forms, such as *irai* 'will go (1sg)', has evolved from the Latin verb *īre* 'to go'. Finally, forms like *vais* 'go (1sg)' or *vont* 'go (3pl)' come from Latin *vādere* 'to go, to walk'.

The idiosyncrasies of languages today can often be explained by looking at the languages of yesterday. In certain cases, such as with catch~caught or think~thought and other verbs like them in English, it is most convenient to speak of partial suppletion.

In these cases, the initial phoneme or phonemes of the word remain the same, but there is both internal change and change to the end of the word (loss of segment (loss of segments and addition of a past tense indicator [t]).

Topic- 060: Syncretism & Typology

Having inventoried a variety of ways in which inflection is expressed cross-linguistically, we now turn to **syncretism**. We speak of syncretism when a single inflected form corresponds to more than one set of morphosyntactic features (this definition is paraphrased from Spencer 1991: 45). In the Bulgarian imperfect and aorist paradigms, the second person singular and third person singular forms are identical. Aorist (/ˈeɪərɪst/; abbreviated AOR) verb forms usually express perfective aspect and refer to past events, similar to a preterite. ... Because the aorist was the unmarked aspect in Ancient Greek, the term is sometimes applied to unmarked verb forms in other languages, such as the habitual aspect in Turkish. (The aorist is a past tense.) We show this in (19) for the verbs *krad* ‘steal’ and *igráj* ‘play’ (these are the citation forms of the lexemes):

(19)

Imperfect	2sg <i>krad-é-š-e</i>	<i>igrá-e-š-e</i>
	3sg <i>krad-é-š-e</i>	<i>igrá-e-š-e</i>
Aorist	2sg <i>krád-e</i>	<i>igrá</i>
	3sg <i>krád-e</i>	<i>igrá</i>

In (19), *-š* is a preterite suffix and *-e* is 3sg agreement. We see here that a single inflectional form such as *kráde* may express more than one set of morphosyntactic features: 2sg aorist or 3sg aorist. Romanian also displays widespread syncretism. For many verbs, the first person singular in the present indicative paradigm is identical to the third person plural form. We show this in (20). Both 1sg and 3pl forms are in boldface (data from Stump 2001: 213–14):

(20) Romanian present indicative paradigms

a umplea a face a ști ‘to fill’ ‘to do’ ‘to know’

2 3 4

1sg **úmpl-u** **fác** **stí-u**

2sg úmpl-i fác-i stí-i

3sg úmpl-e fác-e stí-e

1pl úmple-m fáce-m stí-m

2pl úmple-ti fáce-ti stí-ti

3pl úmpl-u fác stí-u

In (20), the infinitives are put in small caps because they are used as the citation forms of the lexeme. The numbers (2, 3, 4) under the glosses are the numbers of the conjugations these three verbs fall into. Syncretism in the present indicative does not take place in Romanian verbs of the first conjugation.

Typology

The term **typology** refers to a classification based on the comparative study of types, and morphological typology was the first systematic method used by linguists in the nineteenth century to compare the structures of different languages. While other sorts of typology flourish today, especially syntactic typology, morphological typology has languished since it was criticized by the first American structuralists, especially Edward Sapir in his *Language* (1921). Still, the traditional terms are used often enough to warrant mention, and the distinctions, while they may not be valid for entire languages, are still useful for describing individual morphological phenomena.

The basic typology has to do with a scale running from **analytic** to **synthetic** languages, which encodes the degree to which the individual meaningful elements in a language are expressed separately. At the analytic end we have the **isolating** languages, of which Vietnamese is the prototypical example, because the only morphology it has is compounding. It has no derivational or inflectional processes of any kind. The next type is **inflective**, of which the more analytic subtype is **agglutinating**. An agglutinating language like Turkish or Hungarian has affixes, but they are strung out quite separately, each expressing a single notion, and easily identified. Consider the following simple table of Hungarian words:

‘The accusative case marker is *-at* (the vowel is deleted after a stem-final vowel), while the plural marker is *-ak* (with the vowel again deleting after a stem-final vowel). When a word is both accusative and plural, both affixes appear one after the other. Compare a **fusional** language, like Latin, shown in the next table: In these Latin forms, the same four slots, singular and plural nominative and accusative, are filled by four distinct suffixes: *-s*, *-m*, *-ī*, and *-ōs* (here, the stem vowel deletes before the suffix vowel), so we say that the two morphosyntactic features in each of the cells of the table (e.g. Nominative singular) are fused. Latin is actually much more complicated, since these two nouns represent only one of five main types (declensions), each of which has a distinct set of forms.

The last stop on this continuum is **polysynthetic** languages. The languages of this type cited most often come from North America. One example of a polysynthetic language is Nuuchahnulth (called Nootka in earlier literature), a language spoken on Vancouver Island in British Columbia (see Stonham 2004, who reanalyzes data originally published by Sapir 1921 and recasts them according to modern transcription conventions).

In Nuuchahnulth and other polysynthetic languages, speakers can build complex words that express what a speaker of English or Vietnamese would express using several words or even an entire, multiword sentence. Following is an example from Nuuchahnulth (Stonham 2004: 65). /kw/ is a labialized velar stop; /m/ is a glottalized labial nasal consonant; /h/ is a pharyngeal fricative. The final /a/ is a variable-length vowel (see Stonham 2004: 24–6), although not marked as such here. (21) ʔinkw-‘il ~-mĩnh -ʔis-(m)it-ma fire-in.house-pl.dim.past-3ind ‘several small fires were burning in the house’

Even English can have one-word sentences. *Go!* is a perfectly well formed such utterance. But compared to English, polysynthetic languages are able to express much more complex notions using a single word, including subject, verb, object, and other information. The English gloss of this single Nuuchahnulth word, for example, contains a complex subject noun phrase, a progressive verb, and a prepositional phrase expressing location.

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Lesson 11**LEXICAL SOURCE OF ENGLISH WORD FORMATION****Topic- 061: Introduction**

England was ruled for a long period after 1066 by a monarch and a nobility whose native language was a variety of French; this ruling group gradually switched to English for everyday purposes; French a language of law, administration, and culture every educated person learnt it.

It is not surprising, then, that the vocabulary of English contains a high proportion of words borrowed from French than other Germanic languages. French is one of the so-called Romance languages, descended from Latin, along with Portuguese, Spanish, Italian and Romanian. Latin has had a more direct influence too official language of the western half of the Roman Empire and the vehicle of a huge and varied written literature, second only to Greek The liturgical language of all West European Christians until the Protestant Reformation 16 century, and of Catholics till 1970s

Many words were adopted into English from Latin directly, rather than by way of French. The Romans preferring instead to create new Latin terms to translate Greek ones. The possibility of direct Greek influence on English did not arise, however, until 16 c Western Europeans began to learn about Greek culture for themselves in the fifteenth century. In word formation, influence of Greek has been in the invention of scientific and technical words. The variety of the sources that have contributed to the vocabulary of English came in whole or in part, from the same ancestral morpheme in the extinct Proto-Indo-European language from which Greek and the Romance and Germanic languages. A striking feature of these words is that the inherited Germanic forms, and in the forms borrowed from Latin, French or Greek the cognitive roots are bound. In borrowing words, English speakers borrowed not only the roots and affixes but also the pattern of word formation that they conform to --a pattern which does not allow roots to appear naked, so to speak, unaccompanied by some derivational or inflectional affix. Some borrowed roots are free, and a few inherited ones are bound. Most of the roots that are bound in all contexts (that is, most of the roots that have no free allomorphs) do not belong to the vocabulary that English has inherited from its Proto-Germanic ancestor.

Topic- 062: Rarity of borrowed Inflectional Morphology

If English borrows a foreign pattern of word formation, it should be expected to borrow inflectional affixes that conform to that pattern, as well as roots and derivational affixes. English does not use French or Latin inflectional affixes on verbs borrowed from those languages, A language acquires lexemes through borrowing rather than individual word forms.

If English speakers import a new verb V from French, they will not import just its past tense form since we expect to be able to express in English not only the grammatical word ‘past tense of V’ but also the grammatical words ‘third person singular present of V’, ‘perfect participle of V’, and so on

It is much more convenient to equip the new French-sourced verb with word forms created in accordance with English verbal inflection – specifically, the most regular pattern of verbal inflection (suffixes -s, -ed and -ing). The only condition under which English speakers are likely to borrow foreign word forms along with the lexemes if the grammatical words are few in number(not hard to learn), and if their

functions in English and the source language correspond closely. This condition is fulfilled with nouns. The foreign plurals are all vulnerable, however.

Phenomena and data seem solidly established, but for the others not: schemas, cactuses, formulas, cherubs and kibbutzes. Even data tends to be accommodated to English morphology, but by a different method: The usual plural suffix in both medieval and modern French is -s, just as in English. Just a few French borrowings sometimes retain, in formal written English, an idiosyncratic plural suffix -x, e.g. tableaux, plateaux.) The effect of these borrowings is to divide the class of nouns with irregular plurals into two classes: nouns that belong to everyday vocabulary and whose irregular plural survived for frequent use (e.g. teeth, children, mice), and relatively rare or technical survived as a badge of learning or sophistication. What we do not find are irregular plurals that fall between these extremes, in nouns that are not particularly common but do not belong to technical or learned vocabulary either. But what about oxen

Topic- 063: Reduction in Inflectional Morphology

Modern English nouns have no more than two inflected word forms: singular and plural. In Old English, there was a contrast of case, like that found in modern English personal pronouns (nominative we versus accusative us etc.), Old English nouns could distinguish also a genitive case, and a dative case whose meanings included that of modern to in Mary gave the book to John.

These two numbers and four cases yielded a pattern of eight grammatical words for each noun lexeme. Neither name nor stone had eight distinct word forms, one for each grammatical word; instead, they display different patterns of syncretism. All Old English nouns had more than two forms that are available in modern English. If nouns distinguished four cases in Old English, it is reasonable to guess that pronouns should have done so too; and that guess is correct. Pronouns had five cases, including an instrumental. Adjectives and determiners had same forms. Old English verbs displayed a similar inflectional luxuriance.

Most modern English verbs have four distinct forms (e.g. perform, performs, performed, performing), while some common verbs have five (e.g. speak, speaks, spoke, spoken, speaking). Besides the imperative forms ('help!'), or the verbal adjective helpende, which, just like other adjectives in Old English, had forms that distinguished three genders, two numbers and four cases.

Why did English lose this wealth of inflection? The temporary eclipse of English by French as the language of culture and administration after 1066. Partly also it is due to dialect mixture.

The examples of 'Old English' given here come from the dominant dialect of written literature, that of south-western England.

But this was not the dialect of London, which became increasingly influential during the so-called 'Middle English' period (from about 1150 to 1500), and established itself as the main variety used in printing. English inflectional morphology was already by 1600 almost the same as in 2000, so that modern readers of Shakespeare encounter only a few obsolete inflected forms such as 'helpst' and he helpeth, that preserve two Old English suffixes

Topic- 064: Characteristics of Germanic and Nongermanic derivations

The inherited Germanic root heart is free while the cognate roots cord- and card-, borrowed from Latin and Greek, are bound, and the same applies to inherited bear by contrast with borrowed -fer and -pher. If this kind of contrast is general, then it has implications for inherited and borrowed affixes too. We will expect that native Germanic affixes should attach to free bases, while the affixes that attach to bound bases should generally be borrowed. And this turns out to be correct. An example is the suffix -ment, as in development, punishment, commitment, attainment – though it is sometimes found with a bound base, as in the nouns compliment and supplement.

Another example is the prefix de-, as in deregister, delouse and decompose. This tolerance for free bases is surely connected with the fact that, in the terminology de- is formally and semantically rather regular, and can readily be used in neologisms (e.g. de-grass in The courtyard was grassed only last year, but now they are going to de-grass it and lay paving stones). For an affix restricted to bound bases, such a neologizing capacity would be scarcely conceivable in a language where, as in English, most bases are free.

Topic- 065: Fashions in Morphology

The title highlights a respect in which morphology differs from syntax. It makes sense to ask whether a certain word formation process (a particular affix, let's say) is in or out of fashion. Syntax is stable in a way that morphology is not. The fact that many morphological processes are haphazardly 'gappy' (that is, they may not be formally general even if they are formally and semantically regular), whereas few if any syntactic constructions are 'gappy' in this way. In morphology, gaps get filled, or else gappy processes lose their regularity and survive only in a few lexically listed lexemes, like the process of forming abstract nouns by suffixing -th to adjectives, while other processes become increasingly regular to replace them.

A systematic study of morphological fashions belongs to a historical study of English word formation rather than to an introductory survey such as this. Two fashions are important that manifested themselves in the last half of the twentieth century, against trends of the last couple of centuries.

The first is a fashion for certain Latin- and Greek-derived prefixes; the second is a fashion for a certain kind of headless compound. Conscious borrowings from Latin and (to a lesser extent) Greeks were fashionable in certain literary styles of the sixteenth and seventeenth centuries, because of a perceived need to enrich the English vocabulary.

But such borrowings, often obscure and even incomprehensible to ordinary readers, were also attacked as 'inkhorn terms' – mere products of the some scholars' desire to show off his knowledge of Latin. The result is that the Latin- and Greek-derived element in the vocabulary of English has, since the eighteenth century, been pruned rather than increased. A fashion for a certain kind of headless compound or exocentric compound. For example: redhead, lazybones, pickpocket, longneck, climbrock, etc.

Topic- 066: History and Structure

Characteristics of a language that are due purely to historical accident are the characteristics that, in principle, are least likely to interest a general linguist. The Norman Conquest in 1066 is just such an

accident, so its consequences for the vocabulary of English from French may seem to deserve a place only in histories of the English language, not in books about its morphological structure.

But there is more to it than that. If it had not been for the Norman Conquest and its aftermath, English morphology would not have acquired the at first sight rather bewildering mix of characteristics discussed so far. What's more, one cannot dismiss characteristics acquired through the Latin lexical intake as 'unproductive' and therefore not truly part of modern English morphology; Some Latin-derived processes, such as suffixation of -ion and -ence, are in limited domains just as formally regular as processes such as adverb formation with -ly. If the history of the community of English speakers in the British Isles had been otherwise, the English language would be considerably different today not just in its repertoire of lexical items but in how its words are structured.

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Lesson 12

MORPHOLOGICAL PRODUCTIVITY AND THE MENTAL LEXICON**Topic- 067: What is Morphological Productivity?**

A given morphological pattern is more productive than another is to say that there is a higher probability of a potential word in the first pattern being accepted in the language than there is of a potential word in the second pattern. English nouns make their plural in a number of different ways, as can be seen in the following set of words (Bauer 2001):

cats, dogs, horses, oxen, deer, mice, hippopotami, cherubim

What does a speaker of English do when confronted with novel words, such as: ?

(3) argaz ‘crate of specific style’

smick ‘type of cracker biscuit’

brox ‘piece of computer hardware’

ceratopus ‘type of dinosaur’

cheppie ‘type of antelope’

Chances are that, despite the existence of other plural formations, an English speaker will mark all of these words as plural by suffixation of /z/: argazes, smicks, broxes, ceratopuses, and cheppies. The only likely exception is ceratopus, ceratopi, on the basis of Latin as alumni, foci, and nuclei. The word argaz ‘crate’, being a Hebrew noun. It would therefore be possible to pluralize it as argazim, on the pattern of cherubim, seraphim, and kibbutzim.

brox as broxen

broxes and Vaxes

One term potential word. A potential word could be a word but isn’t. One example is Mugglehood ‘the state of being a non-wizard’, which we created from Muggle ‘non-wizard’, a word found in J. K. Rowling’s Harry Potter books, and the English suffix -hood. In 2004, only three instances of Mugglehood, but also numerous instances of Muggledom and Muggleness. Searched for it again on January 3, 2010, showed 292 hits. Perhaps someday it will be included in the dictionary.

The suffix -th creates nouns from adjectives (e.g., deep → depth, wide → width). a meaning that is similar to -ness. Length means the same thing that longness would mean; decidedth, mean the same thing as decidedness. But only -ness can be called productive. Studying productivity, we study phenomena and distinctions like these. One question we need to ask about productivity is whether it is part of linguistic competence. Competence is Chomsky’s (1965) term for the knowledge that speakers and hearers have of their language. Some people would say that productivity is not part of linguistic competence either, because, in order for something to be considered part of competence, it must be structural and ‘all or

none'. Productivity is a probabilistic notion, and some linguists believe that if something is probabilistic, it is not structural and hence is not part of the grammar.

Under this view, productivity would have to be treated as a phenomenon that is related to a speaker's competence, but not part of it.

Topic- 068: Productivity and Structure: Negative Prefixes in English

Productivity and Structure Negative Prefixes in English One reason that it is likely that productivity can tell us something about language structure is that the more productive a morphological derivational process is, the more likely it is to have a compositional one whose meaning is transparently predictable from the meaning of its input. The converse is also true: the less productive a derivational process, the more likely it is to result in a non-compositional, semantically idiosyncratic, non-transparent output.

To illustrate this point, consider negative prefixes in English. Zimmer (1964) looked at three of them, non-, un-, and in-, and discovered that the most productive of the set, non-, also has the most semantically transparent derivatives. This is shown by the contrast between the two:

non-Christian unchristian

non-human inhuman

What's the difference?

Non-Christian means 'not Christian'. unchristian can mean same and 'not behaving in a Christian manner' or even 'uncivilized and barbaric'. Likewise, non-human simply means 'not human', while inhuman refers to the absence of human qualities like pity or kindness. A person can be both human and inhuman, but not human and nonhuman. The non-words in general simply negate their bases, the in- and un- words have the meaning 'completely opposite to X', where X is the meaning of their bases, in the way that east and west or long and short are opposed. To put it in a more technical way, non- is a logical or contrary negator. Using logical notation, we could represent non-Christian as in, where $\neg$ means 'not': $\neg$ Christian. Un- and in- are contradictory negators, whose addition to a word X results in a new word meaning 'opposite of X'. Zimmer's observation extends to other derivational affixes. The suffix -ness is more productive in English than -ity (Aronoff 1976). Consider the pair collectivity~collectiveness.

While both may mean 'the quality or condition of being collective', only collectivity has the additional meaning 'the people considered as a body or whole'. Overall, when we compare many such pairs, the -ness derivative has more transparent semantics. Sometimes the -ity derivative sounds or looks odd, while the -ness derivative is pretty much always acceptable. Compare conduciveness with conductivity. Most English speakers would say that the former is more acceptable than the latter.

We can even go beyond morphology to make the observation that syntax, which is always productive, is by definition compositional.

Topic- 069: Degrees of Productivity

With two observations:

- a. Though many things are possible in morphology, some are more possible than others.
- b. Though there are infinitely many potential words in a language, some are more likely to become actual words than others. Morphological processes do not fall into two neat categories, productive and unproductive. They are best seen as being spread out along a scale, with some more productive than others. There are a number of types of constraints that limit productivity and that contribute to or even determine to what degree a particular formation is productive.

Morphological constraints

Productivity can also be limited by morphological constraints. a morphological process may require the base to have a certain structure, to belong to a certain morphologically or syntactically defined class, such as a gender, or to end in a particular affix. verbs.

Semantic constraints

Productivity can also be constrained by semantic factors. Barker (1998) studies the English suffix -ee found in words like advisee, addressee, enlistee, or employee using a large set of naturally occurring examples such as the following:

To summarize, if a language has two ways of doing something, one of which is less productive, the less productive rule or process has a linguistic purpose. The less productive rule may result in a word that stands out, useful in newspapers, magazines, and advertisements, or it may result in a form that sounds more technical or learned.

Topic- 070: Testing Productivity

Using a tool to come to a better understanding of how speakers and hearers use and process language. Morphological productivity offers many opportunities for researchers with this kind of philosophy. Linguists, in recent years, have used many different tools to come to a better understanding of how we produce complex words such as: standard dictionaries, rhyming dictionaries, and large corpora, technologically sophisticated techniques (positron emission tomography (PET)).

English suffixes

Aronoff and Schvaneveldt (1978) conducted an experiment to verify that productivity figures in individuals' linguistic competence and to judge its consistency across speakers and words. The experiment focused on -ness, a native English suffix, and -ity, of Latin-Romance origin, which often attach to the same morphological and semantic classes of words. We see this in triplets like the following, where all three members can be found in a dictionary:

immense immenseness immensity

scarce scarceness scarcity

exclusive exclusiveness exclusivity

porous porousness porosity

The two suffixes differ, however, in that -ness is more productive overall, especially with certain types of stems like those of the shape X-ive. Speakers were presented with three sets of words: (i) actual words, like activity or assertiveness, where actual means listed in Webster's Collegiate Dictionary;

(ii) possible words, like effervescivity or affirmativeness, where X-ive occurs in the dictionary, but not the -ness or -ity suffixed form; and

(iii) non-words.

Here neither X-ive nor X is listed in the dictionary. Examples of non-words are remortiveness and lugativity. Over the course of the experiment, 141 subjects were asked to judge 40 words, 100 possible words, and 40 non-words, which were presented in randomized lists.

For each subject, half of the words were suffixed in -ity and half in -ness. The 40 actual words were the same across all subjects, but possible words and non-words were counterbalanced so that half of the subjects got, for example, effervesciveness and elaborativity, and the other half elaborativeness and effervescivity. This was done to ensure that judgments would be based on the fecility of the suffix and not some peculiarity of the stem it attached to.

The final variable in the study involved the instructions given to the subjects, who were divided into three groups of 47 each. One group was asked to judge whether the items were in their vocabulary; The second group was asked whether the items were English words; The third group was asked whether the words were meaningful. As it happened, the instructions had little effect on the subjects' judgments. It turned out that English speakers preferred the actual words in -ity (they were balanced for frequency). When it came to the non-words, they didn't care, and on the potential words, they preferred the -ness words. This shows that speakers can tell the difference between a more productive and less productive rule. Any given study can advance our understanding of morphological productivity without being the last word on the subject. There is a great deal of value in studies that ask new questions or try out new ways of testing hypotheses. Design errors may or may not influence a given set of results. When they do, they can often be corrected in follow-up studies, which are just as essential to advancing our knowledge.

Topic- 071: The Mental Lexicon

Studies of the sort just discussed have goals beyond linguistic description and aim rather to describe what is actually going on inside the head of a language user when they use complex words. Psycholinguistics studies this phenomenon, though when the research involves measuring neurological activity, as much modern research does, then it may also be termed neurolinguistics or cognitive neuroscience.

The goal of psycholinguistic and neurolinguistic research on morphology is to understand exactly how the mental lexicon works, the lexicon that all language users store in their brains.

Pinker (1999) is a very lively and informal presentation of the major issues in the field. Aitchison (2003) is a more general introduction. It is important to remember in to distinguish the lexicon itself from how it is used by the speaker or hearer. Most studies of the mental lexicon are concerned with word recognition,

a classic problem in psychology: How do people associate the speech or writing signal with the entries in their mental lexicon?

Until very recently most word recognition studies dealt mostly with written words, not speech.

Psychologists were interested in how people recognize words, both their sound and their meaning, when they read. Only in the last twenty years or so have psycholinguists turned their attention to spoken word recognition, which, for linguists at least, is a much more interesting and certainly more basic problem than reading. After all, as linguists have known for over a century, the primary form of language is spoken (or signed) and written language is secondary.

The basic and universal task of word recognition is thus spoken word recognition. From an experimental point of view, it has always been much easier to deal with written text than with speech, which is why it is only recently, with the advent of tools for regularizing the speech signal, that those psycholinguists have begun to turn to spoken language recognition.

The other side of accessing the mental lexicon, and in some sense the opposite or reverse of word recognition, is word production. When a person speaks, how does that person know what words to use, the words that will eventually come out of his or her mouth?

This is a much more difficult problem than word recognition and has received much less attention from psycholinguists (the classic presentation is Levelt 1989). Most psycholinguists assume that there is only one mental lexicon and that both word recognition and word production access this same lexicon, though in different ways. From an experimental perspective, though, the mental lexicon is not something that we can study directly.

We study it rather only through the intermediaries of production and recognition, both of which must access this dictionary hidden deeply within our heads.

Topic- 072: Psycholinguistics, and Neurolinguistics

To understand what sorts of issues might arise in the study of the mental lexicon, let's look at some of the aspects of Kujamaat Jóola. Consider vowel harmony first. The vowels of Kujamaat Jóola fall into two sets, tense and lax, and all the vowels in a word must be either tense or lax.

When a suffix is added to a word, the vowel of the suffix may be either tense or lax, depending on the vowels of the stem. If a suffix contains a tense vowel, then the vowels of the stem must also become tense. Overall, tense vowels win: as long as a complex word contains one tense vowel, all its vowels must become tense.

There are several ways in which those stems and suffixes that alternate between tense and lax could be represented in a person's mental lexicon. Since morphemes with only tense vowels never alternate, it is to assume that these morphemes have only one form in the mental lexicon.

But other morphemes sometimes have lax vowels and sometimes tense vowels. We could say that each of these morphemes has two alternative forms (allomorphs) in the mental lexicon and that the form with lax

vowels is selected just in case there are no tense vowels in the word. We could say that there is a mental operation that makes lax vowels tense if there are tense vowels in the same word.

A psycholinguist or neurolinguist would ask how these different systems might be encoded in a person's brain and would want to devise experimental tests and measures for determining whether a particular mechanism is actually at work

We certainly know that speakers and signers do not hesitate when speaking or signing, no matter how complex the morphology of their language. Semantics alone thus provides some information about which forms must be stored not tell us how they are stored. Most linguists: they are stored as complex structures, but no purely linguistic evidence to tell us whether that is so.

Similar questions can be asked about all the inflectional forms of a single verb or noun. Are they all stored? And if so, are they stored together or separately and are they stored with or without structure? As it happens, a fairly broad consensus has emerged among psycholinguists in the last decade that "morphological considerations need to be introduced into any model of the mental lexicon. In other words, regardless of whether any particular words are stored or not and regardless of whether related words are stored together or separately, they do have morphological structure.

Whether related words are stored together or separately, we just don't know. Even if everyone agrees that all words in the mental lexicon have morphological structure, many other issues remain undecided. One of the most central is that of the number of different mechanisms speakers use to access their mental lexicons, either in recognition or in production.

The most obvious answer is that there is only one mechanism, but in fact, of the theories of accessing the mental lexicon that are out there in the literature, we can identify two broad types, called the single-route and dual-route models. A very accessible account of both model types is provided by Pinker (1999). It is well known that the more frequent or familiar a word is, the faster a speaker can recognize it (Aitchison 2003).

An English speaker will recognize clean more quickly than glean, even though the two differ phonologically in only one distinctive feature, because the first is about thirty times more frequent than the second. Most of the 500 most frequent words in English are simple, but even if a word is morphologically complex and perfectly regular, it may still be common.

Unfortunately, almost all the psycholinguistic and neurolinguistics research that has been done to date is based on English or other European languages whose inflectional and derivational morphology is fairly simple.

Lesson 13**MORPHOLOGY AND SYNTAX****Topic- 073: Introduction**

Already seen, inflection and syntax are intimately related to one another. Inflection is the realization of morphosyntactic features through morphological means. Morphosyntax deals with the relationship and interactions between morphology and syntax. A variety of topics in morphosyntax, includes morphologists' and syntacticians' definitions of inflection, structural constraints on morphological inflection, inflection and universal grammar, and grammatical-function-changing morphology.

Since our target audience consists of students whose only exposure to syntax comes from an introductory course in general linguistics, we avoid bringing in advanced syntactic analyses. According to the traditional view, the relation between morphology and syntax is the following: while morphology builds up word forms—typically by combining roots with other roots and with affixes, but also by applying other operations to them.

Syntax takes fully inflected words as input and combines them into phrases and sentences. The division of labour between morphology and syntax is thus perfect: morphology only operates below the word level whereas syntax only operates above the word level. These two components of grammar are ordered in strict sequence, such that the syntax takes over after the morphology has done its work.

Topic- 074: Morphological vs. Syntactic Inflection

We begin by distinguishing between two applications of the word inflection, one found chiefly in the morphological literature and the other in syntactic literature. For a morphologist, the presence of inflectional morphology in a language depends on the existence of multiple forms of a lexeme. If a lexeme has only one form, then there can be no morphological inflection.

In syntax, there is no such requirement. Chinese lexemes have only one form, abstracting away from phonologically determined alternations (mostly changes in tone). While Chinese has a few clitics or particles, including one that expresses past tense, these are generally not considered affixes. The same is true of Vietnamese, though the two languages are unrelated. For the morphologist, therefore, these two languages have no inflection.

From a syntactician's point of view, whether or not Chinese and Vietnamese have inflection is an entirely different matter. Even if a language does not express a particular notion such as number or case, it is typically assumed to be present in the syntax. Likewise, a syntactician may argue that a verb always agrees with its subject in an abstract sense.

This abstract agreement is considered just as real in Chinese, where the verb form never depends on its subject, In Russian, where the form of the verb changes depending on the person (first, second, third) and number (singular or plural) of the subject. In sum, morphological inflection is realized overtly, where syntactic inflection may or may not be.

The existence of zero allomorphs does not contradict this definition of morphological inflection. Zero allomorphs is contrasted with non-zero allomorphs. In other words, in some instances the inflectional feature encoded by the zero allomorph is realized overtly. Another difference between the morphological and syntactic usage of the term inflection is that morphologists speak of inflection only when dealing with bound forms. The reason for this is clear inflection informally is as “bending” of a lexeme.

English has a syntactic category of modals, or modal auxiliaries, used to accompany other verbs to indicate that the action or state described by the sentence is something other than simple fact. Should we consider these models to be morphological inflection? No. In order to be classified as morphological inflection, a syntactic category must be expressed through bound forms. In the case of the English modals, we are not dealing with bound forms, but rather with separate words. Again, syntacticians differ from morphologists on this point. Most would treat auxiliaries as part of the inflectional system of a language. Morphologists are not denying the validity of this treatment, only distinguishing the full word modals (i.e., syntactic inflection) from morphological inflection.

Topic- 075: Structural Constraints on Morphological Inflection

Cross-linguistically, constraints on the realization of inflectional morphology. In Russian, for example, verbs show gender agreement with their subject (feminine, masculine, or neuter) only in the past tense. This fact about Russian has a historical explanation: the past tense form was originally an adjective, and Russian adjectives agree in gender with the nouns they modify.

In Modern Hebrew, verbs agree in gender with their subjects only in the present tense. Historically, Hebrew present forms were originally participles. Gender agreement is not optional in Hebrew and Russian. Instead, its morphological realization is context-dependent.

It is surprisingly easy to find languages where verb inflection is obligatory in some contexts but impossible in others. a few such All depend on syntactic context, rather than on tense, which is expressed as an inflectional part of the verb itself in Russian and Hebrew.

In the Kujamaat Jóola the subject agreement is expressed obligatorily except in the past subordinate and positive imperative forms of the verb. In some related languages, like Balanta verbs agree with their subject only in certain syntactic contexts. In Balanta, verbs may be marked for subject agreement, but generally only in the absence of a subject noun phrase. When a subject noun phrase is present, a subject prefix on the verb does not express agreement. Instead, it indicates that the subject is focused:

Arabic (Semitic): Verbs agree with their subjects for number when the word order is SV but not when it is VS. When the word order is VS, the verb bears default 3sg agreement. • Balanta (Atlantic): Verbs agree with their subjects only when there is no subject NP or the subject NP is focused (emphasized).

Topic- 076: Inflection and Universal Grammar

Universal Grammar is the theory developed by Noam Chomsky that states that all languages are identical at some level of analysis. Tremendous influence on the field of linguistics, and most linguists agree with Chomsky that language has an innate component. A key phrase in the definition of Universal Grammar that we have provided is “at some level of analysis.” What is the level of analysis at which languages are identical? At which levels do languages differ? More specifically, are inflectional categories universal? In one sense, inflectional categories are universal. A mistake to say that the realizations of inflectional categories are stable cross-linguistically.

Let’s consider gender. Gender is highly problematic from a universalist point of view. The number of noun classes in the languages of the Atlantic family vary widely.

Kujamaat Jóola 19,

Gombe Fula 25, Serer 16,

Wolof 10

Manjaku 14,

Balanta seven.

Some Atlantic languages have multiple dialects, and dialects do not necessarily share the same number of noun classes. To other language families of the world, we find variation not only in the number of genders but also in the entire organization of the gender system. Despite differences between gender systems some similarities do emerge.

German has three genders called masculine, feminine, and neuter: Masculine Feminine Neuter Mann ‘man’ Frau ‘woman’ Parlament ‘parliament’ Masculine, feminine, and neuter are obligatory inflectional categories of German. This means that every noun in the language, including borrowings like Parlament ‘parliament’, must belong to a gender.

A noun cannot be genderless.

Furthermore, gender is obligatory in that a noun cannot simply carry it around: its gender category must be expressed through agreement. In many languages gender is an obligatory inflectional category. Every noun must have a gender, and that gender must be expressed in the morphology. The gender categories themselves and the number of them are completely

different. Many languages, have certain types of inflectional categories appear over and over again. For example, nouns are regularly inflected for case, number, and gender. For verbs, the picture is similarly limited. Verbs might inflect for tense, aspect, mood, voice, or agreement, but you generally do not find languages where verbs inflect for other categories. Although the inflectional categories themselves may not be universal, universal principles govern what is inflectionally possible and what is not.

Topic- 077: Grammatical Function Change

The governor broke the law.

The law was broken by the governor.

The law was broken

There are times in life when the passive is convenient.

In English we can also say:

Solomon made the governor break the law

Sentences like the one are called causative because they usually express the meaning ‘cause to do something’, or sometimes ‘allow, persuade, help to do something’. In English we can also say:

The governor broke the law for Smith

This sentence introduced another participant, Smith, the person for whom the governor broke the law. We could discuss the morphology of break in the English sentences but its forms are fairly limited: broke, broken, break. None of these forms is limited to expressing a passive, causative, or ‘for X’ interpretation. (The -en of broken is sometimes considered a passive morpheme, but it is not limited to passive sentences. We could also :

The governor has broken the law(which is active).

If we look at other languages, we often find that the passive, causative, and other types of grammatical-function-changing phenomena are associated with particular morphology. Grammatical function change refers to “alternations in the grammatical encoding of referential expressions,” to use the definition presented by Baker (1988: 1).

We see that the agent can be encoded as a subject or object, depending on the form of the verb used: broke or was broken. Passive, causative, and other phenomena are grammatical-function changing phenomena because they can be seen as triggering the encoding change. There are various types of grammatical-function-changing phenomena that are found cross linguistically.

Topic- 078: Productivity in Grammatical Function Change

Passive, causative, and other phenomena are grammatical-function changing phenomena shows productivity of syntax. We do not analyze them, beyond presenting basic definitions, because to do so would require us to go too deeply into syntax. Grammatical-function-changing phenomena involve morphology–syntax interactions at their most intimate.

Passive

Construction in which a grammatical subject of the verb is subjected to or affected by the action of the verb. The verb's agent may be expressed as an adjunct (in English, a by phrase). Example: His memoirs were written by a ghost writer. Antipassive Construction in which the object of the verb is expressed in an oblique case or becomes null. The following is an example of the antipassive voice:

"Mary-erg eats pie-abs." → "Mary-abs eats." "He-erg is speaking the truth-abs." → "He-abs is speaking."

Causative Construction in which the subject causes an event. The causative may be an auxiliary verb or an inflectional morpheme attached to a verb, but the term is also applied to verbs that inherently express a relationship whereby the subject causes a given event (to kill is to cause to die). Sam makes us smile. Applicative Construction in which the addition of a morpheme allows a verb to take an additional object.

This object is often understood as benefiting from or being adversely affected by the action of the verb.

Noun incorporation Morphological construction in which a noun stem is compounded with a verb stem and yields a derived verb stem. Example: Siberian Koryak qoya- 'reindeer' combines with the verb stem -nm- 'to kill', yielding the derived form qoyanm- 'to reindeer-slaughter. Although incorporation does not occur regularly, English uses it sometimes: breastfeed, and direct object incorporation, as in babysit.

Lesson 14**MORPHOLOGY AND PSYCHOLINGUISTICS****Topic-079: Morphology and Mind**

How morphological knowledge (knowledge of complex words and of morphological rules) is represented in the human mind and how it is used in language processing. This is an important topic because the mental representation of morphological knowledge is a battle ground for different theories about the nature of linguistic rules. This is an important topic because the mental representation of morphological knowledge is a battle ground for different theories about the nature of linguistic rules.

A related topic is the balance between storage and computation. Knowledge in a particular domain of human cognition always comprises both storage of information and the ability to compute new information. For instance, when we want to use the word books, we have two ways to do this: either we retrieve this plural form of book from our lexical memory, or we create it on line, by adding the plural suffix -s to the stem book. What determines the choice between these two routes?

Empirical research of such questions may thus help us to get a better understanding of the nature of human cognitive capacities. Morphology may therefore be qualified as a window on the human mind. The empirical domains of psycholinguistic research will be reviewed: the mental lexicon ('the dictionary in your head'), the acquisition of morphological knowledge, and the role of this knowledge in language perception and production.

In the previous chapter, on morphological change, we saw that language change also provides information about the mental representation of morphological knowledge. There are a number of ways in which we can find out how morphological knowledge is represented in the mind and used in language processing: by studying naturalistic data (corpora of language use, speech errors, effects of language impairment, etc.) and experimental data (lexical decision tasks, production tasks, etc.).

Morphology is a battlefield for competing models of linguistic knowledge, and for discussion on the nature of linguistic rules. The frequency of words in actual language use correlates with their activation level in the mental lexicon. Statistical and probabilistic data are therefore relevant for adequate models of morphological knowledge.

Topic- 080: The Use of Mental Lexicon

Lexical knowledge of a native speaker does not have the format of a dictionary. First, the number of lexical entries in a good dictionary is much higher than that in our individual mental list of words. There are many words that most speakers do not know. You can compute the size of your

own vocabulary by checking, on the basis of a representative sample of words from a good dictionary of your native language

How many of the words in that sample do you know ?Adult speakers of English with a higher education might know up to 50,000 word types, The whole English vocabulary comprises hundreds of thousands of words.

This estimate concerns the passive vocabulary, the number of words that you understand.

The active vocabulary, the set of words one uses in language production is much smaller.

A dictionary is conservative by nature, and hence it will contain words from the past that nobody uses any more. Mental lexicon will always be ahead of the dictionary, and contains a substantial number of words that are not listed in dictionaries. New words (neologisms) are coined continuously, and dictionaries always lag behind.

A second difference between a dictionary and the mental lexicon is that words in the mental lexicon bear a number of relationships to each other. Words with similar meanings or similar phonological forms appear to be related in the mental lexicon, as can be concluded from speech errors. Someone talking about the sympathy of a musical conductor used the word symphony instead of sympathy.

In a dictionary, semantic relations between words are usually not expressed directly. The mental lexicon as a multidimensional web of words, with all kinds of connections between those words: semantic similarities, phonological similarities, and morphological relationships. In a dictionary, words have one type of relationship only, that of alphabetical ordering. This ordering is based on the degree of orthographical similarity between words.

A third difference between a dictionary and a mental lexicon is that the latter also stores information about the frequency with which you come across a word. Linguists may compute the frequency of words on the basis of large corpora of actual language use. Frequency counts indicate how many tokens are found for each word type in a particular corpus. Function words such as determiners have a relatively high frequency. Within the set of English adjectives, the word nice is used far more frequently than the word opaque.

A high frequency word is more easily recognized.

Frequency of exposure to a word correlates with the activation level of that word in the mental lexicon. A word with a higher activation level is activated faster in language processing. A related effect is the cumulative frequency effect. The frequent singular form of a noun will also have a frequency effect on its plural form. The token frequencies of related inflectional word forms contribute to the activation level of each of them.

This is called the base frequency effect. Another example is the cumulative root frequency effect: the summed frequencies of all words with the same root also appear to affect response latencies of individual complex words containing that root.

Topic- 081: Acquisition of Morphology

How does a child acquire the morphological system, the set of morphological rules of his or her mother tongue? Morphological rules have to be discovered on the basis of words that are formed according to these rules. The first stage: the acquisition of individual complex words.

The child uses morphologically complex words correctly without making use of morphological knowledge, by retrieving them from memory. Next, he may discover certain recurring properties e.g. plural nouns in English, the ending -s (with the allomorphs [s], [z], and [ɪz]) to the stem. Rules with exceptions go a further refinement. Three stages of morphological acquisition (Berko, 1958).

Firstly, children learn a number of past tense forms of English verbs by rote producing correct past tense forms of both regular and irregular verbs (asked, went). Secondly, children acquiring English have discovered the rule for past tense forms, but will also apply the rule to the class of irregular verbs. They will produce the correct form asked, but the incorrect form goed instead of the correct went, a case of overgeneralization.

Third stage, both the rule and the set of exceptions have been acquired, correct asked and went. So the learning process has the shape of a U-curve: the number of correct past tense verbs in stage I decreases in stage II, and increases again in stage III. This learning curve has been found for many languages. In Romance languages, with a number of different conjugations for verbs, it is the default conjugation that is overgeneralized. In languages with case marking, children will use case-marked word forms from very early on because it is words in their surface forms (not stems) that form the basis of acquisition. For nouns, the nominative form is usually the default case form, used in all syntactic contexts. Children can discover the role of case marking in syntactic contexts at a very early age. Russian children already use accusative nouns after transitive verbs before the age of 2 (Clark and Berman 2004). In the domain of word-formation children discover the building principles for complex words quite soon. able to coin new words themselves at a very early age.

Children coin words for concepts for which there are already established words. For adults, established words have priority above new coinings, in accordance with the principle of conventionality that gives priority to established words, and thus blocks the creation of new words that are synonymous to existing ones. Since children discover morphological processes on the basis of the words they are exposed to, they must be able to compare and analyse words, and assign them morphological structure if possible. Children appear to analyse words consciously, and ask questions like: Does cornflakes have corn in it? (Clark 1993: 40).

Topic- 082: The Sources of Evidence

How morphological information is represented in the mind? Sources of evidence: experimental and naturalistic data. Psychologists use experiments, and in the realm of morphology lexical decision tasks are the most widely used way of probing into the mental representation of morphology. Lexical decision tasks are often used in combination with priming, the prior presentation of another word.

This means that the subject who has to make a lexical decision first receives some other information. An example of priming is the following: exposure to one stimulus influences a response to a subsequent stimulus, a case of identity priming.

A higher level of activation by previous access to the same word. Other ways of priming are phonological priming (the prime word is phonologically similar to the target word), and semantic priming (with a semantically similar word as the prime). If these primes reduce response latencies for the target word, one can conclude that words in the mental lexicon are connected to words that are either phonologically or semantically similar.

In morphological priming, a word that is morphologically related to the target word is used as a prime. For instance, calculate will function as a prime for calculation. Phonological and semantic priming, and not an independent phenomenon. Naturalistic data evidence for theories of morphology in the mind. Data concerning the actual behaviour of language users. Children coining new words or word forms may be interpreted as evidence for the children's having acquired morphological rules.

Speech errors

Take the freezes out of the steaker (take the steaks out of the freezer). The roots freeze and steak exchanged erroneously. Exchange of roots bears on the issue of the lexical representation of morphological structure. The word freezer a complex word is stored as such in the mental lexicon, given its specific conventional meaning shows that its internal morphological structure is still accessible.

Complex words can be stored in lexical memory with their morphological structure still present. The language behaviour of people with a language deficit, such as aphasia, may throw light on the mental representation of morphology.

Aphasics that suffer from agrammatism (the inability to use rules), usually referred to as Brocaaphasics, may not be able to produce correct plural forms of nouns, except for nouns with a high frequency plural form such as the word eyes (Aitchison 2004). Three German aphasic patients with agrammatism it was found that their syntactic abilities were severely impaired, whereas their ability to use inflectional morphology was still intact. This fact can be interpreted as evidence for the position that syntactic rules belong to a different module of the grammar than the rules of inflectional morphology.

Topic- 083: Models of Morphological Knowledge

The observed ability of native speakers to coin new words or word forms is the basic argument for assuming that morphological knowledge encompasses more than storage of the complex words that language users are exposed to. There must be mechanisms in the mind that enable us to extend the set of complex words in a language.

Languages with a rich system of inflectional morphology such as Turkish, it would even be quite absurd to assume that all these forms are memorized as such. The memory load for such languages can be reduced considerably by making use of rules. A possible model is that language users acquire morphological rules by constructing abstract rules or templates on the basis of their lexical knowledge.

There are also morphological patterns that are not productive, but can still be called patterns e.g. English past tense verb forms. Regular past tense forms are created by suffixing -ed to the stem. On the other hand there are a number of stem-changing verbs where the past tense form and the past participle are marked by having a different vowel than the present tense form e.g. an -ing/k -ang/k-ung/k pattern for verbs: ring, sing, spring, drink, shrink, sink, stink

The class of verbs with this kind of vowel alternation will normally not be extended. Yet, it is clear that native speakers of English are able to recognize the abstract pattern involved. Some speakers tend to also inflect the verb to bring as brang and brung.

If people are asked to make the past tense forms of non-existing verbs with a similar phonological make-up as the words spling, they tend to come up with either splang or splung. Some verbs once regular have changed into irregular ones: dive (with the past tense form dove not dived in American English). These facts have led a number of linguists to defend a dual system model of morphological knowledge.

The past tense and participial forms are not stored in lexical memory, but always created by rule. The irregular forms are stored in memory. These stored irregular forms are linked to each other in an associative way, and thus the language user will be able to discover similarity patterns such as -ing/k -ang/k -ung/k. This explains why the pattern may be extended incidentally to similar verbs such as to bring. The dual system theory has also been defended based on a type of evidence: neurological evidence. Jaeger et al. (1996) have argued that regular and irregular verbs in English are processed by different neural systems. The processing location determined by PET-scans.

A second type of rule-less approach is the model proposed by Bybee (1988, 1995, 2001). In her model, individual inflectional forms of words are listed in the lexicon. This also applies to past tense verb forms, both regular and irregular ones.

The regularities in the formal relationship between present tense and past tense forms are captured by abstract schemas that can be constructed on the basis of sets of words.

A third single system approach without rules is the analogy-based model. To be sure, analogy has been recognized as a source of new complex forms in all kinds of morphological models.

In the case, a new word formed on the basis of a related existing word, without an abstract morphological template being involved. The coinage of the English word seascape is based on the word landscape land : sea = landscape :

This survey of different models of morphological knowledge, and the debate about these models illustrates that morphological phenomena play an important role in the debate on the nature of linguistic knowledge. Morphology is an important window on the mental representation of natural language and the human mind.

Topic- 084: Morphological Processing

The insight that complex words are often stored as such in the lexicon raises the question of how they are processed. There are two ways in which a complex word can be processed, the direct route, and the indirect route. When perception of complex words is involved, the direct route means that we do not first parse the complex word, but go directly to its representation in the lexicon, in order to access its meaning.

The indirect route means that a complex word is parsed into constituent morphemes, and that its meaning is computed after we have gained access to its constituent morphemes and their meanings. There are data that might be interpreted as showing that language users try to parse words into morphological sub-constituents.

For instance, when speakers of Dutch are confronted with non-words in a lexical decision task, it takes more time to make the correct decision when that word begins with a syllable that could have been a prefix, compared to words where this does not apply.

Dutch words aderibag, afgeblar, afbegepakt

1 no potential prefix

2, begins with two prefixes (af-, ge-),

3. begins with three prefixes (af-, be-, ge-).

Correlating with this difference, the last nonword has longest response latency, the language users try to strip prefixes from words that are not in their lexicon.

The existence of two routes has led to the assumption of morphological race models. In such models, both routes will be used for word recognition, and compete with each other. If the word is not stored in lexical memory, the indirect route is the only one that will lead to recognition, on the basis of parsing. If the word is stored, the direct route will be faster than the indirect route if

the stored complex word has a high frequency, that is, a high level of activation. For a complex word with a low frequency, on the other hand, parsing is the obvious route.

Both routes are followed, and one of them will turn out to be the fastest, depending on the level of activation of the complex word.

The parsing of complex words is also affected by relative frequency: the frequency of the complex word compared to that of its base word. If the complex word has a lower frequency than the base word, this will make parsing of that complex word easier and more efficient than direct retrieval.

If the inverse situation obtains, direct access to the complex word will be the most efficient route. *dazzlement* will be parsed than its base *dazzle*, *government* will be accessed as a whole than that of its verbal base *govern*. Note that the meaning of *government* is not completely compositional, unlike that of *dazzlement*. The more complex words with a certain affix will be parsed, the more productive the corresponding word-formation process parsing the affix will increase its activation level. The morphological race model can also be used as a model of word production.

It helps us to understand why it is irregular forms of high frequency that are maintained in a language, whereas low-frequency irregular forms disappear more easily. *Went* is a high-frequency form, this route will be faster than the indirect route where the past tense form has to be created by rule, resulting in the incorrect *goed*.

Lesson 15

MORPHOLOGY AND LANGUAGE CHANGE**Topic- 085: Nature of Language Change: Internal**

Geoffrey Chaucer's famous Canterbury Tales

(1) When that Aprill with his shoures soote

“When that April with his showers sweet”

The droghte of March hath perced to the roote

“The drought of March has pierced to the root”

And bathed every veyne in swich licour

“And bathed every vine in such liquid”

Of which vertu engendred is the flour;

“Of which virtue engendered is the flower”

Middle English as used in these lines is different from Modern English. If we assume that it is still the same language, English, we have to conclude that languages can change. But why do they? The statement that languages change is in fact metaphorical in nature. It presupposes that we conceive of a language as an organism that grows, changes, and sometimes dies. Languages have a mode of existence outside their users. A language primarily exists in the minds of its speakers. That is why we say that a language has died when its last speaker has died. So it is speakers that change their language while using it in language perception and production. Why do language users change their language? In fact, ‘change’ is not always the appropriate word for what is going on. It would be better to speak of construction or innovation of language.

What does change when new words or new meanings of words get established is the lexical norm of that language, not the system behind it. The main reason for changing the lexical norm of a language is that language users need expressions for new concepts, or new things. One way of meeting this need is extending the meaning of existing words.

mouse acquired

hard disk has-been

anti-virus-program

Lexical gaps can also be filled through borrowings. Dutch, for example, has borrowed many words from English recently e.g. printer drukker for “printer” .

The change of the lexical norm consists of two steps. The first step is that an individual language user coins a new lexical unit (a word or a phrase), borrows one from another language, Or gives an existing word a new interpretation. Supra-individual lexical norm: change accepted by many users Lexical innovation: users need to express new concepts. The study of language change has a psychological and a social dimension. A second source of linguistic innovation besides changing the lexical norm is reanalysis. Language users cannot grasp the system behind a language in a direct fashion. The only evidence they have are outputs of the system, concrete cases of language use. This opens up the possibility that a language user reconstructs the system underlying the perceived outputs in a slightly different way than previous users. Children do the same by using goed and bringed. In sum, language users may be involved in language change at different stages of their lives.

Topic- 086: Nature of Language Change: External

Language change may also be effected by external causes, in particular language contact. This is called external change. Language contact may lead to borrowing of words from other languages. Most Germanic languages borrowed many words from French, of culture, science, and diplomacy. English was influenced much earlier and more strongly by French due to the Norman conquest of 1066.

Borrowing had its impact on the morphology of Germanic languages. In Dutch, a number of affixes of French origin are used productively in combination with non-native stems, and sometimes even with native stems (as in flauw-iteit “silly joke” derived from the native adjective flauw “silly”). Borrowing of inflectional forms may also take place. English has Greek and Latin plurals, such as criteria, schemata, and data. These learned plurals came to be replaced with regular plurals such as schemas and museums. Lack or rarity of morphology is considered by some linguists to be typical for pidgin and creole languages.

From recent research: both pidgins and creoles do have morphology, both inflection and word-formation (derivation, compounding, reduplication). Language contact as a source of loss of inflectional morphology can be observed in the development of Afrikaans: mother language Dutch. Afrikaans has an extremely simple kind of inflectional morphology, with a present tense form that is identical to the stem, without person and number marking, and a periphrastic form for the non-present tense. Morphological change is the erosion of inflectional systems in the Germanic and Romance languages, referred to as deflection. English has the poorest inflectional system of all.

Most Germanic languages have lost their case system and the three-gender system. Language change is obviously not restricted to the morphological system of languages. Quotation above from Chaucer also shows phonological, syntactic, and semantic changes. Phonological change is exemplified by the loss of the final vowel of roote that has become root in Modern English. The word order noun–adjective in shoures soote corresponds with the Modern English word order

adjective–noun: sweet showers. An example of semantic change is that the present-day meaning of liquor is no longer “liquid”, but denotes alcoholic liquids only.

Topic- 087: The Historical Sources of Morphology

Morphological systems not only erode, they may also arise. An example is the emergence of nominal compounding in Germanic languages that arose from noun phrases of the type:

[N-GEN N]NP

Harry’s book

The pre-nominal noun with genitive case functions as a modifier of the head noun. Such noun phrases were reinterpreted as compound nouns. Univerbation, the reinterpretation of phrases as words, does not necessarily lead to language change. In linguistics, univerbation is the diachronic process of combining a fixed expression of several words into a new single word. Some English examples include always from all [the] way (the s was added later), onto from on to, albeit from all be it, and colloquial gonna from going to.

These changes imply the addition of lexemes to the lexicon, but do not affect the morphological system. Although this kind of univerbation did not lead to new morphological rules, : when a word is multi-morphemic, this does not necessarily imply that it has been created by a morphological process. The frequency plays a role in grammaticalization: defined as ‘the process by which a frequently used sequence of words or morphemes become automated as a single processing unit’ (Bybee 2003: 603).

[G]rammaticalization is that subset of linguistic changes through which lexical item in certain uses becomes a grammatical item, or through which a grammatical item becomes more grammatical’ (Hopper and Closs Traugott1993: 2).An example of the change from lexical to grammatical item is the development of verbs into auxiliaries.

In English the verb to have not only functions as a main verb, with the meaning “to possess”, but also as an auxiliary in perfect tense forms. The verb can has lost its status as lexical verb completely, and functions as a modal auxiliary only. In these examples grammaticalization does not create morphology. This does happen if a (lexical or grammatical) morpheme becomes an affix. The phenomenon of degrammaticalization has raised the question whether grammaticalization is subject to the condition of unidirectionality. Can a grammatical morpheme become less grammatical, or even lexical as well. This depends on one’s definition of grammaticalization. If unidirectionality is one of its defining properties, changes into the opposite direction simply require another descriptive term, for instance degrammaticalization. The reinterpretation of the English genitive case morpheme -s as a clitic is a restricted form of degrammaticalization.

It is a restricted case because the suffix does not end up as a full lexical word. In Early Modern English it has also been interpreted as the weak form of the possessive pronoun *his*, that is, as a full word. Shakespeare writes: the count *his* galleys “the count’s galleys”. This change is related to the fact that English lost its case system and this made a reanalysis of the genitive suffix *-s* necessary. Degrammaticalization is restricted in nature. Drammaticalization is a pervasive phenomenon in natural languages.

Topic- 088: Changes in Morphological Rules

Erosion of inflectional morphology has taken place in most the Germanic and Romance languages. This means that some inflectional rules have disappeared from the grammar of those languages. Dutch, for instance, has lost its morphological cases. The case system has only survived in a number of fixed expressions, prepositional in origin. The prepositions in these phrases govern a particular case.

Some of these frozen expressions are considered as words, others are still spelt as phrases. The case endings are in italics: *te-gelijker-tijd* “at the same time, simultaneously” *met dien verstande* “with that understanding” *in dier voege* “in such a manner”

It is quite a complex system for many non-native speakers. No wonder this system broke down in Afrikaans, a creolized variant of Dutch. In the new system, the adjectives are divided into two classes: either they always carry the inflectional schwa in attributive position, or they never do. The adjectives that do take the schwa are morphologically complex adjectives, and simplex adjectives that exhibit stem allomorphy. For instance, the adjective *sag* “soft” has the stem allomorph *sagte*, as shown in the inflected form *sagte*. Hence, it will have the form *sagte* in attributive position, whereas *sag* is used in predicative position. What we can learn from these facts is that relics of a previous stage of the language, “historical junk” in the words of Lass (1990), might be kept, and even re-used in a different way. That is, these words have lost their morphological structure. Another type of possible change in derivational rules is a change in the category of the base words to which they apply. Very productive rules tend to extend their domain of application to new categories. The productive diminutive suffix *-tje* of Dutch, for instance, is no longer restricted to the domain of nouns, but also attaches to adjectives, verbs, and adverbs:

strijk “to stroke” *strijk-je* “small string orchestra” *blond* “blond” *blond-je* “blond girl” *uit* “out” *uit-je* “outing”

The changes of the French suffix *-age* also illustrate category change (Fleischmann 1977). The suffix *-age* entered many European languages through borrowing of French words in *-age*. English has extended its use to native, Germanic nominal, and verbal stems,

base N: *foot-age*, *front-age*, *mile-age*, *shipp-age*, *wreck-age*, *yard-age*

base V: *break-age*, *brew-age*, *cover-age*, *drain-age*, *leak-age*, *sew-age*.

The word shortage shows that -age has even been extended to adjectival bases. In Dutch we find both denominal and deverbal nouns in -age; as in English, it has also been attached to native stems, an indication of its productivity.

Topic- 089: Changes in Word Structure

Complex words, once they are coined, may be subject to reanalysis and reduction. In reanalysis, words receive a different structural interpretation. A classical example is the reinterpretation of the word hamburger. This word, derived from the base noun Hamburg, and denoting a specific kind of food originating from that city, received the following structural reanalysis:

(24) [[hamburg]er] > [[ham][burger]]

Reanalysis can only be observed when the reanalysed structure serves as a model for new words. Many new words in -burger have been coined, such as beefburger, cheeseburger, fishburger, turkeyfurter on the model of frankfurter. The piece burger itself has been reinterpreted as a noun, witness the brand name Burger King. So in fact we cannot conclude that a new suffix -burger has developed in English. Burger is a new noun denoting a particular kind of fast food, and can be used as the head of nominal compounds. Semantic reinterpretation may also lead to a new set of words.

This has been the case for -gate, as used in Watergate (the name of an apartment building in Washington, DC, that was burglarized by order of President Nixon). The morpheme -gate received a new interpretation, “political scandal”, thus leading to many new words such as Monicagate and Irangate. This suffix also acquired the more general meaning of “scandal”. Reanalysis may also have the effect of a sequence of affixes becoming an affix. A morphological structure [[[x]A]B] can be reinterpreted as [[x]AB].

An example from Dutch is the suffix -erij, originally a combination of the suffixes -er and -ij: base noun in -er noun in -ijbak “to bake” bakk-er “baker” bakk-er-ij “bakery” > bakk-erij. This phenomenon of two affixes becoming one is referred to as affix telescoping. Prefixes may also lose their morphological status. This is quite clear in English borrowings from Latin such as abortion and adoption. prefixes ab- and ad- respectively. The syllabifications these words are a.bor.tion and a.dop.tion respectively. The Dutch prefix ge-, as in geloof “to believe” is no longer productive as a verbal prefix. In Afrikaans, the verb geloof has become a simplex verb, glo, with deletion of the schwa.

Lesson 16**TRADITIONAL GRAMMAR: CATEGORIES & FUNCTIONS****Topic- 090: Introduction**

What is grammar - the meanings of the notion of grammar. An informal and broad definition of grammar is that grammar is a branch of linguistics that studies the form and structure of words (morphology) and the way in which the units of sentence structure are combined in sentences (syntax).

Etymology

The word grammar is derived from Greek, which means "art of letters", from γράμμα (gramma), "letter", itself from γράφειν (graphein), "to draw, to write". The same Greek root also appears in graphics, grapheme, and photograph. In linguistics, grammar (from Greek: γραμματική) is the set of structural rules governing the composition of clauses, phrases, and words in any given natural language. The term refers also to the study of such rules, and this field includes phonology, morphology, and syntax, often complemented by phonetics, semantics, and pragmatics.

Speakers of a language have a set of internalized rules for using that language and these rules constitute that language's grammar. The vast majority of the information in the grammar is – at least in the case of one's native language – acquired not by conscious study or instruction, but by observing other speakers. Much of this work is done during early childhood; learning a language later in life usually involves a greater degree of explicit instruction. Thus, grammar is the cognitive information underlying language use.

The term "grammar" can also be used to describe the rules that govern the linguistic behaviour of a group of speakers. The term "English grammar", therefore, may have several meanings. It may refer to the whole of English grammar, that is, to the grammars of all the speakers of the language, in which case, the term encompasses a great deal of variation. Alternatively, it may refer only to what is common to the grammars of all, or of the vast majority of English speakers (such as subject–verb–object word order in simple declarative sentences). Or it may refer to the rules of a particular, relatively well-defined variety of English (such as standard English for a particular region).

A specific description, study or analysis of such rules may also be referred to as a grammar. A reference book describing the grammar of a language is called a "reference grammar" or simply "a grammar". A fully explicit grammar that exhaustively describes the grammatical constructions of a language is called a descriptive grammar. This kind of linguistic description contrasts with linguistic prescription, an attempt to discourage or suppress some grammatical constructions, while promoting others. For example, preposition stranding occurs widely in Germanic

languages and has a long history in English. John Dryden, however, objected to it (without explanation), leading other English speakers to avoid the construction and discourage its use.

Outside linguistics, the term grammar is often used in a rather different sense. In some respects, it may be used more broadly, including rules of spelling and punctuation, which linguists would not typically consider to form part of grammar, but rather as a part of orthography, the set of conventions used for writing a language. In other respects, it may be used more narrowly, to refer to prescriptive grammar only and excluding those aspects of a language's grammar that are not subject to variation or debate. Jeremy Butterfield claimed that, for non-linguists, "Grammar is often a generic way of referring to any aspect of English that people object to."

Topic- 091: Some Authorities of Grammatical Tradition

The Babylonians made some early attempts at language description, but the first systematic grammars, of Sanskrit, originated in Iron Age India, with Yaska (6th century BC), Pāṇini (6-5th century BC) and his commentators Pingala (c. 200 BC), Katyayana, and Patanjali (2nd century BC). Tolkāppiyam, the earliest Tamil grammar, is mostly dated to before the 5th century AD.

In the West, grammar emerged as a discipline in Hellenism from the 3rd century BC forward with authors like Rhyanus and Aristarchus of Samothrace. The oldest known grammar handbook is the Art of Grammar (Τέχνη Γραμματική), a concise guide to speaking and writing clearly and effectively, written by the ancient Greek scholar Dionysius Thrax (c. 170–c. 90 BC), a student of Aristarchus of Samothrace who established a school on the Greek island of Rhodes. Dionysius Thrax's grammar book remained the primary grammar textbook for Greek schoolboys until as late as the twelfth century AD. The Romans based their grammatical writings on it and its basic format remains the basis for grammar guides in many languages even today. Latin grammar developed by following Greek models from the 1st century BC, due to the work of authors such as Orbilius Pupillus, Remmius Palaemon, Marcus Valerius Probus, Verrius Flaccus, and Aemilius Asper.

A grammar of Irish originated in the 7th century with the Auraicept na n-Éces. Arabic grammar emerged with Abu al-Aswad al-Du'ali in the 7th century. The first treatises on Hebrew grammar appeared in the High Middle Ages, in the context of Mishnah (exegesis of the Hebrew Bible). The Karaite tradition originated in Abbasid Baghdad. The Diquduq (10th century) is one of the earliest grammatical commentaries on the Hebrew Bible. Ibn Barun in the 12th century compares the Hebrew language with Arabic in the Islamic grammatical tradition.

Belonging to the trivium of the seven liberal arts, grammar was taught as a core discipline throughout the Middle Ages, following the influence of authors from Late Antiquity, such as Priscian. Treatment of vernaculars began gradually during the High Middle Ages, with isolated works such as the First Grammatical Treatise, but became influential only in the Renaissance and Baroque periods. In 1486, Antonio de Nebrija published Las Introducciones Latinas Contrapuesto el romance al Latin, and the first Spanish grammar, Gramática de la lengua castellana, in 1492.

During the 16th-century Italian Renaissance, the *Questione della lingua* was the discussion on the status and ideal form of the Italian language, initiated by Dante's *de vulgari eloquentia* (Pietro Bembo, *Prose della volgar lingua* Venice 1525). The first grammar of Slovene language was written in 1583 by Adam Bohorič.

Grammars of non-European languages began to be compiled for the purposes of evangelization and Bible translation from the 16th century onward, such as *Grammatica o Arte de la Lengua General de los Indios de los Reynos del Perú* (1560), and a Quechua grammar by Fray Domingo de Santo Tomás.

From the latter part of the 18th century, grammar came to be understood as a subfield of the emerging discipline of modern linguistics. The *Deutsche Grammatik* of the Jacob Grimm was first published in the 1810s. The *Comparative Grammar* of Franz Bopp, the starting point of modern comparative linguistics, came out in 1833.

Topic- 092: Concept of Parts of Speech is Different in Different Languages

In traditional grammar, a part of speech is a category of words (or, more generally, of lexical items) which have similar grammatical properties. Words that are assigned to the same part of speech generally display similar behavior in terms of syntax—they play similar roles within the grammatical structure of sentences—and sometimes in terms of morphology, in that they undergo inflection for similar properties. Almost all languages have the word classes noun and verb, but beyond these two there are significant variations among different languages.

For example, Japanese has as many as three classes of adjectives, where English has one (not to be confused with the seven types of English adjectives, or the fact that English adjectives can modify both nouns and pronouns). Chinese, Korean, Japanese and Vietnamese have a class of nominal classifiers; and many languages do not distinguish between adjectives and adverbs, or between adjectives and verbs. Because of such variation in the number of categories and their identifying properties, analysis of parts of speech must be done for each individual language.

Nevertheless, the labels for each category are assigned on the basis of universal criteria.

Western tradition

A century or two after the work of Nirukta, the Greek scholar Plato wrote in his *Cratylus* dialog that "... sentences are, I conceive, a combination of verbs [rhêma] and nouns [ónoma]".

Aristotle added another class, "conjunction" [sýndesmos], which included not only the words known today as conjunctions, but also other parts (the interpretations differ; in one interpretation it is pronouns, prepositions, and the article).

By the end of the 2nd century BC grammarians had expanded this classification scheme into eight categories, seen in the *Art of Grammar*, attributed to Dionysius Thrax:

- Noun (ónoma): a part of speech inflected for case, signifying a concrete or abstract entity
- Verb (rhêma): a part of speech without case inflection, but inflected for tense, person and number, signifying an activity or process performed or undergone
- Participle (metokhê): a part of speech sharing features of the verb and the noun
- Article (árthron): a declinable part of speech, taken to include the definite article, but also the basic relative pronoun
- Pronoun (antōnymíā): a part of speech substitutable for a noun and marked for a person
- Preposition (próthesis): a part of speech placed before other words in composition and in syntax
- Adverb (epírrhēma): a part of speech without inflection, in modification of or in addition to a verb, adjective, clause, sentence, or other adverb
- Conjunction (sýndesmos): a part of speech binding together the discourse and filling gaps in its interpretation

The Latin grammarian Priscian (fl. 500 AD) modified the above eightfold system, excluding "article" (since the Latin language, unlike Greek, does not have articles), but adding "interjection".

The Latin names for the parts of speech, from which the corresponding modern English terms derive, were *nomen*, *verbum*, *participium*, *pronomen*, *praepositio*, *adverbium*, *conjunctio* and *interjectio*. The category *nomen* included substantives (*nomen substantivum*, corresponding to what are today called nouns in English), adjectives (*nomen adjectivum*) and numerals (*nomen numerale*).

This is reflected in the older English terminology noun substantive, noun adjective and noun numeral. Later the adjective became a separate class, as often did the numerals, and the English word noun came to be applied to substantives only.

Topic- 093: Parts of Speech in English

Works of English grammar generally follow the pattern of the European tradition as described above, except that participles are now usually regarded as forms of verbs rather than as a separate part of speech, and numerals are often conflated with other parts of speech:

- noun
- verb
- adjective
- adverb
- pronoun
- preposition
- conjunction
- interjection

article or (more recently) determiner

Functional classification

Linguists recognize that the above list of eight- or nine-word classes is drastically simplified. For example, "adverb" is to some extent a catch-all class that includes words with many different functions. Some have even argued that the most basic of category distinctions, that of nouns and verbs, is unfounded, or not applicable to certain languages. Modern linguists have proposed many different schemes whereby the words of English or other languages are placed into more specific categories and subcategories based on a more precise understanding of their grammatical functions.

In English grammar, open class refers to the category of content words—that is, parts of speech (or word classes) that readily accept new members. Contrast with closed class. The open classes in English are nouns, lexical verbs, adjectives, and adverbs. Within a given category, subgroups of words may be identified based on more precise grammatical properties. For example, verbs may be specified according to the number and type of objects or other complements which they take. This is called subcategorization.

Many modern descriptions of grammar include not only lexical categories or word classes, but also phrasal categories, used to classify phrases, in the sense of groups of words that form units having specific grammatical functions. Phrasal categories may include noun phrases (NP), verb phrases (VP) and so on. Lexical and phrasal categories together are called syntactic categories.

Word classes may be either open or closed. An open class is one that commonly accepts the addition of new words, while a closed class is one to which new items are very rarely added. Open classes normally contain large numbers of words, while closed classes are much smaller. Typical open classes found in English and many other languages are nouns, verbs (excluding auxiliary verbs, if these are regarded as a separate class), adjectives, adverbs and interjections. Ideophones are often an open class, though less familiar to English speakers, and are often open to nonce words (A nonce word (also called an occasionalism) is a lexeme created for a single occasion to solve an immediate problem of communication. For instance, in child language testing, examples of such words include "wug" and "blicket). Typical closed classes are prepositions (or postpositions), determiners, conjunctions, and pronouns.

The open–closed distinction is related to the distinction between lexical and functional categories, and to that between content words and function words, and some authors consider these identical, but the connection is not strict. Open classes are generally lexical categories in the stricter sense, containing words with greater semantic content, while closed classes are normally functional categories, consisting of words that perform essentially grammatical functions. This is not universal: in many languages verbs and adjectives are closed classes, usually consisting of few members, and in Japanese the formation of new pronouns from existing nouns is relatively common, though to what extent these form a distinct word class is debated.

Words are added to open classes through such processes as compounding, derivation, coining, and borrowing. When a new word is added through some such process, it can subsequently be used grammatically in sentences in the same ways as other words in its class. A closed class may obtain new items through these same processes, but such changes are much rarer and take much more time. A closed class is normally seen as part of the core language and is not expected to change. In English, for example, new nouns, verbs, etc. are being added to the language constantly (including by the common process of verbing and other types of conversion, where an existing word comes to be used in a different part of speech). However, it is very unusual for a new pronoun, for example, to become accepted in the language, even in cases where there may be felt to be a need for one, as in the case of gender-neutral pronouns.

The open or closed status of word classes varies between languages, even assuming that corresponding word classes exist. Most conspicuously, in many languages verbs and adjectives form closed classes of content words. An extreme example is found in Jingulu, which has only three verbs, while even the modern Indo-European Persian has no more than a few hundred simple verbs, a great deal of which are archaic. (Some twenty Persian verbs are used as light verbs to form compounds; this lack of lexical verbs is shared with other Iranian languages.) Japanese is similar, having few lexical verbs. Basque verbs are also a closed class, with the vast majority of verbal senses instead expressed periphrastically.

In Japanese, verbs and adjectives are closed classes, though these are quite large, with about 700 adjectives, and verbs have opened slightly in recent years. Japanese adjectives are closely related to verbs. By contrast, Japanese pronouns are open class—if they can even be considered a class—and nouns become used as pronouns with some frequency.

Topic- 094: Prescriptive or Normative Grammar

The term prescriptive grammar refers to a set of norms or rules governing how a language should or should not be used rather than describing the ways in which a language is actually used.

Contrast with descriptive grammar.

Also called normative grammar and prescriptivism.

In a prescriptive grammar there is right and wrong language. Rules of “good” or “proper” usage, which dictate what is “good grammar” and what is “bad grammar” Example:

(1)She doesn't know him.

(2)She don't know him.

Besides there are other differences of the following types:

Case of pronoun: never use an object case after the verb 'to be'.

- It is me.
- Instead say; It is I.
- Mary runs faster than me.
- Say: Mary runs faster than I.

Similarly don't say: who are you talking to?

- Say: whom are you talking to?
- Never end a sentence with a preposition.
- Whom are you talking to?
- Say: to whom are you talking.
- Don't use ain't
- Don't use seen as the past tense of see (as in I seen him at the party last night).
- Don't use contractions
- Don't use sentence fragments
- Don't use dangling participles
- Coming to college, there were many tall trees.

Many of these rules are essentially taken from Latin> Latin was the classical language known by all educated people and one regarded as model for all languages. However, there is no reason why English should follow the Latin rules. For example English has no case endings for noun (except in genitive case). So why to follow case?

Topic- 095: Problems of Prescriptive Grammar to Answer Certain Questions

Although everyone knows or thinks they know what a word is and what a sentence is, both terms defy exact definition. The sentence as a linguistic concept has been defined in over 200 different ways, none of them completely adequate. Here are the most important attempts at defining the sentence: The traditional, or common sense definition states that a sentence is a group of words that expresses a thought. The problem comes in defining what a thought is. The phrase an egg expresses a thought but is it a sentence? A sentence like I closed the door because it was cold expresses two thoughts and yet it is one sentence.

Another definition is that a sentence is a group of words expressing a topic (old information) and some comment (new information) about that topic: John left. The problem with the topic-comment definition is that many sentences have no clear topic and comment structure: It's raining. Many sentences have no clear topic and comment structure: It--is raining. (The word it here is the so-called dummy it used to fill the subject slot for impersonal verbs in English. In polysynthetic languages the single word serve as a complete sentence much more frequently. In such languages, morphology rather than syntax usually expresses the distinction between subject and predicate.

Structural ambiguity

- Flying plan can be dangerous.
- The king is eager to please.
- The king is easy to please.

Words and phrases can be grouped according to their sentence building functions. Not all languages have the same parts of speech. Many languages have postpositions rather than prepositions. A common assertion is that all languages have at least nouns and verbs. It is true that all languages have some means of conveying information as a concept or as an event, but what a noun or verb is differs from language to language. In the Salishan languages of the Puget Sound, a single word can be translated into English as village and a village exist or there is a village; in other words, morphemes denoting stationary concepts are often bound roots that require verbal affixes to stand as words. So parts of speech--even nouns and verbs-- turn out to be at best fuzzy categories across languages, not identical or even present in every language.

Lesson 17

WHAT IS SYNTAX?

Topic- 096: Concept of Sentence in Prescriptive Grammar

A sentence is a group of words that are put together to mean something. A sentence is the basic unit of language which expresses *a complete thought*. It does this by following the grammatical rules of syntax. For example: ‘Sana is the most intelligent girl in the class.’ A complete sentence has at least a subject and a main verb to state (declare) a complete thought. Short example: ‘Walkers walk’.

In English and many other languages, the first word of a written sentence has a capital letter. At the end of the sentence there is a full stop or full point (American: 'period'). A Sentence typically contains a subject and predicate, conveying a statement, question, exclamation, or command, and consists of a main clause and sometimes one or more subordinate clauses. A phrase or clause is part of a sentence. This is an example of a sentence: ‘The dog is happy.’ In this sentence, 'The dog' is the subject, and 'is' is the verb. This is an example of a phrase: ‘The happy dog’. There is no verb, so we do not know anything about what the happy dog is doing. Therefore, it is not a sentence. A simple sentence has only one clause. For example: ‘The cat is sleeping.’

Notional definition: a sentence expresses a complete thought.

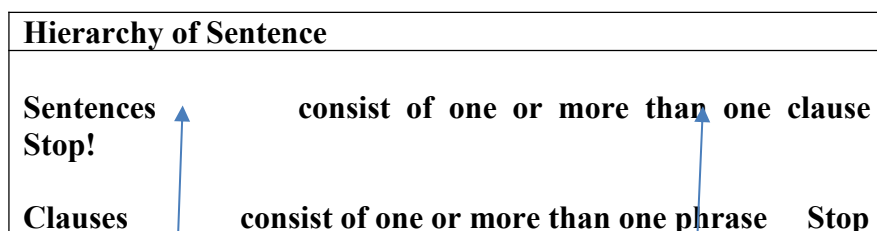
- A cabbage (It is not a complete thought)
- A man (It is not a complete thought)
- If it rains, I shan't come. (It is a complete thought)

A complete thought is a pair of subject and predicate. For example: ‘It's raining.

After discussing on what is a sentence and what is not a sentence and how we can understand the concept of a sentence, not focusing on a complete thought, Palmer, in his book ‘Grammar’, says that ‘*The sentence is a linguistic unit that has the grammatical structure we assign to it.*’

Topic- 097: Hierarchy of Sentence Grammar

Words, phrases, clauses, and sentences constitute what is called the GRAMMATICAL HIERARCHY of a sentence.



Phrases stop	consist of one or more than one word
Words stop	are separated by space before and after

Hierarchy of Sentence

Sentences are at the top of the hierarchy, so they are the largest unit. Words are at the lowest level, though again, some grammars go below the word to consider morphology. At the clause level and at the phrase level, two points should be noted: Although clauses are higher than phrases in the hierarchy, clauses can occur within phrases, as:

- The man *who lives beside us* is ill

Here we have a relative clause *who lives beside us* within the NP the man who lives beside us. Clauses can occur within clauses, and phrases can occur within phrases.

Topic- 098: Descriptive Grammar

The term descriptive grammar refers to an objective, nonjudgmental description of the grammatical constructions in a language. It's an examination of how a language is actually being used, in writing and in speech. Specialists in descriptive grammar examine the principles and patterns that underlie the use of words, phrases, clauses, and sentences. Kirk Hazen notes, "Descriptive grammars do not give advice: They detail the ways in which native speakers use their language. A descriptive grammar is a survey of a language. For any living language, a descriptive grammar from one century will differ from a descriptive grammar of the next century because the language will have changed." ("An Introduction to Language." John Wiley, 2015)

"Descriptive grammar," Edwin L. Battistella notes in "Bad Language," "is the basis for dictionaries, which record changes in vocabulary and usage, and for the field of linguistics, which aims at describing languages and investigating the nature of language." The term descriptive is a little bit misleading, as descriptive grammar does provide analysis and explanation of the language's grammar and not just description of it.

Contrast the type with prescriptive grammar, which notes how something should or should not be used, what is right and wrong. Prescriptive grammarians attempt to enforce rules concerning "correct" or "incorrect" usage. According to Donald G. Ellis, "All languages adhere to syntactical rules of one sort or another, but the rigidity of these rules is greater in some languages. It is very important to distinguish between the syntactical rules that govern a language and the rules that a culture imposes on its language. People spoke long before there were linguists around

to uncover the rules of speaking.... Prescriptive grammars, on the other hand, are the stuff of high school English teachers.

- The sentence "I ain't going,"
- *Who did you arrest?
- Whom did you arrest?
- *What are you waiting for?
- For what are you waiting?
- *I saw several octopuses at the beach.
- I saw several octopi at the beach.

Descriptive grammar, consists of rules that all speakers of a speech community adhere to, barring speech errors. Linguists are usually more interested in such rules, as they are reflective of the properties of actual language, not ideal language.

Topic- 099: Sentence Structures

Simple Sentences

A simple sentence contains a subject and a verb, and it may also have an object and modifiers. It contains only one independent clause. Examples:

- She read.
- She completed her literature review.
- He organized his sources by theme.
- They studied APA rules for many hours.

Compound Sentences

A compound sentence contains at least two independent clauses. These two independent clauses can be combined with a comma and a coordinating conjunction or with a semicolon. The following are the seven coordinators which are in the phrase FAN BOYS:

For And Nor But Or Yet So

For	Our team did not play well, for they did not practiced hard (the second clause is the reason for the first clause).
And	Our teachers are very diligent, and our management too is excellent. (The two clauses express equal, similar ideas).
Nor	Good students study well, nor do they waste their time. (nor means “and not” it joins two negative independent clauses. Notice that question word order is used after nor).

But	Other team players may exercise, but they may not exercise as Ali does. (The two clauses express equal, contrasting ideas).
Or	All students should take healthful diets, or they will risk getting some disease. (the two clauses express alternative possibilities).
Yet	Sportsmen give a little time to their studies, yet in the long run, they are better off than their peers. (The second clause is a surprising or unexpected contrast to the first clause).
Semicolon	He organized his sources by theme; then, he updated his reference list.

Complex Sentences

A complex sentence contains at least one independent clause and at least one dependent clause. Dependent clauses can refer to the subject (who, which) the sequence/time (since, while), or the causal elements (because, if) etc. of the independent clause. Here are a few examples:

- Although she completed her literature review, she still needed to work on her method section.
- They studied APA rules for many hours as they were so interesting.

Compound-Complex Sentences

Sentence types can also be combined. A compound-complex sentence contains at least two independent clauses and at least one dependent clause. Using some complex-compound sentences in writing allows for more sentence variety. Pay close attention to comma usage in complex-compound sentences so that the reader is easily able to follow the intended meaning.

- A simple sentence consists of one independent clause, and a compound sentence consists of two or more independent clauses while a compound -complex consists of two or more independent clauses and at least one dependent clause.

Topic- 100: Structures of a Noun Phrase

A noun phrase may be a single word working as noun or a group of two or more words that is headed by a noun (a person, place, thing, etc.) that includes modifiers (e.g., 'the,' 'a,' 'of them,' 'with her'). A noun phrase plays the role of a noun. In a noun phrase, the modifiers can come before or after the noun. Often a noun phrase is just a noun or a pronoun:

- *People* like to have *money*.
- *I* am tired.
- *It* is getting late.

or a determiner and a noun ...:

- *Our friends* have bought *a house* in *the village*.
- *Those houses* are very expensive.

... perhaps with an adjective:

- *Our closest friends* have just bought *a new house* in the village.

Sometimes the noun phrase begins with a quantifier:

- *All those children* go to school here.
- *Both of my younger* brothers are married
- *Some people* spend a lot of money.

Sometimes the noun phrase begins with numbers:

- He has *two cars*.

Quantifiers come before determiners, but numbers come after determiners:

- *My four children* go to school here. (All my children go to school here.)
- *Those two suitcases* are mine. (Both those suitcases are mine)

So, the noun phrase is built up in this way:

- Noun: people; money
- Determiner + noun: the village, a house, our friends; those houses
- Quantifier + noun: some people; a lot of money
- Determiner + adjective + noun: our closest friends; a new house.
- Quantifier + determiner + noun: all those children;
- Quantifier + determiner + adjective + noun: both of my younger brothers

The noun phrase can be quite complicated:

- a loaf of nice fresh brown bread
- the eight-year-old boy who attempted to rob a sweet shop with a pistol
- that attractive young woman in the blue dress sitting over there in the corner

Some words and phrases come after the noun. These are called postmodifiers. A noun phrase can be postmodified in several ways. Here are some examples:

with a prepositional phrase:

- a man with a gun
- the boy in the blue shirt
- the house on the corner

with an -ing phrase:

- the man standing over there
- the boy talking to Angela

with a relative clause:

- the man we met yesterday
- the house that Jack built
- the woman who discovered radium
- an eight-year-old boy who attempted to rob a sweet shop

with a that clause. This is very common with reporting or summarizing nouns like idea, fact, belief, suggestion:

- He's still very fit, in spite of the fact that he's over eighty.
- She got the idea that people didn't like her.
- There was a suggestion that the children should be sent home.

with a to-infinitive. This is very common after indefinite pronouns and adverbs:

- You should take something to read.
- I need somewhere to sleep.
- I've got no decent shoes to wear.

There may be more than one postmodifier:

- an eight-year old boy with a gun who tried to rob a sweet shop
- that girl over there in a green dress drinking a coke

There are four complex noun phrases in this section:

The accident happened at around 3:00 pm on Wednesday. A man climbing nearby who saw the accident said "It was the most amazing rescue I have ever seen." 42-year-old Joe Candler saw Miss Johnson's fall along with his partner Fay Hamilton. The rescue is the latest in a series of incidents on High Peak. In January last year two men walking on the peak were killed in a fall when high winds blew them off the mountain.

Topic- 101: Structures of Verb Phrase

A verb phrase is a syntactic unit and may consist of a single word or an auxiliary (helping) verb or modal verb preceding the main verb. The verb heads a verb phrase. The verb can be preceded by the negation word *not*. Verbs have distinctive word-shapes (morphologies). English verbs have two distinctive morphological characteristics.

i) the shape of the verb can encode the tense of the sentence;

- He eats meat.
- He ate meat.

Present and past tense is coded by the morphology of the verb.

ii) The second special morphological characteristic is that an English verb can agree with its subject if the verb is present and the subject is third person singular.

- She plays piano well.

The first grammatical form that can appear in a verb phrase in English is the auxiliary verb. In English, the twelve auxiliary verbs are have, be, do, and the nine modal verbs (can, could, may, might, must, shall, should, will, and would). Auxiliary verbs perform the functions of progressive, perfect, passive, operator, and modal within verb phrases.

Progressive Auxiliary Verb | Verb

- The dog was barking.

Perfect Auxiliary Verb | Verb

- The boys have completed their work.

Passive Auxiliary Verb | Verb

- The letter has been written.

Operator Auxiliary Verb | Verb

- Please do call me

Modal Auxiliary Verb | Verb

- They will study today before the test.

Progressive, perfect, passive, and modal auxiliary verb may appear with other progressive, perfect, passive, and modal auxiliary verbs.

The order in which auxiliary verbs can appear together is Modal-Perfect-Passive-Progressive. For example: Perfect | Progressive | Verb

had | been | sleeping

Perfect | Passive | Verb

has | been | broken

Modal | Progressive | Verb

could | be | swimming

Modal | Perfect | Passive | Verb

might | have | been | swindled

Modal | Perfect | Passive | Progressive | Verb

should | have | been | being | watched

A verb phrase often contains complements such as the following:

Subject	Verb	Complements				
		Indirect object	Direct object	Subjective complement	Objective complement	Adverbial
Ali	searched		the room			
He	had given	the girl	an apple			
They	make		him		the chairman	every year
His brother	grew			happier		Gradually
The girl	is			student		at a university
It	rained					steadily all day

Topic- 102: Structures of Adjective Phrase

What is an adjective? An adjective describes or modifies noun/s and pronoun/s in a sentence. Words like small, blue, sharp etc. It normally indicates quality, size, shape, duration, feelings, contents, and more about a noun or pronoun.

We cannot say whether a word is an adjective by looking at it in isolation. The form does not necessarily indicate its syntactic function. Some suffixes are indeed found only with adjective, e.g.: -ous, but many common adjectives have no shape e.g. nice, good, sharp etc.

The major functions of Adjective:

(1) Attributive

This premodifies a noun phrase

- A beautiful painting
- His main argument

(2) Predicative

Predicative adjective can be

(a) subject complement

- Your son is intelligent.

(b) objective complement

- He made his family proud.

(3) Postposition

Follow the item they modify (usually regarded as reduced relative clause) especially with indefinite pronouns ending in –body –one, -thing, -where e.g.

- I want to try on something larger (ie ‘which is larger’)

Structure of Adjectives Phrase

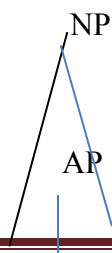
Obligatory postposition adjectives have different sense when used attributively	
• The president elect	‘Soon to take office’
• The city of London proper	‘as strictly defined’
In several compounds –mostly legal or quasi-legal – the adjective is postposition.	
Attorney general, body politic, court martial, heir apparent, notary public	
Some a-adjectives and concerned, involved	
• The house ablaze is next door to mine.	
• The people involved were not found.	

(4) Head of a noun phrase

In some cases, an adjective is used as head of the noun phrase while the noun is taken as understand or implied. It then does not inflect for number or for genitive case and must take a definite determiner before it.

- *The extremely old* need a great deal of attention. (here the noun people is implied)
- *The young* in spirit enjoy life. (here the noun people is implied)
- *The rich* will help only *the humble poor*. (here the noun people is implied)

Adjective phrase in the noun phrase



art A N

The white house

We can add quite to this phrase

- The quite white house
- *the white quite house
- *the quite house

Structure

The head of an adjective phrase is an adjective which may be

- Simple: big
- Comparative: bigger
- Superlative: biggest

The premodifiers of adjective are always adverbs:

- *extremely* hot

Post modifiers can be either adverbs:

- Very tall *indeed*
- Nice *enough*

Or preposition phrase

- M M A PP
- Rather too *hot* (for Comfort)

Lesson 18

PHRASE STRUCTURE IN PRESCRIPTIVE GRAMMAR

Topic- 103: Structures of Adverb Phrase

An adverb is a word that is used to change, modify or qualify several types of words including an adjective, a verb, a clause, another adverb, or any other type of word or phrase, with the exception of determiners that directly modify nouns. A good way to understand adverbs is to think about them as the words that provide context.

Specifically, adverbs provide a description of *how, where, when, in what manner and to what extent* something is done or happens. Normally, we can spot an adverb by the fact that it often ends in -ly, but there are lots of adverbs that don't end in this way e.g. fast, well, etc. Moreover, adverbs can be used in many combinations with each other *very nicely, extremely fast*, etc. In a sentence, adverbs answer the following questions:

- **When?** She always arrives early.
- **How?** He drives carefully.
- **Where?** They go everywhere together.
- **In what way?** She eats slowly.
- **To what extent?** It is terribly hot.

The phrase structure rule for a noun phrase says that an adjective phrase can be contained in a noun phrase. In contrast, an adverb phrase cannot, as is seen below.

- (5) The [AP quite quick] swimmer
- (6) *The [AdvP quite quickly] swimmer

Adverb phrases most characteristically roam free in the sentence and without very major effects on the meaning of the sentence.

Quite quickly I saw the swimmer.

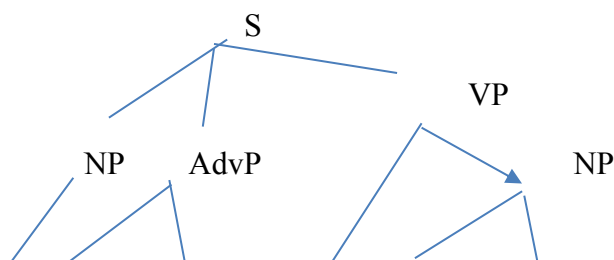
I *quite quickly* saw the swimmer.

I saw *quite quickly* the swimmer.

*I saw the *quite quickly* swimmer.

The adverb phrase cannot roam into the noun phrase.

I saw the swimmer *quite quickly*.

Structure of the adverb phrase in tree diagram

N deg Adv V art N
 I *quite quickly* saw the swimmer.

Two syntactic functions adverb

(1) Adverbial

(2) Modifier of adjective phrase and adverb

Adverb as an adverbial	
Adjuncts:	Integrated with the structure of the clause to some extent
	• They are waiting outside • He spoke about it briefly
Disjuncts	Not integrated within the clause. Semantically express an evaluation of what is said.
	• Frankly, I am tired • Fortunately, no one complained • They are probably at home
Conjuncts	Have connective function. Indicate the connection what is said and what was said before.
	• If you open all the window, then I am leaving • I have not looked into his qualification. He seems very intelligent, though

Adverb as modifier	
Modifier of an adjective	• That was a <i>very</i> funny film
Modifier of adverb	• There I have seen <i>so</i> very many letters like that one
Modifier of prepositional phrase	• The nail went <i>right</i> through the wall • His parents are <i>dead</i> against the trip
Modifier of determiner, predeterminer, Postdeterminer	• <i>Nearly</i> everybody came to our party (intensifying adv. premodifying pronoun) • <i>Over</i> two hundred survivors were reported • I paid more <i>than</i> ten pounds for it

Modifier of noun phrase

Sometimes an adverb phrase may also modify a noun phrase especially with:

(i) intensifiers like *quiet, rather* especially in British English.

- It was *rather* a mess.

- He was quite some player

(ii) Predeterminer: such

- I never heard such wickedness

(iii) Exclamatory: what

- What a fool he is!

Topic- 104: Minimal Noun Phrase

A single written sentence may stand for two different sentence structures, each having a different meaning. The structures differ because the words are grouped into different phrases. Phrases can be discovered by replacement and movement of sequences of words.

MEANING AND PHRASE STRUCTURE

- I was reading the letter to Hasan.

The question is how one sequence of words produce two different and alternative meanings. The ambiguity is based on two different structures for the same sequence of words. This ambiguity is called a STRUCTURAL AMBIGUITY.

To understand how this string of words can produce two different meanings we must say that they are invisibly organized into groups. A box is called a phrase

• I	was reading	a letter to Hasan	
• I	was reading	a letter	Hasan

REPLACEMENT AND PHRASE STRUCTURE

A sentence has a syntactic structure which is hidden from view but which can be revealed by various tests.

TEST: REPLACEMENT BY IT

If a sequence of words can be replaced by the word *it* without significantly changing the meaning, then that sequence of words is a phrase.

• I	was reading	a letter to Hasan	
• I	was reading	It	
• I	was reading	a letter to	Hasan
• I	was reading	it	

So, we are replacing like for like, with the word *it* being a **minimal** or **stripped-down** noun phrase standing for a more **filled-out noun phrase**.

Movement Test

The following movement tests help us know that certain group of words forms possible phrase boxes.

Movement Test		
I	was reading	the letter to Hasan
The letter to Hasan	was being read	
The letter	was being read	to Hasan

It is called a PRONOUN, though it should more properly be called a pro-noun-phrase because it substitutes for a noun phrase.

Topic- 105: Tree Structure of a Sentence

In descriptive grammar we define a sentence in the following way: a sentence represents an EVENTUALITY. An eventuality is an action or an event or a state of affairs: something that happens or something that is. The sentence represents an eventuality by separating out the type of eventuality from the abstract or concrete things which are involved in the eventuality. The type of the eventuality is prototypically represented by the verb, and the abstract or concrete things involved in the eventuality are prototypically represented by noun phrases. Look at the following sentence:

- The man in the hat casually gave a slimy fish to Toby.

Here the eventuality type is an *action of giving*, represented by the verb *gave*. The things involved in the eventuality are represented by the three noun phrases, *the man in the hat* (the giver), *a slimy fish* (the thing given), and *Toby* (the person/thing given to).

Apart from the type of eventuality and the things which participate in it, a part of the sentence can also express the circumstances of the eventuality. The adverb phrase *casually* expresses the circumstances here. The component parts of the sentence thus contribute distinct components of meaning to the overall meaning of the sentence.

Step 1 Look for the main verb and write V above it.

V

The man in the hat casually gave a slimy fish to Toby.

Step 2 A verb is always inside a verb phrase, so draw a line up from V to VP.

VP

V

The man in the hat casually gave a slimy fish to Toby.

Step 3 A verb phrase is usually inside a sentence, so draw a line up from VP to S. S is the root of the tree, the node which contains everything else.

S

VP

V

The man in the hat casually gave a slimy fish to Toby.

Step 4 Now identify the noun phrases by the substitution test. A noun phrase might be replaceable by him or her or it. In this sentence, *the man in the hat* is replaced by *he*, *a slimy fish* is replaced by *it* and *Toby* is replaceable by *him* so here are three noun phrase.

S

VP

V

The man in the hat casually gave a slimy fish to Toby.

He

casually gave

it

to him

Step 5 Underline the NPs and write NP above each of them.

S

NP

VP

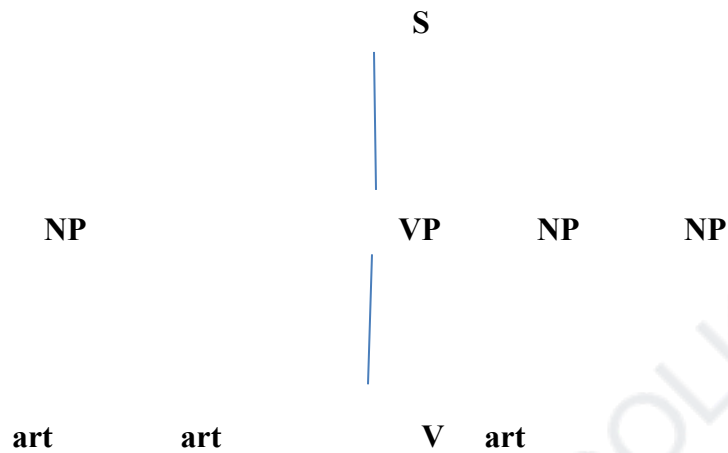
NP

NP

V

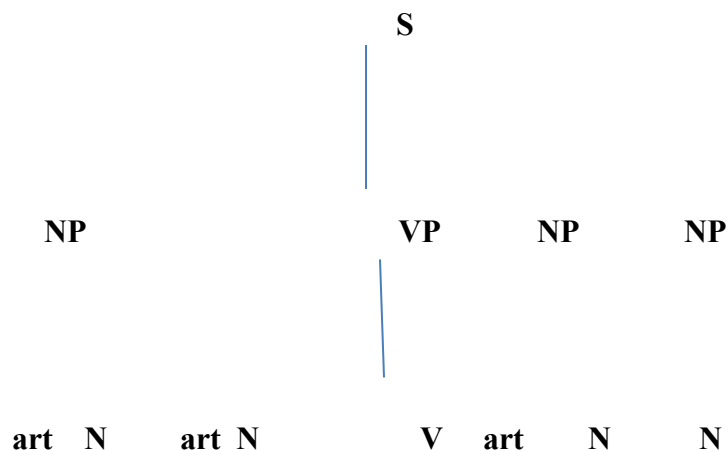
The man in the hat casually gave a slimy fish to Toby.

Step 6 Identify any articles in the sentence and write art above them.



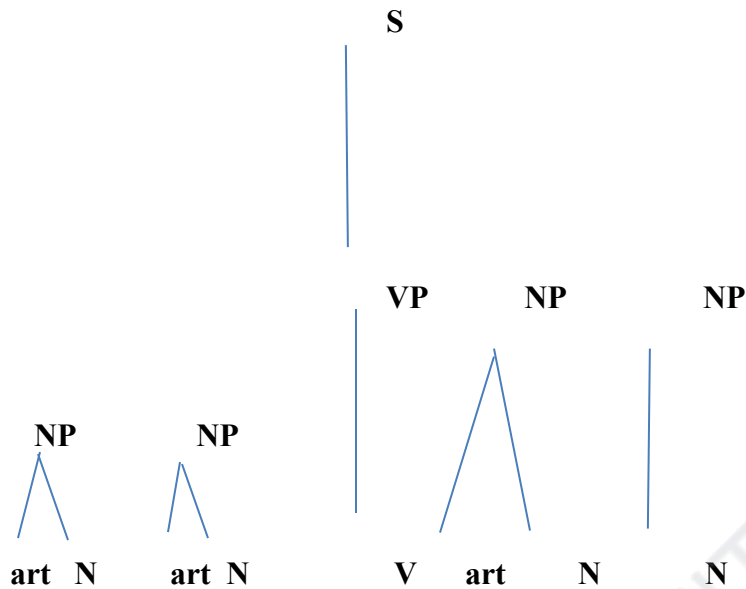
The man in the hat casually gave a slimy fish to Toby.

Step 7 Identify any nouns in the sentence, and write N above them. You know a noun can be pluralized and this is true of *man*, *hat*, and *fish*. You also know that names are nouns, so Toby is a noun.



The man in the hat casually gave a slimy fish to Toby.

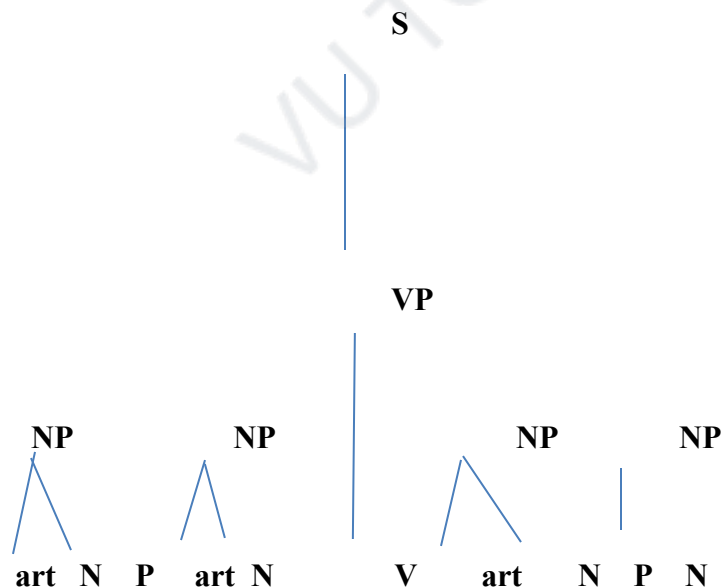
Step 8 An article is always immediately under NP, and N is always immediately under NP. Now work right to left.



The man in the hat casually gave a slimy fish to Toby.

In the first noun phrase we actually have two noun phrases.

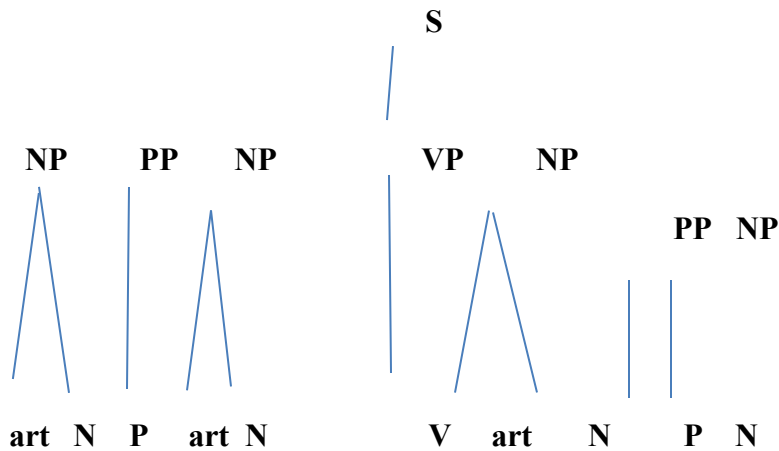
Step 9 The next stage is to identify the prepositions. Write P above the prepositions.



The man in the hat casually gave a slimy fish to Toby.

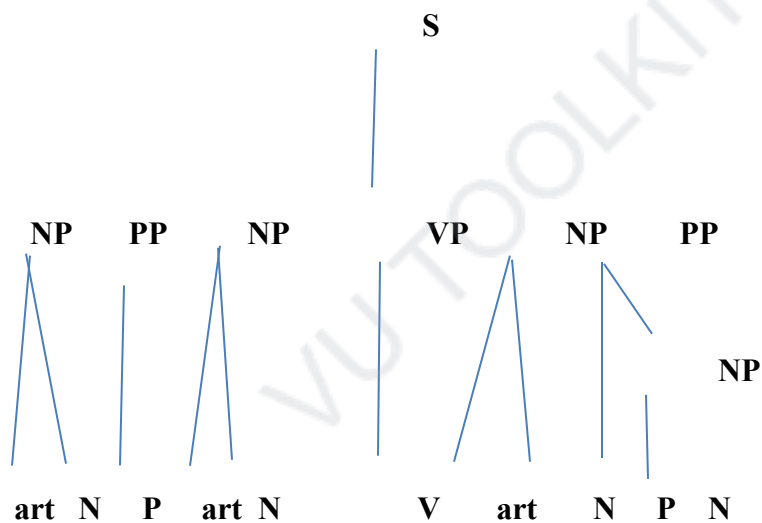
Step 10 A preposition is always in a preposition phrase, so write PP above each

preposition and join it up.



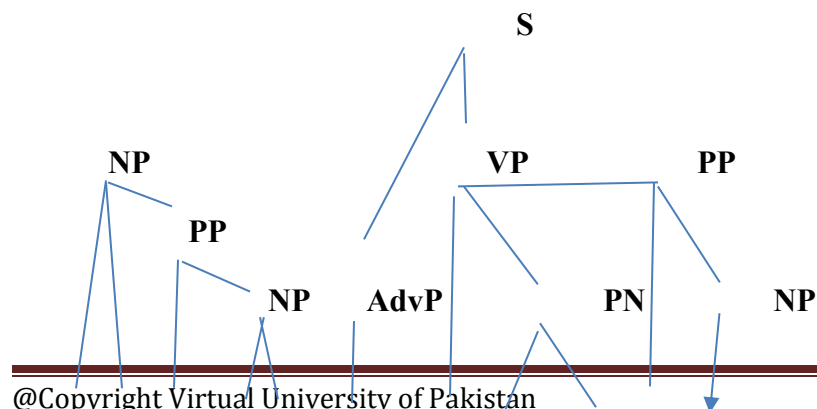
The man in the hat casually gave a slimy fish to Toby.

Step 11 Start putting together



The man in the hat casually gave a slimy fish to Toby.

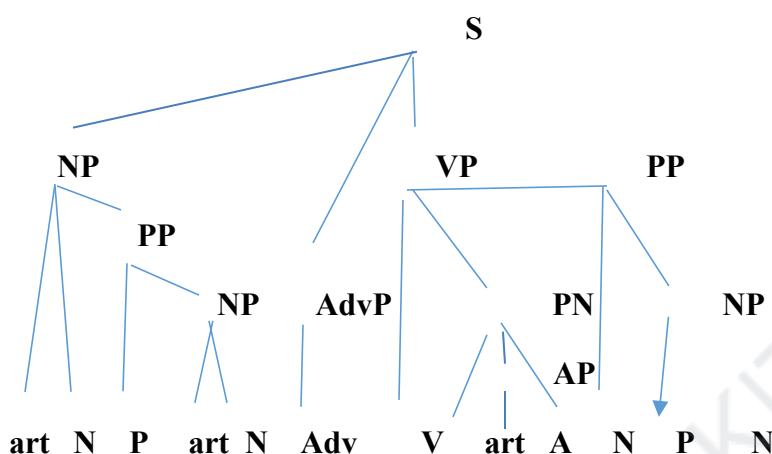
Step 12 The next stage is to identify casually.



art N P art N Adv V art N P N

The man in the hat casually gave a slimy fish to Toby.

Step 13 The final step is to identify slimy. It is an adjective. It has a morphological marker 'y'.



The man in the hat casually gave a slimy fish to Toby.

Causally is not part of the preceding noun phrase as adverb phrases are not part of the noun phrases.

Here are some of the principles and rules we followed in working out the tree structure.

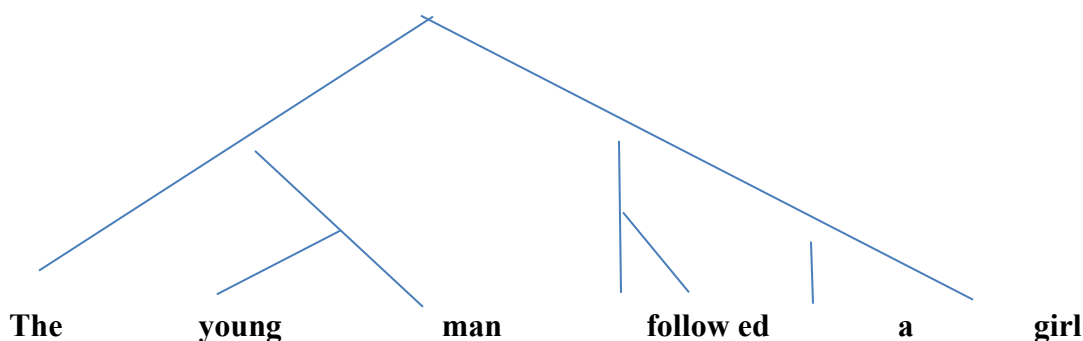
- Identify the verb first, using the tense test if necessary.
- Next identify the noun phrases, using substitution by a pronoun and any other appropriate tests. Once you have identified verb and noun phrases, you have a good sense of the overall shape of the sentence.
- Assume that every noun, verb, adjective, adverb and preposition is in its own phrase. So if you have two nouns you have two noun phrases, though one might be inside another, as we saw.
- If you find a preposition, look to see whether it combines with the following NP to form a PP.
- Once you have a PP, look to see whether it combines with the preceding noun to form a larger NP. In the sentence we looked at, one of the PPs does this and the other does not.
- An article is always in its own NP, and is usually at the beginning of it.

- An adverb (phrase) can usually be moved freely throughout the sentence. It rarely appears directly inside a noun phrase.
- Work from right to left across the sentence. This isn't a strict rule, but I think it makes the task easier.
- Remember that tests don't always work. Substitution by a pronoun did not identify the hat as a noun phrase, and instead we identified.

Topic- 106: IC analysis

In the last few decades syntax has undoubtedly become the most intensively studied topic within linguistics. Many new ideas and new linguistic models have been put forward. Even the best descriptive accounts have been written in conservative terms. These models have partly been influenced by modern theoretical proposal. The structuralists (American of 1950s) discovered and defined morpheme, but needed to establish what other grammatical units were and how they were distributed. Their approach was to divide the sentence up into IMMEDIATE CONSTITUENTS or ICs. Their aim was to discover relevant linguistic element. They were not clear whether there were any definable units beyond morpheme, apart from sentence. The principle of IC analysis is to cut a sentence into two and then to cut those parts into two and to continue with the segmentation until the smallest indivisible unit, the morphemes are reached. As a general principle the division is binary:

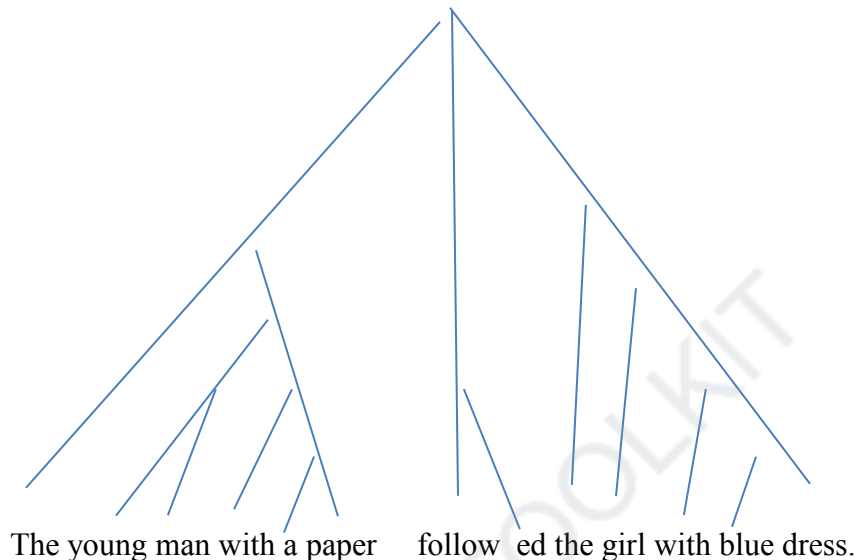
- The young man followed a girl.
- The young man + followed a girl.
- Followed + a girl
- a + girl
- The young man = the + young man
- Young + man
- Follow + ed
- The || young ||| man | follow ||| ed || a ||| girl.
((The (((young) (man))) ((followed) ((a) (girl)))).



This how IC analysis describes the constituents of a sentence.

Topic- 107: Methods of Display

The best method of display is to use a tree diagram where branching shows where to make the first cut? The answer lies in the notion of expansion. A sequence of elements said to be an expansion of another, if it can be substituted for it. Substitution is the basic procedure in all structuralist analysis. The first ICs are the following:



Reasons:

Firstly, *The young man with a paper* is an expansion of John as

John followed the girl with blue dress

Similarly

followed the girl with blue dress can be an expansion of arrived.

- The young man with a paper arrived.

On the basis of our arguments we can conclude: the ICs of our longer sentence are *The young man with a paper* and *followed the girl with blue dress*. Expansion being technical indicates the substitution of one sequence of element with another. Look at the following:

- Children
- American children
- Three American children
- Three American children with a dog
- Those three American children with a dog

Expansions can be of different types

Endocentric construction

An endocentric construction is one in which there is expansion of the more literal kind, and contains an element, or a word for which it can be substituted. This word is known as Head; like *children* in above example.

Exocentric construction

An exocentric construction cannot be substituted for any of its element and thus has no head. For example, *with a paper* is not an expansion of *with* or *a* or *paper*

It is not an expansion of any of its parts. An endocentric construction cannot be substituted for any of its element and thus has no head. Because of endocentricity that we can talk of noun phrase, verb phrase for these are endocentric constructions with nouns, verbs etc., as their heads.

Topic- 108: Some Flaws of the Theory

Distribution in IC analysis is a criterion (of structuralism). Same construction occurs in different places and help us with identification.

- *To go*
- *To go* is fun
- *wants to go*
- It is *wants* and *to go*
- Not *wants to* and *go*

Similarly

- The king of England is divided
- The and king of England
- Not *the king* and *of England*

In view of

- He became king of England

Simply dividing a sentence into ICs does not provide much information. Nevertheless, it can sometimes prove illuminating. It shows a certain type of ambiguity or difference of meaning which is related to hierarchical structure of the IC analysis.

Ambiguity can be clearly seen by paraphrasing the following

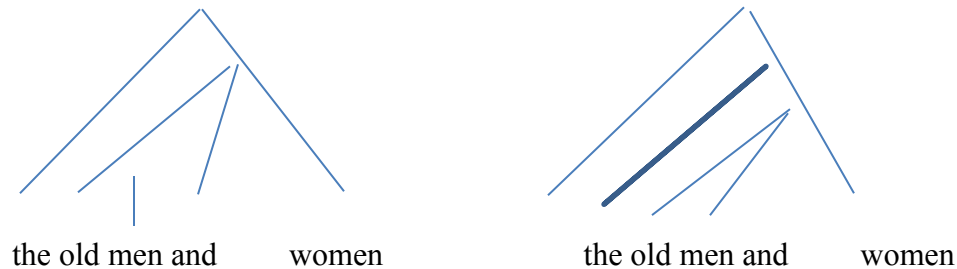
- The old men and the women

Is it 'the old men and the women

Or

the old men and the old women

we would have two different analyses by the tree:



A complication

With a conjunction it is difficult to provide a justification for a binary cut. We cannot make preference like the following:

Men and and women

Or men and *and women*

Similarly

- The Egyptian cotton shirt

First cut after Egyptian (a cotton shirt made in Egypt)

Or after cotton (a shirt made of Egyptian 'cotton')

Dividing actually occurring sentences out into their constituents' part is unsatisfactory for several reasons,

Firstly, it does not indicate what kind of elements those constituent parts are; it does not even identify except by implication.

Secondly, it does not show clearly that noun phrases are built on nouns, verb phrases on verbs etc.

Thirdly, IC analysis does not tell us how to form new sentences, i.e. to produce sentences that have not already been attested in some corpus of data.

Lesson 19

GRAMMATICAL THEORIES I

Topic- 109: Phrase Structure Rules

So after IC analysis what was needed was a model that would not merely segment the constituents, but would also identify them in grammatical terms, would, in addition, change the emphasis of, the linguistic analysis from mere description of sentences that have actually occurred to the specification of what sentences are possible in a language. Such a model is generative; it is said to GENERATE all the grammatical sentences of a language. The model is associated with transformational generative grammar, but since it deals with the constituency structure and not the transformational aspects of that grammar. IC analysis uses brackets, or bracketing. The bracketed elements can, however, be given grammatical identification by what is aptly known as LABELLED BRACKETING.

Let us consider the sentence:

- The man followed a girl.
- a can be identified as determiners,
- man and girl as nouns
- and followed as a verb

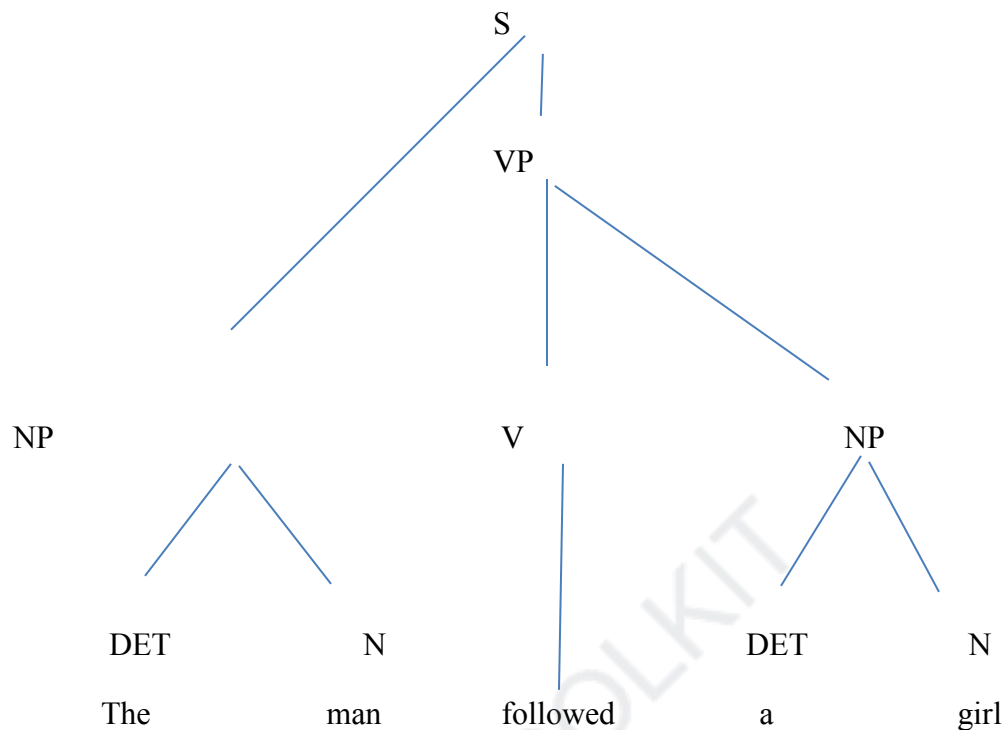
Further as *follow-* and *-ed*, are as the past tense of FOLLOW, but this is not wholly relevant and be ignored). Moreover, the man and a girl are also constituents of the same, but larger type, noun phrases, and if our IC analysis is correct, *followed a girl* is also a constituent, a verb phrase, while the whole is a sentence or clause.

- DET for determiner
- N for noun
- V for verb
- VP for v and noun phrase
- VP for verb phrase
- S for sentence

A label bracketing for the sentence

- [S [NP [DET the] N. man]] [VP [V followed] [NP [DET a] N. girl]]]

The representation of phrase structure is known as Phrase Marker or P-Marker



- Nodes: points that are joined by the lines
- Terminal nodes: nodes on the bottom line
- Dominate: one node is higher in tree than the other
- Immediately dominates: joined by a single line
- Dominance: a larger constituent may consist of one more of smaller kind.
- The tree structure preserves the linear order of constituents just like IC analysis does.
- The first noun phrase precedes the Verb phrase
- The verb phrase precedes the second noun phrase, the determiner precedes the noun.
- Precedence like dominance is clearly shown in the tree diagram.
- Label brecking and phrase structure trees provide much more information than IC analysis,
- Except implication does not state how new sentences can be generated

This can be done with the use of Phrase Structure Rules or PS Rules:

(1) $S \rightarrow NP - VP$

(2) $VP \rightarrow V - NP$

(3) $NP \rightarrow DET - N$

(4) V → followed

(5) DET → the, a

(6) N → man, girl

PS rules are different from traditional grammar. These rules can generate more than one sentence.

- A girl followed the man.
- The girl followed a man
- A man followed a man.

Topic- 110: Disadvantages of Phrase Structure

A phrase structure grammar is essentially a grammar of segmentation and categorization. It is a taxonomic model - a grammar of lists, an inventory of elements, and of class of sequences of elements. It is very strong in giving structural description of the language, but is deficient in generative capacity. Besides, it is incapable of accounting for all the intentions of native speakers. It fails to disambiguate all the ambiguities and understand all the synonymies. The processes that pose problems to PS grammar are:

1. ambiguities
2. synonymies
3. permutations
4. discontinuous constituents (e.g. particles)
5. remote relationship (e.g. those of cases)
6. concord phenomena
7. co-ordination.

A PS grammar runs into difficulties in describing syntactic structures of questions, negatives, passives, relatives, etc. easily. It fails to capture the deep meaning. It cannot discover the crucial notions, nor can it prevent the assignment of false, ungrammatical structure. PS rules are incapable of - except by having recourse to very arbitrary solutions - of accounting for the multiplicity of relations existing either between elements in the same sentence, or between different sentences.

For example:

1. The police diverted the traffic.
2. The traffic was diverted by the police.

3. The traffic was diverted by a country road.

PS rules fail to show the relationship that connects 1 to 2. In sentence 2 'by the police' will be shown as a prepositional phrase consisting of a preposition, a determiner and a noun, and in sentence 3 'by a country road' too will be shown as a prepositional phrase (prep + NP). Thus, it would ignore semantic considerations and case relations. PS grammar does not have the precision, simplicity, elegance, power, insight, and competence of the TG grammar. For sentences like the following both IC analysis and PS theses are identical.

- John is eager to please
- John is easy to please

Phrase Structure Rules cannot help distinguish among ambiguous sentences.

Topic- 111: The Great Automatic Grammatizator

Every normal human being can talk. For that reason the average person tends to think that there is little or nothing mysterious about language. As Noam Chomsky has pointed out: We lose sight of the need for explanation when phenomena are too familiar and 'obvious'. Easily to assume that explanations must be transparent and close to the surface . . . As native speakers, we have a vast amount of data available to us. The trap of believing is that there is nothing to be explained. Nothing could be further from the truth. (Chomsky 1972a: 25–6)

But the mysterious nature of human language becomes more apparent when one realizes that no one has yet managed to simulate the language ability of a human being. Computers can play chess, sort bank statements, and even talk about limited topics such as cubes, squares and cones. But we are far from producing a 'great automatic grammatizator' which could unaided hold conversations on any topic. Why is this? Perhaps we should think about language more carefully. When people start thinking about language, the first question which often occurs to them is this: is language natural to humans? – in the same way that barking comes naturally to dogs. Clearly, in one sense, children 'learn' whatever language they are exposed to, be it Chinese, Nootka or English. So, no one would deny that 'learning' is very important. But the crucial question is whether children are born with 'blank sheets' in their head as far as language is concerned – or whether humans are 'programmed' with an outline knowledge of the structure of languages in general.

This question of whether language is partly due to nature or wholly due to learning or nurture is often referred to as the nature–nurture controversy, and has been discussed for centuries. This particular issue resurfaced in linguistics in 1959 when the linguist Noam Chomsky wrote a devastating and witty review of Verbal Behavior, a book by the Harvard psychologist B.F. Skinner (Skinner 1957; Chomsky 1959). This book claimed to 'explain' language as a set of habits gradually built up over the years.

According to Skinner, no complicated innate or mental mechanisms are needed. All that is necessary is the systematic observation of the events in the external world which prompt the speaker to utter sounds. Skinner's claim to understand language was based on his work with rats and pigeons. He had proved that, given time, rats and pigeons could be trained to perform an amazing variety of seemingly complex tasks, provided two basic principles were followed. This type of 'trial-and-error' learning was called operant conditioning by Skinner, which can be translated as 'training by means of voluntary responses' (the word 'operant' means a voluntary response rather than an automatic one). Skinner suggested that it is by means of this mechanism that the vast majority of human learning takes place, including language learning.

Recent work has shown that the methods can be extended to human behaviour without serious modification. (Skinner 1957: 3). All one needed to do in order to understand language, he said, was to identify the 'controlling variables', which would enable us to predict specific utterances.

'Please pass the bread and butter.'

Or the presence of a beautiful painting might call forth the exclamation, 'Oh how beautiful.' Or a bad smell might cause one to exclaim 'Oh what a terrible smell.' In theory, Skinner saw no difficulty in linking up any particular set of words which a human might wish to produce with an identifiable external happening. In practice, the matter is far from simple, as Chomsky pointed out. Chomsky made two major criticisms of Skinner's work. First, the behaviour of rats in boxes is irrelevant to human language. Second, Skinner fundamentally misunderstood the nature of language.

Topic- 112: Behaviouristic View: The Irrelevance of Rats

Chomsky pointed out that the simple and well-defined sequence of events observed in the boxes of rats is just not applicable to language. The terminology used in the rat experiments cannot be re-applied to human language without becoming hopelessly vague. For example, how do you know that someone is likely to say 'Oh what a beautiful picture' (when looking at a beautiful painting)? They might say instead,

- 'It clashes with the wallpaper',
- 'It's hanging too low',
- 'It's hideous.'

Skinner would say that instead of the utterance being 'controlled' by the beauty of the picture, it was 'controlled' by its clash with the wallpaper, or hanging too low, its hideousness. But this reduces the idea of 'control' to being meaningless, because you have to wait until you hear the utterance before you know what controlled it. This is quite unlike the predictable behaviour of rats which could be relied upon to respond to certain stimuli such as a flashing light with a fixed response.

Another problem

The rats were repeatedly rewarded. Children do not receive pellets of food when they make a correct utterance. The idea of reward or reinforcement (since it reinforces the behaviour that is being learned) can in humans be naturally extended to approval or disapproval. Skinner suggested that in these cases children automatically ‘reinforce’ themselves because they know they are producing sounds which they have heard in the speech of others. A poet who is uttering words aloud in an empty room will be ‘reinforced’ by the knowledge that others will be influenced by the poetry. So, reinforcement seems a very woolly notion, since an actual reward need not exist, it need only be imagined or hoped for. A notion is certainly not comparable to the food pellets given to rats when they make a correct response.

Studies by Roger Brown and his associates provided even more problems for Skinner’s notion of reinforcement. After observing mother–child interactions they pointed out that parents tend to approve statements which are true rather than those which are grammatically correct. So, a boy who said ‘Teddy sock on’ and showed his mother a teddy bear wearing a sock would probably meet with approval. But if the child said the grammatically correct utterance ‘Look, Teddy is wearing a sock’, and showed his mother a bear without a sock, he would meet with disapproval.

In other words, if approval and disapproval worked in the way Skinner suggested, you would expect children to grow up telling the truth, but speaking ungrammatically. In fact the opposite seems to happen (Brown et al. 1968). Chomsky used these and similar arguments to show the irrelevance of Skinner’s experiments to the problem of understanding language. Perhaps ‘irrelevance’ is too strong a word, since there are areas of language where habit forming works.

Of course, just because Skinner’s ideas were over-simple does not automatically mean that Chomsky’s ideas were right. Maybe both Skinner’s and Chomsky’s views are outdated. Now, in the twenty first century, we know a lot more about language and its special qualities, partly because Chomsky in particular inspired so many to take language seriously as a key to understanding the human mind, to work on it further.

Topic- 113: Is Language Restricted to Humans?

Judging by newspapers and popular books, there appear to be a vast number of animals which ‘talk’ – talking budgerigars, talking dolphins– even a talking fish, etc. Clearly, the word ‘talk’ can be used in two totally different senses.

1. ‘to utter words’, as in ‘Archibald’s got a talking parrot which says Damn if you poke it.’
2. ‘to use language in a meaningful way’.

Animals such as budgerigars can ‘talk’ in the first sense of the word.

Psycholinguists: Can animals ‘talk’ in the second sense also.

Interested in this problem because they want to know the answer to the following question: are we the only species which possesses language? If so, are we the only species capable of

acquiring it? Are we biologically singled out as ‘articulate mammals’ or not? Do animals talk naturally? In order to answer this question, we must compare human language with animal communication. A comparison presents a number of perhaps unsolvable problems. The first problem is this: are we comparing systems which differ quantitatively or qualitatively? On the one hand, human language may have gradually evolved from a more primitive animal means of communication in a continuous line of growth – a viewpoint sometimes known as a ‘continuity’ theory

On the other hand, human language may be something quite different from our basic animal heritage, and superimposed on it - a ‘discontinuity’ theory. The second major problem we face is that it is not always easy to decide what counts as communication in animals. As one researcher notes: Students of animal behaviour have often noted the extreme difficulty of restricting the notion of communication to anything less than every potential interaction between an organism and its environment. (Marshall 1970: 231)

A useful first step might be to attempt to define ‘language’. This is not as easy as it sounds. Many definitions found in elementary textbooks are too wide. For example: ‘A language is a system of arbitrary vocal symbols by means of which a social group cooperates’. This definition might equally well apply to a pack of wolves howling in chorus. A superficially promising approach was that suggested by the linguist Charles Hockett in the 1960s: the various ‘design features’ which characterize language.

These are: use of the vocal-auditory channel, arbitrariness, semanticity, cultural transmission, spontaneous usage, turn-taking, duality, displacement, structure-dependence, creativity, ability to read intentions. Finally, we come to birds. They also have failed to give any evidence of creativity. We might expect them to communicate about a multiplicity of situations, since the individual notes of a bird’s song can be combined in an indefinite number of ways. But as far as researchers can judge, birdsong deals above all with just two aspects of life: courting a mate, and the marking of territory (Nottebohm 1975; Marler 1991).

A final, crucial feature of language has come to the forefront in recent years. This is intention-reading (Tomasello 2003), or mind-reading. Some limited awareness of it has been detected among apes, especially chimps. But humans are the best at this skill.

Teaching sign language to apes: Washoe and Nim. Over the past 50 or so years, several attempts have been made to teach human language to chimpanzees. Chomsky may be right, therefore, when he points out that the higher apes ‘apparently lack the capacity to develop even the rudiments of the computational structure of human language’ (Chomsky 1980: 57). Chomsky (1968) claimed that language is specific to humans as only humans possess a language acquisition device to acquire language – the universal grammar.

Topic- 114: Biological evidence for innate language capacity

If an animal is innately programmed for some type of behaviour, then there are likely to be biological clues. It is no accident that fish have bodies which are streamlined and smooth, with fins and a powerful tail structurally adapted for moving fast through the water. Biological clues are not essential. It is impossible to guess from their bodies that birds make nests, and, sometimes, animals behave in a way quite contrary to what might be expected from their physical form. There are sometimes inexplicable divergences between species obvious behaviour.

Such unpredictability is not universal, and need not discourage us from looking for biological clues connected with speech – though we must realize that we are unlikely to find the equivalent of a large box labelled ‘language’. Changes in the form of the body or structural changes are the most direct indications of innate programming. But we must also take into consideration physiological adaptations – changes in the bodily functions, such as rate of heartbeat and breathing. If we look at the organs used in speech, humans seem to be somewhere in the middle between the obvious structural adaptation of birds to flying, and the apparent lack of correlation between birds and nest building.

The human brain and vocal tract have a number of slightly unusual features. By themselves, these features do not sufficiently indicate that people can talk. But if we first assume that all humans speak a language, then a number of puzzling biological facts fall in place. They can be viewed as partial adaptations of the body to the production of language. Human teeth are unusual compared with those of other animals even in height, and form an unbroken barrier.

Another important difference between humans and monkeys concerns the larynx, which contains the ‘voice box’ or ‘vocal cords’ simpler in structure than that of other primates. This is an advantage as air can move freely past and then out through the nose and mouth without being hindered by other appendages. Our brain is very different in appearance from that of other animals, heavier, with more surface folding of the cortex, and the outer layer of ‘grey matter’ which surrounds the inner core of nerve fibres – though grey matter is actually pink in live humans, it goes grey after death. Superficially, the brains of a chimp and a human have certain similarities. A further related discovery is that the location of speech centres in the left hemisphere seems to be linked to right-handedness. That is, most humans are right-handed, and most people’s speech is controlled by the left hemisphere.

This lateralization or localization of language in one half of the brain is a definite biological characteristic of the human race. Humans are physically adapted to language in a way that snails, sheep and even apes are not. Their vocal organs, lungs and brains are ‘preset’ to cope with the intricacies of speech in much the same way that monkeys are pre-set to climb trees, or bats to squeak.

Lesson 20**GRAMMATICAL THEORIES II****Topic- 115: A Preordained Language Programme**

Language emerges at about the same time in children all over the world. ‘Why do children normally begin to speak between their eighteenth and twenty-eighth month?’ asks one researcher: Surely it is not because all mothers on earth initiate language training at that time. No evidence that any conscious and systematic teaching of language takes place, just as there is no special training for stance or gait (Lenneberg 1967: 125). This regularity of onset suggests that language may be set in motion by a biological clock, similar to the one which causes kittens to open their eyes when they are a few days old. Until relatively recently, few people had considered language within the framework of biological maturation.

The characteristics of biologically triggered Behaviour

Behaviour that is triggered off biologically has a number of special characteristics. If it can be shown that speech, like sexual activities and the ability to walk, falls into the category of biologically scheduled behaviour, then we shall be rather clearer about what is meant by the claim that language is ‘innate’. Exactly how many ‘hallmarks’ of biologically controlled behaviour should itemize is not clear. The six listed below were obtained mainly by subdividing Lenneberg’s four:

1. The behaviour emerges before it is necessary.
2. Its appearance is not the result of a conscious decision.
3. Its emergence is not triggered by external events (though the surrounding environment must be sufficiently ‘rich’ for it to develop adequately).
4. Direct teaching and intensive practice have relatively little effect.
5. There is a regular sequence of ‘milestones’ as the behaviour develops, and these can usually be correlated with age and other aspects of development.
6. There may be a ‘critical period’ for the acquisition of the behaviour.

Although children differ enormously in their ability to knit or play the violin, their language proficiency varies to a much lesser extent.

‘Tomorrow I am going to start to learn to talk.’ Children acquire language without making any conscious decision about it. Children begin to talk even when their surroundings remain unchanged. Most of them live in the same house, eat the same food, have the same parents, and follow the same routine. No specific event or feature in their surroundings suddenly starts them off talking.

Children do not manufacture any new brain cells after birth. They are born with millions, perhaps billions. At first the cells are not all interconnected, and the brain is relatively light

(about 300g). From birth to around 2 years, many more cells interconnect, and brain weight increases rapidly. By the age of 2, it weighs nearly 1000g (Lenneberg1967). Although no external event causes the behaviour, the surrounding environment must be sufficiently ‘rich’ for it to develop adequately.

‘Direct teaching and intensive practice have relatively little effect.’ In activities such as typing or playing tennis, a person’s achievement is often directly related to the amount of teaching they receive and the hours of practice they put in. Not ‘naturally’ superb athletes can win. We may conclude, then, that mere practice – in the sense of direct repetition and imitation – does not affect the acquisition of language in a significant way. The following chart gives an overview of the likely linguistic progress of children.

Child’s likely progress chart	
Language stage	Beginning age
• Crying • Cooning • Babbling • Intonation patterns • One-word utterances • Two-word utterances • Word inflection • Questions, negatives • Rare or complex constructions • Mature speech	• Birth • 6 weeks • 6 months • 8 months • 1 year • 18 months • 2 years • 2 1/4 years • 5 years • 10 years

Topic- 116: Conceivable Innate Grammatical Information

It is relatively easy to show that humans are innately predisposed to acquire language. The hard part is finding out exactly what is innate. People have indulged in speculation about this for centuries. The Egyptian king Psammetichus experimented to know the divine language. Nobody takes Psammetichus’s theory seriously today – especially as the few reliable accounts we have of children brought up without human contact indicate that they were totally without speech when they were found.

Although the speculations of Psammetichus can safely be ignored, the ideas of Noam Chomsky on the topic of innateness were for a long time taken seriously. He claimed that for language acquisition to be possible, a child must be endowed with a ‘rich internal structure’. His ideas cannot be summarily dismissed. Chomsky’s notion of a rich innate schema contrasted strongly with the point of view popularly held earlier in the century that children are born with ‘blank sheets’ as far as language is concerned. Consequently, some people considered Chomsky as someone who had set out to shock the world with outrageous and novel proposals.

Chomsky denied and pointed out that he was following in the footsteps of eighteenth-century ‘rationalist’ philosophers, who believed in the existence of ‘innate ideas’. Such philosophers held that ‘beyond the peripheral processing mechanisms, there are innate ideas and principles of various kinds that determine the form of the acquired knowledge in what may be a rather restricted and highly organized way’ (Chomsky 1965: 48).

Chomsky gave an explicit account of his early views in his (now outdated) linguistic classic *Aspects of the Theory of Syntax* (1965), though he has repeated them in a number of other places with minor variations. Recently he has changed his mind on various points, quite fundamentally. His later views were set out in *Knowledge of Language: Its Nature, Origin and Use* (1986), later ones still in *The Minimalist Program* (1995) and further views in *On Nature and Language* (2002). He regards his previous ideas unsatisfactory, and outlines his more recent ideas. Nevertheless, his ideas were specific enough to be interesting.

Chomsky pointed out that children are to some extent in the same situation as a linguist faced with an unknown language. Both child and linguist are surrounded by a superficially unintelligible confusion of sound which they must somehow sort out. According to Chomsky, children would ‘know’ in advance how their internalized grammar must be organized. They must have sets of the following rules:

- SEMANTIC RULES
- SYNTACTIC RULES
- PHONOLOGICAL RULES

He assumed that children were endowed with an innate hypothesis-making device, which enabled them to make increasingly complex theories about the rules which would account for the language they heard going on around them. In making these hypotheses, children were guided by an inbuilt knowledge of language universals. These provided a ‘blueprint’ for language, so that the child would know in outline what a possible language looked like. This involved, first, information about the ‘building blocks’ of language, set of sounds. Second, it entailed information about the way in which the components of a grammar were related to one another, and restrictions on the form of the rules.

Chomsky argued that children automatically knew that language involved two levels of syntax – a deep and a surface level, linked by ‘transformations’. And (as he later argued) children also knew about some innately inbuilt constraints on the form sentences could take. With this help a child could speedily sift through the babble of speech he heard around him, and hypothesize plausible rules which would account for it. Children needed to be equipped with this information, he claimed, because the ‘primary linguistic data’ (the data children are exposed to) was likely to be ‘deficient in various respects’ (1965: 201). According to Michael Tomasello, ‘how children learn language is not a logical problem but an empirical problem.’ (Tomasello 2003: 328).

In his opinion, we need to turn to a usage-based approach, one which explores how human children combine inherited talents and learned skills as they acquire language. 'The human capacity for language is best seen as a conspiracy of many different cognitive, social-cognitive, information-processing, and learning skills, some of which humans share with primates and some of which are unique products of human evolution'.

Topic- 117: Transformational Grammar

In linguistics, transformational grammar (TG) or transformational-generative grammar (TGG) is part of the theory of generative grammar, especially of natural languages. It considers grammar to be a system of rules that generate exactly those combinations of words that form grammatical sentences in a given language and involves the use of defined operations (called transformations) to produce new sentences from existing ones. The theory largely originated and developed by Noam Chomsky.

It is without question the most influential theory of linguistics in modern times, and one that no serious scholar can afford to neglect. But the student of TG will often experience frustration and puzzlement, for it is a curious mixture of quite brilliant insights and of complex, technical but often seemingly artificial devices. The primary literature, apart from Chomsky's *Syntactic Structures*, the first published work, is often difficult to read and its full intelligibility depends on knowing all the background.

The notion of PS grammar, although much of what was said comes from Chomsky's works, but his fame does not rest on it. Much more important is the notion of transformation, together with ideas that are associated with it, or have developed out of a transformationally based theory. *Although Syntactic Structures* (1957) first introduced the world to this important theory, *The Logical Structure of Linguistic Theory*, which was not published until eighteen years later, was written some years before it.

The theory was, undoubtedly, revolutionary and even greeted with total skepticism by many scholars, but, as with all revolutions, some of it had been foreshadowed in earlier works, particularly in the writings of Chomsky's own teacher Zellig Harris. Chomsky has considerably modified his ideas since 1957, and in his philosophical views has vastly distanced himself from the structuralists, and indeed from the whole of the empiricist tradition. Undoubtedly the best known, in some ways neatest and most discussed theoretical position is that of *Aspects of the Theory of Syntax*, a position that Chomsky himself has called the 'Standard Theory'. This introduced and expounded the notion of Deep Structure, an essential part of the theory. Several important changes were made in the 1970s, to produce the 'Extended Standard Theory'(EST). Chomsky's position at the beginning of the 1980s is to be found in *Lectures on Government and Binding*.

Development of basic concepts

Though transformations continue to be important in Chomsky's current theories, he has now abandoned the original notion of Deep Structure and Surface Structure. Initially, two additional levels of representation were introduced (LF — Logical Form, and PF — Phonetic Form), but in the 1970s, Chomsky sketched out a new program of research known as Minimalism, in which Deep Structure and Surface Structure are no longer featured and PF and LF remain as the only levels of representation. To complicate the understanding of the development of Chomsky's theories, the precise meanings of Deep Structure and Surface Structure have changed over time.

By the 1970s, they were normally referred to simply as D-Structure and S-Structure by Chomskyan linguists. In particular, the idea that the meaning of a sentence was determined by its Deep Structure (taken to its logical conclusions by the generative semanticists during the same period) was dropped for good by Chomskyan linguists when LF took over this role. Previously, Chomsky and Ray Jackendoff had begun to argue that meaning was determined by both Deep and Surface Structure.

Topic- 118/119: The Founder and the Theory

Avram Noam Chomsky (born December 7, 1928) is an American linguist, philosopher, cognitive scientist, historian, political activist, and social critic. Sometimes called "the father of modern linguistics", Chomsky is also a major figure in analytic philosophy and one of the founders of the field of cognitive science. He holds a joint appointment as Institute Professor Emeritus at the Massachusetts Institute of Technology (MIT) and laureate professor at the University of Arizona, and is the author of over 100 books on topics such as linguistics, war, politics, and mass media.

Born to middle-class Jewish immigrants in Philadelphia, Chomsky developed an early interest in linguistics. He began studying at the University of Pennsylvania at age 16, taking courses in linguistics, mathematics, and philosophy. From 1951 to 1955, he was appointed to Harvard University's Society of Fellows. While at Harvard, he developed the theory of transformational grammar; for this, he was awarded his doctorate in 1955.

Chomsky began teaching at MIT in 1957 and emerged as a significant figure in the field of linguistics for his landmark work *Syntactic Structures*, which remodelled the scientific study of language. From 1958 to 1959, he was a National Science Foundation fellow at the Institute for Advanced Study. Chomsky is credited as the creator or co-creator of the universal grammar theory, the generative grammar theory, and the Chomsky hierarchy. He also played a pivotal role in the decline of behaviorism, being particularly critical of the work of B. F. Skinner. One of the most cited scholars in history, Chomsky has influenced a broad array of academic fields. He brought about a major revolution in the human sciences, contributing to the development of a new cognitivist framework for the study of language and the mind.

Topic- 120: Deep and Surface Structure

The structuralist approach to language was corpus bound as both phoneme and morpheme are units of form rather than of meaning. Their method was inductive and it also failed to account for the ambiguities in sentences and their interrelatedness.

(a) To answer these questions, an adequate grammar was needed that is productive, complex and arbitrary. It must establish the relationship of various sentences and account for the deep and surface structure of the sentences. It must explain the relationship between sets of sentences:

(i) He wrote a letter.

(ii) A letter was written by him.

(i) Why did you go there?

(ii) You went there for some reason.

(i) Where do you live?

(ii) You live somewhere.

TG grammar is both transformational and generative. It goes a step further from the structural grammar. It analyses the sentences, divides them into parts and shows the functions of various parts and rearranges them and shows the inter-relatedness between sentences. It takes up the basic or kernel sentence first. A basic or kernel sentence is simple, assertive, and active in form. Here are some examples:

Kernel sentences	Hasan is playing football. I wrote a letter. You spoke the truth.
Non-kernel sentences:	• Is Hasan playing football? (Interrogative) • Hasan is not playing football. (Negative) • A letter was written by me. (Passive) • Who wrote the letter? (Question form) • Why did you speak the truth? (Question form) • You spoke the truth when you were forced. (Complex sentence) • You did not speak the truth. (Negative) • You spoke the truth but told a new story. (Compound sentence)

All the non-kernel structures or complicated structures can be thought of as having been derived from their kernel forms with the help of some transformations. A sentence like: Where did you go yesterday? can be derived from the kernel sentence: You went somewhere yesterday by applying certain transformations.

Transformational analysis not only shows the inter-relatedness between sentences but also explains the ambiguities between sentences that appear identical but are transforms from different kernels. Here is, for example, a sentence:

Flying planes can be dangerous.	(a)Some people fly planes. This can be dangerous. (b)Planes fly. They can be dangerous.
(a)I expected Hasan to hit a six. (b)I asked Hasan to hit a six.	(a)(i)I expected (something). (ii)Hasan hits a six. (b)(i)I asked Hasan (something). (ii)He hits a six.
(a)Hasan is eager to please. (b)Hasan is easy to please.	(a)(i) Hasan is eager (for something). (ii)He pleases (someone). (b) (i)It is easy. (ii)Someone please Hasan.

A kernel sentence is deep structure to which a string of transformation rules is applied to generate the surface structure.

Lesson 21

CONSTITUENTS I: WORDS ORDER

Topic- 121: Grammatical Categories

In traditional grammar, words are assigned to grammatical categories (called parts of speech) on the basis of their: semantic properties (i.e. meaning), morphological properties (i.e. the range of different forms they have) syntactic properties (i.e. word-order properties relating to the positions they can occupy within sentences): A set of words which belong to the same category thus have a number of semantic, morphological and syntactic properties in common. There are traditionally said to be two different types of word, namely content words/connectives (= words which have substantive lexical content) on the one hand, function words/functors (= words which essentially serve to mark grammatical properties) on the other.

The differences between contentive like *car* with a functor like *they*: a noun like *car* has substantive lexical content in that it denotes an object which typically has four wheels and an engine, and it would be easy enough to draw a picture of a typical car. By contrast, a pronoun such as *they* has no descriptive content (e.g. you can't draw a picture of *they*), a functor which simply marks grammatical (more specifically, person, number and case) properties in that it is a third person plural nominative pronoun.

Because they have lexical semantic content, content words often (though not always) have antonyms e.g. the adjective *tall* has the antonym *short*, the verb *increase* has the antonym *decrease*, the preposition *inside* has the antonym *outside*: by contrast, a typical function word like e.g. the pronoun *me* has no obvious antonym. Corresponding to these two different types of words content and function, there are two different kinds of grammatical category – namely lexical/substantive categories (= categories whose members are content words) functional categories (= categories whose members are function words).

Topic- 122: Lexical and Substantive Categories

The main lexical/substantive categories found in English are noun, verb, adjective, adverb and preposition conventionally abbreviated to N, V, A, ADV and P in order to save space. Following are some of the characteristics of noun.

Nouns (= N)	
Semantic property	denote entities: bottle (since it denotes a type of object used to contain liquids), water (since it denotes a type of liquid) Hasan (since it denotes a specific person).
Distinct subtypes of noun	Count noun: like chair (cf. one chair, two

	chairs . . .), Mass noun: like furniture is a mass noun in that it denotes an uncountable mass (*one furniture, *two furnitures).
Common and proper nouns	Boy a common noun like boy (which can be modified by a determiner like the – as in The boy is lying) Andrew (which cannot be used in the same way in English, as *The Andrew is lying).
Number	Count nouns generally have the morphological property that they have two different forms: a singular form (like horse in one horse) used to denote a single entity, a plural form (like horses in two horses) used to denote more than one entity.
Common nouns have the syntactic property	Only (an appropriate kind of) noun can be used to end a sentence such as They have no . . . (car, or a plural count noun like friends or a mass noun like money, but not other types of word (e.g. not see or slowly or up, as these are not nouns).

Verb (= V).	
semantic property	denote actions or events: so, eat, sing, pull and resign are all (action-denoting) verbs.
Syntactic properties	From a syntactic point of view, verbs have the property that only an appropriate kind of verb (in its uninflected infinitive form) can be used to complete a sentence such as They/It can . . . So, words like stay, leave, hide, die, starve and cry are all verbs words like apple, under, pink and if aren't.
Morphological point	regular verbs like cry in English have the property that they have four distinct forms: e.g. alongside the bare (i.e. uninflected) form cry we find the present tense form cries, the past tense/perfect participle/passive participle form cried and the progressive participle form crying.

Adjective (= A).

semantic property	denote states or attributes (cf. ill, happy, tired, conscientious, red, cruel, old etc.).
Syntactic properties	Can occur after be to complete a sentence like They may be . . . (as with They may be tired/ill/happy etc.), denote a gradable property varying degrees fast, faster, fastest
Modified by a degree word	very/rather/somewhat (cf. She is very happy). Many (but not all) adjectives have the morphological property that they have comparative forms ending in -er and superlative forms ending in -est (cf. big /bigger/biggest).

Adverb (= ADV)	
Semantic property	denote the manner in which an action is performed (as with well in She sings well).
Morphological property	Regular adverbs have the that they are formed from adjectives by the addition of the suffix -ly. the adjective sad we have the adverb sadly
Syntactic property	an adverb (like e.g. badly) is the only kind of word which could be used to end sentences such as She behaved . . . , He treats her . . . or He worded the statement . . .

Preposition (=P)	
Semantic property	of marking location (cf. in/on/off/inside/outside/under/above/below).
Syntactic property	with the appropriate kind of meaning can be modified by right in the sense of 'completely', or by straight in the sense of 'directly' down in He fell right down the stairs to in He went straight to bed.
Morphological property	they are invariable/uninflected forms (e.g. the preposition off has no past tense form *offed, no superlative form *offest and so on).

Topic- 123: Functional Categories

Following are functional categories. Determiner (= D) traditionally said to include the definite article *the* and the demonstrative determiners *this/that/these/those*.

Determiner (= D)	
Includes	definite article the and the demonstrative determiners this/that/these/those
Semantic property	determine specific semantic properties of the noun expression that they introduce, marking it as a definite referring expression Shall we take the car? (specific) car which is assumed to be familiar to the hearer/addressee.
Function	
Indefinite article	

Quantifier (= Q),	
Function	denoting expressions of quantity, such as some/all/no/any/each/every/most/much/many.
Indefinite article	the indefinite article a to be a quantifier – one which quantifies over a single entity a car

Pronouns(=PRN)	
Items	said to ‘stand in place of’ (the meaning of the prefix pro-) or ‘refer back to’ noun expressions.
Types	a number of different types of pronoun found in English and other languages He/his/him
The word one	Hasan has a red car and Jim has a blue one has no lexical semantic content of its own, rather takes its content from its antecedent (i.e. one refers back to the noun car and so one is interpreted as having the same meaning as car.
Morphological perspective	pronoun one behaves like a regular count noun in that it has a plural form ending in -s (as in I’ll take the green apples if you haven’t got any red ones)
Status of one	N-pronoun /or pronominal noun
Some	sentence like Many miners were rescued, but some died, the word some seems to function as a Q-pronoun (i.e. a pronominal quantifier)
Those	These apples are ripe, but those aren’t, the word those seems to be a D-pronoun (i.e. a pronominal determiner).
Personal pronoun as D- pronouns	I/me/we/us/you/he/him/she/her/it/they/them the rationale: We republicans don’t trust you democrats,

	we could be argued to be a determiner modifying the noun republicans, you could be seen as a determiner modifying the noun democrats
--	---

Auxiliary (verb).	
Semantic property	marking grammatical properties such as tense, aspect, voice or mood
Syntactic property	unlike lexical/main verbs they can be inverted with their subject in questions (so that corresponding to a statement like It is raining we have the question Is it raining? (the auxiliary is has moved in front of the subject it and is said to have been inverted).

The items italicized in (2) below (in the use illustrated there) are traditionally categorised as auxiliaries taking a [bracketed] complement containing a bold-printed verb:	
(a) He has/had [gone]	have/be in (2a,b) are (perfect/progressive) aspect auxiliaries
(b) She is/was [staying at home]	
(c) They are/were [taken away for questioning]	be in (2c) is a (passive) voice auxiliary,
(d) He really does/did [say a lot]	do in (2d) is an expletive or dummy auxiliary (i.e. one with no intrinsic lexical semantic content)
(e) You can/could [help us]	can/could/may/might/will/would/shall/should in (2e–h) are modal auxiliaries
(f) They may/might [come back]	
(g) He will/would [get upset]	
(h) I shall/should [return]	

What auxiliaries in sentences like those above have in common is the fact that they inflect for present/past tense. Hence, in work in syntax over the past ten years or so, they have been said to belong to the category T (= tense-marked auxiliary). To tense-marking auxiliaries in work over the past thirty years or so is the infinitive particle *to*, in sentences such as:

- They are now expecting the president *to* be impeached tomorrow

The infinitival *to* seems to have future time-reference. The act of impeachment will take place at some time in the future. Why we can use the word tomorrow in the *to*-clause. infinitival *to* seems to have much the same function as the auxiliary will in 'They are now expecting that the president will be impeached tomorrow'. The infinitival *to* is an infinitival tense marker, and so belongs to the same category T as present/past tense auxiliaries such as *is/was*. The difference

between auxiliaries and infinitival *to* is that most auxiliaries overtly inflect for present/past tense (though this is not true of the invariable auxiliaries *must* and *ought*), whereas infinitival *to* is invariable in form. We can thus say that an auxiliary like *will* is a finite T constituent, whereas infinitival *to* is a nonfinite T.

Topic- 124: Complementisers

Look at the following sentences:

- I think [*that* you may be right]
- I doubt [*if* you can help me]
- I'm anxious [*for* you to receive the best treatment possible]

Each bracketed clause is a complement clause, the complement of the word immediately preceding it (think/doubt/anxious); The italicized (*that, if, for*) word which introduces each clause is a complementiser (= C). Complementisers are functors in the sense that they encode particular sets of grammatical properties. Complementisers encode whether they are finite or nonfinite.

For examples		
I think [that you may be right]	I doubt [if you can help me]	I'm anxious [for you to receive the best treatment possible]
That is inherently finite in the sense that it can only be used to introduce a finite clause	If is also inherently finite in the sense that it can only be used to introduce a finite clause.	For is an inherently infinitival complementiser, and so can be used to introduce a clause containing infinitival to
That introduces a declarative clause (a statement),	If introduces an interrogative clause (i.e. a question)	For Introduces an irrealis clause (i.e. one relating to a hypothetical event)
That is a finite declarative complementiser	If is a finite interrogative complementiser	For is an infinitival irrealis complementiser

Using the set of syntactic categories outlined above, we can employ the traditional labelled bracketing technique to categorise words in a way which describes how they are being used in a particular sentence.

- **The president is clearly feeling angry that Congress has refused to negotiate with him**
- **(b) [D The] [N president] [T is] [ADV clearly] [V feeling] [A angry] [C that] [N Congress] [T has] [V refused] [T to] [V negotiate] [P with] [PRN him]**

Grammatical differences between individual words belonging to the same category are traditionally described in terms of sets of grammatical features, and these features are enclosed in square brackets.

- She invited us to dinner.

For example, both *she* and *us* are pronouns.

She carries the features [third-person, singular-number, feminine-gender, nominative-case],

Us carries the features [first-person, plural-number, accusative-case].

Traditional grammarians are also concerned to describe the grammatical functions.

- Hasan laughs
- The president laughs
- The president of Utopia laughs
- The former president of the island paradise of Utopia laughs

Hasan reads novel S V C(DO)	• The complement novels describe the entity on which the act of reading is being performed • A verb which has a noun or pronoun expression as its direct object complement is traditionally said to be transitive. • From a semantic perspective, subjects and complements generally represent entities directly involved in the particular action or event described by the predicate: • we can say that subjects and complements are arguments of the predicate with which they are associated. • Predicates may have one or more arguments
Hasan reads English novels S V C(DO)	
Hasan reads English novels S V imported from England C(DO)	
Hasan reads English novels S V C(DO) imported by a friend of his from England	

(a) [John] resigned	A predicate like resign in (a) which has a single argument is said to function as a one-place predicate
(b) [John] felt [remorse]	(b) which has two arguments is a two-place predicate;
(c) [John] sent [Mary] [flowers]	(8) which has three arguments is a three-place predicate.

In addition to predicates and arguments, sentences can also contain adjuncts, as we can illustrate in relation to below:

- The president reads a novel after dinner
- The president reads a novel in his office

after dinner /in his office serve to provide additional information and are both adjuncts.

We also find complex sentences which contain more than one clause

- Mary knows Hasan smokes
- The press clearly thinks the president deliberately lied to Congress.

Topic- 125: Types of clauses

Finite and non-finite clauses:

Finite clauses must contain a verb which shows tense. They can be main clauses or subordinate clauses: Is it raining?

Non-finite clauses contain a verb which does not show tense. We usually use non-finite verbs only in subordinate clauses.

	Finite and non finite verbs	
She <i>writes</i> a letter.	Bare infinitive form	does show tense/present time
She <i>wrote</i> a letter.	Past Tense form	does show tense/past time
*She <i>to write</i> a letter.	Infinitive form/ Infinitival To	does not show tense
*She <i>writing</i> a letter.	Present participle form	does not show tense
*She <i>written</i> a letter.	Past participle form	does not show tense

In linguistics, grammatical mood is a grammatical feature of verbs, used for signaling modality. It is the use of verbal inflections that allow speakers to express their attitude towards what they are saying (e.g. a statement of fact, of desire, of command, etc.). A clause containing a verb in the indicative mood denotes a real (or realis) event or state occurring at a specific point in time; A Subjunctive clause by contrast denotes a hypothetical or unreal (= irrealis) event or state which has not yet occurred and which may never occur.

Types of clauses	
(a) She was glad <i>that he apologised</i>	The verb <i>apologised</i> is finite by virtue of being inflected for past tense and indicative mood, and by virtue of having a nominative subject (he)
(b) She demanded <i>that he apologise</i>	The verb <i>apologise</i> is finite by virtue of being inflected for subjunctive mood (and perhaps present tense), and by virtue of having a nominative subject (he).
(c) I can't imagine him <i>apologising</i>	Nonfinite clause/ a tenseless and moodless gerund form, and has an accusative subject him
(d) It would be sensible <i>for him to apologise</i>	Nonfinite clause/a tenseless and moodless infinitive form/an accusative subject him
(e) It's important <i>to know when to apologise</i>	Nonfinite clause/a tenseless and moodless infinitive form a 'silent' (implicit) subject it

A finite clause determines the kind of subject it can have. Finite clauses can have a nominative pronoun like *he* as their subject, but nonfinite clauses cannot. Whether a particular clause is finite or not is to see whether it can have a nominative pronoun (like *I/we/he/she/they*) as its subject. In addition to being finite or nonfinite, each clause within a sentence has a specific force.

Specific force	
(a) He went home	Declarative in force, in that it is used to make a statement.
(b) Are you feeling OK?	Interrogative in force in that it is used to ask a question.
(c) You be quiet!	Imperative in force, by virtue of being used to issue an order or command.
(d) What a great idea that is!	Exclamative in force, in that it is used to exclaim surprise or delight.

In complex sentences, each clause has its own force	
(a) He asked where she had gone	The main (asked) clause is declarative, whereas the complement (gone) clause is interrogative.
(b) Did you know that he has retired?	The main (know) clause is interrogative, whereas the complement (retired) clause is declarative.
(c) Tell her what a great time we had!	The main (tell) clause is imperative, whereas the complement (had) clause is exclamative.

From the perspective of traditional grammar, the syntax of a language is described in terms of a taxonomy (i.e. a classificatory list) of the range of different phrase-, clause- and sentence types found in the language. This reflects the fact that the primary goal of traditional grammar is description rather than explanation.

Lesson 22

CONSTITUENTS II: WORDS ORDER**Topic- 126: Introduction**

In linguistics, syntax is the set of rules, principles, and processes that govern the structure of sentences in a given language, usually including word order. The term syntax is also used to refer to the study of such principles and processes. The goal of many syntacticians is to discover the syntactic rules common to all languages. ‘Syntax’ also means ‘sentence construction’: how words group together to make phrases and sentences. For some people the term grammar means syntax. Most linguists follow the more recent practice whereby the grammar of a language includes all of its organizing principles: information about the sound system, about the form of words, how we adjust language according to context, and so on; syntax is only one part of this grammar.

The term ‘syntax’ is also used to mean the study of the syntactic properties of languages. In this sense, it’s used in the same way as we use ‘stylistics’ to mean the study of literary style. The scope of syntax study includes the classification of words, the order of words in phrases and sentences, the structure of phrases and sentences, and the different sentence constructions that languages use. One basic description of a language's syntax is the sequence in which the subject (S), verb (V), and object (O) usually appear in sentences.

Over 85% of languages usually place the subject first, either in the sequence SVO or the sequence SOV. The other possible sequences are VSO, VOS, OVS, and OSV, the last three of which are rare. In most generative theories of syntax, these surface differences arise from a more complex clausal phrase structure, and each order may be compatible with multiple derivations. Usually, we think of syntax as "word order." However, syntax is also achieved in some languages such as Latin by inflectional case endings. A sentence could make no sense and still be correct from the syntax point of view as long as words are in their appropriate spots and agree with each other. Here is a classic example by Noam Chomsky, a linguist that illustrates a case in which a sentence is correct but does not make sense:

- Colorless green ideas sleep furiously.

In a language such as English, the main device for showing the relationship among words is word order; e.g., in “The girl loves the boy,” the subject is in initial position, and the object follows the verb. Transposing them changes the meaning. In Latin, for example, “The girl loves the boy” may be *puella puerum amat* with “the girl” in initial position, or *puerum puella amat* with “the boy” in initial position, or *amat puella puerum*, *amat puerum puella*, or *puella amat puerum*.

The meaning remains constant In the sentence “My dog is playing in the yard” there is a closer relationship between the words “is playing,” which together form the verb, than between the

words “playing in the,” which form only part of the verb and part of the phrase indicating the location of the playing. The study of syntax also includes the investigation of the relations among sentences that are similar, such as “John saw Mary” and “Mary was seen by John. “Syntax received much attention after 1957, when the American linguist Noam Chomsky proposed a radically new theory of language, transformational grammar.

Topic- 127: Some Concepts and Misconceptions

It is a misconception that children are taught the grammar of their native language by their parents or by teachers at school. All developing children in every culture learn their native language or languages to perfection without any formal teaching. The simple exposure is required to ordinary, live, human language within a society. No culture or cultures teach their children ‘grammar’.

All children must be capable of constructing a mental grammar of their native languages without any formal instruction. Most linguists now believe that human infants are born pre-programmed to learn language the way as they are pre-programmed to walk upright. Children are not taught their native grammar the way they are not taught by their parents how to walk.

All that’s needed for language to emerge is appropriate input data – hearing language (or seeing it; sign languages are full languages too) and taking part in interactions within the home and the wider society.

So if you weren’t taught the grammar of your native language, what was it you were being taught when your parents tried to get you not to say things like *I ain’t done nowt wrong, or He’s more happier than what I am*, or when your school teachers tried to stop you from using a preposition to end a sentence with? In just the same way, some people have the idea that certain forms of language are more beautiful, or classier, or are simply ‘correct’. But the belief that some forms of language are better than others has no linguistic basis. Since we often make social judgements about people based on their accent or dialect, we tend to transfer these judgements to their form of language.

We may then think that some forms are undesirable, that some are ‘good’ and some ‘bad’. For a linguist, however, dialectal forms of a language don’t equate to ‘bad grammar’. Some people think examples I didn’t do nothing wrong are non-standard English are sloppy speech, or perhaps illogical. Middle English (the English of roughly 1100 to 1500) also had a double negative. Ironically for the ‘logic’ argument, the variety of French that has the double negative is the most formal and prestigious variety, whereas colloquial French typically drops the initial negative word.

Socially stigmatized forms of language are potentially just as ‘logical’ as standard English. Speakers of non-standard dialects are, of course, following a set of mental rules, in just the same way that speakers of the most prestigious dialects are. The various dialects of a language in fact share the majority of their rules, and diverge in very few areas, but the extent of the differences tends to be exaggerated because they arouse such strong feelings. In sum, speakers of prestige dialects may feel that only their variety of English is ‘grammatically correct’, but these views cannot be defended on either logical or linguistic grounds.

Topic- 128: Use of Linguistic Examples

Throughout the course we have used examples from a wide variety of languages, including English. It is difficult to study examples from unfamiliar languages, and perhaps you wonder why we don’t just use examples from English. Two main reasons for using foreign - language examples:

- to learn about the differences between languages,
- to learn about the similarities between them.

First, languages don’t all look the same, and examining just English language and its immediate relatives doesn’t show how much languages can differ. Imagine two languages, English and German, two closely related Germanic languages from northern Europe. Look at a word - for - word translation.

- der schone Wasserfall (German)
- the pretty waterfall (English)

You might imagine that the translation of this phrase would look the same in any language: first a word for ‘the’, then a word for ‘pretty’ or ‘beautiful’, then a word for ‘waterfall’. But this is not so.

In Spanish, for instance, we’d get:

- la cascada Hermosa
- the waterfall beautiful
- ‘the beautiful waterfall’

The world’s languages have many interesting and important syntactic features that you should know about. English has some but not all of these features, so if we only looked at English you’d miss out on the rest.

In the following we see one example, from Spanish:

Es nuevo. (Spanish)

is new

It's new.

The above example has no word for 'it'. It literally means 'is new' – an impossible sentence in English. Spanish typically drops the subject pronoun meaning 'it' in such examples; for this reason, it's known as a pro-drop language. Many languages have examples parallel to this, but confining the discussion to English would never reveal that. In yet other languages, such as Arabic and Indonesian, the three-word English sentence *It is new* translates as 'It new'. These simple examples show that we can't expect sentences in other languages to be word-for-word translations of English sentences. The second reason is that linguists want to discover the common properties that languages share – their homogeneity or sameness.

One of the most important discoveries of modern linguistics is that languages don't vary from each other at random, but are remarkably alike. Most linguists want to uncover the central patterns common to all languages. Although specific constructions are not universal (= common to all languages), all languages use a sub-set of the same basic tools of grammar. Each language has a wordlist or lexicon which all its speakers share, and that wordlist always contains words from several different classes. All languages combine these words into phrases and sentences. Languages manipulate the order of the phrases for various purposes –perhaps to ask questions, or to emphasize different parts of a sentence, or to show who's doing what to whom. This is syntax, and it forms the subject matter of our discussion.

Topic- 129: Why Languages have Syntax

Speakers manipulate sentences in all sorts of ways because they're trying to convey different meanings. Syntax allows speakers to express all the meanings that they need to put across. In the simplest cases, this might mean altering the basic word order of a sentence, to emphasize or downplay a particular phrase, or to ask a question, or else grouping words together in different ways to modify the meaning. A preliminary idea of in languages demonstrates that languages really do have syntactic structure.

Each of the word orders in the following sentences is attested (= found) among the world's languages, though some are much more common than others. The three most common basic word orders in languages other than English are those of (a, b and (c).

Kim drank the tea.		
Kim, the one drinking the tea;	Drank, the verb, which expresses what Kim did;	The tea, expressing what is being drunk.
	a. Kim drank the tea.	
	b. *Kim the tea drank.	Japanese has this basic word order of (b).
	c. *Drank Kim the tea.	Welsh has this basic order of (c).
	d. *Drank the tea Kim.	Malagasy, spoken in

		Madagascar, has this basic order in(d).
	e. *The tea drank Kim.	Two word orders in (e and f) are the rarest basic word orders in the languages of the, found in the Carib language family of the Amazon basin. For example, Hixkaryana
	f. The tea Kim drank.	

It is possible to determine the basic, neutral word order in a language, but the flexibility or rigidity of the basic word order differs widely among the world's languages. English has a fixed basic word order, Russian a very flexible word order, but Japanese allows many different orders.

- *Drank Kim the tea.

In English, this ungrammatical word orders might be permissible in poetry, but not in the spoken language or in prose.

- f The tea Kim drank.

The above sentence *f* may sound odd but it can be used to focus on what it was that Kim drank; the phrase the tea is fronted from its usual position. Let's try adding a bit of context: Kim visits an eccentric aunt who makes tea and beer out of strange garden plants: The tea, Kim drank ____, but the home-made nettle beer, she really hated ____.

The gaps are used to show the normal position of the fronted phrases.

- The last exercise, a stall while climbing, I didn't do ____ well. Object-fronting is, in fact, quite rare in English.

It's known as a marked (= unusual) construction, while the usual basic word order as in (a) is termed unmarked. Often there are stylistic reasons for changing basic word order. The fronted phrase in it is rather long, and sounds clumsy in the usual object position: *I didn't do the last exercise, a stall while climbing, well.*

A different kind of word order change, which involves breaking up a rather long phrase by moving part of it to the right.

- Estimates ____ vary greatly about the number of fluent speakers (i.e. of Esperanto).

This avoids the clumsiness of a long initial phrase estimates about the number of fluent speakers before the short phrase vary greatly. (Compare the normal word order in Estimates about this vary greatly.)

As showed, English has a generally inflexible word order in the sentence, but optional modifying phrases can be reordered quite easily, as is the case for the about ... phrase which modifies (= expands on) the word estimates.

Topic- 130: All Languages have Structure

All languages, whether living or dead, have syntactic structure, including sign languages (such as British Sign Language). A language doesn't just consist of strings of words, but that the words group together to form phrases, and the phrases group together to form larger phrases and sentences. Linguists describe this phrases-within-phrases pattern as hierarchical structure.

One kind of hierarchical structure is seen in embedded sentences, a construction in which sentences occur within other sentences, such as Chris told Lee [Kim couldn't swim]. This property is known as recursion. Here, the sentence in brackets – Kim couldn't swim – is the embedded sentence. It serves to tell you what it was that Chris told Lee. More examples from English are given in below: the embedded sentences are again in square brackets.

- I wonder [if Lee will arrive late].
- The claim [that she doesn't like Kim] is very surprising.
- [That we've no coffee left] isn't my fault.
- We asked [how to get to the station].

There is no limits to the number of embedded sentences that can be strung together.

- Kim couldn't swim
- Lee thought that Kim couldn't swim,
- I said that Lee thought that Kim couldn't swim.

Let's look at the following sentences:

- a) I charged up the battery.
- b) I charged up the street.

At first glance these sentences appear to be structurally identical. Of course, you might be aware when you read them that charge means something different in (a) and (b), but otherwise, the only difference seems to be that street replaces battery. And yet the syntactic behaviour of the two sentences is entirely different. As always, the asterisks indicate ungrammatical sentences:

- a. I charged the battery up.
- b. *I charged the street up.
- a. *It was up the battery I charged.
- b. It was up the street I charged.
- a. *I charged up Lee's battery and (then) up Kim's too.

Since native speakers of English agree about where the asterisks showing an ungrammatical sentence should be placed, we must all share an unconscious knowledge of the sentence structure

of English. Even though pairs of sentences like those in (48) look the same, they in fact have different structures. Different sets of words group together to form phrases in each case, and linguists represent this using brackets:

- a. I [charged up] the battery.
- b. I charged [up the street].

In (a), the brackets show that there's a phrase charge up. This makes sense if you think that the only thing you can do with a battery is charge it up; you can't charge it down, over, across, or anything else. So up belongs with charge in (a). In (b), it doesn't: instead, there's a syntactic unit up the street, which can be moved around the sentence for focus, as for example in *It was up the street I charged (not the corridor)*, *It was up the street I charged*. And in *I charged [up the street]*, up can be replaced with a number of other words: *I charged down the street/over the street/across the street* and so on. The very fact that up forms a unit with charge in (a) but with the street in (b) is responsible for the patterns of (un)grammaticality in **I charged the street up* to **I charged up Lee's battery and (then) up Kim's too* and **It was up the battery I charged (not the engine)*.

As a second demonstration of syntactic structure, possessive -'s as in *Lee's friend*. You might assume at first that this possessive ending simply attaches to a noun, a word such as *Lee* or *government*, as in the *government's dilemma*.

But consider:

- a. I saw the woman next door's children.
- b. What was that guy who retired last month's name?
- c. The student I lent the book to's roommate said she'd left.

-'s actually attaches to the end of a whole phrase, not to the single noun at the end of the phrase.

When, as linguists, we try to figure out the syntactic structures of a language, we rely on the judgements of native speakers to tell us whether our example sentences are possible or impossible (the latter being starred). These grammaticality judgements, along with examples that are collected from a spoken or written corpus of the language, form the data of the science of linguistics. It doesn't matter that native speakers usually can't tell us why they feel that a particular sentence is good or bad; the very fact that they have these intuitions shows up the structural differences and similarities between sentences.

Topic- 131: Promotion and Demotion Processes

The syntactic variations involved simply reordering the elements of a sentence. Syntactic changes can have much more radical results than this. Let's discuss the idea of promotion and demotion processes –making a word or phrase more prominent or less prominent in the sentence.

In English we can say either (a) or (b) – they alternate freely. In (b), the word Hasan appears earlier in the sentence; let's say that in (b), Hasan has been promoted in the sentence:

a. Ali sent the letter to Hasan. (Indonesian)

b. Ali sent Hasan the letter.

In Indonesian, we find the same alternation, shown in following example. The main 'foreign' feature in following sentence is *surat itu* 'letter the' where English has the word order 'the letter'; otherwise, the word order in the two Indonesian examples is the same as that of the two English examples given above a. Ali menga-kirim surat itu kepada Hasan.

Ali send letter the to Hasan

'Ali sent the letter to Hasan.'

b. Ali menga-kirima-kan Hasan surat itu.

Ali send Hasan letter the

'Ali sent Hasan the letter.'

In (b), we find an ending *-kan* on the word for 'send': this ending indicates in Indonesian that the word Hasan has been promoted. English has no equivalent to *-kan*.

Now look again at the English sentence below:

a. Ali sent the letter to Hasan. (Indonesian)

b. Ali sent Hasan the letter.

When Hasan is in the promoted position in (b), we can promote it to an even higher position in the sentence, giving below. We indicate the position that Hasan is understood to have moved from with the gap _____. In the following sentence, there is also a change from *sent* to *was sent*, which signals the further promotion of Hasan. To understand why a language would need to indicate this promotion of some part of the sentence, think about the difference in meaning between *Hasan sent the letter* and *Hasan was sent the letter*.

Hasan was sent _____ the letter by Ali.

If we start with first sentence (a), however, where *Hasan* is not in a promoted position, then trying to promote it from there directly to the very highest position in the sentence would give the following sentence: again, I show the position the word Hasan has moved from with the gap. But the following sentence is not a possible sentence of English (as indicated by the asterisk):

*Hasan was sent _____ the letter by Ali.

Here is another construction involving promotion and demotion – the passive – in English and Japanese. The passive is illustrated in bold type in (1) and (2), and is an extremely common construction in both spoken and written English.

(1) The women and boys with crates converged on the boats and their catch was counted out by the market boss.
(2) His normal work was filing girls' teeth to points, although <i>pointed gnashers were considered a bit old-fashioned by the girls here.</i>
Compare these passive constructions with the sentences in (3) and (4), which are their counterparts in meaning, but are both active constructions:
(3) The market boss counted out their catch.
(4) The girls here considered pointed gnashers a bit old fashioned.

Before reading further, please try to figure out what properties differentiate the active from the passive constructions. Use the correct technical terms where you know them. Start by deciding on the grammatical role of the phrases in bold type in (3) and (4), and see what role these same phrases have in the passive constructions.

What purposes do the two different construction types seem to serve? Can you describe any additional grammatical features? First, assume that active sentences are the more basic; they are learnt much earlier by children than are passives. Two properties of the passive occur in any language which has the construction:

- (i) the passive involves promotion of an object phrase to a new position in the sentence, known as the subject position, and
- (ii) the phrase that used to be in the subject position undergoes demotion.

Demotion means that they are consigned to a by - phrase, outside the core of the sentence. The by-phrase is entirely optional. It could be omitted.

- Their catch was counted out.

Compare

- The market boss counted out their catch.

Both the subject the **market boss** and the object **their catch** are core elements of the sentence, and neither can be omitted.

Lesson 23

UNIVERSAL GRAMMAR

Topic- 132: Introduction

Chomsky takes a cognitive approach to the study of grammar. For Chomsky, the goal of the linguist is: to determine what it is that native speakers know about their native language which enables them to speak and understand the language and how this linguistic knowledge might be represented in the mind/brain: In studying language, linguists are studying a specific kind of cognition (i.e. human knowledge).

In a fairly obvious sense, any native speaker of a language can be said to know the grammar of his or her native language. For example, any native speaker of English can tell you that the negative counterpart of I like syntax is I don't like syntax, and not e.g. *I no like syntax: in other words, native speakers know how to combine words together to form expressions (e.g. negative sentences) in their language. Likewise, any native speaker of English can tell you that a sentence like She loves me more than you is ambiguous and has two interpretations which can be paraphrased as 'She loves me more than she loves you' and 'She loves me more than you love me'

Native speakers also know how to interpret (i.e. assign meaning to) expressions in their language. Grammatical knowledge of how to form and interpret expressions in your native language is tacit (i.e. subconscious) rather than explicit (i.e. conscious): so, it's no good asking a native speaker of English a question such as 'How do you form negative sentences in English?' 'since human beings have no conscious awareness of the processes involved in speaking and understanding their native language. To introduce a technical term devised by Chomsky, we can say that native speakers have grammatical competence in their native language: tacit knowledge of the grammar of their language – i.e. of how to form and interpret words, phrases and sentences in the language. And performance: what people actually say or understand by what someone else says on a given occasion.

Competence: is 'the speaker-hearer's knowledge of his language',

- A grammar of a language tells you what is needed to know in order to have native-like competence in the language (i.e. to be able to speak the language like a fluent native speaker):
- grammar is concerned with competence rather than performance.

Performance: is 'the actual use of language in concrete situations'

Very often, is an imperfect reflection of competence

- Misproductions and misinterpretations are performance errors, attributable to a variety of performance factors like tiredness, boredom, drunkenness, drugs, external distractions and so forth.
- Performance is more properly studied within the different – though related – discipline of psycholinguistics, which studies the psychological processes underlying speech production and comprehension.

The study the grammatical competence of a native speaker is the study of a cognitive system internalized within the brain/mind of native speakers of English which is the product of a ‘cognitive organ’ which is ‘shared among human beings and in crucial respects unique to them’ (Chomsky 2006, p. 1). In the terminology adopted by Chomsky (1986a, pp.19–56), our ultimate goal in studying competence is to characterize the nature of the internalized linguistic system (or I-language). According to Chomsky a grammar of a language is ‘a theory of the I-language . . . under investigation’. Attempting to uncover the internalized linguistic system (= I-language) possessed by native speakers of e.g. English. Chomsky’s ultimate goal is to devise a theory of Universal Grammar/UG: ‘The theory of human I-languages . . . that identifies the I-languages that are humanly accessible under normal conditions’. In other words, UG is a theory about the nature of possible grammars of human languages: hence, a theory of Universal Grammar answers the question: ‘What are the defining characteristics of the grammars of human I-languages?’

Topic- 133: Universal Grammar

There are a number of criteria of adequacy which a theory of Universal Grammar must satisfy. One such criterion is universality. A theory of UG must provide us with the tools needed to provide a descriptively adequate grammar for any and every human I-language since the ultimate goal of any theory is explanation.

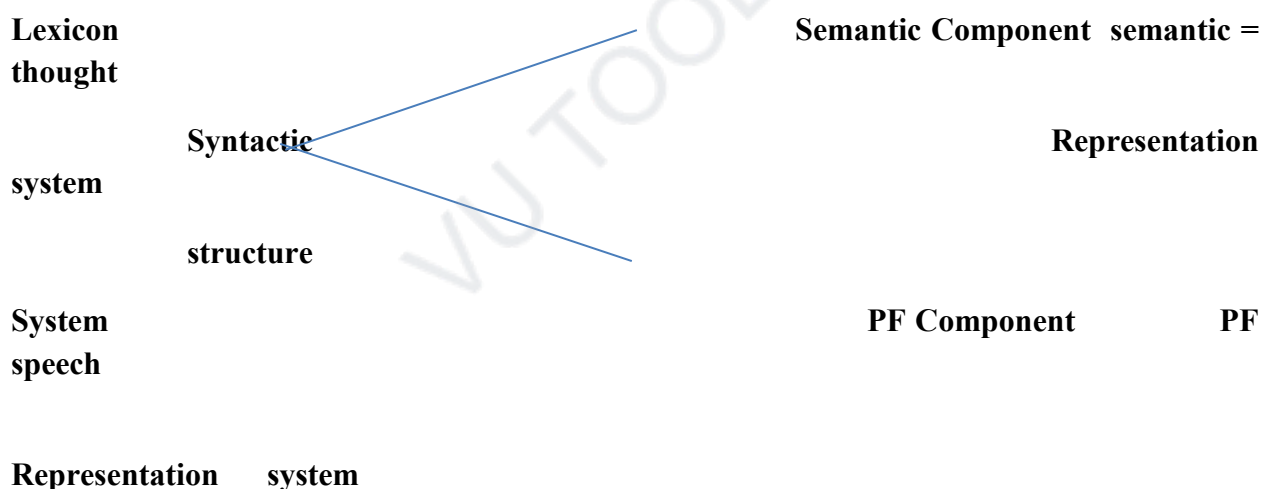
So, a key question for any adequate theory of UG to answer is: ‘Why do grammars of human I-languages have the properties they do?’ The requirement that a theory should explain why grammars have the properties they do is conventionally referred to as the criterion of explanatory adequacy. ‘What are the defining characteristics of human I-languages which differentiate them from artificial languages like those used in mathematics and computing or from animal communication systems?’

A third condition which we have to impose on our theory of language is that it be maximally constrained: it should provide us with technical devices which are so limited in their expressive power that they can only be used to describe natural languages, not appropriate for the description of other communication systems. A theory which is constrained in appropriate ways should enable us to provide a principled explanation for why certain types of syntactic structure and syntactic operation simply aren’t found in natural languages. A related requirement is that

linguistic theory should provide grammars which make use of the minimal theoretical apparatus required: in other words, grammars should be as simple as possible

To make this discussion rather more concrete, let's look at the internal organisation of the grammar of a language. One component of a grammar is a lexicon (=dictionary = list of all the lexical items/words in the language and their linguistic properties), and in forming a given sentence out of a set of words, we first have to take the relevant words out of the lexicon. Our chosen words are then combined together by a series of syntactic computations in the syntax (i.e. in the syntactic/computational component of the grammar), thereby forming a syntactic structure. This syntactic structure serves as input into two other components of the grammar. One is the semantic component which maps (i.e. 'converts')

the syntactic structure into a corresponding semantic representation (i.e. into a representation of linguistic aspects of its meaning): the other is a PF component, so called because it maps the syntactic structure into a PF representation (i.e. a representation of its Phonetic Form, giving us a phonetic spell out for each word, telling us how it is pronounced). The semantic representation interfaces with systems of thought, and the PF representation with systems of speech – as shown in diagrammatic form below:



Chomsky (2005b, p. 3) refers to the interface with thought systems as the 'conceptual-intentional interface (CI)', and to the interface with speech systems as the 'sensory-motor interface (SM)'. A fourth condition which a linguistic theory must meet is that of learnability: it must provide grammars which are learnable by young children in a short period of time. The desire to maximise the learnability of natural language grammars provides an additional argument for minimizing the theoretical apparatus used to describe languages, in the sense that the simpler grammars are the simpler it is for children to acquire them.

Topic- 134: Language Faculty

Mention of Learnability leads us to consider the related goal of developing a theory of language acquisition. An acquisition theory is concerned with the question of: How do children acquire grammars of their native languages? Children generally produce their first recognizable word (e.g. Mama or Dada) by around the age of twelve months. For the next six months or so, there is little apparent evidence of grammatical development in their speech production.

Children linguistic development	
By twelve months	• first recognizable word (e.g. Mama or Dada)
For the next six months	• little apparent evidence of grammatical development in their speech production.
Single-word stage	• productive vocabulary increases by about five words a month until it reaches around thirty words at age eighteen months. • evidence of the acquisition of grammar • No use of inflections • don't productively combine words together to form two- and three-word utterances

However, it should be noted that perception experiments have suggested that infants may acquire some syntactic knowledge even before one year of age.)

Children linguistic development	
At around the age of eighteen months	• start to make productive use of inflections (doggy, doggies) • inflected verb forms like going/gone alongside the uninflected verb form go • three-word utterances such as <i>Want Teddy, Eating cookie, Daddy gone office etc.</i>
age of around thirty months	• able to produce adult-like sentences such as <i>Where's Mummy gone? What's Daddy doing? Can we go to the zoo, Daddy?</i>

The central phenomenon which any theory of language acquisition must seek to explain is this: how is it that after a long-drawn-out period of many months in which there is no obvious sign of grammatical development, around 18 months there is a sudden spurt as multiword speech starts to emerge. A phenomenal growth in grammatical development then takes place over the next twelve months?

This uniformity and rapidity in the pattern of children's linguistic development of first language acquisition, Chomsky maintain, is determined by a biologically endowed innate Faculty of Language/FL (or language acquisition program in the brain. This program provides children with a genetically transmitted algorithm (i.e. Set of procedures) for developing a grammar, on the

basis of their linguistic experience. Listening to people around constitute the child's linguistic experience of the language.

Chomsky notes, language acquisition is an ability which all humans possess, entirely independently of their general intelligence: The apparent uniformity in the types of grammars developed by different speakers of the same language suggests that children have genetic guidance in the task of constructing a grammar of their native language. The child's linguistic experience is often degenerate, since it is based on the linguistic performance of adult speakers, and this may be a poor reflection of their competence. A further argument Chomsky uses in support of the innateness hypothesis relates to the fact that language acquisition is an entirely subconscious and involuntary activity. A critical period for the acquisition of syntax, before puberty help children generally achieve native competence in it.

After the age of nine or ten years a child rarely manages to achieve native-like syntactic competence. Accordingly: development of language in the individual must involve three factors: (1) genetic endowment, which sets limits on the attainable languages, thereby making language acquisition possible; (2) external data, converted to the experience that selects one or another language within a narrow range; (3) principles not specific to FL. The 'third factor principles' referred to under (3) 'enter into all facets of growth and evolution' and include 'principles of efficient computation'.

Topic- 135: Principles of Universal Grammar

If (as Chomsky claims) human beings are biologically endowed with an innate language faculty, an obvious question to ask is: What is the nature of the language faculty? An important point is children can in principle acquire any natural language as their native language. It follows that the language faculty must incorporate a theory of Universal Grammar/UG which enables the child to develop a grammar of any natural language on the basis of suitable linguistic experience of the language.

Experience of a particular language L serves as input to the child's language faculty which incorporates a theory of Universal Grammar providing the child with a procedure for developing grammar of L. If so, it follows that certain aspects of child (and adult) competence are known without experience, and hence must be part of the genetic information about language with which we are biologically endowed at birth.

Such aspects of language would not have to be learned, precisely because they form part of the child's genetic inheritance. Those aspects of language which are innately determined will also be universal. Thus, in seeking to determine the nature of the language faculty, we are in fact looking for UG principles which determine the very nature of language. Thus, detailed analysis of one grammatical construction in one language could reveal evidence of the operation of principles of UG. By way of illustration, let's look at question-formation in English.

Speaker a: He had said someone would do something.

Speaker b: He had said who would do what?

The type of question produced by speaker b is called an echo question. However, speaker b could alternatively have replied with a non-echo question like that below:

Who had he said would do what?

This involves two movement operations which are not found in b.

(a) He had said who would do what? (= echo question)

(b) Who had he said would do what?

B involves proposing the first wh-word *who* and the first auxiliary *had*, and that this results in a grammatical sentence.

(c) *Who would he had said do what? C involves proposing the first wh-word *who* and the second auxiliary *would*;

(d) *What had he said who would do? d involves proposing the second wh-word *what* and the first auxiliary *had*;

(e) *What would he had said who do? e involves proposing the second wh-word *what* and the second auxiliary *would*.

A theory of grammar which posits that grammatical operations are constrained by innate principles of UG offers the important advantage that it minimises the burden of grammatical learning imposed on the child. The UG provides a straightforward account of the rapidity of the child's grammatical development, since it posits that there is a universal set of innately endowed grammatical principles which determine how grammatical operations apply in natural language grammars. It also (correctly) predicts that there are certain types of error which children will not make – e.g. producing sentences such as (c–e).

Topic- 136: Parametre

The language faculty incorporates a set of universal principles which guide the child in acquiring a grammar. However, all aspects of the grammar of languages are universal; if this were so, all natural languages would have the same grammar and there would be no grammatical learning involved in language acquisition. No need for children to learn anything about the grammar.

Only lexical learning and their idiosyncratic linguistic properties, e.g. whether a given item has an irregular plural or past tense form needed to be learnt. Some universal principles do determine the broad outlines of the grammar of natural languages. Some language-particular aspects of grammar which children have to learn as part of the task of acquiring their native language. Thus,

language acquisition involves not only lexical learning but also some grammatical learning. Grammatical learning will be limited to those parameters of grammar which vary from one language to another called parametrised aspects of grammar. The obvious way to determine just what aspects of the grammar of their native language children have to learn is to examine the range of parametric variation found in the grammars of different (adult) natural languages.

Contrast between the English and its Italian counterpart:

(a) Maria thinks that *(they) speak French

(b) Maria pensa che parlano francese

‘Maria thinks that speak French’

Since the verb *parlano* speak has no overt subject, it must have a **null subject** which can be thought of as a silent or invisible counterpart of the pronoun they which appears in the corresponding English sentence. This null subject is conventionally designated as **pro**, so that Italian sentence has the following structure:

Maria pensa che pro parlano francese

‘Maria thinks that pro speak French,’ where pro is a **null subject pronoun**.

We can describe the differences between the two types of language by saying that Italian is a **null-subject language**, whereas English is a **non-null-subject language**. A more familiar aspect of grammar is parametrised relates to word order. One type of word-order variation can be illustrated in relation to the following contrast between English and Chinese questions:

(a) What do you think he will say?

(b) Nǐ xiàng tā huì shuō shénme

You think he will say what?

In simple wh-questions in English (i.e. questions containing a single word beginning with wh-like what/where/when/why) the wh-expression is moved to the beginning of the sentence, as is the case with what in (a). By contrast, in Chinese, the wh-word does not move to the front of the sentence, but rather remains in situ (i.e. in the same place as would be occupied by a corresponding on-interrogative expression), so that shénme ‘what’ is positioned after the verb shuō ‘say’ because it is the (direct object) complement of the verb, and complements of the relevant type are normally positioned after their verbs in Chinese.

Let’s now turn to look at a rather different type of word-order variation, concerning the relative position of **heads** and **complements** within phrases. It is a general (indeed, universal) property of phrases that every phrase has a headword which determines the nature of the overall phrase.

A rather different type of word-order variation, concerning the relative position of heads and complements within phrases	
students of philosophy	<i>of philosophy</i> functions as the complement of the noun students
stay with me	Verb Phrase which comprises the head verb stay and its complement <i>with me</i>
fond of fast food	the head adjective fond with its complement <i>of fast food</i> .
English head-first language	Korean head-last language
Close the door	C V Muneul dadara (door close)
desire for change	C P N (H) byunhwa-edaehan galmang

The language faculty allows only a binary set of possibilities –namely that a given kind of structure in a given language is either consistently head-first or consistently head-last. Generalising, it seems possible that all grammatical variation between languages can be characterized in terms of a set of parameters, and that for each parameter, the language faculty specifies binary choice of possible values for the parameter.

Topic- 137: Parameter Setting

The theory of parameters has important implications for a theory of language acquisition. If all grammatical variation is a series of parameters with binary settings, it follows that the only grammatical learning by children is to undertake the syntactic properties of the relevant class of constructions and to determine which of the two alternative settings for each parameter is the appropriate to acquired. So, for example, whether the native language they are acquiring is a null subject language or not, whether it is a wh-movement language or not, and whether it is a head-first language or not . . . and so on for all the other parameters along which languages vary.

Children also face the task of lexical learning what kinds of structures they can be use and so on. On this view, the acquisition of grammar involves the twin tasks of lexical learning and structural learning (with the latter involving parameter-setting). The central task which the child faces in acquiring a language is to construct a grammar of the language. The innate Language Faculty incorporates

- (i) a set of universal grammatical principles, and
- (ii) a set of grammatical parameters which impose severe constraints on the range of grammatical variation permitted in natural languages. Since universal principles don't have to be learned, the

child's syntactic learning task is limited to that of parameter-setting. For obvious reasons, the theory outlined here (developed by Chomsky) is known as **Principles and-Parameters Theory/PP**. One of the questions posed by the **parameter-setting model** of acquisition outlined here is just how children come to arrive at the appropriate setting for a given parameter, and what kind(s) of evidence they make use of in setting parameters.

There are two types of evidence which we might expect to be available to the language learner in principle, namely positive evidence and negative evidence. Positive evidence: a set of observed expressions illustrating a particular phenomenon: for example, if children's speech input is made up of structures in which heads precede their complements, this provides them with positive evidence which enables them to set the Head Position. Negative evidence of two kinds – direct or indirect. Direct negative evidence could come from the correction of children's errors by other speakers of the language. Correction plays a fairly insignificant role in language acquisition, for two reasons.

Firstly, correction is relatively infrequent:

Secondly, children are notoriously unresponsive to correction:

Child: Nobody don't like me

Adult: No, say: Nobody likes me

Child: Nobody don't like me

Rather than say that children rely on direct negative evidence, we might instead imagine that they learn from indirect negative evidence (i.e. evidence relating to the non-occurrence of certain types of structure). Suppose that a child's experience includes no examples of structures in which heads follow their complements (e.g. no Prepositional Phrases like *dinner after). On such indirect negative evidence (such structures never occur in English), the child might infer that English is not a head-last language.

Learnability considerations such as these have led Chomsky to conclude that 'There is good reason to believe that children learn language from positive evidence only'. The claim that children do not make use of negative evidence in setting parameters is known as the **No-Negative-Evidence Hypothesis**; it is a hypothesis which is widely assumed in current acquisition research.

Lesson 24

SENTENCE PRESENTING

Topic- 138: Introduction

Syntax, or syntactic analysis, may be defined as:

- (a) determining the relevant component parts of a sentence
- (b) describing these parts grammatically.

The component parts of a sentence are called constituents. Syntax involves the two closely related tasks of:

- (a) breaking down the sentence into its constituents
- (b) assigning some grammatical label to each constituent, stating what type of constituent it is, and what grammatical function it has.

This definition of syntax implies that we start from what is regarded as the largest unit of syntactic description -the sentence -and proceed until we arrive at the smallest meaningful unit. This is called a 'top to bottom' analysis. The units smaller than the sentence will be referred to as clauses, phrases, words and morphemes respectively. We might also look at the sentence the other way round - that is. 'from bottom to top' - and say that constituents at different levels can combine to form increasingly larger units: we proceed then from the morpheme to the sentence as a whole.

Constituents are like building blocks which pattern in certain ways to form larger and larger units, the largest unit being the sentence. Each constituent (except the smallest) can be broken down into its component parts. The purpose of doing syntax is to discover the ways in which constituents combine to form the structure of sentences. We adopt the (traditional) hierarchy of sentence constituents, as shown in the following diagram:

The hierarchical scale of constituents.

The four double-pointed arrows indicate that it may be read 'from left to right', or 'from right to left.'

SENTENCE ↔ CLAUSE ↔ PHRASE ↔ WORD ↔ MORPHEME

The purpose of doing syntax is to discover the ways in which constituents combine to form the structure of sentences. The arrows pointing to the right indicate that a sentence may consist of one or more than one clause.....Conversely, as indicated by the arrows pointing to the left, we

might also say that one or more than one morpheme may constitute a word. A clause may contain one or more constituent clauses, and that a phrase may contain one or more constituent phrases or clauses.

To illustrate the hierarchical structure of sentences, let us consider sentence

The snake killed the rat and swallowed it.

Clause 1

Clause 2

Two coordinate clauses

The first clause is: The snake killed the rat,

The second is: swallowed it.

The second clause has a reduced form.

Its complete form would be: ... it swallowed it.

The snake killed the rat (two phrases)

swallowed it (one phrase)

One noun phrase

Two verb phrase

Each phrase is made up of words. Sentence (1) contains eight words, including the coordinator and. Each word consists of one or two morphemes: the, snake, rat, and and it are one-morpheme words, whereas killed and swallowed are both two-morpheme words. The, snake, rat, etc. are full words and morphemes at the same time: the word and morpheme boundaries coincide. The two morphemes of killed and swallowed are kill and -ed, and swallow and -ed.

Topic- 139: Structure

The syntactic structure of sentence may be represented provisionally by marking off each constituent from sentence level to word level with a square brackets: []. To simplify matters, we shall ignore the morpheme boundaries here. This convention of bracketing yields the following analysis, which looks rather daunting at first sight:

(1 [[[[The][snake]] [killed] [[the] [rat]]]] [and] [[[swallowed] [[it]]]]]

Sentence-

.(2a) [The snake killed the rat and swallowed it]

Clauses:

(2b) [[The snake killed the rat] and [swallowed it]]

Phrases:

(2c) [[[The snake] [killed [the rat]]] and [[swallowed [it]]]]

Words:

(2d) [[[[The][snake]] [[killed] [[the] [rat]]]] [and] ([[swallowed][[it]]])]

Analysis (2d) is of course identical with (1) above.

In (1) above we can see that word and morpheme boundaries may coincide: the, snake, rat, etc. are all one word and one-morpheme, as opposed to killed and swallowed, which are words consisting of two morpheme each. Words and phrases may also coincide, as in:

(3) John laughed.

In this sentence John is both a phrase (a noun phrase) and a word (a noun); laughed is also both a phrase (a verb phrase) and a word (a verb). The units sentence and clause also coincide in (3). Bracketing from sentence level to word level yields (4):

(4) [[[[John]] [[laughed]]]]

Word, phrase, clause and sentence in (9) may coincide, as in run:

(5) Run!

The bracketing of sentence (5) is as follows:

(6) [[[[Run]]]]

It is structurally one sentence, one clause, one phrase and one word (also one morpheme). The above examples show that a sentence is not necessarily longer than a clause, a clause not necessarily longer than a phrase, and a phrase not necessarily longer than a word (in general, we shall not go beyond the level of the word in our analyses). We shall see that sentences may vary in length and complexity from one clause to indefinitely many clauses, clauses from one phrase to indefinitely many phrases, and phrases from one word to indefinitely many words.

Topic- 140: Labeled Bracketing

The system of bracketing used so far is not very satisfactory. It is difficult to see which brackets go together to mark off a constituent. The notation introduced could be improved by adding an appropriate grammatical label to each pair of square brackets. Let us consider again the structure of the sentence: The snake killed the rat and swallowed it. (1 above)

We shall use square brackets with the label S (short for 'sentence') to mark off the boundaries of the whole sentence, as follows:

(7) [s The snake killed the rat and swallowed it]

The category label is inserted in the bottom corner of the left-hand bracket. This sentence, as we have seen, consists of two clauses: the clause

The snake killed the rat and the clause (*j swallowed it. The two clauses are gathered by the coordinator and The clauses can be bracketed and labelled as follows:

(8) [s1 [s2 [NP The snake] [VP killed [NP the rat]]] and [s3 [NP [it]] VP swallowed [NP it]]]

The two clauses are labelled S 2 and S 3 here, and the whole sentence is labelled S,. S 2 and S 3 are clauses inside S, (the numbers 1,2,3 etc. are added here and elsewhere just for convenience: they enable us to refer unambiguously to the different Ss). We use the label S for sentences as well as clauses, since clauses can be defined as Ss inside an S or inside a phrase. The clauses in (12) can be further analysed into phrases as follows:

(9) [s1[s2 [NP [DET The](N snake)]] [vp [v killed][and][NP [DET the] [N rat]]]]
[and] [s3[NP(it)] [VP [v swallowed] [NP [N it]]]]

The label NP stands for noun phrase, and the label VP for verb phrase. The snake and the rat are NPs, because their most important element is a noun (N); it, which replaces an NP, is also an NP. The VPs in (9) consist of a verb (V) followed by an NP. The verbs are killed and swallowed The NPs the rat in killed the rat and it in swallowed it are parts of the VPs, and act as Complements of the verbs killed and swallowed respectively.

The label Det stands for determiner, the label N for noun, and the label V for verb. Noun and verb are major word classes, and Det is a collective term for various items preceding the noun, e.g.: the, a, that, this, some, any. The only difference between the two is that pairs of brackets are labelled here.

The syntactic analysis of the sentence does not usually go below the level of the word or even the phrase, but occasionally it will be useful to mark off the morpheme structure of a given word, for example the structure of the past tense forms of verbs, or of the plurals and genitives of nouns. This, too, can be done by means of labelled bracketing, as, for example, in the case of killed:

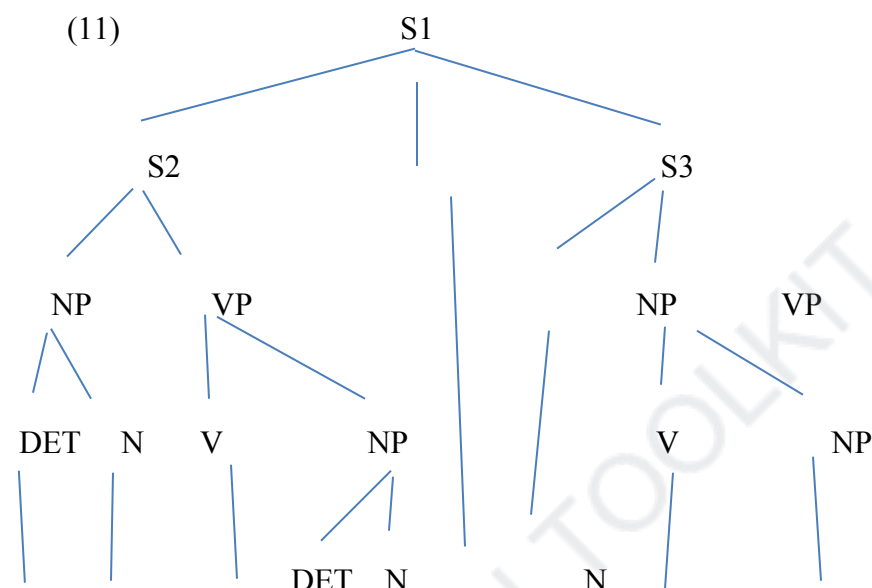
(10) [v [B kill] [suff ed]]

This convention is called labelled bracketing.

Topic- 141: Tree Diagram and Phrase Structure Rules

Labeled bracketing is one of the most common ways of representing the constituent structure of sentences. There are many other methods of marking diagrammatically what elements in a sentence go together and what elements do not. One other very common representation is the tree diagram. The tree diagram is a notational device which is entirely equivalent to labelled bracketing. Although different, it provides the same information about the syntactic structure of a sentence. Consider the following tree diagram of the sentence:

The snake killed the rat and (it) swallowed it.



The snake killed the rat and (it) swallowed it.

The tree diagram provides the analysis of sentence (1) down to word level (determiner, noun, verb, etc.); It would also be possible to stop at phrase level (NP, VP, etc.), or to go beyond word level and indicate the morphological structure of each of the words. You can make your grammatical analysis as detailed as you like, or as is necessary for a specific purpose. The syntactic information contained in diagram (11) is essentially the same as that provided by the labelled bracketing in (8).

It is largely a matter of taste or practical convenience whether one chooses the notation of bracketing or tree diagram. To read tree diagram (11) we require some additional terminology. For ample, we say that S, (the whole sentence) is expanded as S and S (two clauses), which are coordinated by and S, is said to contain as its immediate constituent S2, S3 thereby the VP-node. The NP immediately dominated by the VP-node is, also dominated by S/S3, but it is not immediately dominated by S/S;- 'T important to distinguish between dominance and immediate dominance. In the latter case there must be no further nodes intervening between the nodes considered. The lexical items (i.e. the words) the, snake, killed, etc., are attached to the so-

called terminal nodes of the tree diagram, i.e. the bottom nodes. The other nodes in the tree diagram are non-terminal.

There is a convention which is generally used to sum up the system expending one unit into other. It is a set of instructions called phrase structure rules (PS rules).

(12a) $S \rightarrow \text{-cood- } S$

(12b) $S \rightarrow \text{NP-VP}$

(12c) $\text{VP} \rightarrow \text{V- NP}$

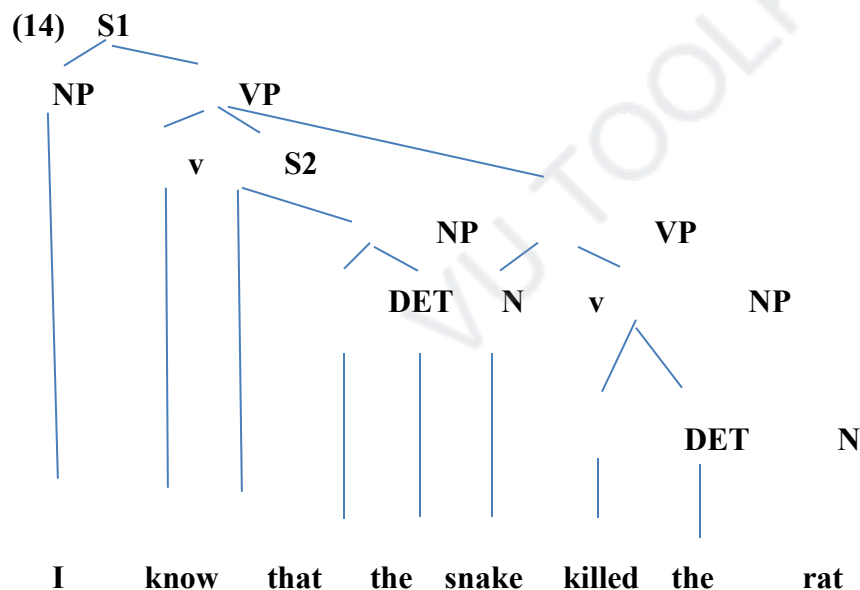
(12d) $\text{NP} \rightarrow \{\text{DET-N} \}$

N

Topic- 142: Subordination in PS rules

Subordination of a clause

(13) I know [s that the snake killed the rat] may be represented as follows:



The tree diagram (14) is provisional.

It has a sequence

that – the snake killed the rat

The PS rules cannot fully describe structure like (14)

Rule (12a) $S \rightarrow \text{-cood- } S$ not applicable to all sentence

We start the rewriting operation of (14) at rule (12b).

Instead of writing VP

as V-NP we write VP in (14) V-S. The PS rule (12c) must adapt it in the following way

(12c) $VP \rightarrow v- \{NP\}$

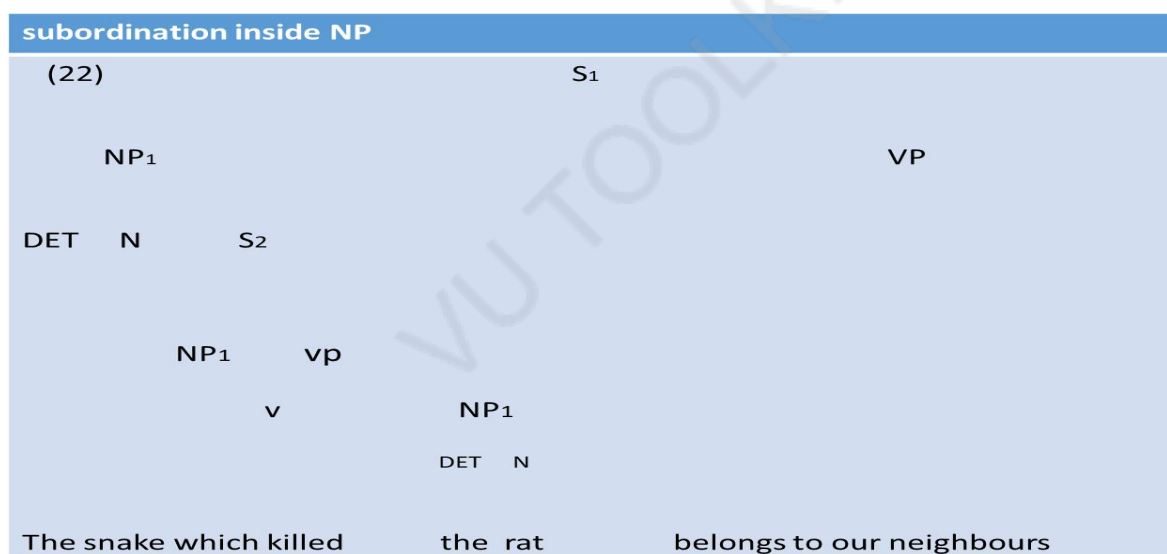
The S may be expanded into NP-VP. It is possible to have S embedded inside another S. The embedding or subordination may be repeated many times. For example;

(15) I know that you think that she hopes that you will say to her that you love her.

In diagram (19) S2 is a subordinate clause, which functions inside S1: S2 is dominated by S1 and immediately dominated by VP. Ss (clauses) may also appear inside NPs as in:

(16) [The snake [which killed the rat]] belongs to our neighbours.

We have in this sentence subordination not in the VP as in (18) and (20) but subordination inside NP.



In (21) we need to adapt the PS rules given in (17).

We formulate rule (17d) as follows:

$NP \rightarrow \{ DET-N-(s) \}$

N

Round bracket means S is optional.

We may choose to rewrite

NP as Det-N-S or

As Det-N

Or may write NP as just N

Topic- 143: Sentence and Discourse

The sentence is regarded as the largest or highest unit on the hierarchical scale of constituents; it is the largest unit of syntactic description. Sentences do not normally occur in isolation. They usually form part of a larger text (or discourse) which is also organised in a particular way. The sentences of a text follow each other in some 'logical' order, and reflect a certain sequence of thoughts or events. There are often elements in a sentence which mark its relationship with the context.

Texts are structured, and are more than just a random collection of sentences.

Consider, for example, the text 1.2.4, consisting of only four sentences: The internal structure of this text is shown by a number of factors. For example:

Soon after breakfast Mary Ann brought in The Times. Mr. Carey shared it with two neighbours. He Had it from ten till one, when the gardener took it over to Mr. Ellis at the Limes, with whom it remained till seven. Then it was taken to Miss Brooks at the Manor House, who, since she got it late, had the advantage of keeping it.

**Adapted from W. Somerset Maugham,
Of Human Bondage,**

(a)The order in which the four sentences occur is not accidental; it reflects the order in which the events described in the text occur.

(b)The words it, he and she relate to constituents in previous sentences. Check which elements these words are related to in the text above.

(c)There is a link between 'two neighbours' in the second sentence and 'Mr. Ellis' and 'Miss Brooks' in the two following sentences.

(d)The use of then at the beginning of the fourth sentence also indicates that there is a certain temporal sequence of events. Like sentences, texts have an internal structure.

The structural organisation of texts is different from that of sentences. The rules conditioning the organisation of texts are the rules of text grammar or discourse grammar, whereas the rules which determine the structure of sentences make up sentence grammar or syntax. The rules of text grammar often require an appeal to notions which are not syntactic. We shall only be concerned with the description of sentences. The analysis of longer stretches of discourse will generally be left out of consideration. We shall usually only pay attention to the structure of the individual sentences in the text.

Occasionally, we shall go beyond the level of the sentence, if this is necessary for the clarification of some point. The emphasis of this course is on formal structural properties of sentences rather than on their pronunciation, their meaning or their use in communication: it is a book on English syntax, not on phonology, semantics, or pragmatics. We use the terms 'grammar' or 'syntax', 'grammatical' and 'syntactic' interchangeably. The term 'grammar' is also sometimes used in a very general sense to include syntax, phonology, semantics, etc.

VU TOOLKIT

Lesson No. 25

CONSTITUENCY TESTS**Topic- 144: Introduction**

Text consists of sentences and that each sentence/clause may be broken down into the constituents NP and VP. VP consists of a V, which may be followed by one or more constituents. The immediate constituents of the sentence/clause (NP-VP) and those of the VP are considered to be the main constituents of syntactic analysis. They may be involved in a number of processes which change the appearance of a basic sentence pattern.

These processes always affect completion constituents, and thus enable the student of language to discover what the constituents of sentences are. Let us return to one of our example sentences:

The snake killed the rat and (it) swallowed it.

We have argued that sentences have structure, and are not just strings of words which occur in a random order. The words do not just follow each other like the beads on a string. The words of a sentence are strictly organised internally: there is an underlying pattern. For example, the goes with snake rather than with killed, and it is more closely associated with swallowed than with and, killed or snake. The items in a text are organised and have a certain structure. Sentences are hierarchically organised into different constituents: sentences into clauses, clauses into phrases, phrases into words, and words into morphemes

Topic- 145: Variation on Basic Sentence Patterns

Most of the silk we see in Britain comes from silkworms, but wild silk moths in countries like India and Japan also produce it. They all spin cocoons and, although their silk is not as fine as silkworms', several species are cultivated.

The two most important wild silks are tasar and muga. Tasar is produced in a humid and dense belt of tropical forest in India. Once the cocoons have been spun, the largest are used for future breeding. When the moths emerge, the females are tethered by thread tied round the base of their wings to prevent them flying away.

Muga silk is produced exclusively in Assam by the assamiensis caterpillar. The silk varies in colour and texture, depending on their diet. Those fed on the young leaves of the majankori tree produce pale silk. - almost creamy white - while those fed on older leaves produce silk of a golden colour.

Adapted from Observer (Colour supplement) 14 March 1982.

Fixed sentence pattern

Word order in English is fixed to a large extent, and if a given word order is disrupted the sentence may become less acceptable or even ungrammatical. Some most important sentence processes applied to the sentences in this text. We shall deal briefly with operations called:

- Clefting
- Pseudo-clefting
- Passivisation
- Pronominalisation
- Fronting
-

Topic- 146: Clefting

Cleft sentences are one way to add emphasis to what we want to say. A cleft sentence is a sentence in which some part is moved from its normal position into a different place to give it more emphasis. The first sentence of text contains two coordinate clauses. The first clause is:

(1) Most of the silk we see in Britain comes from silkworms.

The constituent from silkworms in (1) may be given more emphasis by placing it at the beginning of the sentence and 'surrounding' it by the words it was/ was that, as follows:

(2) It is from silk worms that most of the silk we see in Britain comes.

(1) and (2) have a different outward appearance (or form), but their content (or meaning) is more or less the same.

The only difference in meaning between them is that in (2) the element from silkworms is very emphatically contrasted with something else in the context (wild silk moths, in the second coordinate clause): this contrast is more clearly marked in (2) than in (1).

The operations of fronting a constituent like from silk worms and surrounding it by it was/ was ...that are part of a process known as clefting. If we represent the string from silk worms by X, we could summarise the process as follows:

CLAUSE → It is X that CLAUSE

[With X] [Without X]

So:

(a) Most of the silk we see in Britain comes from silk worms.

(b) It is from silk worms that most of the silk we see in Britain comes.

Consider also: Jane gave this book to Bill on Saturday. Again we show how one can extract a constituent and put it in a more prominent position by applying the process of clefting:

(a) X = on Saturday	It was on Saturday that Jane gave this book to Bill.
(b) X = to Bill	It was to Bill that Jane gave this book on Saturday.
(c) X = this book	It was this book that Jane gave to Bill on Saturday.
(d) X = Jane	It was Jane that gave this book to Bill on Saturday.

Sentences resulting from the clefting process are called cleft sentences. It is not possible to produce a cleft sentence by putting just anything in the position for X. For example:

(3)* It was Jane gave that this book to Bill on Saturday.

Clefting cannot affect the string Jane gave, because this is not a constituent in the sentence.

Clefting is a useful way of discovering which words in a sentence form constituent and which do not:

Jane is a constituent but Jane gave, for example, is not.

English is very rich in cleft constructions. Below are examples of other types of clefts found in English

- **It-cleft:** It is Jaime for whom we are looking.
- **Wh-cleft/Pseudo-cleft:** [1] What he wanted to buy was a Fiat.
- **Reversed wh-cleft/Inverted pseudo-cleft:** A Fiat is what he wanted to buy.
- **All-cleft:** All he wanted to buy was a Fiat.

However, not all languages are so rich in cleft types as English, and some employ other means for focusing specific constituents, such as topicalization, word order changes, focusing particles and so on (see Miller 1996).

Topic- 147: Pseudo-clefting

A pseudo-cleft sentence is a kind of cleft sentence in which the subordinated clause is a relative clause headed by an interrogative pro-form. In English they are of the form:

wh-relative clause + be + X

X can be a constituent of one of many varieties.

The so-called 'inverted' pseudo-cleft sentence reverses the order of the two constituents:

X + be + wh-relative clause

Examples:

Pseudo-cleft Inverted pseudo-cleft

- What John gave to Mary were flowers.
- Flowers were what John gave to Mary.
- (4)What Jane did was give this book to Bill on Saturday.
- (5)What Jane gave to Bill was this book.
- (6)This book is what Jane gave to Bill on Saturday.

These three sentences can all be related to the sentence Jane gave this book to Bill on Saturday. Pseudo-clefting affects whole constituents, e.g. gave this book to Bill on Saturday and this book.

Topic- 148: Passivisation

Most of our example sentences above express an activity performed by a person and affecting some other person, animal or thing which undergoes the activity. The person who performs the activity is the Agent, and whoever or whatever undergoes it is called the Patient. Agent and Patient are roles.

In the (a)-sentences in (7) and (8) we first mention the Agent, then the Patient. In the (b) sentences it is the other way round. The change from (a) to (b) is passivisation.

(7a) Wild silk moths in countries like India and Japan also produce it.

(7b) It is also produced by wild silk moths in countries like India and Japan.

(8a) People cultivate several species.

(8b) several species are cultivated.

Like clefting, passivisation is a way of rearranging the information in a sentence: if for some reason the Patient is to be made more prominent, we may use a passive sentence. In passives the Agent role is no longer obligatorily expressed (8b) above). In active sentences, normally, the Agent precedes and the Patient follows. In passive sentences the Patient precedes and the Agent, if any, follows. Passivisation also affects complete constituents. Sentence (7b), for example, shows that wild silk moths in countries like India and Japan and it are both constituents. In English grammar, passivization is the transformation of a sentence from an active form to a passive form. Verb: passivize. Also known as raising. Alternate spellings: passivisation (chiefly British).

Through the process of passivization, the direct object of an active declarative sentence can become the subject of a passive sentence. The opposite of passivization is activization. Both terms were coined by linguist Noam Chomsky.

Topic- 149: Pronominalisation & Fronting

They all spin cocoons and, although their silk is not as fine as silkworms', several species are cultivated. Consider the sentence of text. It begins with they and it also contains the word their. What do they and their relate to? To answer this question, we must look at the preceding sentence. The two words they and their have the same reference as wild silk moths in countries like India and Japan: they and there are co-referential with the N P.

Since they can be substituted for an NP, they are called pronouns, or more generally pro-forms. Substitution by pronouns is called pronominalisation. Like clefting and passivisation, pronominalisation can affect constituents only. In traditional transformational grammar, pronominalization is a rule replacing lexical items with a pronoun, whereas later approaches analyzed the pronouns as being generated in the base.

FRONTING

The fourth sentence of text is as follows:

- (9) Tasar is produced in a humid and dense belt of tropical forest in India.

This sentence contains the string in a humid and dense belt of tropical forest in India, which may be moved to the beginning of the sentence, as_ follows:

In a humid and dense belt of tropical forest in India, tasar is produced.

We call this process fronting (or preposing). Only certain sentence elements can be fronted, and these elements must be constituents.

Lesson 26

CONSTITUENTS I: WORDS ORDER

Topic- 150: Sentences: Simple Declarative & Interrogative

Coroner : The collision occurred at the junction of High Street and Church Road. How far away were you?

Witness: I was 54 yards, 2 feet, 6 inches away from that spot.

Coroner: You are very precise. Did you measure the distance?

Witness: Yes. I measured it carefully.

Coroner: Why did you measure it?

Witness: I knew that some idiot would ask me.

Adapted from Christian Brann, Pass the Port

This conversation clearly involves an exchange of information: the coroner asks questions, and the witness tries to answer them. Sentences can generally be used to give information, and they can be used to ask for information. Identify the questions in the above text. Questions like Did you measure the distance? ask for the answer Yes or No.

They are said to be Yes-No questions. If you ask a Yes-No question, you want to know whether something did or did not happen. If you know that something happened, you might want to hear about the circumstances of the event (e.g. the time, place, reason of the event), or you might want to know more about the participants (the people involved)..

When the coroner in the text above asks Why did you measure it?, he knows that the witness measured the distance, because the witness has just said so. But the coroner does not know the reason for this (why); consequently he asks the witness to supply the missing information. Questions which ask for a particular piece of information, rather than for simply Yes or No, are called wh-questions. We call them wh-questions because they are often introduced by a word (or phrase) beginning with wh-.

wh-questions

- (1) Where did John hide the money?
- (2) What did she say?
- (3) When did he come back?
- (4) Which girl did Bill like best?
- (5) Who(m) did you meet there?

However, the wh-word does not always open the question. It may also appear inside the first constituent of the sentence:

Inside wh-question

- (6) For what reason did you leave early?
- (7) In what way did John offend her?
- (8) At what distance were you from that spot?

These are also wh-questions. In what way can be replaced by how, and at what distance by how far. How and constituents containing how are also regarded as wh-elements.

How-question

- (9) How did you get to know him?
- (10) How far away were you from that spot?
- (11) How old are you?
- (12) How much money do you earn?

Sentences used to ask questions are interrogative sentences; Sentences used to supply information are declarative sentences.

Topic- 151: Compound & Complex Sentence

Mary Ann's stories of the sea touched Philip's imagination. One evening he asked whether he might go home with her. His aunt feared that he might catch something. His uncle said that evil communications corrupted good manners.
Adapted from W. Somerset Maugham, *OJ Human Bondage*.

Mary Ann's stories of the sea touched Philip's imagination. One evening he asked whether he might go home with her; but his aunt feared that he might catch something, and his uncle said that evil communications corrupted good manners.
Adapted from W. Somerset Maugham, *OJ Human Bondage*.

Of the two texts, the B-version approximates the original passage in Somerset Maugham's novel more closely. How many sentences are there in the A-text? How many do you find in the B-text? Note that the second, third, and fourth sentences of A have been joined together in B to form one long sentence. The boundaries of the sentences in A are still visible in B: the place where the linking has taken place is indicated by means of special punctuation (a comma or a semi-colon) and by means of coordinators such as and, but and or

A clause which is coordinated with another clause in this way is called a conjoin: Thus, the second sentence of B consists of three conjoins which have been joined to form one sentence.

The three conjoins are

- (I 3a) One evening he asked whether he might go home with her.
- (I 3b) (but) his aunt feared that he might catch something.
- (I 3c) (And) his uncle said that evil communications corrupted good manners.

A sentence which consists of a string of coordinated clauses (conjoins) is called a compound sentence.

- (14) George cheated at Harvard and Jane cheated at Yale.
- (15) Sue went to London and stayed there for a week.
- (16) John studied very hard last year but he failed his test.
- (17) Paul will go to the cinema tonight, or he will stay at home to read a book.

In examples (14)-(17) simple sentences have been coordinated to form compound sentences. A third category is that of complex sentences. To define these, let us consider the sentences:

- (18) I know that the snake killed the rat.
- (19) The snake which killed the rat belongs to our neighbours.

In (18) the string that the snake killed the rat is a clause, but it is not co-ordinated with another clause. Rather, it forms part of another clause; it is embedded in another clause. The other clause (the main clause) is: I know that the snake killed the rat. The embedded clause is a constituent of the VP of the main clause (see 2.4.3). In (19) the string which killed the rat is also a clause. This clause forms part of: The snake which killed the rat, which is an NP (see 2.4.2). The clause is embedded in the NP; the embedded clause is a constituent of the phrase in which it occurs. Sentences which contain embedded clauses of this kind are said to be complex. We shall return to the syntactic differences between compound and complex sentences below.

There are also mixed types of sentences, which involve both co-ordination and subordination: there may be subordinate clauses within conjoins and subordinate clauses may also be coordinated.

- (20) One evening he asked whether he might go home with her; but his aunt feared that he might catch something, and his uncle said that evil communications corrupted good manners.

The following are examples of coordination of subordinate clauses:

- (21) I know that the snake killed the rat and (that it) swallowed it.
- (22) The snake which killed the rat and (which) swallowed it belongs to our neighbours.

These mixed types of sentences are called compound-complex sentences. There are important differences between coordinate and subordinate clauses. For example:

Coordinate clauses are introduced by coordinators (and, but, or, for, etc.) and subordinate clauses by subordinators (when, before, because, if, since, although, that, etc.). Compare:

- (23a) The snake killed the rat and it swallowed it.

(b) The coordinator always appears in a position between two co-ordinated clauses, whereas the subordinator is part of the subordinate clause. As a result, it is impossible to place the second conjoin together with the coordinator in front of the first conjoin. Compare:

- (24a) The snake killed the rat and it swallowed it.
- (24b) *And it swallowed it, the snake killed the rat.

On the other hand, subordinate clauses and the subordinators which introduce them can be fronted together:

- (25a) The snake killed the rat, before it swallowed it.
- (25b) Before it swallowed it, the snake killed the rat.

The subordinator and the clause form one constituent; the coordinator and the second conjoin are two separate constituents, which cannot be moved together. This difference between coordination and subordination of clauses may be represented as follows:

Coordination: [S1 [S2] coord [S3]]

Subordination: [S1 [S2 subord]]

(c) Coordination with and, but and or allows deletion of the subject of the second conjoin, if it is co-referential to the subject of the first conjoin. Deletion of the subject is generally impossible in subordinate clauses.

- (26a) The snake killed the rat and it swallowed it.
- (26b) *The snake killed the rat and before it swallowed it.

Topic- 152: Clauses

Clause' are constituents of other sentences or of phrases. A constituent sentence (or clause) may be coordinated with one or more other constituent sentences to form a compound sentence, or it may be embedded in another sentence or in a phrase to form a complex sentence. In simple sentences the boundaries of sentence and clause coincide: a simple sentence is a one- clause sentence, whereas compound and complex sentences always contain at least two clauses.

Let us return to the compound sentence.

In order to show its structure, we shall now bracket each of the conjoins and label them by means of S (for sentence/clause) in the left-hand corner, as follows:

- (1a) [s One evening he asked whether he might go home with her;]

But

- (1b)[s his aunt feared that he might catch something,]

and

- (1c) [5his uncle said that evil communications corrupted good manners.]
- (1b) as a separate independent sentence. If we passivise this sentence, we get:
- (2) That he might catch something was feared by his aunt.

Passivisation affects the string and tell that this string is a sentence Constituent. The same is confirmed by pseudo-clefting and by Pronominalisation (use of it, that, what, etc.):

Pseudo-clefting and Pronominalisation

- (3) What his aunt feared was that he might catch something.
- (4) His aunt feared {it, that}
- (5) What did his aunt fear?

Bracketing of this constituent provisionally gives us the following result:

- (6) [s His aunt feared [that he might catch something]]

It is clear that the string he might catch something is also an S, but what about the element that? Sentence (6) is to be bracketed as follows:

- (7) [s1 His aunt feared [x that [s2 he might catch something]]]

However, passivisation, psudo-clefting and pronominalisation (see (2)-(5) above) affect the constituent labelled X in (7), not just S. The role of that in (7) is that of linking S2 (the embedded sentence) to the sentence inside which it occurs: that is the subordinator. The constituent X comprising the subordinator and S2, is often labelled s (S-bar) or S' (S-prime). It is a unit containing the subordinator and the subordinate

S that→S

The element that occupies the so-called, complementiser slot (COMP slot in front of the sentence S2. Our bracketing of (1b) may now be adapted as follows:

- (9) [s1His aunt feared [s (that [s2he might catch something]]]

Let us now try to analyse the third conjoin in (1c) above, repeated here as (10):

- (10) His uncle said that evil communications corrupted good manners.

Following the procedure, we have used above, we find that (10) has essentially the same structure as (1b). Moreover, in both sentences that can be omitted (or deleted):

- (11a) His aunt feared ---he might catch something.
- (11b) His uncle said ---evil communications corrupted good manners.

(in the passive variants of these sentences that is not deletable)

This string whether + he might go home with her is an embedded yes-No question. When Yes-No questions are embedded, the subordinator is whether or if, not that. The sentence (13) may be bracketed as f(16)[S1 One evening he asked [s { whether/if*that} [s2 he might go home with her]]] (COMP slot can be filled by whether follows:

Notice that whether and if cannot be deleted, unlike that in (11a) and

- (11b).

wh- questions:

- (17) Why did you measure the distance?
- (18) What did you see there?

Such wh-questions may also be embedded in other sentences, but we need no special subordinator to do that.

- (19) [s1 The coroner asked [s- why [s2the witness had measured the distance]]]
- (20) [s1 The policeman asked me [S what [s2 I had seen there]]]

Now also compare (21) and (22) below (22) is more acceptable American English than in Standard British English. Is there any difference in meaning between them? And what about their form?

- (21) The new law requires that seat-belts should be worn by all passengers
- (22) The new law requires for seat-belts to be worn by all passengers.
- (21) and (22) have the same meaning, and the same structure. we may passivise them:
- (23) That seat-belts should be worn by all passengers is required by the new law.
- (24) For seat-belts to be worn by all passengers is required by the new law.

pseudo-clefting to both:

- (25) What the new law requires is that seat-belts should be worn by all passengers
- (26) What the new law requires is for seat-belts to be worn by all passengers

What conclusion can we draw with respect to the constituents of these sentences? It seems that (21) requires the following (partial) bracketing

- (27) [s1 The new law requires [s that [s2 seat-belts should be worn by ...]]]

The string for seat belts to be worn by all passengers in (22), as we have seen, structurally resembles the S- that seat-belts should be worn by all passengers. We shall assume, by analogy with (27), that for also occupies the COMP slot, and that seat -belts to be worn by all passengers is an embedded sentence/ clause. The analysis, then, of (22) is as follows:

Note that (21) allows a great deal of variation in front of be worn.

We could replace should by must or shall, and should be worn by are worn.

In fact, should, must or shall may also be absent.

- (29) The new law requires that seat-belts {{(should/must/shall/etc.}} be} worn...

The sequence to be worn is called a to infinitive of the verb wear.

If we replace seat-belts by a pronoun, we find another contrast.

- (30) The new law requires that they should be worn...
- (31) The new law requires for them to be worn . . .

Non-finite clauses may also contain a bare infinitive, or they may contain an -ing participle or an -ed participle. Consider, for example:

- (32) We saw Mary leave.
- (33) We saw Mary leaving
- (34) I do not mind Mary's leaving (gerund clause)
- (35) We found all the seats occupied

English also has verbless clauses. (usually a form of to be) and sometimes other elements have been deleted. Consider, for example:

- (36) John believes the prisoner innocent.

In this sentence the italicised sequence is a verbless clause, which we assume is a reduced version of the to-infinitive clause in (37):

- (37) John believes the prisoner to be innocent.
- The following sentences contain further examples
- (38) He considered the girl a good student.
- (39) Whenever in trouble, Bill rang his girl-friend.
- (40) He married her when a student at Harvard

Non-finite clauses and verbless clauses are always embedded / subordinate.

Topic- 153: Phrases:

Let's decide the types of sentences in the following text (simple, compound, complex, or compound-complex):

Text

The tramp read the diary. He laughed. He turned a page, he read it and he laughed again. He leaned towards the German girl and said a few words to her. The Egyptian was clowning; the noise in the room continued. Soon the young German girl was offering chocolate for the second time. Her voice was very soft.

The tramp was unfolding his magazine slowly. He stopped suddenly, he looked at the chocolate. But she had given him no chocolate. He unfolded his magazine. Then he destroyed it.

V.S. Naipaul, In a Free State

Try to isolate as many phrases as you can in each sentence above. You will see that these constituents fall into various grammatical categories (or constituent types). The different categories of phrases.

NOUN PHRASES

The constituents such as the tramp, the diary, he, a page, it, the German girl, a few words, her, the Egyptian, the noise, the young German girl, chocolate, etc.

- Noun Phrases
- (1a) { The tramp }
- (ab) { The young German girl }
- (1c) { The Egyptian }
- read { his diary
his magazine
a page }
- (2a) { The girl }
- (2b) { The very young girl }
- (2c) { The girl from Berlin }
- offered { chocolate
no chocolate
hot chocolate }

Constituents which can appear in the same environments (or contexts) said to have the same distribution. The Constituents have as its most important element a word of the class of nouns (N): tramp, diary, page, chocolate . . .

NP chocolate

Elements before the Head (a, the)

Tramp read diary. .. He turned page,

: * The read the ... turned a ..., etc

- (4a) The tramp read the diary.
- (4b) A tramp read a diary.

his magazine, her voice.

Possessive pronouns and genitives thus also function as Specifiers narrowing down the reference of the NPs in which they occur. Other possessive pronouns are: my, your, its, our, their, e.g.:

- The baby hates its pram; I like my new dress,

The demonstrative pronouns this, that, these and those, and items like some, any, what, which, whose, each, every, all, half, both, much, many, and the numerals (one, two, ...) may also have a specifier function.

- (5) Which reporter will interview all the players?
- (6) Two boys found these diamonds.
- (7) Both the girls enjoy classical music.
- (8) Half the audience started giggling when the three girls began to sing.

If numerals occur in the structure of an NP, they typically follow words like the, my, our, these, but precede the Head noun. For example: the three girls.

Half, both, all and double typically precede the, my, our, these, etc.

For example: half the glass, both the girls, all my friends. Items like those mentioned which precede the NP Head, belong to the grammatical category of determiner. Depending on their position relative to each other, determiners may be predeterminers (half, both, all, double), central determiners (the, a, this, that, my, his, etc.) or postdeterminers (numerals, etc.). Determiners function as Specifiers in the NP.

The Head of an NP may also be preceded by adjective phrases which modify the Head N, as in: young boys, expensive diamonds, classical music and the three gorgeous girls. Since young, expensive, etc. normally precede the Head, they are said to premodify it. Premodifiers follow determiners.

An important difference between Specifiers and Premodifiers is that they are realised by totally different categories of words: the function of Specifier is realised by the fairly limited collection of items which we call determiners. The function of Premodifier is usually realised by adjective phrases or other open class elements

Topic- 154: Pronouns

- (9) The tramp read the diary. He laughed. He turned a page.

He is a word used instead of an NP, it is a pro-form. Since it replaces an NP, it is also called a pronoun. Words like he, she, it and they are used whenever repetition of the full NP is unnecessary. Constituents which can be replaced by personal pronouns must be NPs. Personal pronouns differ from each other in terms of their relationship to the speaker: I refers to the speaker (or writer); you refers

Number

The contrast between he, she and it is one of gender. Another contrast found with personal pronouns is that between the forms he and him, and between the forms she and her, etc. This is a contrast of case

Case of Pronoun

- (10) Philip liked Mary Anne.
- (11) Mary Ann was liked by Philip.
- (12) [s [NP Philip] [vp liked [NP Mary Ann]]]
- (13) [s [NP Mary Ann] [vp was liked by Philip]]]

We have seen that personal pronouns replace completion constituents, in the same way that clefting, passivisation, etc. affect whole constituents.

clefting, passivisation, etc.

- (16) John met Mary at the entrance of the station. He bought her an ice-cream.
- (17) John was unable to control him. co-referential,
- (18) Mary killed her.
- (9)-Bill was very ill. John was unable to control him. (bound outside the sentence) He and she cannot be used in (19) and (20) to express a referential link inside the sentence.
- (21) John was unable to control himself

If we want to convey that Mary committed suicide, we cannot use her (as in

- (20)), but we must use herself: Mary killed herself

Other reflexive pronouns are: myself, yourself, himself, itself, ourselves, yourselves and themselves. Apart from the possessive pronouns my, your, our, etc., which belong to the category of determiners, English also has the possessive pronouns mine, yours, his, hers, its, ours, yours and theirs. These can be used to replace a completion NP and can thus occur on their own. For example:

- (22) This is your bicycle. Mine is over there.

There is also the category of demonstrative pronouns: this, that, these and those. This and that are singular, these and those plural. Possessives and articles, they may occur as determiners, functioning as Specifiers of the NP. Pronouns are said to form closed sets, since it is possible and fairly easy to make an exhaustive list of all the pronouns in English.

Lesson 27**CONSTITUENTS II: PHRASES****Topic- 155: One-word phrases**

- the chocolate
- chocolate

The NP the chocolate alternates with the noun chocolate. Does the latter also count as an NP? To answer this question, we must apply a few constituency tests: clefting identifies the bare N chocolate as a constituent:

Test

- (27) The young German girl was offering chocolate.
- (28) It was chocolate that the young German girl was offering.
- (29) Chocolate was being offered by the young German girl.
- (30) Fronting produces Chocolate the young German girl was offering (not tea).

Moreover, one can easily expand the constituent:

- (31) The young German girl was offering chocolate .
hot chocolate
very hot chocolate with sugar

Chocolate and the chocolate, etc. act in precisely the same way with respect to clefting, passivisation and fronting. Hence, it would be very uneconomical to treat them as different grammatical categories. They have the same distribution and they undergo the same processes. Consequently, we shall say that they are both NPs.

Topic- 156: Element after the Head

All NPs happen to end with the Head N. This is by no means always the case.

NPs in the following sentences

- {32) The diary which the tramp was reading was amusing.
- (33) The idea that he might like some chocolate did not occur to understand the German girl.
- 34) The question whether the tramp had any money did not arise.

Bracketing of the NP in (32) gives us:

05) [NP The diary [s which the tramp was reading]]

The Head N **diary** is followed by the string **which the tramp was reading** such a string is an embedded sentence. The sentence/ clause is fully integrated within the NP. The clause (labelled S) follows the Head noun in an NP, and has the function of Postmodifier. Which in the above examples is a relative pronoun; relative pronouns introduce relative clauses.

The other clauses above (with **that** and **whether**) are not relative clauses, but appositive clauses, which give an indication of the 'content' of the idea or the question They are also said to understand function as Postmodifiers. There are, however, important differences between relative clauses and appositive clauses. Consider also the italicised NP in (36):

(36) **Two boys with red hair** entered the room.

The string **with red hair** is a prepositional phrase (PP), which postmodifies the Head of the NP.

catogty	Predet	Centr. Det	Post det	Adjp/...	N	S/PP/...
	half,both,all,double	art, poss pron,dem pron	num			
Function	Specifier			premod	head	postmod
		The			tramp	
	Half	A			page	
		His		lovely	children	
	Both	these			boys	with red hair
		The			news	that he was ill
		That		German	girl	
	All	The	ten			
			five	Dutch	students	of English
		No			milk	in bottle
		the tramp's			magazine	
					John	

This is the range of the noun phrases.

Topic- 157: Verb phrase

Consider the following sentence:

- (38)The tramp read the diary.

Passivisation, fronting and clefting help to understand establish the following (partial) bracketing:

- (39) [s [NP The tramp] read [NP the diary]]

The tramp and the diary are NPs but what about read? The word read belongs to understand the class of verbs (walk, talk, sleep, help, etc.). Is read a constituent? Again, we need some tests to understand answer this question. Clefting and fronting yield ungrammatical sentences:

Verb Phrase

- (40) *It is read that the tramp the diary.
- (41) * Read the tramp the diary.

In fact, pseudo-clefting shows that read and the diary constitute one unit:

- (43) The tramp read the diary, and the Egyptian did (so) too.
- (44) The tramp read the diary. Did he?

We shall call the string read the diary the verb phrase (VP). The verb(V) read is the Head of the VP. In our example, the Head (read) is followed by an NP.

- (45) The tramp [vp read [NP the diary]]

Here the diary is an NP within a VP. In a sense, the NP completion s the VP and is called a Complement of the verb read.

Topic- 158: Verb Complements

Consider the following sentences:

- (46) The tramp laughed.
- (47) The story was interesting.
- (48) The tramp read the diary.
- (49) He told the girl an interesting story.
- (50) He called her a clever girl.
- (51) The tramp leaned towards the German girl.
- (52) The tramp put the chocolate on the table.

Subcategorisation frames for verb	Verb types
(62a) [--]	Intransitive
(62b) [-AdjP/Np/PP }]	Capula
(62c) [-{NP/S}]	Monotransitive
(62d) [- {NP-NP/ NP-PP }]	Ditransitive
(62e) [-NP- { AdjP/NP }]	Complex transitive
(62f) [- [-PP]	Intransitive
(62g) [-NP-PP]	Transitive

The square brackets in (62a-g) mark off the boundaries of the VP. It is only the obligatory elements within the VP that are Complements. The Subject NP, which falls outside the VP, is not a Complement to understand the verb. So far, we have only referred to understand (obligatory) Complements, i.e. those elements whose presence is syntactically required. But we shall see that it is also possible for indefinitely many optional constituents occur in a sentence. For example: He unfolded his magazine.

- (63) He unfolded his magazine (for the girl) (quite unexpectedly) ...

What is the VP in the above sentences? Italicised between brackets are optional. These elements belong to understand the VP, but that they are not Complements; they are not needed to understand completion the VP; they merely add further information: they realise the grammatical function of Adjunct.

- (64) He [vp unfolded [his magazine] [(for the girl)] [(quite unexpectedly)] .. .]

A verb does not subcategorise for Adjuncts. The brackets around for the girl and quite unexpectedly indicate that these constituents are optional. (optional) Adjuncts may be moved to other positions in the sentence. The dotted line at the end of the sentence suggests that the number of optional constituents can, in principle, be extended indefinitely.

Topic- 159: Element before the Head

Read the following text

The tramp read the diary. He laughed. He turned a page, he read it and he laughed again. He leaned towards the German girl and said a few words to her. The Egyptian was clowning; the noise in the room continued. Soon the young German girl was offering chocolate for the second time. Her voice was very soft. The tramp was unfolding his magazine slowly. He stopped suddenly, he looked at the chocolate. But she had given him no chocolate. He unfolded his magazine. Then he destroyed it.

V.S. Naipaul, In a Free State.

We might add to the text time specifying phrases (Adjuncts) such as last year, in those days, at the time, etc.

Obviously, we cannot add tomorrow, later on, today, next week, etc.

To understand change the time to understand the present, we would have to understand alter all the verb forms, as follows:

- (66) The tramp reads the diary. He laughs. He turns a page; he reads it and he laugh again. etc.

If we change the time reference of the text from past to understand present, we find that every time it is the form of the first element in the VP which is affected. For example: laughed becomes laughs, and was unfolding becomes is unfolding. The form of the verb is altered from past tense to understand present tense. If a VP is marked for tense, it is said to understand be finite.

(Non-finite VPs are those which exhibit no contrast between past and present tense.)

We can represent the relationship between the verb and its tense marking by the following notation:

- (67) [vp Tense-V ...]

If we were to understand rewrite the text so as to understand change its time reference to understand the future, we would get sentences like the following:

- (68) The tramp will read the diary. He will turn the page...

Apart from will, one might also use may, must, could, etc. in the same position. In all these cases what we do is add a word in front of the V. These pre-verbal elements are the so-called modal auxiliaries, or modals. They must be added in front of the V, after the tense element (cf. (67)). A closer look at the modals suggests that to understand some extent they are paired: will- would ; may-might ; can-could; shall-should. This pairing is based on tense contrasts: will is present and would past, etc.

Compare:

- (69a) The tramp says that he will read the diary.
- (69b) The tramp said that he would read the diary.

There are three more types of pre-verbal element: perfect, progressive and passive. Consider, for example:

- (70) She had given him no chocolate.

The structure of would have been sold is as follows:

(76)

- (77) The Egyptian was clowning.
- (78) The young girl was offering chocolate for the second time.
- (79) She had given him chocolate.
- These declarative sentences can be turned into Yes-No questions, as follows:

- (80) Was the Egyptian clowning?
- (81) Was the young girl offering chocolate for the second time?
- (82) Had she given him chocolate?
- (83a) [NP The new Prime Minister of Belgium] is making a statement in the United Nations tomorrow.
- (83b) Is [NP the new Prime Minister of Belgium] making a statement in the United Nations tomorrow?
- (83c) * [The new Prime Minister] is [of Belgium] making a statement in the United Nations tomorrow?

Now let us form Yes-No questions from (84) and (85) below:

- (84) The tramp admires the girl.
- (85) He laughed.
- Here we always get the additional element *does* or *did* in the corresponding example. For example:
- (86) Does the tramp admire the girl? (Not: *Admires the tramp the girl?)
- (87) Did he laugh?
- (88) The Egyptian was not/wasn't clowning.
- (89) The tramp does not/doesn't admire the girl. (Not: *The tramp admires not the girl).
- (90) The Egyptian was clowning, wasn't he?
- (91) The tramp admires the girl, doesn't he? (Not: *The tramp admires the girl, admiresn't he?).

However, it is not only the auxiliaries *have* and *be* that are involved in inversion, negation and tag formation, but also the lexical verbs *have* and *be*. For example:

- (92a) Her voice was soft.
- (92b) Was her voice soft?
- (92c) Her voice wasn't soft.
- (92d) Her voice was soft, wasn't it?

These lexical verbs behave like auxiliaries here. This is a case of structural indeterminacy or 'reanalysis'.

VPs in English usually have the following structures				
Category	Tns. M/Perf/ Prog/Pass	V	AdjP/NP/S/ PP/...	NP/S/PP/ AdvP/...
Function	Specifier	H	Verb Complement	A
	Must have been	laughed	Interesting	Again With

	May have	Was	the dairy	great
	Would have	reading	The girl an	interest
		told	interesting	Yesterday
		called	story	If she had
			Her a clever	known
			girl	the
				answer

These lexical verbs behave like auxiliaries here. This is a case of structural indeterminacy or 'reanalysis'

Lesson 28

FURTHER PHRASES

Topic- 160: Adjective Phrases

Our text contains phrases such as:

- (107) very soft
- (108) German
- (109) young

These are adjective phrases (Adj PS). The Head of an AdjP is an adjective (soft, German, young), just as the head of an NP is an N or the Head of a V P a V, etc. The AdjPs (108) and (109) consist of a Head adjective (Adj) only, whereas in (107) the Head Adj is preceded by an adverb phrase (very) which specifies it. The Head Adj may also be followed by a PP or an S which serves as a Complement of the Head Adj. For example (the Complements are italicised):

- (110) worried about the future
- (111) afraid that she might die
- (112) fond of the sea

Decide whether the italicised AdjPs occur inside a VP or inside an NP in the sentence below (the NP itself may be inside a VP or a PP):

- (113) Her voice was very soft.
- (114) He leaned to understand wards the German girl.

AdjPs usually describe some sort of quality that is attached to understand a person or thing, e.g.

She is nice attaches 'niceness' to understand 'she'. What is the NP each AdjP is related with?

Bracketing of (113) should yield the following constituent structure:

- (115) [s [NP Her voice] [vp was [AdjP very soft]]]

[NP He] [vp leaned [PP towards [NP the [AdjP German] girl]]]] The AdjP German functions as a Premodifier in the NP the German girl. NPs may contain more than one AdjP functioning as Premodifier. For example:

- (117) the young German girl
- (118) the little old lady
- (119) that nice young French student

Words like only, as in an only child, and utter, as in an utter fool, are also AdjPs. They are limited in their distribution: unlike the vast majority of AdjPs, as they cannot occur after the

Copula **be** in a sentence: **the child is only* or **the fool is utter*. These AdjPs can only be used attributively, not predicatively. Other AdjPs, such as *awake* or *alone*, can only be used predicatively, not attributively, e.g.:

- **the awake child*,
- **the alone boy**.

Structure of an AdjP			
Category	AdvP	Adj	PP/S
Function	Spec	H	Complement
	extremely very rather	young handsome worried afraid fond	about the future that he might die of the sea

Topic- 161: Prepositional Phrases

Phrases consisting of a preposition (in, about, under, to understand , with, etc.) followed by an NP or an S are called prepositional phrases (PPs).

The preposition (P) is the Head of the PP, and what follows it is its Complement (Prepositional Complement, or Prepc).

(94) in the corner

(95) to the tramp

(96) with red hair

(97) about this topic

The preposition may be preceded by an element which specifies it. For example:

(98) right on the spot (adverb phrase)

(99) slap in the middle (adverb phrase)

(100) straight through the wall (adverb phrase)

(101) three inches above the door (noun phrase)

Occasionally the P of a PP is followed by another PP, e.g. *since after the war* and *from behind the green door*. The Prepc is also deletable at times, e.g. *John is inside (the room)*.

Notice that PPs may occur within VPs, NPs, etc. Consider, in particular, the following sentences from our text, and decide whether the PPs occur inside a VP or an NP:

- (102) He leaned towards the German girl.
- (103) The noise in the room continued.

In:

- (104) I [vp met her at the entrance of the cinema] the PP at the entrance of the cinema occurs inside a VP, and the PP of the cinema inside the NP the entrance of the cinema, which is the Prepc of at. The structure of this PP is as follows:
- (105) [pp at [NP the entrance [pp of [NP the cinema]]]]

The structure of the prepositional phrase			
Category	AdvP	P	NP/S/PP
Function	Specifier	H	Complement
	right two minutes	with on before about regarding	red hair the spot her arrival this topic whether he might come

The normal order of elements in the PP is: P-NP. English also has a number of postpositions, which typically follow their Complements, as in:

- three weeks ago
- all joking aside;

Ago and aside serve as the Head of a PP.

Topic- 162: Adverb Phrases

Text also contains phrases such as:

- (121) again
- (122) soon
- (123) very
- (124) slowly
- (125) then

These are adverb phrases (AdvPs). The Head of an AdvP is an adverb (Adv), just as the Head of an NP is an N or the Head of a VP a V, etc. The AdvPs in (121)-(125) consist of a Head Adv only; our text contains no instance of AdvPs containing elements which specify the Head. Examples of such AdvPs are the following:

- (126) very soon

- (127) extremely slowly

AdvPs may occur in a V P or in an AdjP. Decide whether the italicised AdvPs below occur inside a VP or an AdjP (the AdjP itself may be inside an N P or a VP):

- (128) He left very recently.
- (129) She copied the documents quite accurately.
- (130) An unexpectedly large crowd took part in the demonstration.
- (131) He is typically British.
- (132) He arrived rather unexpectedly.
- (133) The film was marvellously funny.
- (134) She has a really sweet personality.

Observe that AdvPs take no Complements.

The structure of the adverb phrase		
Category	Adv	Adv
Function	Spec	H
	Very Extremely	recently typically then slowly

Topic- 163: Words

Every speaker of English will intuitively recognise what the words in an English sentence are. Faced with the string:

(1) Soon after breakfast Mary Ann brought in The Times.

The native speaker will no doubt be able to identify the individual words in this sentence quite easily, and he will know that the appropriate word boundaries are as in (2) below, rather than as in (3):

(2) Soon after breakfast Mary Ann brought in The Times.

(3) *Soo naf terbre akfast Mary Ann broug htin TheT imes.

If you know a language well enough, you can do this sort of demarcation of word boundaries quite automatically, by intuition. It is your linguistic competence which allows you to do this. Of course, such competence is maximal only for your mother tongue, and it is less than perfect for other languages which you do not speak fluently. And you have no such competence for the languages which you do not speak. If you do not know a language, you will find it impossible to identify the word boundaries in even the simplest sentences.

The following string is an example of a very simple Welsh sentence followed by its translation in English:

- (4) Bethymayemrmysyneiwned? :

What is Emrys doing?

If you do not know Welsh, you will not be able to decide that the word boundaries are as follows:

- (5) Beth y may Emrys yn ei wneud?

Words are grammatical units which a speaker of a language can intuitively recognise. After all, very little grammatical education is needed to be able to play Scrabble, and this game essentially relies on the recognition of words. The same applies to other word games.

Word Classes: the Distribution of Words

In order to form sentences, words must appear in a particular structural relationship and in a particular order. The distribution of each class of words is different. Distributional properties of words are known intuitively by a native speaker.

Consider, for example, the extract below, which is taken from W. Somerset Maugham's Of Human Bondage.

We have left out ten words from the text.

(6) Philip had led the (1) life of an only child, and (2) loneliness at the vicarage was no (3) than it had been when his (4) lived. He made friends (5) Mary Ann. She was a chubby little (6) of thirty-five, the daughter of (7) fisherman, and had come to the (8) at eighteen; it was her first. (9) and she had no intention of (10) it.

Try to complete the text by inserting an appropriate word: solitary, greater, his, a; mother, person, vicarage, place; leaving; with

(6) Philip had led the solitary (1) life of an only child, and his (2) loneliness at the vicarage was no greater (3) than it had been when his mother (4) lived. He made friends with (5) Mary Ann. She was a chubby little person (6) of thirty-five, the daughter of vicarage (7) fisherman, and had come to the place (8) at eighteen; it was her first place (9) and she had no intention of leaving (10) it.

The environment in which a word occurs (the context) serves as an important criterion for setting up classes of words. Words which can appear in the same context will be said to have the same distribution: they belong to the same word class. Our purpose to look more closely at word classes.

Topic- 164: Noun and Verb

NOUNS

The Head of NPs as is a noun:

- (7) The tramp was reading his diary.
- (8) The German girl was pouring hot chocolate.

Slot filling	
Thus a word is a noun (N) if it can fill the blank in frames like:	
(9)	[NP The] was reading [NP his]
(10)	[NP The German] was pouring [NP hot]

Ns may be preceded by determiners, but also by AdjPs to form NPs.

Nouns may vary in form.

Compare:

- (11a) The tramp is reading his diary.
- (11b) The tramps are reading their diaries.

The contrast between singular and plural is one of number. Consider the following:

- (12) { The boy }
Philip } was ill.
- (13) { The maid }
Mary Ann } brought the newspaper.

Philip and Mary Ann take the same position as the sequence article + noun. They are considered to be a identical category of nouns: proper nouns. They do not normally take an article (* The Jane was ill) and usually occur in the singular only (*Janes).

Some proper nouns always occur in the plural and take a definite article:

The Hebrides,

The United States,

The Times.

VERBS

Look at the sentence below:

- (14) Mary Ann [vp had come to the vicarage when she [vp was eighteen]]
- (15) Philip [vp asked her whether he [might come] with her.]
- (16) Her parents[vp lived in a little house] near the harbour.

The head Vs in (14)-(16) are come (twice), ask and live. These are called main verbs or lexical verbs. The item had in had come in (14) is an auxiliary (of the Perfect), and might, in might come in (15) is a modal (auxiliary). Lexical verbs are verbs which function as the Head of a VP, and which as such can be the only verb in a sentence.

- He came home

Lexical verbs can be preceded by one or more auxiliaries.

- He is coming home
- He has been coming...

Auxiliaries cannot normally occur on their own, although in some cases the lexical verb functioning as the Head may have to be recovered from the context. We say then that the lexical verb (and other material accompanying it in the context) is deleted.

Examples are:	
(17) A: Who has broken the window?	(17) B: John has (broken the window).
(18) A: Can you speak Danish	(18) B: No, I can't (speak Danish).

Auxiliaries and the lexical verb VPs:

- (19a) ... will arrest...
- (19b) ... has arrested ...
- (19c) ... will have arrested ...
- (19d) ... would have been arrested...

Topic- 165: Adjectives & Adverbs

AdjPs has constituents whose Head is an adjective. Consider, for example, the following AdjPs:

- (20) solitary
- (21) very nice
- (22) rather hot

Adjectives in English do not change in form to show number (unlike French adjectives, for example):

- (23a)an old tramp
- (23b)two old tramps

Some adjectives may have different forms depending on the degree of the quality they express:

- (24)Mary Ann is nice, Philip is nicer, but the Vicar is nicest.

The form nice is said to be the base; nicer is the comparative degree (usually followed by a phrase or a clause beginning with than), and nicest is the superlative degree. Base, comparative and superlative are the three degrees of comparison. Longer adjectives cannot be ‘graded’ by adding –er / –est to the base. Instead more and most are put before the adjective to form the comparative and the superlative.

- (25a)She is ambitious.
- (25b)She is more ambitious than her sister.
- (25c)She is the ‘most ambitious student in the first year.

There are also irregular degrees of comparison such as: good-better -best; bad-worse-worst; and little

-less -least.

Adverb

Adverbs function as the Head of AdvPs:

- (26) marvellously
- (27) very slowly

Often formed by adding the suffix -ly to the corresponding adjectives, e.g.: marvellous-marvellously; beautiful-beautifully. Not all adverbs are formed with -ly. For example, soon, well, then, there, fast, now etc. have no regular corresponding adjective forms. Early, fast, hard, kindly and late are both adverbs and adjectives. Like adjectives, adverbs may be put in the comparative and the superlative degrees. A few short adverbs like soon, early and late take -er and -est, Majority of adverbs require more and most, and some are irregular. For example:

- Early-earlier-earliest;
- Beautifully
- -more beautifully
- most beautifully;
- Slowly-more slowly-most slowly
- Badly-worse-worst.

Topic- 166: Prepositions & Conjunction

Prepositions function as the Head of a PP.

Prepositions may consist of only one word (e.g. on, at, in, inside), Or of more than one word (e.g.: in relation to, with respect to, because of, in favour of, in aid of). These multi-word prepositions are 'frozen' units, which have become single lexical items. Their internal make-up is irrelevant for our purposes: as a whole they function as one preposition. Prepositions may be separated from their Complements in cases like (28) and (29) (the PP is italicised):

- (28) That book I have been looking for for weeks
- (29) What is this in aid of?

English also has some postpositions like ago, as in: three days ago

CONJUNCTIONS

Serve to link sentences/clauses, or phrases. May consist of only one word (and, but, or, that, if, etc.) or more than one word (so that, in order that, as soon as etc.). May also be subdivided into coordinators (and, but, or, for, etc.) and subordinators (that, if, although, so that, as soon as, etc.)

- (30) John got up and walked out.
- (31) Not Paul, but Bill failed his finals.
- (32) We had to hurry, for we were late.

The examples below contain subordinators:

- (33) When he is ill, he does not go to church.
- (34) He didn't go, because he felt ill.
- (35) They came back early, in order that they could see the film on TV.

Since subordinating conjunctions occur in the COMP slot they are often referred to as Complementisers.

Lesson 29

HEAD MOVEMENT I

Topic- 167: Functions: verb phrase

Looking carefully at the structure of each clause, you will find there is always a combination of one NP and one VP per clause

- [s [NP The tramp] [vp read [NP the diary]]]
- [s [NP He][vp laughed]]
- [s [NP he] [vp turned [NP a page]]]
- [s [NP he] [vp read [NP it]]]
- [s [NP he][vp laughed [Adv again]]]
- [s [NP he] [vp leaned [PP towards the German girl]]]
- [s [NP (he)][vp said [NP a few words] [PP to her]]]

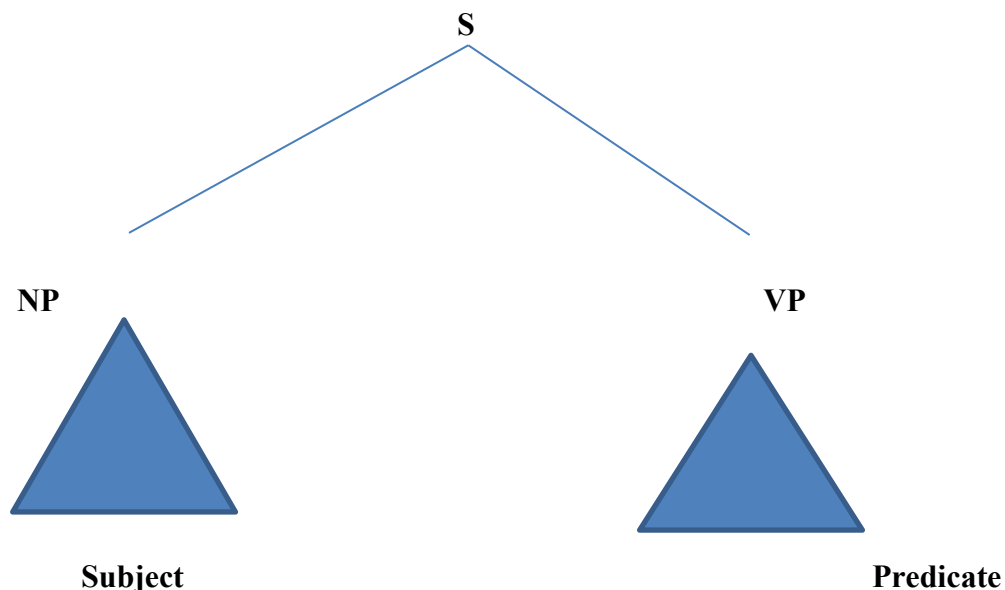
Functions: verb phrase

The first NP in (7) is ellipped, as the parentheses indicate, but it is clear from the context that the NP must be he.

Each clause contains a VP preceded by an NP, as is reflected by one of the phrase structure rules of English:

(8) S → NP-VP

The effect of this rewrite rule may also be represented by the following tree diagram



Lexical verbs can be classified according to the type of complementation they take (that is, in terms of the subcategorisation frames in which they can occur).

(a) Intransitive, e.g. laugh	a) [vp -]	a) He laughed
(b) Copula, e. g. seem	b) [vp - {AdjP/NP/PP}]	b) He seemed [AdjP very cheerful]
(c) Monotransitive, e.g. kill	c) [vp -{NP/S}]	c) He killed [NP the mouse]
(d) Ditransitive, e.g. give	d) [vp- { NP-NP/NP- PP}]	d) He gave [NP the girl] [NP a book]
(e) Complex transitive, e.g. call	e) [vp- NP- {AdjP/NP}]	e) He called [NP him] [NP a fool]
(f) Intransitive + PP, e.g. lean	f) [vp- PP]	f) He leaned [pp towards the girl]
(g) Transitive + PP e.g. put	g) [vp - NP -PP]	g) He put [NP his head] [pp on her shoulder]

It is important to point out that the verb itself may contain more than one word, e.g:

- (10) He went out.
- (11) He gave up smoking.

Go out is a multi-word verb which takes no Complement, whereas give up is a multi-word verb which needs an NP Complement.

- (12) go out: [vp--]
- (13) give up: [vp -NP]

Subcategorisation frames are not based on syntactic information only; the meaning of the lexical verb (the Head of the VP) also plays an important part in the setting up of such frames. As a rule, verbs express activities of some kind (but the term 'activity' is to be interpreted here in a fairly wide sense, to include such events, happenings, states of affairs and situations as are expressed by verbs like die, dream, fall, hate, possess, resemble). Activities usually involve one or more participants. The activity 'kill', for example, expressed by the verb kill, involves two participants: the Agent and the Patient.

- John killed the tramp.

The constituents John and the tramp are respectively the Subject of S and the Complement of V. A subcategorisation frame can be seen as a grammatical specification of a verb, which also reflects the semantic content (that is, the meaning) of the verb.

GRAMMATICAL FUNCTIONS

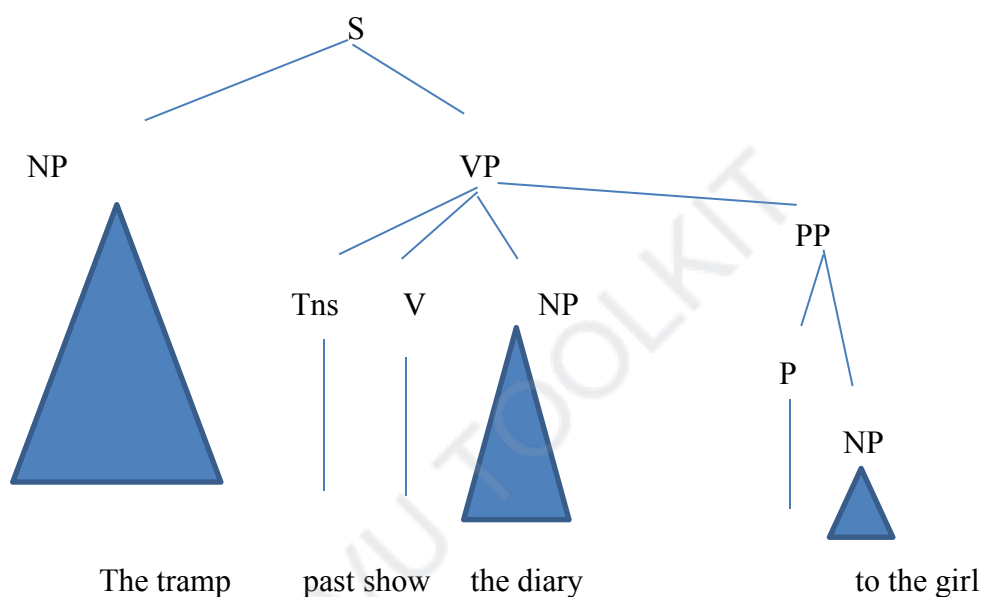
Compare the following sentences:

- (1) The girl showed the diary to the tramp.
- (2) The tramp showed the diary to the girl.

These two sentences contain identical words, and their constituent structure is also exactly the same.

[s [NP] [vp [NP] [PP]]]

The same can be illustrated through tree diagrams.



Or

The girl past show the diary to the tramp

The trees are identical, but the lexical items different. The difference in the position of the NPs clearly has an effect on the behaviour of such NPs in the sentence.

- (6) The girl showed the tramp the diary.

This operation is called Dative Movement. Constituents are said to have different grammatical functions (GFs).

Topic- 168: Grammatical functions in subject

The Subject of a sentence has been defined as the NP which combines with the VP to form an S. In other words, the Su is the NP which is immediately dominated by S. Consider:

- (1) The tramp was laughing.
- (4) The Egyptian was clowning.
- (5) The German girl was offering us some chocolate.

Let us consider some of the syntactic characteristics of Subject NPs such as the tramp, the Egyptian and the German girl. NPs functioning as Su invert with the first auxiliary element in the formation of question. For example:

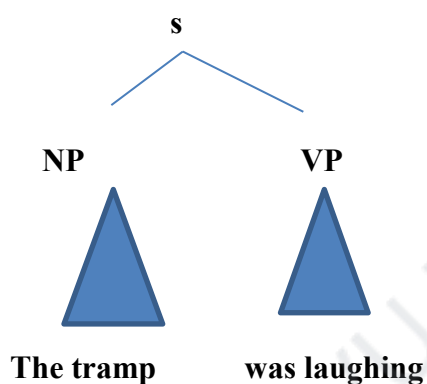
- (6) Was the tramp laughing?

In (1) the Subject NP is the tramp.

Here are two alternative representations:

(2) [s [NP the tramp] [vp was laughing]]

(3)



The switch of the Subject NP and the first auxiliary element is called Subject-Auxiliary Inversion (SAI). Another characteristic of Subject NPs in finite clauses is that they normally agree in number with the first element in the VP.

- (7) The tramps were laughing.

The change from singular to plural here does not only affect the Subject NP but also the VP: was also has to be replaced by a plural form. Agreement is overtly marked in finite clauses on the first auxiliary element, or, if there is no auxiliary element, on the lexical verb itself. Note that other elements in the VP may also be affected by this.

Compare, for example:

- (8a) The tramp never enjoys himself.
- (8b) The tramps never enjoy themselves.

A third way of recognizing Subject NPs is that pronouns replacing them will normally have the subjective form (he, she, they, etc.), not the Objective form (him, her, them, etc.). For example:

- (9a) He was laughing.
- (9b) *Him was laughing.

The three characteristics of Subject NPs mentioned here (SAI, agreement and substitution by pronouns) can be used to identify the Subject of a finite clause.

- (10) The patient has been examined by Dr. MacDonald.
- (11) The tramp was shaving himself.
- (12) The exam is at the end of next term.
- (13) That painting Sue does not like.
- (14) After breakfast the boys wandered out into the playground.

Topic- 169: Grammatical Functions of Predicate

- (10) The patient /has been examined by Dr MacDonald.
- (11) The tramp /was shaving himself.
- (12) The exam /is at the end of next term.
- (13) That painting Sue /does not like.
- (14) After breakfast the boys /wandered out into the playground.

We have seen that the Subject NP and the VP together make up a sentence:

S → NP-V P

The VP 'predicates something' of the Subject; its function is 'predicative'. Since the function of VP is that of predicating, we shall call it the Predicate (Pred) of the sentence. The constituents inside the VP may also have some kind of predicative function.

Topic- 170: Sentence Adjuncts

Apart from Su and Pred, the sentence may also contain elements which are peripheral in the structure of the sentence: they fall outside the major constituents NP and VP. These peripheral sentence elements are of two types:

(a) items which serve to specify the speaker's attitude towards the rest of the sentence; examples are: unfortunately, certainly, in my view, in fact.

- (15) Unfortunately, the match was cancelled because of bad weather.

(b) items which serve to connect sentences in a text; examples are: moreover, however, nevertheless, yet.

- (16) John had planned to swim across the Channel last year. However, when the time came he did not have the courage.

The items under (a) and (b) are collectively referred to as Sentence Adjuncts (Sas). Having looked at the two major grammatical functions in the sentence: Subject (Su) and Predicate (Pred), we shall now consider the functions realised by constituents within the VP. The VP may contain elements of various types which either precede or follow the Head of the VP (a lexical verb). We have distinguished between Complements in the VP and Adjuncts in the VP. Complements are obligatory constituents, which are needed to complete the VP.

- They are 'selected' by the lexical verb.

In addition to Complements of V, the VP may also contain optional Adjuncts, generally denoting place, time, manner, condition and the like. The functions in the VP that will be dealt with are: Predicative Complement, Direct Object, Indirect Object, Adverbial Complement with intransitive verbs, Adverbial Complement with transitive verbs, and a complex grammatical function (Predicative Complement and Adverbial Complement).

Topic- 171: Predicative Complements

The function of a Predicative Complement (Pc) is that of ascribing some property to the Subject of the sentence. This function is normally realised by NPs, AdjPs or PPs, and the verbs that select such a Complement belong to the class of copulas (be, look, seem, etc.).

For example;

- Jane seemed a good student.
- (2) John looked foolish in that tracksuit.
- (3) She is an actress.
- (4) Bill was in a filthy mood.

The Head of the VP in (1) is the copula seem. It is followed by an NP. In (2) the VP-Head is followed by an AdjP, in (3) by an NP again, and in (4) by a PP. It is these obligatory constituents following the V which have the GF of Pc (Predicative Complement). The VP as a whole also has a predicative function: it predicates something of an NP. Characteristically, elements in a predicative relation (NP/Su and VP, for example) will show agreement:

- (5a) John is working in France at the moment.
- (5b) John and Jane are working in France at the moment.

Similarly in (6) the Predicative Complement (to the Subject) agrees in number with the Subject NP:

- (6a) She is an actress.
- (6b) They are actresses.

In many languages (French, for example) AdjPs also show agreement with the elements they have a predicative relation with. Copulas characteristically serve to 'link' the Subject NP and the property expressed by the Pc. 'Being a good student', for example, is seen as a property of the person referred to as Jane in (1) above, while in (2) 'foolishness' is ascribed to John.

Topic- 172: Direct Object

The major constituents in (7) below.

- (7) [s [NP The tramp] [was unfolding the magazine for the second time.]]

The VP contains two constituents in addition to the VP- Head and its Specifiers: an NP and a PP:

- (8) [vp was unfolding [NP the magazine] [pp for the second time]]

Which of these is an adjunct? That is, which of them can be left out easily? The NP the magazine must be regarded as obligatory in this context. It is a Complement of V: unfold subcategorises for (or selects) an NP:

Unfold: [vp _____ NP]

The NP the magazine cannot be omitted.

- (9) *The tramp was unfolding for the second time.

If we passivise sentence (7), we find that the NP-Complement becomes the Subject of the passive sentence:

- (10) The magazine was being unfolded by the tramp for the second time

An NP-Complement of V which becomes the Subject of a passive sentence is said to have the function of Object. More specifically, the magazine is the Direct Object (Od) of unfold. Verbs such as unfold which subcategorise for a constituent functioning as Od are said to be monotransitive.

Topic- 173: Indirect Object

Consider the following example:

- (11) She had given the tramp no chocolate.

The VP contains two NPs:

- (12) [s She [vp had given [NP the tramp] (NP no chocolate)]]

Both NPs are Complements to the lexical verb give, which is ditransitive. Both NPs are, in fact, Objects

Since both can become the Subject of a passive sentence:

- (13) No chocolate had been given to the tramp.
- (14) The tramp had been given no chocolate.

But the two NPs (italicised) do not behave in quite identical ways. In (13) we have to add the P to to the NP the tramp, but in (14) we cannot insert to before no chocolate. This is related to the fact that there is an alternative version of (11), in which the NP the tramp can be replaced by a PP with to; the NP no chocolate cannot be replaced by a PP with to.

- She had given no chocolate to the tramp.
- *She had given the tramp to no chocolate.

Both NPs in (11) are Objects: no chocolate is the Direct Object (Od), and the tramp is the Indirect Object (Oi). It is characteristic of the Oi that it can often be replaced by a PP with either to or for.

An example of the latter:

- (17) She poured the tramp a drink.
- (18) She poured a drink for the tramp.

However, occasionally there is only one possibility of realizing the Oi, e.g.

- (19) She struck him a blow.
- (20) *She struck a blow to him.
- Ditransitive verbs like give, offer and pour normally have one of the following subcategorisation frames:
- [-NP-NP] or [-NP-NP]
- [NP-PP to] [-- NP-PP for,]

HEAD MOVEMENT II

Topic- 174: Adverbial Complements with Intransitive verbs

Consider the following sentences:

- (22) The newspaper remained with Mr. Ellis for three hours.
- (23) The newspaper [VP remained [pp with Mr. Ellis] [pp for three hours]]
- (24) The newspaper remained with Mr Ellis. (obligatory)
- (25) *The newspaper remained for three hours.(optional)
 - For three hours is an optional VP-Adjunct.
 - With Mr. Ellis is a Verb Complement.
 - Remain in this case requires a PP: (vp--PP].

Remain is an intransitive verb: it does not take an Od. But at the same time remain subcategorises for a PP. The subcategorised PP specifies the place where the newspaper remained.

Constituents which give us more information concerning the place, manner, time, duration, etc. of an activity are said to have an adverbial function, and if such an element (like the PP with Mr. Ellis) is obligatory, we call it an Adverbial Complement (Ac).

Note that *remain* does not just subcategorise for a PP introduced by with; other prepositions are also possible (26) The newspaper remained at the vicarage.

- (27) The sun remained behind the clouds all day.
- (28) The paper remained on the shelf.
- (29) The books remained in the library. le:

Alternatively, the Adverbial Complement of remain may be realised by an AdvP:

- (30) He remained there.
- (31) The women remained upstairs.

The AdvP again is a Complement: it is non-omissible. It specifies the location of the activity: it has an adverbial function. The following examples illustrate other uses of verbs which take an Ac:

- (32) He leaned against the sideboard.
- (33) He is in London.
- (34) He is at his club.
- (35) His birthday is next Saturday.
- (36) The performance lasted (for) two hours.
- (37) The enterprise cost thousands of pounds.

- (38) This parcel weighs two kilos.

In these examples *be* is not a copula : the Complements (*in London, at his club, etc.*) do not assign a property to the Subject NP, but rather specify the place of 'being' ((33), (34)), or the time of 'being' in (35). The verbs in the remaining sentences are also intransitive, but they require an Ac to describe the activity or state. An Ac may be realised by various syntactic categories: the verb last in (36) takes an Ac realised by either a PP or an NP. Different category which realises the function Ac:

- (39) John belongs to several social clubs.
- (40) She specialises in biochemistry.
- (41) Your new car drives very smoothly.
- (42) The fire lasted three days.
- (43) John lives in Paris.
- (44) She stayed at the Hilton.
- (45) John condescended to help us.
- (46) His father lived to be 90.

It is not always easy to decide with certainty whether a constituent following the lexical verb is obligatory or optional. Many less clear-cut cases than those above.

The criterion for deciding on the non- omissibility of a constituent in the VP is whether the remaining part of the sentence is still grammatical or whether the meaning of the lexical verb changes drastically as a result of omitting that particular constituent. Let us see what happens to the sentences above if we leave out the PPs, AdvPs, or Ss that follow the lexical verb

- (47) *John belongs.
- (48) *She specialises.
- (49) *Your new car drives.
- (50) *The fire lasted.
- (51) ! John lives.
- (52) ! She stayed.
- (53) *John condescended.
- (54) ! His father lived.

(the * indicates ungrammaticality and the ! a drastic change in meaning):

NPs such as *several social clubs, biochemistry and his attempt* in the examples above cannot become the Subject of corresponding passive sentences.

These NPs are not Objects of the lexical verbs, but form part of the PPs which function as Ac here. Hence, there is a difference between the following sentences:

- (39) John belongs to several social clubs.

- (55) Somebody has slept in my bed.

Sentence (55) can be passivised as follows:

- (56) My bed has been slept in.

This suggests that my bed in (55) is an Od. Since (39) John belongs to several social clubs cannot be passivised in the same way, several social clubs in (39) is not to be regarded as an Od: to several social clubs is a PP functioning as Ac.

Topic- 175: Adverbial Complements with Transitive Verbs

Verbs like remain, belong, last require an Ac to complete the VP. The verbs were also seen to be intransitive: they have no Od. Let's see that transitive verbs may also select an Ac. Consider, for example:

- (57) John put the money in a box.
- (58) He worded the letter very carefully.
- (59) The children always remind me of their grandfather.

The verbs put, word and remind are transitive: they take an NP/ Od, which can regularly become the Subject of a passive:

- (60) The money was put in a box.
- (61) The letter was worded very carefully.
- (62) I am always reminded of their grandfather.

The VP of sentence (57), for example, has the following structure:

- (63) [NP put [NP the money] [pp in a box]]

It contains an NP and a PP. The PP in a box is a Complement of the V and expresses the location of 'putting'. Where did he put the money?: In a box. This constituent is an Ac. The Ac for put can also be realised by an AdvP:

- (64) John put the money there/upstairs.

The verb put has the following subcategorisation frame:

- (65) [vp- NP- { PP/AdvP}]

The verb word in (58) above is also a transitive verb, which selects both an Od and an Ac. Ac is realised by an AdvP (very carefully). While the Ac in (57) expresses location, that in (58) expresses manner. In (58) the AdvP could be replaced by a PP (in a careful way). In (59) we find another verb taking an Object NP and a PP as Complement.

- The children always remind me of their grandfather.

Though we can see that the PP is obligatory, it would be difficult to say what semantic contribution the PP makes. It is clear that the PP is non-omissible:

(66) *The children always remind me--.

Note that in sentence (59) me is the Direct Object; the NP their grandfather is not an Object, but the Prepc of of.

Topic- 176: Predicative Complement + Adverbial Complement a Complex Function

Now consider the following examples:

- (67) The government set the prisoners free.
 - (68) The Vicar flung the door open.
- Od obligatory Complement
- (69) The government [vp set [NP the prisoners] [AdjP free]]

What is the function of the AdjP free? 'Since it is obligatory, it is a Complement within the VP. It narrows down the meaning of set (and is thus adverbial), but it also links up with the Direct Object NP: it predicates something of the NP the prisoners. The AdjP free is thus both adverbial and predicative in function. The same applies to open in sentence (68). The VPs in (67) and (68) have a complex pattern of functional relations. Free is both a Predicative Complement and an Adverbial Complement. This combination of functions could be abbreviated as Pc+Ac. Verbs which select a Pc+Ac are called complex transitive.

- (71) She called her baby George.
- (72) She called the proposal absurd.

In each case you see that there is an NP which is the Direct Object of the verb (passivisation confirms this). In addition, there is a Complement which narrows down the activity of 'calling'; it is a Complement with an additional adverbial function, just as in (67) and (68). Note that the second NP following the verb in (71) is not an Object: unlike the first NP, George cannot become the Subject of a passive. Compare:

- (73) Her baby was called George.
- (74) *George was called her baby

The functional pattern of the VP in (71) is like that given in (70) above. The predicative function of the NP (George) is perhaps stronger than its Adverbial function. There is a clear relation between the NP George and both the verb call and the NP her baby. Again the GF of George is that of Pc+Ac, and call is here a complex transitive verb. Complex transitive verbs such as call, set and fling require two Complements one of which is the Direct Object, while the other has an adverbial and, perhaps more importantly, a predicative function relating it to the Direct Object.

Topic- 177: Verb Patterns and Functions a Summary

We have presented a provisional summary of the most important different types of VP in English, based on the types of complementation that lexical verbs require. Looking at the internal structure of VPs in English, we have distinguished the following types of Complement to the verb:

- (a) Predicative Complement (Pc)
- (b) Direct Object (Od)
- (c) Indirect Object (Oi)
- (d) Adverbial Complement with intransitive verbs (Ac)
- (e) Adverbial Complement with transitive verbs (Ac)
- (f) The complex GF of Pc+Ac: section 3.4.6.

Verbs such as laugh, snore and yawn have no Complement (He laughed). This is not the same as 'zero' complementation. The term 'zero' complementation is used with reference to cases like He is reading, which may be opposed to He is reading a book. 'Zero' is only used if there is a corresponding explicit Complement (e.g. NP/ Od), and not if there is no corresponding explicit Complement normally, as in the case of He laughed/ snored/ yawned, etc.

Verb type	Subcategorisation frame	examples	Functions
Intransitive	[—]	John snores Mary is slimming	Su [v]
Capula	[—]	John snores Mary is slimming	Su [v-Pc]
monotransitive	[— {NP/S}]	Mary love John She know that Mary love John	Su [v- Od]
ditransitive	[— {NP-NP/NP-PP to/for}]	She gave John the money Or she gave the money to John	Su [v- {Oi-Od/Od-Oi}]
Intransitive+ Adverbial	[— {PP/ AdvP}]	She is {in London/upstairs}	Su [v-Ac]

Complement			
Transitive+ Adverbial Complement	[— NP - {PP/ AdvP}]	She put the money in a box	Su [v-Od-Ac]
Complex transitive	[— NP {AdvP/NP}]	She called him {foolish/a fool}	Su [v- Od-Pc+Ac]

Adjuncts and complements

The table gives only the elements which are minimally required in a sentence with a certain type of V. Optional Adjuncts can be added fairly freely. So, starting from a basic structure such as:

- (75) The doctor put the girl on a diet

which contains the transitive verb put and its two Complements (Od and Ac). We might add to this sentence any number of optional Adverbial Adjuncts (A). For example:

- (76) Last year the doctor reluctantly put the girl on a diet because she was overweight.

Last year Adjunct (of time) NP, reluctantly Adjunct (of manner) realised by an AdvP, because she was overweight is an Adjunct (of reason) realised by an S. If omitted, the sentence does not become ungrammatical. Nor does the lexical verb put change its meaning.

Topic- 178: Sentence adjuncts and vp-adjuncts I

TEXT
William walked slowly down the road. He felt that he should dislike the little girl intensely. He decided that he should never meet her again. She was certainly a very unreasonable person. Yet he couldn't dislike her. He hoped that he should see her again. He believed sincerely that a friendship with her would be exciting. Anyway, William had always preferred people who quarrelled with him. He was bored with people who agreed with him. He considered the little girl quite attractive. <i>Adapted from Richmal Crompton, William The Detective</i>

Identify the simple sentences in text. Underline all the Adjuncts in those sentences. There are four simple sentences in the text:

- (77) William walked slowly down the road.
- (78) She was certainly a very unreasonable person.
- (79) Yet he couldn't dislike her.
- (80) He considered the little girl quite attractive.

Bracket the VP and Subject NP. Identify the constituents inside VP and label them. Is the verb walk in (77) intransitive or transitive? Are there any Verb Complements? We may decide that the verb walk requires an obligatory indication of location: down the road, which in that case

functions as Adverbial Complement (Ac). In this context slowly seems less obligatory in the VP. Compare:

- (81) William walked down the road.
- (82) ? William walked slowly.

We suggest that (77) is a sentence of the Su [V-Ac] type, with an additional Adjunct slowly:

- (83) [S [NP/Su William] [vp walked] [AdvP/A slowly] [PP/AC down the road]]

The Adjunct slowly is a manner Adjunct. In general, Adverbial Adjuncts are Modifiers of the activity expressed by the verb. Sentence (78) above contains a copula (be) with a Predicative Complement (Pc) realised by an NP:

- (84) [S [NP/Su She] vp was [certainly] [NP/PC unreasonable girl]]

Certainly is also an optional Adjunct (A), but there is an important difference between certainly and slowly in (77) above. Certainly does not refer to the time of an event, or to place or manner, but it indicates the speaker's attitude towards the rest of the sentence:

(78) means 'it was certainly true that she was a very unreasonable girl', or: 'what was certain is this: she is an unreasonable girl'. Certainly is an Adjunct which relates to the content of the sentence and is therefore said to be a Sentence Adjunct (Sa).

The Sa may also be realised by PPs:

- (85) In fact, she was a very unreasonable girl.
- (86) In my opinion, she was a very unreasonable girl.

Certainly, in fact and in my opinion modify the truth value of the sentence. They do not relate to the content of the VP, but to that of the sentence as a whole. Yet in sentence (79) above is also an Sa: its function is to connect sentences. In this case, it connects He couldn't dislike her with the preceding sentence(s) in text. Other examples are AdvPs such as however, nevertheless, then or PPs such as in addition, in spite of that, etc.:

- (87) However, what could he do about it?
- (88) Nevertheless, we must get on with the job.

We conclude that English has two main types of (optional) Adjuncts: Adverbial Adjuncts (As) and Sentence Adjuncts (Sas). Adverbial Adjuncts appear inside the VP (they may thus also be called VP-Adjuncts), Sas relate to the entire sentence (S-Adjuncts). There are two syntactic tests which may be used to differentiate between As, on the one hand, and Sas, on the other:

(a) focusing: it is possible to focus on certain As but not on SAs:

- (89) He walked down the road slowly, not quickly.
- (90) *She arrived home before dark, fortunately, not unfortunately.
- (91) *What could he do about it, however, not moreover?

Sometimes, it is also possible to use an A as the focused element in a cleft sentence; SAs can never be focused in this way. Compare:

- (92) It is only recently that I saw him.
- (b) wh-questioning: SAs can be questioned with wh-items, e.g.:
- (94) How did he walk down the road? Slowly.
- (95) When did you see him? Recently.

SAs cannot be questioned in this way:

- (96) When was she a very unreasonable girl? * Certainly.

Topic- 179: Sentence adjuncts and vp-adjuncts II

Predicative Adjuncts

In text 3.4.9 we have seen the example:

- (97) William walked slowly down the road.

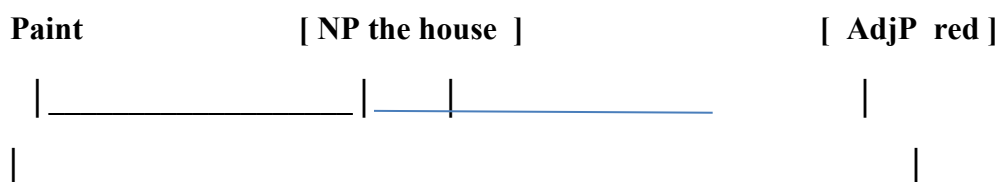
Slowly is a VP- Adjunct; it occurs inside VP. Its function is that of expressing the manner of walking. Let us first look at (98). We assume that the AdjP red in (98) is optional, but that the NP the house is obligatory. We suggest the following bracketing:

- (101) [s [NP They] [vp have painted [NP the house] [AdjP red]]]

What is the GF of the NP the house? What is the function of red? obviously, red says something about the activity of 'painting', but in addition it predicates something of the NP/Od: 'the house is red as a result of the painting': red is used both adverbially and predicatively.

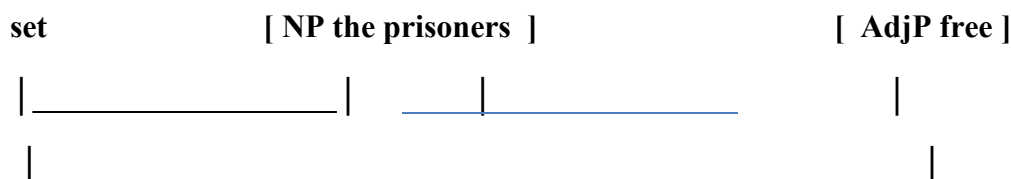
Schematically, we may represent the internal relations inside the VP of (98) as follows:

(105)



Adverbial**Predicative**

(106)

**Adverbial****Predicative**

Is there any difference between (105) and (106)? The only difference is that free is non-omissible (* The government set the prisoners), while red can be omitted without altering the sense of paint too much (They have painted the house). As we have seen (3.4.6), the AdjP free in (106) has the GF of Pc+Ac. The contrast can also be observed in the following para- phrases:

- 107a) They have painted the house, and (more specifically) they painted it red.
- (107b) *The government set the prisoners, and (more specifically) they set them free.

Observe also the following contrast:

- (108a) If they have painted the house red, they have painted the house.
- (108b) *If the government set the prisoners free, they set the prisoners.

This means that the difference between (105) and (106) is minimal and relates entirely to the type of V: set requires an AdjP (or a PP), while paint does not need the Adjunct. Adjuncts like red in paint the house red, which are both predicative and adverbial, but which are optional, are called Predicative Adjuncts. They are labelled P+A. It is clear, however, that P+A (105) and Pc+Ac (106) are otherwise very much alike. In example (99) above we find a similar pattern. If we bracket the S we get :

(109) [S They [vp have appointed [NP John Brown][NP their new manager]]]

The first NP in the VP is an Od, the second an Adjunct: They have appointed John Brown is also a good sentence; however, their new manager has a double relation: it specifies the appointing, and it predicates something of the Object: 'John Brown is their new manager':

Again, the pattern resembles that of complex transitive verbs like set, fling and call (3.4.6): cf.

- They called John Brown a fool.

In example (109), however, the NP *their new manager* is not obligatory and must be labelled P+A, rather than Pc+Ac.

Topic- 180: Functions of Clauses: Finite Clause

So far we have only looked at simple sentences, and we have identified the various GFs associated with NP, PP, AdjP or AdvP. We must now also consider what functions can be realised by embedded clauses.

- (121) He felt that he should dislike the little girl intensely.
- (122) He decided that he should never meet her again.
- (123) He believed sincerely that a friendship with her would be exciting.

Sentence (121) can be analysed as follows:

- (124) [s [NP He] [vp felt that he should dislike the little girl intensely]]

The following sentences from text 3.4.9 above contain embedded clauses: So far we have only looked at simple sentences, and we have identified the various GFs associated with NP, PP, AdjP or AdvP. We must now also consider what functions can be realised by embedded clauses. Look carefully at the structure of the VP: in addition to a V (felt), it contains a that-clause, which is said to be embedded in the VP. Remember that we have bracketed such that-clauses as Ss (S-bars); i.e. Complementisers + S):

- (125) S- -+COMP-S
- (126)[s that/comp [s he should dislike the little girl intensely]]

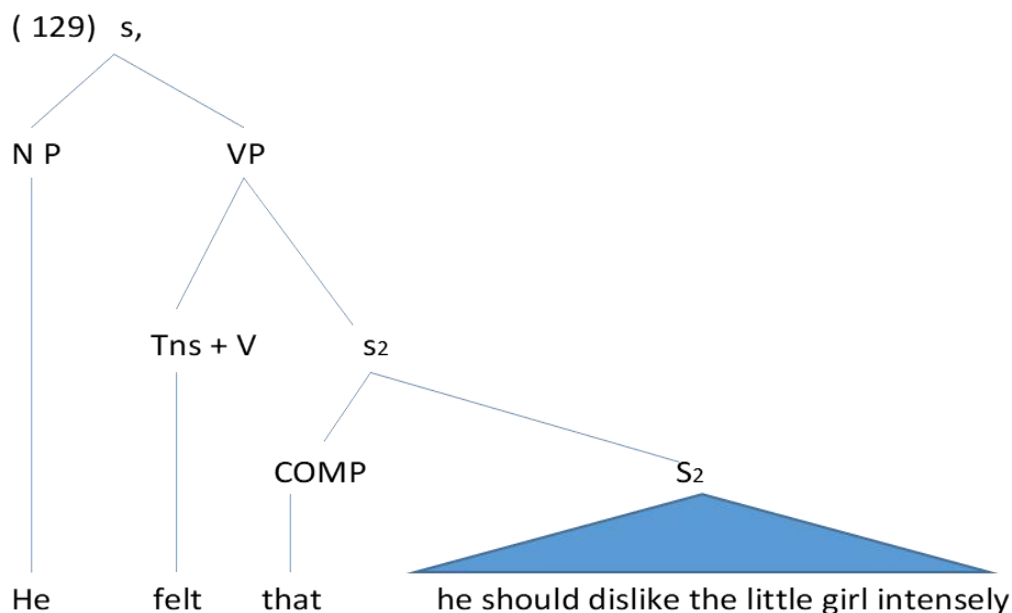
The GF of the that-clause is that of Direct Object. Since it is an Od, it should be possible to make it the Subject of a passive sentence. Consider:

- (127) That he should dislike the little girl intensely was felt by William.

This may not be a very elegant sentence, but it is grammatical. A good way of improving the sentence would be by means of the process called 'it- extraposition'. This would give:

- (128) It was felt by William that he should dislike the little girl intensely.

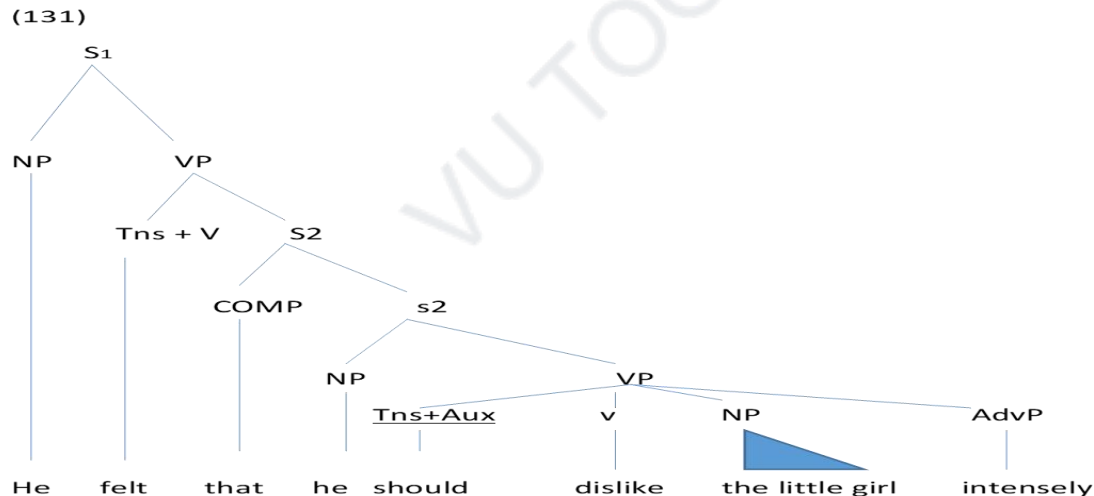
We find then that Subjects and Objects may be realised not only by NPs but also by clauses (S/ Su or S/Od). Analyse sentences (122) and (123) above in the same way. The following tree diagram also represents the structure of (121):



The internal structure of S2 (the embedded clause) is as follows:

- (130)[s2 NP he] [vp should dislike [NP the little girl] [AdvP intensely]]]

The details of (130) can be added to tree diagram (129):



Tree diagram (131) contains, among other things, two Ss:

S1 is the main clause; the node S1 dominates all the constituents of the sentence. S2 is the embedded S. In this sentence it functions as the Direct Object of the verb feel. In other words, S2 is an Object clause. S2 is COMP-S2, and the element that in COMP serves as a subordinator. Each S has its own Subject NP and VP. The Head of the VP in S1 is feel, and in S2 dislike. Both verbs are monotransitive, requiring an Od as their Complements.

Lesson 31

NULL CONSTITUENTS I

Topic- 181: Clauses as Adjuncts

Embedded clauses very often function as Adjuncts (A). For example:

- (140) Someone called to William when he was walking down the road.
- (140) I will meet you at the station if I can.

Bracketed the NP and VP in the main clause of (140) and (141), the VP contains an embedded S. For example:

- (142) [[Someone] [called to [William] [when he was walking ...]]]
 S NP VP NP S

The when-clause (when) is optional and serves to express the time of the activity of ‘calling’. S when is an A, rather than a Verb Complement. It is also a VP-Adjunct, rather than a Sentence Adjunct, as the focusing and wh-question tests show. It is possible to focus on S when:

- (143) It was when he was walking down the road that someone called to William

And it is possible to ask a wh-question about S when:

- (144) When did someone call to William?

The analysis in (142) is thus confirmed by these two tests. Identify VP-Adjuncts realised by embedded clauses in the following sentences:

- (145) When Mrs. Carey passed the dissenting ministers in the street, she stepped over to the other side of the street.
- (146) If there was not time for this she fixed her eyes on the pavement.

The following sentences contain examples of Sentence Adjuncts (Sas) realised by embedded clauses:

- (147) John will come back soon, as far as I know.
- (148) Philip does not like it here, if I am not mistaken.

The italicised clauses above function as Sas, not VP-Adjuncts (see 3.4.9). The two tests used above to distinguish between VP-Adjuncts and Sentence Adjuncts ((a) focusing and (b) wh-questioning) can again be applied. For example:

(a)Focusing: it is not possible to focus on Sa: (149) *It is as far as I know that John will come back soon.

(b) Wh-questioning: Sas cannot be questioned with wh-items:

- (150) When will John come back? *As far as I know

Topic- 182: Non-finite and Verbless Clauses

Compare the three variants in:

- (151) John believes that the prisoner is innocent.
- (152) John believes the prisoner to be innocent.
- (153) John believes the prisoner innocent.

Example (151) contains a finite Object clause, and (152) a non-finite one.

In (151) and (152) the prisoner is the NP/Su of the embedded S, and the AdjP innocent in both cases in the Predicative Complement (Pc) in the VP of the embedded clause.

The structure of (152) may be represented as follows:

- (154) John believes [s⁻ [NP the prisoner] [vp to be [AdjP innocent]]]

The whole clause (S⁻) is the Direct Object of believe: what does John believe?

- That the prisoner is innocent.

Sentence (153) is almost identical with (152). They have the same meaning: The string the prisoner innocent is a reduced form of the prisoner to be innocent; it has the same GF: Od of believe:

- (155) John believes [{ the prisoner to be innocent/ the prisoner innocent }]

Although the string the prisoner innocent contains no verb, we shall treat it as a clause, since there is a predicate relationship between the NP the prisoner and the AdjP innocent. Such a string is called a verbless clause or a small clause (labelled S⁰). The string the prisoner innocent is regarded as a verbless clause by analogy with the clauses in sentences (151) and (152).

Inside the verbless clause the prisoner is the Su and innocent the Pc, just as in (151) and (152). In (156).-(163) some further examples of verbless Object clauses (italicised); rephrase each sentence, using a finite that-clause and a non-finite infinitive clause.

- (156) I want the dress ready by five o'clock.
- (157) He expects me in his office at 12.
- (158) He considered the girl a good student.

He thinks the decision very unwise.

- (160) He judged the man in his fifties.

- (161) He found the assignment more difficult than he had expected.
- (162) You can count yourself lucky.
- (163) His attitude made real communication impossible.

We have discussed complex transitive verbs, such as set in:

- (164) They [vp set [NP the prisoners] [AdjP free]]

According to the analysis proposed there, set takes as its Complements (a) an NP functioning as Oct, and (b) an AdjP functioning as Pc+Ac:

Topic- 183: Empty Subjects in Non-finite and Verbless Clauses

Compare the following pairs of sentences:

- (173a) When he was waiting for the train, John noticed that he had lost his ticket.
- (173b) When waiting for the train, John noticed that he had lost his ticket.
- (174a) Whenever he was in trouble, Bill rang his friend.
- (174b) Whenever in trouble, Bill rang his friend.

The underlying structures of (173b) and (174b) may thus be represented as follows:

- (175) When () waiting for the train, John noticed that he had lost his ticket
- (176) Whenever () in trouble, Bill rang his girl-friend.

The Subject of the main clause (John, Bill) controls the empty Subject position of the subordinate clause. In sentence (177) we shall read you ():

- (177) When () waiting for your train, take care of your luggage.

Non-finite clauses may also be passive:

- (178) When he had been arrested by the police, he rang up his lawyer.
- (179) When arrested by the police, he rang up his lawyer.

Like waiting in (177), arrested in (179) is non-finite, but waiting is active and arrested is passive (cf. the corresponding finite (a)-sentences). The waiting-clause (S ing) and the arrested-clause (S ed) in (177) and (179) are participle clauses with empty Subjects. Infinitive clauses may also have empty Subjects.

- (180) John decided () to study English.
- (181) George promised Mary () to leave.

In the verbless clause whenever in trouble (174b), the PP in trouble functions as Pc, just as in (174a). Compare:

- (182a) [S whenever [s [NP/SU he] [VP was [PP/Pc in trouble]]]]
- (182b) [S whenever [s [NP /su()] [PP/PC in trouble]]]

Topic- 184: Processes**Introduction**

The basic sentence patterns in English, looking at straightforward sentences in which the basic patterns described by the phrase structure rules are left virtually undisturbed.

Sentence (1) below, for example, is such a 'basic sentence':

(1) Jane gave this book to Bill on Saturday.

Starting from a set of PS rules (2a), and after insertion of the appropriate lexical material (chosen from the set in (2b)), we arrive at the underlying structure or deep structure (2c):

(2a) S ---+NP-VP

NP ---+{DetN }

VP ---+ Tns-Y-NP-PP-PP

Tns ----+ Past

PP ---+ P-N P

(2b) N : Saturday, Bill, Jane, book

Det: this

V : give

P: on, to

(2c) [s [NP Jane [vp [Tns Past] [vp give J [NP this book] [pp to [NP Bill]] [pp On [NP Saturday]]]]

All we need to do now is to ensure that the past tense marker ends up on the right of the verb give, producing gave. The step from the underlying structure (2c) to (1) is the operation of a morphological rule which ensures proper attachment of affixes. It is possible to vary the basic patterns of sentences by performing additional operations on the underlying or deep structure. Starting from (2c) once again we may arrive at quite different patterns:

(a) by forming a Yes-No question:

Did Jane give this book to Bill on Saturday?

101 (b) by forming a wh-question:

What did Jane give to Bill on Saturday?

To whom did Jane give this book on Saturday?

- (c) by fronting the Adjunct PP:

On Saturday Jane gave this book to Bill.

- (d) by passivising the sentence:

This book was given to Bill (by Jane) on Saturday. Bill was given this book (by Jane) on Saturday.

- (e) by clefting:

It was Jane who gave this book to Bill on Saturday.

It was to Bill that Jane gave this book on Saturday.

It was on Saturday that Jane gave this book to Bill.

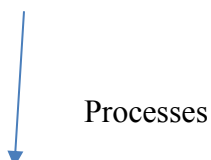
- (f) by pseudo-clefting:

What Jane did was give this book to Bill on Saturday.

Example (1) as well as the examples listed under (a)-(f) above are all related to the string (2c).

Underlying structures are defined as structures resulting from lexical insertion (2b) in the structures described by the PS rules (2a). Surface structures result from applying one or more operations to the underlying structures. In cases like example (1) only one morphological operation has been performed (attachment of the affix), but often more drastic changes will occur (cf. (a)-(f) above). Schematically our description is organised as follows:

- (a) underlying structure



- (b) surface structures

Let us return once more to example (1). Instead of inserting Ns and Det (as in (2b)) into the NP slots generated by the PS rules (2a), we may also choose to insert pro-forms:

(3a) $[s [_{NP} She] [_{VP} [_{Tns} Past] [_{V} give] [_{NP} it] [_{PP} to [_{NP} him]]] [pp On Saturday]]]$

(3a) is again an underlying structure. After attaching the past tense affix to V, we arrive at the surface structure (3b): She gave it to' him on Saturday. Inserting pronominal elements rather than full lexical NPs is referred to as pronominalisation.

Topic- 185: Leftward Movement

Questions are of different types. Earlier we have noted that English has various types of questions. We have distinguished between:

- (a) Yes-No questions, e.g.: Did you measure the distance?
- (b) Wh-questions, e.g.: Why did you measure it?

and between:

- (c) Direct questions, e.g.:

May I go home with you?

- (d) Indirect questions, e.g.:

One evening he asked whether he might go home with her. All these interrogative sentences are related to basic declarative structures, e.g.

- You measured the distance,
- I may go home with you, etc.

We shall consider three main operations which relate questions to basic sentence patterns.

The three operations are:

- Subject-Auxiliary version,
- do- insertion
- and wh-movement.

Topic- 186: Subject-auxiliary Inversion

The following text has three instances of direct Yes-No questions. Let's identify them.

One morning the elder girl hung back in my room. She had something to say. She said: Shall I show you my rude drawings? ; ' I was interested. She showed me the drawings: a child's view of unclothed dolls. I was greatly, moved. She said: 'do you like my rude drawing' 'I like your drawings, Yvonne.' 'I will show you some more tomorrow. Would you like to keep these?' 'I'd rather you kept them, Yvonne.' 'No, you can have these. I can always do some for myself.'

V.S. Naipaul, The Mimic Men.

The direct Yes-No questions in the text are:

- Shall I show you my rude drawings?
- Do you like my rude drawings?
- Would you like to keep these?

What distinguishes them from the basic patterns to which they are related is the order of the first two elements in the sentence: the auxiliary and the Subject NP. Compare (1) and (3) with (4) and (5) below:

- (4) I shall show you my rude drawings.
- (5) You would like to keep these.

The operation which changes the order of elements in a sentence (for example, to get from (4) to (1) or from (5) to (3) is called a transformation.

declarative: [s [NP I] [VP [shall] [show you my rude drawings]]

+Present

interrogative: [s [Shall] [NP I] [vp show you my rude drawings]]

+Present

In order to form Yes-No questions like (1) and (3) we move the first auxiliary to the left and place it in front of the Subject. The moved auxiliary may be a modal auxiliary (shall, will, must, can, etc.), perfect have, progressive be, or passive be; all these auxiliaries are marked for either present or past tense. The following is a more abstract formulation of the rule involved in Yes- No question formation. The movement transformation works on basic patterns as follows:

underlying structure: [S [NP] [VP [aux(n't) :] V ...]]

+Tense

surface structure: [s [aux (n't)] [NP] [VP V ...]]

+Tense

Here are some examples involving the application of these rules for deriving the surface structure of sentence from their kernels:

Surface structures: is he sleeping

The kernel sentence of the deep structure in the case of this sentence is:

He is sleeping

Analysis: He - pres, -be -ing sleep.

Interr: Pre – be –he –ing –sleep.

Affix: be – pre – he – sleep – ing
(is) (sleeping)

Has Ali been playing a match? (Surface structure)

Ali has been playing a match. (Deep structure)

Analysis: Ali – pres – have – en – be – ing – play - a match.

Interr: Pres. – have – Ali –en – be – ing- play- a match.

Affix: have – pres. – Ali – be – en – play – ing – a match.
(has) (been) (playing)

The transformation affects only the tensed auxiliary. If a contracted negator -n 't is attached to the tensed auxiliary, it must be moved along to the left (see (10))

- (10) John hadn't seen his friend so angry before.
- (11) Jane hasn't left home yet.

The morpheme -n't is not an independent word; it must attach to the auxiliary. The uncontracted form not, on the other hand, is independent and does not move along with the auxiliary. The operation is generally referred to as Subject-Aux Inversion, or SAI for short.

Lesson 32

NULL CONSTITUENTS II

Topic- 187: Do-insertion

(2) Do you like my rude drawings?

(2) Do you like my rude drawings?

We may assume that this corresponds to the basic pattern in (13):

(13) You like my rude drawings.

In (13) the VP contains no auxiliary element apart from the obligatory tense element: there is no modal auxiliary, or have or be.

The underlying structure of (2) is as follows (cf. (13)):

(14) [s [NP You] [vp [Tense] like my rude drawings]]

In such cases there is not only Subject-Aux Inversion, but also do⁻ insertion. The transition from (13) to (2) involves two changes, instead of one:

(a)SAI: the tense element is moved out of the VP and placed in front of NP/ Su;

(b)do is inserted.

We have seen that SAI affects the first auxiliary and the present or past tense element associated with it. If there is no auxiliary in the sentence, SAI affects only tense, which is also an auxiliary element. But it is clear that in such cases we cannot stop after SAI, or else we would end up with an isolated tense affix in front of the Subject N P:

(15) [s [Tense] [NP you] [vp like my rude drawings]]

Tense can never occur on its own; it must always be attached to a verb or an auxiliary. One way of 'saving' this isolated (or stranded) tense-marker is to insert a prop-word, which has the function of carrying tense: this support-word is the auxiliary do (step (b) above: do-insertion).

Do⁻ insertion attaches do to the stranded tense element.

(16) [s [DO] [NP you] [vp like my rude drawings]]

+Tense

Another way of looking at do-insertion also called do-support

Surface structure: Did the girl see him?

Kernel sentence: The girl saw him.

Analysis: the girl – past – see – him

Interr: past - the girl – see – him

Do-support: past – do – the girl – see – him

Now, one can apply the affix-switch rules.

Affix: do-past – the girl – see – him

SAI (followed by do-insertion if necessary) can also occur in cases like the following:

- (17) No sooner had he entered the house than the police arrived.
- (18) Never did I suffer more than on that unfortunate day!
- (19) On no account will he do such a thing again.

SAI and do-insertion are triggered 'by the presence of negative VP- Adjuncts such as no sooner, never and on no account.

Topic- 188: Wh-movement

We have looked at -Yes-No questions and the two orations involved in forming them (SAI and do-insertion). Let us now consider the formation of wh- questions. you will find several instances of such wh-quesuons.

Text 4.2.4.

I was brought up in Cornwall. On one sad night during the war a sentry of the Home Guard shot what he took to be an interloper. At the subsequent enquiry his commanding officer said:

‘Did you challenge him?’

‘Yes, sir’

‘What did you say?’

‘Who goes there?’

‘What did he say?’

‘Friend, sir’

‘What did you do?’

‘I shot him’

‘Then why did you shoot him?’

‘If it had been a real friend, he w have said, Don’t be a silly.’ And I would have let him pass.’

Adapted from Christian Brann, Pass the Port.

Two wh-questions from the text are:

(20) What did you say?

(21) Who goes there?

We shall argue that the structure underlying (20) is roughly:

(22) [s [NP You][vp Past say [NP what]]]

and that leftward movement of what, as well as SAI and do-insertion are required to give us (20). The underlying structure of (21) is as follows:

(23) [s [NP Who] [vp Present go [AdvP there]]]

We shall say that wh -movement also applies here, but since who already appears at the beginning of the sentence, the rule is said to apply vacuously. As we shall see do-insertion is not required in this case. If we form wh-questions from the following sentences, replacing the italicised item by a wh -phrase, it would be example:

- John put his money in the top drawer of his desk.

becomes:

Where did John put his money?

- (24) The wedding is to take place on 13 April.
- (25) - Susan will be wearing her blue dress tonight. (When)
- (26) A man with red hair was waiting for Jane. (for whom)
- (27) I heard the telephone, not the door-bell. (what)
- (28) These are Charles's shoes. (what)
- (29) John hit Bill because he hated him. (why)
- (30) George got here by train. (how)
- (31) The terrorists assassinated the ambassador. (Whom)
- Subject: who/what
- Direct object: Whom/what
- Adverb:
- Place: where
- Time: when
- Manner: how

Topic- 189: Subcategorisation Frames and Wh-movement

We must first justify our choice of (22) and (23) as the structures underlying (20) and (21) above. Our argument is largely based on the subcategorisation of lexical verbs. Subcategorisation frames can be seen as providing the slots into which to insert the verb. Assassinate, for example, ((31) above) has the following frame:

- (32) assassinate

+V, [-- NP]

- (33) [s [NP The terrorists] [VP assassinated [NP the ambassador]]]

Notice that a frame must be fully realised. Otherwise the sentence will be ungrammatical:

- (34) *[s [NP The terrorists] [vp assassinated [NP--]]] (42)

In basic sentences in English the NP/Od typically follows the verb. Consequently, (35) below is ungrammatical:

- (35) *[s [NP The terrorists] [VP [NP the ambassador] assassinated]]

If you form a Yes-No question from the terrorists assassinated the ambassador, you will find that the sentence is grammatical only if the NP/ Od follows the verb, as stipulated by the subcategorisation frame for assassinate.

For example

- (36) [s Did [NP the terrorists] [VP assassinate [NP the ambassador]]]

Not: * Did the terrorists the ambassador assassinate? Now consider the following wh-question:

- (37) Who(m) did the terrorists assassinate?

If you try to fit this into the subcategorisation frame (32) above, you will get the impression that the NP/Od after the lexical verb is not realised:

- (38) Who(m) did the terrorists assassinate [NP--]

But what happens if we insert a constituent in the apparently empty position? We get:

- (39) *Who(m) did [NP the terrorists] [vp assassinate [NP the ambassador]]

We must conclude that the Object position in (38) is somehow already occupied. And indeed, we can easily provide a filler; the sentence-initial wh-phrase who(m) can fill the position:

- (40) [S [NP The terrorists] [vp assassinated [NP who(m)]]]

+WH

Compare now again (41) and (42) below (slightly simplified):

- (41) [NP The terrorists] [vp assassinated [NP who(m)]]

+WH

- (42) [NP Who(m)] did [NP the terrorists] [vp assassinate]
+WH

It would be very uneconomical to devise two separate frames for assassinate: one for (41) and one for (42), as follows:

- (43) [VP - NP]
+WH
- (44) [Vp -]

Let us assume that assassinate always has the frame:

- (45) [VP - NP]
+WH

We start from (41):

- (46) [S [NP The terrorists] [vp Past assassinate [NP who(m)]]]
+WH

We can move the wh -phrase [NP+WH] from its original position , and place it in front of the sentence. Assume that it is placed in a special slot marked as [+WH], which occurs in front of S:

- (47) [+WH] [s....]

The movement of the wh-phrase to the +WH slot (i.e. wh-movement) may be represented schematically as follows:

- (48) [NP Who (m)] [s the terrorists Past assassinate ---]
+WH

What does the linguistics teach you?

Kernel: Linguistics teaches you something

Anaysis: Linguistics – pres – teach you – something

Wh-sub: Linguistics – pres – teach –you - what

Interr: Pres. - -Linguistics - -teach –you – what.

Wh-front: What – pres. Linguistics – teach – you.

Do-supp: What – pres. – do – Linguistics teach – you.

Affix: What – do – pres. –Linguistics – teach – you.

(does)

We have seen that wh-questions are formed by (a) moving a constituent containing a wh-word into the +WH slot in front of the sentence, and by (b) SAI, with or without do-insertion. Operation (a) is the transformation called wh-movement.

Topic- 190: Prepositional Phrases in Wh -questions

Consider the following 'echo-question' containing a [PP/ +WH]:

- (55) John bought the necklace [PP for whom]?

+WH

Wh-movement can operate in two ways:

(a) It may affect the whole PP:

- (56 a) [+WH For who (m)] [S did John buy the necklace [PP---]]
- (b) It may affect only the wh-word whom, which is an NP:
- (56b) [+WH Who(m)] [S did John buy the necklace [PP for [NP---]]]

In both cases the moved constituent matches the type of gap (PP or NP). In (b) it is only the NP following the preposition that is moved. The preposition left behind is said to be stranded. In (a) the moved wh-word (who(m)) trails the preposition along with it. If movement affects more than just a wh -constituent, we say that the other elements are pied-piped. Pied-piping is usually optional with PPs.

In some cases, native Speakers feel that pied-piping is obligatory. Let's produce wh-questions corresponding to (57)-(59) below:

- (57) He offended her in what way?
- (68) You got home by what time?
- (59) She asked you to pick her up at what station?

Preposition stranding yields particularly awkward sentences when the PPs are Adverbial Adjuncts of time or place, as in (59):

- (59) What station did she ask you to pick her up at?

Certain prepositions (e.g. during, notwithstanding) never allow stranding:

* Which holiday did he meet her during?

Topic- 191: Wh-movement of Subjects

Normally, wh-question formation involves:

- (a)wh-movement
- (b)SAI, with or without do-support.

With Subject wh -phrases, this is not true. Consider, for example:

- (60) Who left last night?

Since a Subject wh-phrase occupies a sentence-initial position, we shall say for the sake of generality that wh-movement takes place here as well, but that its effect is not visible.

Wh-movement is said to apply vacuously here.

- (61) [] [s [NP who] [Past leave]]
 +WH +WH
- (62) [NP who] [s [NP---] [Past leave]]
 + WH

There is no SAI. Evidence that the NP/ Su is indeed affected by wh- movement can be found in certain complex sentences. Look for the gap corresponding to the wh-phrase in the following examples:

- (63) Who did you say had left the garage open?
- (64) Which boy did you say has broken the window?

Who the Subject is of had left the garage open, and which boy is the Subject is of has broken the window?

Wh-movement S⁻

We have so far developed a schema for wh-question formation, which is based on the assumption that in underlying structure such sentences contain an interrogative wh-element and are preceded by a slot for wh-items. Wh -movement extracts the wh-phrase from its original position in S, leaves a gap marked by ---, and moves the wh-phrase into the slot marked for receiving it:

- (65) [+WH What] [S John will send --- to Mary]

The presence of what triggers SAI:

- (66) What will John send to Mary?

We shall say that the slot marked for wh-word S at the beginning of the sentence together with the following S again constitute an S⁻ (S-bar);

After wh-movements the wh-phrase will occupy the slot labelled +WH

[S⁻ [+WH What] [s will John send --- to Mary]]

Topic- 192: Indirect Yes-No Questions

Questions may be direct or indirect, for example the following sentence contains the indirect question:

- (68) One evening he asked her whether he might go home with her.

Let us look at the structure of (68). First, we note that the phrase *one evening* has been fronted. It is an Adjunct, whose normal position we have assumed is after *He asked her* (or even at the end of the sentence):

- (69)He asked her one evening whether he might go home with her.
- (70)He asked her whether he might go home with her one evening. (Ambiguous)

Bracketing (69) for main clause constituents yields the following:

- (71) [S [NP He] [VP asked [NP her] [pp one evening]]

[s⁻ whether he might go home with her]]]

Od

In 2.3 we have argued that whether (like that or if) occupies the COMP slot in S^- :

- (72a) He said her [S^- [COMP that her] [s^- ...]
- (72b) He asked her [S^- [COMP if] [s^- ...]
- (72c) He asked her [S^- [COMP whether] [s^- ...]

Unlike that, the Complementisers *whether* and *if* indicate that the subordinate clause is a question.

Whether and if cannot be deleted. Indirect questions (i.e. subordinate clauses) introduced by whether or if correspond to direct questions of the Yes-No type. Compare:

- (73a) He asked her: 'May I go home with you?'
- (73b) He asked her {whether} he might go home with her.

Direct Yes-No questions undergo SAI. Indirect Yes-No questions, on the other hand, are not subject to SAI. Consider:

- (74) ... [[COMP whether [su he] [Aux might] go home with her]]

The normal Su-Aux word order is preserved here, and the presence of the complementiser (whether or if) signals that the subordinate clause is a question.

Lesson 33

NULL CONSTITUENTS III

Topic- 193: Indirect Wh-questions

The following contains a complex sentence:

- (75) Do you know what happens when one of your tail lights fails?

The italicised string is an indirect wh-question, corresponding to the direct question:

- (76) What happens when one of your tail lights fails?

How do we form structures with indirect wh-questions? Let us assume that we start from the basic sentence patterns as. Assume that we have a sentence such as:

- (77) Mary wondered: 'What is John doing?'

The string with quotation marks is an example of direct speech or thought: we report Mary's thoughts or words directly. However, we may also decide to report them indirectly by means of an embedded sentence:

- (78) Mary wondered what John was doing.

Underlying this example, we shall assume the basic pattern:

- (79) Mary wondered [S⁻ [COMP] [S John [VP was doing [NP what]]]]

where the NP what is the Direct Object and follows the verb do. As you see, the COMP slot is left unfilled here. The wh-phrase which we find inside the subordinate clause in (79) is moved out of S into COMP:

- (80) Mary wondered [S⁻ [comp what] [S John [VP was doing [NP_ _ _]]]]

Wh -movement leaves behind a gap bound by the moved element in COMP. With direct questions, as we have seen, Wh-movement is followed by SAI. There is no SAI in embedded wh-questions. Explain what we mean by saying that in examples like (75) movement applies 'vacuously' (cf. 4.2.4). Provide the analysis of (75), and indicate the difference between (75) and (78).

We have seen now that wh-movement and SAI may be used together to form wh-questions, but SAI and wh-movement can also occur independently: they are elementary operations which are not necessarily linked. For example, SAI occurs on its own in:

Yes-No questions: Did you measure it?

Declarative sentences with fronted negative Adjuncts: Never had he felt so happy. Wh-movement occurs without SAI in :

Indirect questions: She wondered what he was doing.

Exclamations: What fools they have been!

wh-movement nor SAI is necessarily linked with question formation: SAI occurs with fronted negative Adjuncts in declaratives and wh-movement occurs with exclamations.

Topic- 194: Long Wh-movement

Let us look at the sentence: .

- (81) Who do they think is spying on them all the time?

This is, of course, a wh-question. Wh-movement has put the wh-word in front position. Where does who come from? Obviously, it is not the Subject of the main clause: they is the Subject NP of the think-clause. Underlying the sentence we assume the basic pattern (82) below:

- (82) They think [S^- [COMP] [S who is spying on them all the time]]

The clause (that) who is spying on them all the time is subordinate; it is an Object clause of think. Wh-movement again takes out who and moves it first to the COMP of the embedded clause (S^-), and then in a second move to a COMP slot in front of the main clause :

- (83) [COMP Who] [do they think [S^- [COMP] [--- is spying on them all the time]]]
- (84) Who did John say he met --- on the bus?
- (85) Who did Mary tell you she had seen --- on the train?
- (86) Who did Bill say --- was asking for me the other day?
- (87) At what time do they think the taxi will reach the airport ---?
- The fronting of who trigger SAI(with do-insertion)
- (84) Who did John say he met --- on the bus?
- (85) Who did Mary tell you she had seen --- on the train?
- (86) Who did Bill say --- was asking for me the other day?
- (87) At what time do they think the taxi will reach the airport ---?

Long wh-movement as in the sentences above need not be restricted to only one clause-leap: the wh-element may move over more than one clause:

- (88) Who did John say that Bill thought that Mary wanted to see ---?

Topic- 195: Relative Clauses

- The man who is sitting there is my brother.

Subordinate clauses introduced by a wh-word which are not questions are relative clauses. The NPs listed below all contain a postmodifying (or relative) clause and are thus complex: the GF of Postmodifier is realized by a clause:

- (91) The day *on which Lawlessness reared its ugly head at Blandings Castle*
- (92) The smooth green lawns *on which it shone*
- (93) Those *who read thrillers*
- (94) The chronicler *who wishes to grip his audience*
- (95) The wave of crime *which was to rock one of Shropshire's stateliest homes to its foundations*
- (96) The persons *whom it involved*

Complex NPs have the following structure:

Category	Det	AdjP	N	S
Function	Spec	Premod	Head	Postmod
	The	handsome	policeman	who directed us to the station
	The	German	Girl	who gave me this book

The NPs listed below all contain a postmodifying (or relative) clause and are thus complex: the GF of Postmodifier is realized by a clause:

- (91) The day *on which Lawlessness reared its ugly head at Blandings Castle*
- (92) The smooth green lawns *on which it shone*
- (93) Those *who read thrillers*
- (94) The chronicler *who wishes to grip his audience*
- (95) The wave of crime *which was to rock one of Shropshire's stateliest homes to its foundations*
- (96) The persons *whom it involved*

Complex NPs have the following structure:

The NPs (91)-(96) above all have the same structure as The German girl who gave me this book.

The Head noun girl is post modified by an S^- : who gave me this book. In each case the Postmodifier is realised by an S^- consisting of a wh-phrase followed by a sentence with a matching gap. Superficially, the pattern of the postmodifying S^- is similar, but not quite identical, to that of indirect wh-questions.

Compare:

- (97) I wonder who(m) John talked to.
- (98) the girl who(m) John talked to

Superficially, the pattern of the postmodifying S^- is similar, but not quite identical, to that of indirect wh-questions.

Compare:

- (97) I wonder who(m) John talked to.
- (98) the girl who(m) John talked to

On the analogy of wh-questions, we assume that (98) has the following underlying structure:

- (99) [NP The girl [s^- [COMP] [s John talked to whom]]]

The element whom is moved to the COMP slot in front of S, and to is stranded:

- (100) [NP The girl [s^- [comp whom] [s John talked to ---]]]

In wh-question formation, as we have seen, there is a choice with respect to the movement of wh-PPs: one can either strand the P and extract the NP only, or one can pied-pipe the P. In the example above we have stranded the P; we could also have pied-piped.

Compare:

- (101) I wonder to whom John talked.
- (102) the girl to whom John talked

Apply wh-movement to the italicised wh-word s in the sentences below. Apply pied-piping whenever necessary (-- indicates COMP):

- (103) the house-----John lives in which
- (104) the book----- Sue is waiting for which
- (105) the girl-----John is in love with whom

In (103)-(105) above pied-piping is optional. Consider now the following NPs:

- (106) the woman--I live in whose house
- (107) the man --I met whose daughter for the first-time last year
- (108) the cathedral -- the cardinal was murdered on the steps of which

How should WH -movement be carried out here? Pied-piping is obligatory

In this case, for we cannot just move whose, which, etc. here. For example:

- (109) *the woman whose I live in --- house

We must pied-pipe in one of the following two ways:

- (110) the woman in whose house I live ---
- (111) the woman whose house I live in ---

As in the case of wh-questions, we shall assume that wh-movement also operates on Subject wh-phrases, although there is no obvious gap in the surface structure: it operates vacuously.

On the analogy of wh-questions, we shall assume that the underlying structure (112) below becomes (113), as a result of wh-movement in the NP:

- (112) [NP the man [S^- [COMP] [s [NP who] carried the gun]]]
- (113) [NP the man [s [coMP who] [s [NP---] carried the gun]]]

We have seen that the postmodifying S^- in a complex NP is linked to the Head N by a wh-word.

The WH -words which relate the subordinate clause and the Head, are relative pronouns.

The relative pronoun in front of S is said to bind a gap inside the relative clause.

Topic- 196: The Meaning of Relative Clauses

Postmodifying (relative) clauses may help to identify the referent of the Head noun: they may narrow down the reference of the Head. For example, in: the man who lives next door, the relative clause who lives next door indicates who exactly the speaker is talking about. The clause provides information that is essential for the identification of the referent. Relative clauses of this kind are necessary if it is not clear from the context who the speaker is referring to. They, are said to be restrictive. Here are some additional examples of restrictive relative clauses (italicised):

- (114) What is the name of the film which is now showing at the ABC?
- (115) The books which I ordered have finally arrived.
- (116) The man who saw the accident was not prepared to testify.

relative clauses may also be non-restrictive, as in:

- (117) The Pope, who was expected to visit Poland again this summer, decided to go to France instead.

Clauses of this kind are said to be non-restrictive because no further identification of the referent (the Pope, in this case) is needed. These modifying clauses provide additional, non-essential information about the Head of the NP. It is worth noting that the distinction between restrictive and non-- restrictive relative clauses also applies to other types of postmodification and premodification.

Topic- 197: Deletion in Relative Clauses**Consider:**

- (118) The film which we saw at the ABC last week has been banned from all official programmes.
- (119) the man whom you see over there is the owner of the company.

The relative pronoun in these sentences can be deleted. We shall use the symbol $\emptyset$ to mark the deleted element, and call it the zero relative pronoun:

(a) if it functions as the Subject of the embedded sentence, e.g.:

- (121a) The man who is standing over there is the Prime Minister of Pakistan.
- (121b) *The man $\emptyset$ is standing over there is the Prime Minister of Pakistan.

(b) if it is preceded by a preposition, e.g.:

- (122a) The man for whom we were all waiting did not come.
- (122b) *The man for $\emptyset$ we were all waiting did not come.

Compare:

- (123) The man {whom/ $\emptyset$ } we were all waiting for --- did not come.

If the preposition is stranded, the moved wh-word on its own can be deleted. Notice the use of the zero relative pronoun in Subject position in structures of the type: There is a gentleman at the door $\emptyset$ wishes to see you and of the type: It is Jane $\emptyset$ gave the book to Bill.

Topic- 198: That in Relative Clauses

One also often finds the complementiser that instead of who(m) or which at the beginning of restrictive relative clauses.

- (124) The young girl that plays the violin so well is Egyptian.
- (125) The man that we liked so much turned out to be a real crook.

The GF of Postmodifier is realised by an embedded clause preceded by that:

- (126) [s⁻ that [s plays the violin so well]]
- (127) [s⁻ that [s we liked so much]]

The sentence (S) together with that is an S⁻. There is a great deal of similarity between these Ss and restrictive relative clauses.

For example:

(a) both occur inside an NP,

(b) both are restrictive Postmodifiers to the Head,

(c) in both cases there is a gap inside the S, e.g.:

- (128) [NP the man [{ that/who(m)/ $\emptyset$ } [s we like - - - so much]]]

That, who(m) and zero are largely interchangeable, except that for stylistic reasons who(m) may be preferred. We assume that the complementiser that here also links the Head noun and the embedded S. Since in other languages and in Middle English one can also find strings of a wh-word followed by that (literally, for example: The dog which that we saw yesterday and not the reverse), we shall assume that the COMP slot has two positions:

- (130) COMP is reserved for wh-elements (a) and for complementisers (b).

(130)

To form a relative clause we now have quite a few options. Take the following underlying structure:

- (131) The woman [s [coMP (a) (b)] [s we invited whom to our party]]

Move whom to position (a) in COMP:

- (132) The woman whom (b) we invited --- to our party.

We may delete whom:

(133) The woman $\emptyset$ (b) we invited --- to our party.

We may also insert the complementiser that at position (b) in COMP:

(134) The woman (a) that we invited --- to our party.

In present-day English, sentences like (129) are ungrammatical. Modern English has a rule that the COMP slot may be filled by one element only, either a **wh-word** or the complementiser that.

Lesson 34

HEAD MOVEMENT I

Topic- 199: Adverbial Relative and Free Relative Clauses

The relative clauses below also contain wh -elements.

- (135) This is the village where I spent my youth.
- (136) Did they tell you the reason why they all left?

Relative clauses introduced by where, why, when, etc. are called adverbial relative clauses. In some cases, the Head noun seems to merge with the wh-word or that into one element.

- (137) {Any person who/whoever} knows him hates him.
- (138) I shall give you {anything that/ what/whatever} you want.

Relative clauses like these, which lack a real Head noun, are called free relative clauses. In structure these clauses resemble full NPs consisting of 'an empty Head noun and a postmodifying clause.

For example, whatever

- (139) [NP $\emptyset$ [S⁻ whatever [s you want ---]]]

Non-restrictive Relative Clauses

Consider:

(140) The Pope, who was expected to visit Poland again this summer, decided to go to France instead.

(141) This car, which I bought only two years ago, is already beginning to show signs of disrepair.

The postmodifying clauses above do not restrict the reference of the Head noun. The clause who was expected to visit Poland again this summer is not meant to identify the Pope, since the reference is sufficiently clear. Under normal circumstances, there can be hardly any misunderstanding between speaker and hearer about the identity of the referent. The above relative clauses are said to be non-restrictive. They provide extra, non-essential information, and can be left out without changing or obscuring the reference of the Head. In non-restrictive relative clauses deletion of the relative pronoun is not possible:

- (142) *The Pope, $\emptyset$ was expected to visit Poland again this summer, ...

Nor does the complementiser that normally occur here:

- (143) *The Pope, that was expected to visit Poland again this summer, ...

The intonation of non -restrictive clauses is different from that used for restrictive clauses. In print, non- restrictive relative clauses are usually marked off by commas.

Topic- 200: Appositive clause

Consider the following sentences containing complex NPs:

What is the structure of the above N Ps?

(144) The news that John had left his job came as a shock to us all.

(145) The rumour that tomorrow is a local holiday has not been confirmed by the media.

(146) The story that petrol prices are going to be decreased has been made up from beginning to end.

(147) The fact that the Prime Minister was involved in a drugs scandal was an embarrassment to the Government.

(148) The assumption that John is coming back soon seems unfounded.

Superficially, the complex NPs in the above sentences have the following structure:

Spec	H	Postmodifier S⁻ (-- that + S)
The	News	[S⁻ that [s John had left his job]]

Is the postmodifying clause a relative clause? We have said that relative clauses are preceded by a relative pronoun which binds the gap inside the relative clause, and we have said that the function of linking the Head and the embedded S⁻ may also be fulfilled by that. For example:

(149) the man {w o(m)} you saw --- there

Can you locate similar gaps in (144)-(148)? You will find that the answer is: No.

Compare these examples with those in (150) -(154) below

- (150) The news that John told his wife gave her a shock.
- (151) The rumour that Anne spread in town was terrible.
- (152) The story that Mr. Adams wrote for his children has recently been published in a collection.
- (153) The news that the Prime Minister gave the reporters did not surprise anyone.
- (154) The assumption that he bases his theory on may be false.

In (150)-(154) we find the expected gap in the embedded S, while in (144)-(148) there is no gap. We must conclude, then, that (144)-(148) are not relative clauses. What are they? Consider the Heads of the Subject NPs in (144)-(148): in some of the examples the Head

noun relates to a verb (report); in other cases it has something to do with 'messages' (news, story, ...), i.e. communication. In fact, the that-clause gives the content of the communication.

For (144), for example, the news was: 'John has left his job'.

Note that which cannot replace that here, and that that fulfils no grammatical function (e.g. Subject or Object) in the clause; moreover, that is obligatory here and this use of that can also occur before non-restrictive clauses:

Your assumption, that there is no life on other planets, is unfounded. In all these respects, the that-clauses in (144)-(148) differ from relative that-clauses. That-clauses like those in (144)-(148), which give the content of nouns such as fact, report, news, etc., are called appositive clauses. Appositive clauses may also be introduced by the complementiser whether: I have no idea whether they will appreciate our help).

Topic- 201: Reduced Relative Clause

Let's look at these sentences:

(155) The first man to arrive at the place of the crime was our local police officer.

(156) The last man to talk to in such circumstances is your father.

(157) I want a tool to fix the sink with.

(158) We all want a decent place to live in.

(159) The first train to leave from platform 3 is the 6.00 for Rawalpindi.

The structure of these NPs is as follows:

Spec	Premod	H	Postmod
the first		Man	to arrive at the place of the crime
A	decent	Place	to live in

The to-constructions serve as Postmodifiers and can be changed into full relative clauses, e.g.:

- (160) The first man who arrived at the place of the crime ...

The to-constructions used in (155)-(159) are called reduced relative clauses. Full relative clauses, introduced by who, which, etc., are finite. The postmodifying strings in (155)-(159), on the other hand, are non-finite (to-infinitive) clauses.

Reduced relative clauses may also have an -ing or -ed VP:

- (161) The train arriving at platform 7 is the 7.03 from Reading.
- (162) The guy wearing those odd clothes is Jane's husband .

- (163) None of the guests invited by John turned up.
- (164) Essays handed in after 1st July will not be marked until the beginning of next term.

The non-finite clauses above are restrictive. Here are two examples of non-restrictive non-finite clauses:

- (165) The Pope, warned not to go to Poland, decided to visit France.
- (166) The Princess of Wales, haunted by the press, found it more and more difficult to lead a normal life.

The structure of reduced relative clauses in more detail. We find that certain syntactic positions which in full relative clauses are usually occupied by lexical material are left empty in reduced relative clauses, i.e. not filled by overt lexical material. Compare, for example,

(155) with its full version (155a): The first man to arrive at the place of the crime ...

- (155a) The first man who arrived at the place of the crime ... The underlying structure of (155a) is as follows:
- (a) before wh-movement:
- (155b) [NP the first man [s COMP [s who [vp arrived ...]]]]
- (b) after 'wh-movement' of the NP/Subject:
- (155c) [NP the first man [s who [s--- [vp arrived ...]]]]

The gap --- is bound by the moved wh-word (who).

For the reduced clause (155) we shall assume an analogous underlying structure, but instead of a finite VP there is a non-finite VP:

- (a) before wh-movement:
- (155d) [NP the first man [COMP [s, [vp to arrive ...]]]]
- (b) after wh-movement:
- (155e) the first man [∅ COMP/ s [s to ----- [VP to arrive ...]]]

(155) differs only minimally from (155a):

- (1) the VP in (155) is non-finite, that in (155a) is finite
- (2) the Subject wh-word in (155a) is an overt element (who), that in (155) is non-overt, i.e. not lexically realised and here indicated by ∅. We assume now that the overt wh-word as well as the non-overt element ∅ move from the Subject position into COMP, leaving a

gap ---. Now let us consider (156). Here, too, we may try to base the structure of (156) on the analogy with its finite full version:

- (156a) The last man that you should talk to in such circumstances. ..

It is immediately clear that there is a gap inside both (156) and (156a) and that the preposition to is stranded: (pp to ---]. In (156a) the complementiser that (in COMP) binds the gap and relates the relative clause to the Head N; in (156) there is no overt element in COMP. Let us thus assume that again there is a non-overt element $\emptyset$ in COMP. Another problem now. What is the Subject of the relative clause in (156a)? Clearly, it is you. In (156) the non-finite relative clause lacks an overt Subject, but we may interpret the Subject as referring, for example, to you, as represented in (156b):

- (156b) The last man [s^- for [s you to talk to]]

We shall thus have to say that (156) also has a non-overt Subject, which we shall indicate by (). We assume then that before wh-movement (156) has the structure:

- (156c)[NP the last man [s^- Comp for [s to () [to talk to $\emptyset$]]]]

and that wh-movement moves the non-overt element $\emptyset$ into COMP:

- (156d) [NP the last man [$\emptyset$ for COMP s^- [s^- to ()] [to talk to ---]]]

In English, the sequence for . . . to is ungrammatical; for must be deleted. Essentially the reduced relative clause in (156) thus has a non-overt Subject as well as a non-overt relative element in COMP (after movement) indicated by $\emptyset$, which binds a gap indicated by ---. All these non-overt positions are assumed on the basis of corresponding overt positions in other parallel clauses.

Topic- 202: Fronting

In Ms. Sgol's article 'Even Accountants Will Buy It', which you published on February B, Durham is quoted as one of a group of universities who 'expect you to concentrate on one or perhaps two subjects from the day on which you start there until your final exams'.

This is inaccurate and profoundly misleading. Quite apart from the fact that Durham offers the degree of BA in General Studies, which is a three subject degree in all three years and the BSc in Natural Sciences, involving over three years a minimum of three and maximum of five subjects, a high proportion of other degree courses offered by Durham "make it possible for candidates to include more than two subjects in their degree courses. To give but one further example, candidates taking honours courses in the Social Sciences have to read three different subjects in their first year.

There is considerable flexibility within the Durham system for changes in courses after the end of the first year. Indeed about 20 percent of candidates graduate with a degree whose title is

different from the one for which they first registered on entry.

Fronting

So far we have seen that auxiliaries, tense and wh-phrases may be moved leftwards. Other constituents may also be preposed. For example:

- (167) Soon after breakfast Mary Ann brought in The Times.

Soon after breakfast has been fronted: it is a PP/ A.

- (168) Mary Ann [vp brought in The Times [PP soon after breakfast]] Compare also:
- (169a) I have never before met a man like John.
- (169b) Never before have I met a man like John.

Fronting of never before in this case triggers SAI. Particles of phrasal verbs (170) and PPs (171) may also be fronted. For example:

- (170) He came out.
- (171) Out he came.
- (171a) He walked into the garage.
- (171b) Into the garage he walked .

Topic- 203: Passivisation

The Process

Read the following text. Bracket the major. Constituents in each sentence, and underline the Head of the VP. Assign GFs.

Text 4.3.1

The eating habits of the Indo- Pakistan subcontinent are influenced by historical and geographical factors. Since the earliest times the sub-continent has been invaded by many tribes from the North. Later on it was occupied by the British. Only recently was the region divided into the two independent countries of India and Pakistan.

The influence of all the different invasions can be found in the culture and the eating habits of the sub-continent. Indian and Pakistani food are very similar, but regional and religious influences can be observe. The consumption of beef is forbidden to the Hindu, and the consumption of pork is not allowed for Muslims.

The art of the presentation of food was also developed in these countries. Silver or gold leaf is often used for decoration. The metal is beaten very fine: it can almost be blown away.

Nowadays Indian food is esteemed all over the world.

*Adapted from A. Hosain and S. Pasricha,
Cooking the Indian Way*

This text contains several instances of passive sentences. Passive sentences have the following pattern:

- [s [NP] [VP be V-ed [pp by...]]]

For example:

[NP The eating habits of the Indo-Pakistan sub-continent] [vp are influenced [pp by historical and geographical factors]]]

- (3) [s [PP Since the earliest times] [NP the sub-continent] [vp has been invaded [PP by many tribes . . .]]]

Normally a passive sentence has a corresponding active sentence. The active sentence is seen as the underlying structure, from which the passive sentence is derived. The active sentence corresponding to (2), for example, is (4):

(4) [s [NP Historical and geographical factors] (VP influence [NP the eating habits of the Indo-Pakistan sub-continent]]]

Sentence (4) contains a transitive V (influence), which requires as its Complement an NP functioning as Od. The structure of the sentence is: NP/ Su (V-NP/Od]. If we compare the passive sentences from text with their active counterparts, we see that the active strings contain an Object NP. The process of passivisation crucially involves moving the Object NP (Od or Oi) from the active VP into the Subject position.

Consider also:

- (5a) The gardener takes the newspaper to Mr. Ellis.
- (5b) The newspaper is taken to Mr. Ellis by the gardener.

If we compare (5a) and (5b), we find that the Object of (5a), the newspaper, is moved to the left, and ends up in Subject position in (5b). The Object NP is assumed to leave a gap --- in the underlying active sentence after passivisation. In order to verify the Subject status of the NP the newspaper in (5b) we can apply various tests, including the agreement test (a) and the SAI test (b):

- (6a) The newspapers are taken ...
- (6b) Is the newspaper taken ...?

The tests confirm that in (5b) the newspaper is the NP/Su. The shift of the Object NP into Subject position is accompanied by another operation: the Subject of the active sentence is removed from its original position and may end up as the Agent by-phrase in the passive

sentence: by the gardener (very often the Agent by-phrase is deleted). In the active sentence (5a) the role of Agent is assigned to the Subject NP.

In the passive sentence (5b) the Agent role is assigned to the by-phrase. In addition, passivisation converts takes into is taken : the auxiliary be is added and the verb is changed into a past participle. The change from active to passive verb form may be represented as follows:

- (7a) present-take : takes
- (7b) present-be-taken: is taken

Schematically passivisation works as follows:

- (8a) [s[NP The gardener][VP takes [NP the newspaper][pp to Mr Ellis]]]
- (8b) [s[NP The newspaper][VP is taken[NP to Mr Ellis][PP by the gardener]]] or more abstractly:
- (9a) active: [s [Su NP1][VP V [Od NP2]]]
- (9b) passive: [s [NP 2] [VP be V-ed [PP by [NP1]]]]

Topic- 204: By-phrase

Consider again the last sentence of the first paragraph of text:

- (10) Only recently was the region divided into the two independent countries of India and Pakistan.

Only recently is a fronted AdvP, originating in VP. The fronting has given rise to SAI Compare the word order of (10) with that of (11):

- (11) The region was divided into the two independent countries only recently.

Try to restore (11) to the active. At first sight, this seems impossible, since there is no by-phrase in (11). We get something like (12), which is ungrammatical:

- (12) *--- divided the region into the two independent countries only recently.

The Agent of the action denoted by the verb (divide) is not explicitly mentioned, although in this case we know that the 'understood' Agent is probably something like The British Government, etc. (cf. the Indian Independence Act, 1947). Sentence (1) requires an explicit Subject (cf. the PS rule).

S→NP-VP

It will be ungrammatical. If one does not know precisely who the Agent of the activity is, or if it is irrelevant or unnecessary to specify who performed the activity. One may choose to use a passive sentence, since in passives the Agent by-phrase is optional. Passivisation allows us to

suppress the Agent. Very often the effect of passivisation is that the text becomes more impersonal and formal.

Agent-less passives are especially characteristic of the style of official documents, textbooks, scientific articles, instructions rules, etc. Examples: '

- (13) You are hereby given leave to enter the United Kingdom for six months.
- (14) Bicycles must not be left in front of the building.
- (15) Smoking in this room is prohibited.
- (16) Contributions are welcomed from linguists in all countries, and not merely from members of the Linguistics Association of Great Britain.

Lesson 35

HEAD MOVEMENT II

Topic- 205: Movement of Focus Position

It-extraposition

Read the text below. Bracket the that-clauses.

It arouses indignation (that some night workers at British Leyland's Rover factory in Solihull are in the regular habit of sleeping for a large part of their shift.)

But on whom should the indignation be vented? From the beginning of time, it has been accepted (that men who are on duty during the night will be tempted to sleep.) It is assumed (that nothing but the most diligent supervision supported by grave penalties will prevent them from going to sleep.)

The fact (that it took a newspaper reporter to unveil this particular manifestation of a normal human phenomenon to the Leyland management) speaks for itself.

It is said (that the erring workers normally finish their assignment by 3 a.m.) and (that they are not due to clock off until seven.) It is possible (that management has given them far too little to do or much too long in which to do it.) For them to remain alert and idle at their posts for four solid hours each night simply out of piety to Sir Michael Edwardes would argue an almost perverted sense of duty.

Adapted from Daily Telegraph, 13 November 1979

IT-extraposition in Passive Sentences

A very common pattern in the text above is the following:

It + passive verb + that-clause.

In the second paragraph, for example, we find:

- (1) From the beginning of time, it has been accepted that men who are on duty during the night will be tempted to sleep.

(2) It is assumed that nothing but ... will prevent them from going to sleep. What is the Subject of (1) and (2)? The Subject of the main clause is obviously it, since SA1 affects the order of it and the auxiliary has or is:

- (3) Has it been accepted that...?
- (4) Is it assumed that...?

Since the verb is in the passive (has been accepted, is assumed), we should be able to reconstruct the underlying active string. Indeed, the active sentences corresponding to (1) and (2) may be something like:

- (5)... everyone has accepted that men who are on duty during the night will be tempted to sleep.
- (6)One assumes that nothing but ... will prevent them from going to sleep.

The choice of the items everyone and one as underlying Subject NPs in (5) and (6) is fairly arbitrary, of course: some other general NP would also have been possible here. The Subject NPs of (5) and (6) are absent from the corresponding passives (1) and (2) above.

Bracketing the constituents in the main clauses of (5) and (6) we can have the GF of the that-clause in (5) and (6) and it is represented by the structure of (5) as follows:

(7) [S ... [NP everyone] [VP has accepted [s⁻ that men who ... will be tempted to sleep]]]

The that -clause (S⁻) is a Complement of the V, and functions as Od. Sentence (7) has the same structure as simple sentences such as (8)

(8) [S [NP Everyone] [vp has accepted [NP the fact]]]

Pseudo-clefting bears out the analysis of (5), showing that the that-clause is integrated in the VP:

(9) What everyone has done is [VP accept that men who are on duty during the night will be tempted to sleep]

It is also possible to use the auxiliary have as a substitute for the whole VP, as (10) illustrates:

(10) Everyone has accepted that men who are on duty during the night will be tempted to sleep, but John hasn't (accepted that men who ...)

If the that-clause is the Od of (5), then under passivisation it should become the Subject of the passive sentence. And this is indeed what happens:

(11) That men who are on duty during the night will be tempted to sleep has been accepted (by everyone).

Sentence (11) can be given the following constituent structure:

(11a) [s [NP That men who are ... will be tempted to sleep] [vp has been accepted]]

However, (11) is not a very elegant sentence; there is something wrong with the balance between Subject and Predicate. The Subject is heavy and the VP following it relatively short. We get rather a lot of information at the beginning of the sentence, while in English the focus of

attention in a sentence is usually towards the end of the sentence. It is for this reason that (11) should preferably be converted into (12):

- (12) It has been accepted that men who are on duty during the night will be tempted to sleep.

Consider again:

- (11) That men who ... will be tempted to sleep has been accepted.

It has been accepted that men who... will be tempted to sleep.

Two main operations are involved in converting (11) into (12):

(a) The Subject clause (with that) is shifted rightwards towards the end of the sentence, leaving a gap ---:

- (13) --- has been accepted [s that men who ... will be tempted to sleep]

(b) It is inserted into the vacated Subject position. It is a dummy element, functioning as a place-holder for the Subject, and is called the grammatical or anticipatory Subject, as opposed to the notional Subject (the that-clause). The anticipatory Subject will be labelled *su*, to distinguish it from 'normal' Subjects (*Su*). The structure of sentence (12) may now be summed up as follows: *It/su [V pass-S⁷/ Su]*.

In (12), the moved that-clause gives content or meaning to it ; the two are linked and are said to form a chain. The movement transformation which shifts the that-clause rightwards in the sentence is called extraposition. The combination of extraposition + *it*-insertion is referred to as *it*-extraposition.

Topic- 206: It-extraposition in Active Sentences

It-extraposition is not confined to passive sentences such as (1) and (2) above. If you read through text again, you will also find active sentences showing the same process.

- (14) It arouses indignation that some night workers ... are in the regular habit of sleeping for a large part of their shift.

If we bracket the sentence constituents, we get the following:

- (15) [s [NP *It*] [vp arouses [NP indignation] [-s-that some night workers ...]]]

The function of the NP indignation is that of *Od*, as passivisation shows:

- (16) [NP indignation] is aroused by (the fact) that some night workers...

What is the GF of the that-clause? It supplies the content for it:

- (17) What arouses indignation is [s that some night workers ... are in the regular habit of sleeping for a large part of their shift]

It in (14) is the grammatical Subject, and the that-clause the notional Subject. (14) is another instance of an it-extraposition pattern. We can, in fact, undo the effect of it-extraposition quite easily:

(a) remove it:

[s--- [vp arouses indignation [s⁻ that some night workers ... are in the regular habit of sleeping ...]]]

(b) replace the displaced that-clause into its original position:

[s [s⁻ That some night workers ... are in the regular habit of sleeping ...] [vp arouses indignation]]

Here is another example of it-extraposition from the text:

- (18) It is possible that management has given them far too little to do.

What is the analysis of sentence (18)? Can you undo the effect of it-- extraposition (i.e. first remove it, then shift the that-clause back to its normal position at the beginning of the sentence)? The effect of it-extraposition is that the heavy Subject clause is moved into focus position at the end of the sentence.

Topic- 207: It-extraposition with Non -finite Clauses

So far, we have concentrated on extraposition involving finite Subject clauses. It-extraposition may also move non-finite clauses. In the last paragraph of the text we find: It-extraposition with Non -finite Clauses

(19) [s [s⁻ For [s them to remain alert and idle at their posts for four solid hours each night simply out of piety to Sir Michael Edwardes]] [VP would argue an almost perverted sense of duty]]

As we can see, the Subject of this sentence is realised by a non-finite clause (S⁻). The complementiser in this clause is for, and for cannot be left out. Sentence (19) has not undergone it-extraposition. The effect of it--extraposition would be as follows:

(a) extraposition:

[s --- would argue an almost perverted sense of duty [s⁻ for [s them to remain alert and idle for four solid hours each night simply out of piety to Sir Michael Edwardes]]]

(b) it-insertion:

[s⁻ it would argue an almost perverted sense of duty [for [s⁻ them to ..]]]

Topic- 208: It-extraposition in Verbless Clauses

Let us now look at the examples below:

- (22) They considered her behaviour with George very foolish.
- (23) The committee found the proposal unacceptable.

Both examples may be analysed as follows :

- (24) NP/Su [V [s,Od NP/Su-AdjP/ Pc]]

In other words: consider in (22) and find in (23) have as their Direct Object a verbless clause (her behaviour with George very foolish and the proposal unacceptable).The verbless clauses in themselves contain a Subject and a Predicative Complement, but there is no verb to link them. Compare (22) with (25) and (26) below:

- (25) They considered her behaviour with George to be very foolish.
- (26) They considered that her behaviour with George was very foolish.

Obviously, nothing would prevent the Subject of the verbless clause in (22) from also being realised by a clause, rather than by an NP. For example:

- (27) They considered that she behaved badly with George very foolish.
- (28) They considered for her to behave badly with George very foolish

In (27) the Subject of the verbless clause is a finite that-clause ; in (28) it is non-finite. But again (27) and (28) sound very awkward; they can be improved if we rephrase them as (29) and (30):

- (29) They considered it very foolish that she behaved badly with George.
- (30) They considered it very foolish for her to behave badly with George.

and (30) are related to (27) and (28) by it-extraposition. In this case, the domain of it-extraposition is the verbless clause functioning as Od. The two familiar steps can again be described as follows:

(a) extrapose the Subject clause of the Object clause:

They considered h... -- very foolish [s that she ...]]

insert it:

They considered [s it very foolish [s that she . . .]]

Or schematically:

U.S.: NP [vp v [s⁻ [s⁻] [AdjP /Pc]]]

(a) NP [vp v [s⁻ --- [AdjP /Pc][s⁻]]]

(b) NP [vp v [it [AdjP/Pc] [s⁻]]]

In (29) and (30), it is the filler of the Subject position inside the verbless Object clause. The notional Subject is the extraposed Subject clause.

Topic- 209: It-extraposition with Object Clauses

Consider the following sentence:

- (31) Mary called William' s behaviour rather odd.

If we adopt the analysis proposed the following pattern emerges:

- (32) NP/Su [V-NP/ Od-AdjP/ Pc+AcJ

The AdjP realises the GF of Pc+Ac. As seen, an Od may also be realised by a clause (an Object clause). For example:

We [vp regretted [s⁻/Od that she arrived so late]]

Consider now also:

- (34) We [vp called [s⁻/Od that she arrived so late] [AdjP/Pc+Ac very regrettable]]
- (34) is, of course, a very clumsy sentence. Its style can be improved if we rearrange it as follows:
- (35) We called it very regrettable that she arrived so late.
- (34) and (35) have the same meaning. If we compare their structures, we see that (35) is the result of it-extraposition.

We have again applied the two-step procedure which we have seen before:

(a) extrapose the (Object) clause, i.e. shift it to the right:

(b) insert it in the vacated slot:

- (37) We called it very regrettable [that she arrived so late]

It is the place-holder for the extraposed Object clause: it is the grammatical or anticipatory Object of called, and the that-clause is the notional Object.

The grammatical Object is labelled *od*, to distinguish it from 'normal' Objects (*Od*). The structure of (35) may be summed up as follows:

- (35a) NP/ Su [V-it/ *od*-AdjP/Pc+Ac-S⁻/ *Od*]

We have pointed out that there are alternative analyses for sentences with so-called complex transitive verbs. One such alternative is to treat (31) as follows:

- (38) Mary [vp called [*sø*/*Od* [NP/Su William's behaviour] [AdjP/Pc rather odd]]]

and there is a similar analysis for (34) above:

- (39) We [vp called [*sø*/*Od* [s⁻/Su that she arrived so late] [AdjP/Pc very regrettable]]]

Under such an analysis (35) would be a straight forward example of it-- extraposition of a Subject clause inside a verbless clause, similar to the cases dealt earlier:

- (40) We [vp called [*sø*/*Od* [NP/su it] [AdjP/Pc very regrettable] [s⁻/su that she arrived so late]]]

It is the place-holder (*su*) for the extraposed Subject clause; the domain of it-extraposition is *Sø*. Rather than treating the AdjP *rather odd* in (31) as merely a phrasal constituent of the entire *S*, we have proposed that it might be seen as part of a verbless clause:

Mary [vp called [NP/*Od* William's behaviour]

[*sø*[NP/SU()] (AdjP/Pc rather odd)]]

Again, this would have consequences for the analysis of (35), which would then have the following analysis:

- (42) We [vp called [NP/*od* it] [*Sø* [NP/Su ()] [AdjP/Pc very regrettable]] [s⁻/*od* that she arrived so late]]

Topic- 210: Subject Verb Inversion

Let us consider the following text.

Blackstable was a fishing village. It consisted of a high street in which were the shops, the bank, the doctor's house, and the houses of two or three coalship owners. Round the little harbour were shabby streets in which lived fishermen and poor people.

Adapted from W. Somerset Maugham, *Of Human Bondage*

Look at a simplified version of the last sentence.

- (43) Round the little harbour were shabby streets.

In what respect is (43) a 'simplified version' of the last sentence of the text? Compare (43) with (44) below:

- (44) The shabby streets were round the little harbour.

This last sentence has the pattern: NP/Su [V-PP/Ac]. The PP round the little harbour has been fronted in (43), and the Subject and the Verb have changed places. (43) and (44) are both grammatical have the same meaning, contain identical constituents with identical GFs, the distribution of these constituents is different. By reorganizing the sentence, the author achieves greater emphasis on the Subject NP, since this is now in sentence-final (i.e. focus) position. The rearrangement is a two-step procedure:

(a) the PP/ Ac is fronted.

(b) the Subject is moved to the right.

Step (a), the fronting of the Adverbial Complement, precedes and triggers step (b), the inversion of the Subject and the verb. Step (b) is called Subject-Verb Inversion.

Let us return to the original sentence in the text:

- (45) Round the little harbour were shabby streets in which lived fishermen and poor people.

What is the Su? Make sure you bracket the full NP/Su. The constituent order of (45) is: Ac-V-Su, which is not a basic pattern. If you restore the sentence to its basic pattern, you will get a sentence of the Su -V-Ac type, but with a heavy Subject NP:

- (46) [s [NP Shabby streets in which lived fishermen and poor people]
[vp were [pp round the little harbour]]]

By fronting the PP round the little harbour and moving the Subject NP shabby streets in which lived fishermen and poor people rightwards, the sentence is made much more balanced. Let us now look at the postmodifying clause in the Subject NP: in which lived fishermen and poor people. The constituent order of this clause is as follows:

- (47) [s⁻ [PP in which] _ [s [vp lived [NP fishermen and poor people]]]]

Is this a basic sentence pattern? If it is not, what arrangements do you think have been made? Of course, one rearrangement is due to wh -movement of in which into COMP. A second rearrangement moves the Subject NP fishermen and poor people to the right, placing it after the verb lived (Subject-Verb Inversion or SVI, for short). We repeat the structure underlying the NP shabby streets in which lived fishermen and poor people:

- (48) [NP shabby streets [s^- [COMP +WH]
[s fishermen and poo, people lived in which]]]

The two rearrangements work as follows:

(a) wh-movement of in which (with pied-piping) results in:

- (49) [NP shabby streets [s^- [COMP +WH in which]
[s [NP fishermen and poor people] lived ---]]]

This movement is obligatory.

(b) Subject-Verb Inversion produces:

- (50) [NP shabby streets [s^- [COMP +WH in which]
[s --- lived --- ,fishermen and poor people]]]]

The inversion is not obligatory. SVI often greatly improves the style of a sentence by creating a better balance between the beginning and the end of the sentence. The result is again that a prominent Subject is moved into the focus position at the end of the sentence. It is often called a 'stylistic inversion'. Like it-extraposition, SVI shifts a constituent towards the end of the sentence. An interesting difference between Subject-Verb Inversion and Subject-Aux Inversion. SVI changes the order of Su and V by moving Su rightwards, placing it after V in the VP, SAI reorders Su and aux by moving aux (+ tense) leftwards out of the VP, placing it in front of Su.

Compare:

SVI: [s NP/Su [vp ... V ...]] => [s--- [vp... V ... NP/ Su]]

SAI: [s NP/ Su [vp Aux (+Tns) V ...]] => [s Aux (+ Tns) NP/ Su [VP V ...]]

Lesson 36

HEAD MOVEMENT III

Topic- 211: There-insertion

Consider example:

- (51) There were three shops in the village.

Bracketing the constituents and identifying the Subject, you will find that there are two constituents which seem to be likely candidates for Subject status:

(a) the word *there*, which seems to occupy the Subject position, as is confirmed by Subject-Aux Inversion:

- (52) Were there three shops in the village?

(b) the NP *three shops*.

Support for this can be found in the fact that this NP and the verb have agreement. Compare:

- (53) There was one small shop in the village

Has this sentence then got two Subjects?

If we look at it more carefully, we find that it corresponds to a more basic pattern:

- (54) Three shops were in the village.

This is an Su (V-Ac] sentence. (51) and (54) can be related by a movement operation, which shifts the Subject NP in (54) rightwards (SVI) to the position after *be*:

- (55) --- were three shops in the village.

Into the vacated slot we then insert a dummy element: *there*. *There*-insertion in (55) gives us (51) above. Like it, *there* serves to announce that the notional Subject is to follow. *There*-sentences, also called existential sentences, are often used to introduce new elements into the discourse, as the following text illustrates:

Text

There was a university in New England where the students operated a bank of term papers and other homework assignments. There were papers for every need: there were papers for an A grade, papers for a B and papers for a C.

In this university there was also a student who had spent the weekend on pursuits other than homework; he went to the bank. He drew out a paper for C, retyped it, and handed his work in. In due course he received it back with comments in red

ink and they were as follows: 'I wrote this paper myself twenty years ago. I always thought it worth an A, and now I am glad to give it one'.

Adapted from Christian Brann, Pass the Port.

There-insertion is usually only possible with indefinite Subjects and the verb used is normally be (either the lexical verb be, or one of the auxiliary uses of be): lexical verb be:

- (56) A mouse is in the kitchen. There is a mouse in the kitchen.

Especially with the lexical verb be the versions with their sound far better than those without there.

progressive be:

A man is waiting for you. There is a man waiting for you.

passive be:

- (58) Some money has been stolen. There has been some money stolen.

Topic- 212: Extraposition from NP

A further method of balancing the sentence is by extraposition from an NP. Above we have seen that heavy Subjects may be moved towards sentence-final position. For example:

- (69) It is a scandal that they have allowed this practice for so long.
- (70) In this area live fishermen, painters, ironmongers, coalminers, and unskilled workers.

The tendency to move heavy constituents to the right is very general in English.

A few more examples below.

Consider, for example, (71) and (72):

- (71 a) Did you see that man with those funny clothes yesterday?
- (71 b) Did you see that man yesterday with those funny clothes?
- (72a) A new book by Professor Winters which deals with the suppression of women in the fifteenth century has just appeared.
- (72b) A new book by Professor Winters has just appeared which deals with the suppression of women in the fifteenth century.

Moving constituents out of an NP is also called extraposition, more specifically extraposition from NP. operation in the (b)-sentences, we see that it splits up the NP into two parts:

(73) [NP that man ... [PP with those funny clothes]]

The PP with those funny clothes is not an independent constituent: it cannot be the focus element in a cleft sentence. We can say that NPs may be discontinuous: their components may be separated by intervening material. To determine GFs in sentences such as (74)-(77), it is always best to restore the sentence to its basic word order first, since separated components of a constituent really make up only one constituent, with one GF. In (74), for example, the Object NP in the basic pattern is:

- (78) [NP an anonymous letter which rather shocked me]

Topic- 213: Heavy object-shift

Now consider (79a) and (79b):

- (79a) You will give him a sleeping tablet every night.
- (79b) You will give him every night two sleeping tablets, a cup of hot chocolate, and a glass of milk:

Sentence (79a) has the following pattern: NP/Su [V-NP/ Oi-NP/ Od- NPIA]. (79b) is very similar, but the Direct Object NP has now been shifted rightwards to sentence-final position. Again this is a stylistic movement. The NP/ Od in (b) is rather long and heavy, and it is exactly such long and heavy Object NPs which tend to be moved rightwards. Here is another example from P.G. Wodehouse, Lord Emsworth and Others:

- (80) Poskitt was a man who brought to the tee the tactics which in his youth had won him such fame as a hammer thrower.

In the following examples, too, the Od has been shifted rightwards. Bracketing the Direct Object NP, note that if you replace the heavy NP by a short one (e.g. a pronoun), there is no need to extrapose it to the end of the sentence. The PP with those funny clothes is not an independent constituent: it cannot be the focus element in a cleft sentence, for example. We can thus say that NPs may be discontinuous: their components may be separated by intervening material.

Topic- 214: Clefting

Clefting is a device which is used to focus on a particular constituent in the sentence. The process of clefting involves extracting a constituent from its basic position in the sentence and putting it in a more prominent position. Consider the ways in which clefting could be applied to sentence (1):

- Jane gave this book to Bill on Saturday.
- The cleft versions of (1) include the following:
 - (2) It was on Saturday that Jane gave this book to Bill.
 - (3) It was to Bill that Jane gave this book on Saturday.
 - (4) It was this book that/which Jane gave to Bill on Saturday.

- (5) It was Jane that/who gave this book to Bill on Saturday.

The focus element may be either an NP ((4) and (5)), a PP ((2) and (3)) or an Adjunct clause ((7)). As the examples show, the focus element is surrounded by: It is/ was. .. That/ who/ which -clause. Although these that/ who/ which -clauses are very similar in appearance to relative clauses, they are not quite the same. One difference is that normal relative clauses usually serve to postmodify Heads of NPs, i.e. Ns, not PPs or Adjunct clauses. The relative clauses in cleft sentences can 'postmodify' PPs and Adjunct clauses.

Another difference is that in the common type of relative clauses the that/ who/ which-element cannot be left out if it is the Subject of the clause, but in cleft sentences that, who, or which can be omitted in colloquial style if it is the Subject of the clause.

- (8a) The girl who gave the book to Bill is my sister.
- (8b) *The girl gave the book to Bill is my sister.
- (9a) It is Jane who gave the book to Bill.
- (9b) It is Jane $\emptyset$ gave the book to Bill.

Topic- 215: Pseudo-clefting

Clefting is a useful way of making clear what you are focusing on, but it is not possible, for example, to make the verb the Head of a cleft construction. An alternative way of focusing on constituents, which is applicable to verbs and their Complements. For example:

- (10) What Jane did was give the book to Bill on Saturday.
- (11) What the waiter did was open the beer-tins first.

We focus on the VP give the book to Bill on Saturday, or open the beer-tins first by putting it sentence-finally and placing the string: what NP/Su did was . . . before it. This process is called pseudo-clefting. Some more examples:

- (12) What Bill will do is have a quick shower before he leaves.
- (13) What Anne did was open the box and take out the gun.
- (14) What I am doing is trying to clear up this mess.

Pseudo-clefting is not only used for focusing on the VP as a whole. Other constituents can also be focused on by pseudo-clefting. For example:

- (15a) I need a sandwich and a coffee most at this time of the day.
- (15b) What I need most at this time of the day is a sandwich and a coffee.

What . . . day in (15b) is a free relative clause functioning as Subject. It may also be a Complement (Pc):

- (15c) A sandwich and a coffee is what I need most at this time of the day.

Topic- 216: Substitution by pro-forms

Constituents as a whole may at times be replaced by substitute forms, also called pro-forms. Consider:

- Bill's sister announced the news of her marriage the day before yesterday.

In this sentence the NP Bill's sister may be replaced by she; the news of her marriage by it ;the day before yesterday by then ;

- (2)[S [She] [vp announced [it] [then]]]

She and it are personal pronouns; then is an adverb replacing a time Adjunct. She, it and then are examples of pro-forms. Interrogative pronouns (who, what, which) and interrogative adverbs (such as when, where, why, how) are also used to replace constituents. Take for example:

- (3) What did Bill's sister announce the day before yesterday?

The answer to this wh-question may be: 'The news of her marriage'. We have seen that wh-pronouns also serve to introduce relative clauses (clauses which postmodify the Head of an NP), as in:

- (4) The girl who announced the news was Jane.

This occurrence of who is also a pro-form. Another important replacive device which may be used to avoid repetition is the string do so, as in :

- (5) Mary bought her wedding dress in London and so did Jane.

The string do so substitutes for the VP bought her wedding dress in London. If the VP contains auxiliaries, these may also be used as substitutes for the entire VP:

- (6) Mary may buy her wedding dress in London and {Jane may too/ so may Jane} .
- (7) Susan may already have left and {Anne may have too /so may Anne }

It is, of course, possible to rewrite the text above, supplying for each pro-form the elements it replaces. However, this would make the text repetitious and over-explicit: there would be a great deal of redundant information. It is a matter of economy to use substitute forms for elements or information already known to the reader or hearer.

Lesson 37**A-MOVEMENT****Topic- 217: Ellipsis**

Not just pro-forms that are used to economise in language. For example, speakers often reduce redundancy by leaving out superfluous information, i.e. information which can easily be recovered from the situation or the context. This phenomenon is called ellipsis. Elliptical sentences depend for their interpretation on what precedes them in discourse.

Thus, in:

- A: When did Susan leave?
- B: At two o'clock.

We may assume that in the 'response sentence' the string Susan left has been omitted. The omission from speech or writing of a word or words that are superfluous or able to be understood from contextual clues. It is very rare for an ellipsis to occur without a linguistic antecedent.

Topic- 218: Subject Deletion

Consider the following sentence:

- (2) The doctor checked the pulse of the patient who was lying unconscious and looked worried.

The sentence is compound-complex: it contains two coordinated clauses, the second of which is looked worried, but this is incomplete as it stands: looked worried is a VP without NP/Su. Can you supply the missing Subject? Intuitively, we know that the Subject of looked worried must be the doctor, or he: ... and he looked worried. It is characteristic of clauses coordinated by and, or and but that the Subject of the second clause may be deleted if it is co-referential to the Subject of the first clause. Other examples are:

- (3) John sat down and (he) told us the news.
- (4) Susan will sing a song or (she) will play the piano.
- (5) Bill arrived at eight but (he) had to leave immediately.

Topic- 219: Verb Gapping & Backward Gapping

We have seen how the second of two identical Subject NPs may be deleted in coordinate clauses. Similarly, a verb may be omitted if it is identical with a preceding verb:

- (6) John was eating cornflakes and Mary was eating porridge.

In (6) was eating is repeated, but it may be deleted:

- (7) John was eating cornflakes and Mary --- porridge.

This kind of deletion is called verb gapping. Note that you can only gap over adjacent clauses:

- (8) John was writing an essay, Bill was reading a novel and Anne a short story.

In (8) the gapped V in the third clause must be was reading ;it cannot be was writing. Let us look at another example with such parallelisms:

- (9) John liked the film about flies and Mary disliked the film about flies.

Here the V differs, but the NPI Od is identical: the film about flies. We can replace (9) by (10):

- (10) John liked, and Mary disliked, the film about Gandhi.

Sentence (10) is an example of gapping in reverse: the NP/ Od is deleted in the first clause in anticipation of the identical NP/ Od in the second clause. This process is called backward gapping. Further examples:

- (11) Mary wrote, and Susan typed out, the essay.
- (12) Jim earned, and his wife spent, the money.
- (13) Mary bought, and Peter immediately sold, the shares.

Apart from reducing redundancies (identical Object NPs), the process of backward gapping also throws light on the contrast between the verbs in the clauses. The process of gapping is not sentence-bound: its domain is not restricted to one clause/sentence, but also involves the immediate context: it is discourse-bound.

Topic- 220: Non-finite clauses raising and control introduction

Already discussed Sentences may be embedded inside other sentences. In (1), for example, there is an Object clause (S⁻) inside an S:

- (1) [s [NP The Vicar] [vp said [that [s evil communications corrupted good manners]]]]

In (1) S⁻ is a finite clause. However, subordinate clauses are not always finite. Compare (2a) and (2b):

- (2a) The Vicar believed that evil communications corrupted good manners.
- (2b) The Vicar believed evil communications to corrupt good manners.

In (2a) there is a finite subordinate clause, as in (1). In (2b) the subordinate clause is non-finite: its VP contains a to-infinitive. The function of Object may also be realised by a clause containing a VP with a bare infinitive, an-ing participle or an -ed participle:

- (3) I heard Mary sing carols.

- (4)I heard Mary singing carols.
- (5)They found their house occupied by squatters.

Finally, an Object clause may also be verbless: in such cases there is a predicate relationship between the clause elements:

- (6)I consider him guilty of this crime.

We shall look more carefully at the structure of sentences with subordinate non-finite clauses.

Topic- 221: Non-finite clauses: raising and control

Let's identify the complex sentences.

(1)The book shows how children believe they are responsible for everything in their lives.

(1)[The book shows] how children believe they are responsible for everything in their lives.

Bracket the main constituents in (1).Is the sentence simple, compound or complex? What is the Od of shows? In (1) the Od is realised by a wh- clause. The wh-element how occurs in COMP. The subordinate clause itself is also complex: it contains another Object clause: the Od of believe is:

[s (that) [s they are responsible for everything in their lives]]

If try diagram going from top to bottom you will find three S nodes:

- the main clause: S1
- the first Object clause: of shows: S2
- the second Object clause: of believe: S3 •

What sentences (4)-(10) have in common is that the superordinate verb in each case (show, believe, think, wish, etc.) takes a finite clause as Od. Finite clauses always have an explicit Subject. We also see that the sub-ordinator that may sometimes be deleted (cf. (1), (5), (7), (10).

Topic- 222: Non-finite subordinate clauses with lexical subjects

Compare (11a) and (11b):

- (11a)But I did not wish (that) the book should become a family saga.
- (11b)But I did not wish (for) the book to become a family saga.

Bracket the main constituent *s* in (11a) and (11b). Notice that *wish* takes a clause as its *Od* in both examples, but in (11a) the Object clause is finite (cf. 5.2.2), while in (11b) it is non-finite. Let's show how the sentences in (11) should be bracketed:

- (12a) [*s* [NP *I*] [vp did not wish [*s*⁻ { that / $\emptyset$ } [*s* the book should become a family saga]]]]
- (12b) [S [NP *I*] [did not wish [*s*⁻ {for/ $\emptyset$ } [*s* the book to become a family saga]]]]

The subordinate clauses in (12a) and (12b) differ in several respects:

(a) The finite subordinate clause in (12a) is introduced by the complementiser *that* in COMP. That is optional (cf. $\emptyset$). The non-finite subordinate clause in (12b) is introduced by the complementiser *for*, which is optional.

(b) The VP of (12a) is finite; that of (12b) is non-finite.

(c) If we insert a pronoun in the Subject position of an embedded finite clause, the pronoun will take the Subject form:

- (13a) I did not wish that they/ she/ he/ it should...

But if we use a pronoun as the Subject of a non-finite clause, the pronoun will take the Object-form:

- (13b) I did not wish (for) them/ him/ her/it to ...

By analogy, the book in (12a) is assumed to be in the Subject form, and the book in (12b) in the Object form. The complementiser *for* in (14) is obligatory, while it is optional in (15): *for* may be inserted.

- (14) People preferred very much for these things to be forgotten.
- (15) I did not intend the book to be taken as an oracle.
- (16) But people do not believe these things to be real.

The non-finite Object clause in (16) can not take *for*:

- (16a) *But people do not believe for these things to be real.

On the basis of these examples we can draw the following conclusions:

(a) Some verbs, like *prefer*, take non-finite clauses with the complementiser *for*;

others, such as *believe*, take non-finite clauses without *for*. In the latter case the COMP of the non-finite clause is empty:

- (16b) But people do not believe [*s*⁻ [*s*⁻ these things to be real]]

(b) The complementiser *for* may be deleted if it is immediately preceded by a *V* such as *prefer*. If *prefer* (or a similar *V* allowing *for*) is separated from the complementiser by intervening material (for example by an Adjunct), then *for* is no longer deletable like the following.

(c) People preferred very much *for* these things to be forgotten.

The non-finite clauses discussed above all occur as Complements of *Vs*; they are all Object clauses.

Non-finite clauses may also be Subjects:

- (17a) *For me to be writing a book on this subject* was an irrational act.
- (17b) It was an irrational act *for me to be writing a book on this subject*.

In (17a) the non-finite clause is the *Su*; in (17b) it is an extraposed *Su*, linked to the Subject position by *it* (*su*). In (17) the non-finite clauses are again introduced by *for* in COMP, but *for* is not deletable here.

- (18a)**Me to be writing a book on this subject* was an irrational act.
- (18b)*It was an irrational act *me to be writing a book on this subject*.

Lesson 38

AGREEMENT, CASE & A-MOVEMENT-I**Topic- 223: Non-finite clauses without lexical subjects**

Now consider (19a) and (19b) below. What is the difference between them?

- (19a) I did not wish (for) the book to become a family saga.
- (19b) I did not wish to write a family saga.

Both clauses have an embedded Object clause, and in each case the embedded clause is non-finite. In (19a) the Subject of the embedded clause is lexically realised by an NP, the book ; the complementiser *for* is optional. In (19b), on the other hand, the embedded clause lacks an overt Subject. The Subject position is not lexically filled, and we shall indicate this by parentheses ().

Furthermore, the COMP in (19b) must not be filled by *for*:

- (19c) *I did not wish *for* to write a family saga.

The structure of (19b) is indicated in (20):

- (20) I did not wish [_S [_S () [vp to write a family saga]]]

What does (19b) mean? Of course, it means 'I did not wish that I should write a family saga'. () in (20) is to be interpreted as 'I'. We shall say that the non-lexical Subject position () is controlled by the Subject of the higher clause. It is an example of control. We return to control later.

Topic- 224: FOR ... TO FILTER

It is not just non-finite Object clauses that may be lacking an overt lexical subject. Subject clauses, too, may have a non realised subject:

- (21a) To be writing a book on this subject was irrational.
- (21b) It was irrational to be writing a book on this subject.

Compare (21a) and (21b) with (17a) and (17b). Observe that there is no overt Subject in the embedded clauses of (21a) and (21b), while there is one in the embedded clauses of (17a) and (17b). We assign the Subject clause of (21a) the following structure:

(22) [_S [_S () [VP to be writing a book on this subject]]]

As in (19b), the complementiser *for* is not allowed to occur here. In other words, the string [_{COMP} *for* () *to*] is not allowed in English: it must be filtered out. The complementiser *for* must

not immediately precede to. The rule which disallows the string for ... to is more general than shown above. Take, for example, the sentence:

- (23a) You wished for which book to become a success?

If wh-movement applies to this sentence we arrive at (23b):

- (23b) *Which book did you wish for --- to become a success?

but (23b) is ungrammatical, since it contains the string for ... to; for must be deleted:

- (23c) Which book did you wish to become a success?

Topic- 225: Raising

Raising with passive verbs

Consider (1a) and (1b):

- (1a) I intended (for) the political parts of the book to reveal many things.
- (1b) The political parts of the book were intended to reveal many things .

The higher clause in (1a) is active (I intended); that in (1b) is passive (were intended). In (1a) the NP the political parts of the book appears as the Su of the embedded S: In (1b) the NP the political parts of the book is the NP/ Su of the higher clause: it is the Subject of were intended. Furthermore, it would seem that the embedded S in (1b), to reveal many things, lacks an overt Subject. (1b) is the passive counterpart of (1a).

Passivisation results in the suppression of the original Subject of intend (I), and the leftward movement of the Su of the embedded non-finite clause (the political parts of the book), which is now placed in the Su-position of the higher clause, leaving a gap at its original position:

For example;

- (3a) [s I [vp intended [s⁻ [s⁻ the political parts of the books to reveal many things]]]]
- (3b) [s [NP The political parts of the book] [vp were intended [s⁻ [s--- to reveal many things]]]]

Again, for must not be present in (3b), since this would result in a string for. . . to. Let us look more closely at the properties of the NP the political parts of the book in (3b). We have seen that it is the grammatical Subject of the higher passive verb (were intended). Secondly, it is also related to the Subject position in the embedded clause

We claim that passivisation triggers movement of the NP/Su out of the embedded clause into the higher clause, leaving a gap ---. The NP/ Su of the higher clause binds the Subject position

(more specifically the gap) in the lower clause, in the same way that an antecedent NP binds a reflexive pronoun. The operation of moving a Subject out of a non-finite clause into the Subject position of the next clause up is called raising.

Topic- 226: Raising II

Let us return once more to (3b) above and consider one further problem.

- (3b)[s [NP The political parts of the book] [vp were intended [s⁻ [s--- to reveal many things]]]]

We know that the function of the NP the political parts of the book is that of Su of the higher clause. Now what is the function of the embedded non-finite S: [s--- to reveal many things]? It is obviously not the Subject of were intended., though it is related to the Subject by the gap---. Structurally the non-finite clause appears in the position of a Verb Complement, parallel to an Object clause (3a)).

- (3a) [s I [vp intended [s⁻ [s⁻ the political parts of the books to reveal many things]]]]

It might thus be justified to treat the clause as a 'retained ' Object clause: the non-finite clause is a remnant of the original Object clause. Passivisation in this case does not affect the whole Object clause, but only the Subject of the Object clause, leaving the rest of the clause in its original position following the verb in the higher clause.

The passives of verbs such as intend, think, expect, say etc. all trigger raising of a NP/ Su from an embedded non-finite S, as can be seen in the following examples:

- (4) They are thought to be acts of the imagination.
- (5) John is expected to be back soon.
- (6) They are said to be leaving at dawn.
- (7) Anne was found to be guilty of high treason.
- (8) This was felt to be the only solution.
- (9) She was considered to be the best player in the team.
- (10) Mary was seen to steal the money.

It is very important to observe that the gap left after movement cannot be filled by another lexical element. Compare the examples in (11) and (12):

It is very important to observe that the gap left after movement cannot be filled by another lexical element. Compare the examples in (11) and (12):

- (11a) The political parts of the book were intended --- to reveal many things.
- (11b) *The political parts of the book were intended various chapters to reveal many things.
- (12a) Who did the terrorists assassinate ---?

- (12b) *Who did the terrorists assassinate the ambassador?

Compare also (13a) and (13b):

- (13a) A conspiracy was believed to exist in the country.
- (13b) There was believed to exist a conspiracy in the country.

In (13a) the Subject position of the main clause is occupied by the NP a conspiracy. As in the examples discussed above, this Subject NP has been raised out of the embedded clause, leaving a gap:

- (14a)[NP A conspiracy] was believed [s⁻ [s --- [vp to exist in the country]]]

In (13b) the GF of Su in the main clause is realised by there. We shall assume, again, that there originates in the lower clause as the Subject of exist, and has been moved into the main clause Subject position:

- (14b)There was believed [s⁻[s ---to exist a conspiracy in the country]]

There binds the gap (---) in the subordinate clause. We have seen before that there can only function as the Subject of existential sentences. In (14b) we see that such a Subject of a non-finite existential sentence may also be raised. So far we have concentrated on raising from non-finite clauses with to- infinitives. The process also applies, however, to clauses with participles in -ing and -ed., and to verbless clauses:

Topic- 227: Raising with intransitive verbs and with adjectives

we have given examples of raising triggered by passive raising verbs:

- (18) John is believed [s [s--- to have left already]]

Certain adjectives in English may also trigger raising. For example: The pattern in (18) can also be found with non-passive, intransitive verbs:

- (19) John seems to have disappeared.

(19) corresponds to (20):

- (20) It seems that John has disappeared.

In (20) John is the Subject of the embedded finite clause that John has

Disappeared. It -extraposition is obligatory here:

*That John has disappeared seems. In (19) we also want to relate John to the Subject position of the sub-ordinate non-finite clause. This can be done if we assume (as in (18)) that the NP John originates as the Subject of the subordinate clause and is then raised to the Subject position of the main clause:

- (19a) --- seems [s [s John to have disappeared]]
- (19b) John seems --- to have disappeared.

Some comments are necessary here.

(a)the embedded clause must not be introduced by the complementiser for, because this would give rise to the sequence for ... to.

(b)(19a) as it stands could not take it as an anticipatory NP/su. If seems takes a non-finite clause, it-extraposition is impossible and raising is obligatory.

Conversely, if seems takes a finite clause, raising is not possible and it-extraposition is obligatory.

(23) John is likely --- to propose to Mary.

Here John is the grammatical Subject of is likely, which binds the Subject position of the subordinate non-finite clause. The clause is interpreted as 'John will propose to Mary'.

Topic- 228: Concluding raising

We have seen that raising verbs and raising adjectives share the following general properties:

(a)Raising only applies to the NP/Su of non-finite clauses.

(b)There is a gap after the V or Adj, which cannot be filled by a lexical NP:

- (27) They seem/ are likely/are said --- to have left.
- (28) *They seem/ are/likely/are said the children to have left.

The gap is bound by the Subject NP of the main clause.

(c)There is usually a paraphrase with a finite clause and extraposition:

- (29) It seems/is likely/ is said [s⁻ that they have left]

(d)In raising patterns the Subject there may be separated from the existential verb it goes with by raising:

there {verb/adj} --- to be/exist/...

- (30) There seem/ are likely --- to be many children with reading problems.

AGREEMENT, CASE & A-MOVEMENT II

Topic- 229: Control: Non-lexical subjects

Look at the examples below:

(1) I did not wish to write a family saga.

We have already pointed out that such sentences contain a non-finite Object clause, whose Subject is not lexically filled. We have adopted the following convention for marking such non-lexically realised Subject positions:

- (2) I did not wish [s^- [s () [VP to write a family saga]]]

We have also seen that non-finite subordinate clauses may have a lexical Subject.

- (3) I did not wish [s^- (for) [s the book to become a family saga]]

The complementiser *for* is allowed in (3), but it is disallowed in (2), because the string *for ... to* is not grammatical in Standard English. In (1), but not in (3), an NP outside the subordinate clause controls the Subject position in the subordinate clause. The non-lexical Subject is controlled by the main clause Su/ NP (I): () is interpreted as 'I'. Reflexivisation can be used to support the analysis proposed here. Consider, for example:

(4) I want [s^- [s () to enjoy myself]]

Myself is a reflexive pronoun and must be bound within its clause. We assume that in (4) myself is bound inside the subordinate clause by (), the non-lexical Subject, which in turn is controlled by the higher Subject I. This may be represented as follows:

(5) I want [s^- () to
enjoy myself

Compare (5) with (6):

(6) I want [s^- you to enjoy {yourself /*myself}

Topic- 230: Control vs raising

It is important to realise that the controller NP in (5) above is both the underlying and the surface Subject of *want*; it is not the case that I has been moved into that Subject position by a raising transformation: *want*, *decide*, *wish*, *hope* are control verbs, not raising verbs. In order to verify this claim, let us compare sentences like (1) above with raising patterns. We have seen that intransitive raising verbs (*seem*, *appear*, etc.) may appear in the string:

NP-seem --- to-infinitive, where --- cannot be filled by a lexical NP:

- (7a) They seem --- to have left.
- (7b) *They seem the children to have left.

Secondly, raising verbs usually have a paraphrase with a finite clause and it-extraposition:

- (8a) They seem to have left.
- (8b) It seems that they have left.

And, thirdly, in there-sentences the Subject there may be separated from its existential verb (be, exist, etc.). This can be accounted for by assuming that there has been raised, and has thus been moved away from the existential verb. For example:

- (9) There seem --- to be many children with reading problems.

Let us see what happens if we apply the same three tests to the string: NP- want-()-to-infinitive, where the empty Subject position () is controlled by the Subject NP. In the first place, we often have the option here of filling or not filling the empty Subject slot:

- (10) I want {()/ You} to leave

A lexical Subject NP may occur in the subordinate non-finite clause. Secondly, there is no paraphrase with it-extraposition:

- (11a) I want to go.
- (11b) *It wants that I should go.

And thirdly, there are no sequences like :

- (12) * There want () to be many children with reading problems.

Let's sum up the results of these tests.

The first column gives the way in which the tests work for raising verbs like seem, the second shows the results of applying the tests to control patterns with verbs such as want:

		Raising	Control
A	lexical Subject in non-finite clause	no	Yes
B	it-extraposition paraphrase	yes	No
C	there V to ..	Yes	No

What raising and control have in common is that both apply to non-finite clauses: raising operates from non-finite clauses, and control into non-finite clauses. Only non-finite clauses may lack an overt NP/ Su (indicated by --- or ())

Topic- 231: Subject and object control

- (13) He promised her to mow the lawn.

- (14) General Dyer told his troops to fire on an unarmed crowd in the streets.

Both sentences have the pattern Su [V-Oi-Od], unlike sentence (1), which has the pattern Su [V-Od]. What is the difference between (13) and (14)? In (13) the non-lexical Subject position in the subordinate clause is controlled by the Subject NP He, not by the NP/Oi her, whereas in (14) it is the NP/Oi his troops which is the controller, not the NP/Su General Dyer. (13) and (14) may thus be bracketed as follows:

- (15) [s [NP He] [VP promised [NP her] [s⁻ () to mow the lawn]]]
- (16) [S [NP General Dyer] [VP told [NP his troops] [s⁻ () to fire on an

unarmed crowd in the streets]]]

The type of verb in the main clause determines whether the NP/Su or the NP/ Oi will be the controller. Verbs like promise have Subject control, whereas verbs like tell, order, ask, force, persuade, allow, encourage have Object control. Here are some example of control patterns like 16

(17) He persuaded her to buy some more books.

(18) The boys did not allow the girls to take part in the games.

19) We encourage them to try again.

(20) The asked him to resign.

Sentences like (17)-(20) can be passivised: the NP / Oi (her, the girls, them, him) can be made the NP/ Su of the passive sentence.

Taking (17) as an example, the passivisation process may be described as follows:

(21) [s [NP He] [VP persuaded [NP her] [() to buy some more books]]]

Passivisation moves the NP/ Oi her, and leaves the subordinate clause unaffected :

(22) [s[NP She][VP was persuaded ---[s () to buy some more books]]]

We still have a control structure here, but now the Subject she controls the Subject position of the subordinate clause.

Topic- 232: Indefinite control

English also has control patterns which have no overt controller. Take, for example:

- (23) {To write/writing} a book on this subject was an irrational act.

The Subject of the non-finite subordinate clause is empty. We analyse (23) as follows:

- (24) [s⁻ [s () {to write/writing} a book on this subject]] was an irrational act.

The controller in this case might be somebody mentioned in the context, or someone unknown or indefinite. We call this indefinite control. Other examples of indefinite control are:

(25) Is () to say No to nuclear weapons a defence policy?

(26) Oxbridge's grip on the top ranks of the civil service should be broken by () encouraging students at provincial universities () to come forward as potential mandarins. The second control site in (26), unlike the first, is not a case of indefinite control. What is the controller of that second control site? Non-finite clauses may have functions other than Od or Su. They may, for example, be Adjuncts or Modifiers:

- (27) I want to go to Paris to buy some new clothes.
- (28) He received a letter ordering him to leave the country.

Both these non-finite clauses lack an overt lexical Subject NP: the empty Subject position in (27) is controlled by I, and that in (28) by a letter.

Topic- 233: Control adjectives

There are also control adjectives in English, not just control verbs. These adjectives are followed by non-finite clauses with controlled Subject positions:

- (29) I am eager [s⁻ [s () to come to your party]]
- (30) He is reluctant [s⁻ [s () to tell her the truth]]

By applying the three tests mentioned earlier, we can determine that (29) and (30) are control patterns, not raising patterns. For example:

- (31) I am eager for Mary to be at the party too.
- (32) *It is eager that I should come to your party.
- (33) *There is eager to be a party next weekend.

Other examples of control adjectives are:

- (34) He is anxious

() to make his new business a financial success.

- (35) We are delighted () to send you our pamphlets.
- (36) She is afraid of

() hurting his feelings.

Topic- 234: Reanalysis a problem of bracketing

Introduction:

Bring in, bring out, etc.

A single word or string of words may be analysed in two ways. Already seen, for example, that the elements *be* and *have* are structurally ambiguous, in that they may behave like lexical verbs or like auxiliaries. Another instance of double structure in English is that of V-Od strings like *kick the bucket*, can regularly occur in the passive (*The bucket was kicked*), but not in their idiomatic sense:

**The bucket was kicked by the old man* = *The old man died*).

It appears that *kick the bucket*, as an idiom, is no longer a V- Od string, but has been restructured into just a composite intransitive verb. A well-known example of ambiguous patterning is of strings such as *depend on*, *look at*, *give in to* and others, which may be structurally interpreted as intransitive verbs followed by a PP, or alternatively as transitive verbs followed by a Direct Object. These kinds of structural ambiguity result from the process of reanalysis (also called 'restructuring').

Reanalysis changes the structure of a string of elements, so that before and after reanalysis the string will pattern differently. We shall compare the properties of verb + preposition strings with those of some other multi-word verbs, and examine the indeterminacy of their structure.

Lesson 40

SPLIT PROJECTIONS I

Topic- 235: Prepositions and particles

Let's read the text.

The sentences have been numbered here for ease of reference.

- (1) Soon after breakfast Mary Ann brought in The Times.
- (2) Mr. Carey shared it with two neighbours.
- (3) He had it from ten till one, when the gardener took it over to Mr. Ellis at the Limes, with whom it remained till seven; then it was taken to Miss Brooks at the Manor House, who, since she got it late, had the advantage of keeping it.

Let's read the text.

- (4) In summer Mrs. Carey, when she was making jam, often asked her for a copy to cover the pots with.
- (5) When the Vicar settled down to his paper,
- (5b) his wife put on her bonnet
- (5c) and went out to do the shopping.

Adapted from W. Somerset Maugham, Of Human Bondage.

For the purpose of our discussion we here present skeleton versions of the sentences (1) and (5) above:

- (1) Mary brought in The Times after breakfast.
- (5a) The Vicar settled down to his paper.
- (5b) His wife put on her bonnet.
- (5c) She went out.

In each of the sentences above, the lexical verb is followed by another element that is closely linked with the verb: in, down, on, out.

The question is: what is the status of in, down, etc.? Are they prepositions which take a Prepositional Complement (as in: in the vicarage). Do these elements belong to the V, or are they perhaps independent units? Example (1) above may be bracketed as follows:

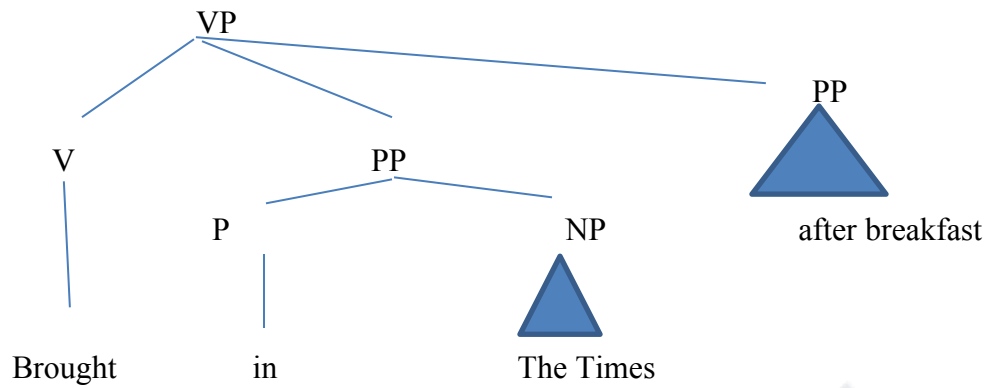
- (6) [s [NP Mary Ann] [vp brought in The Times [pp after breakfast]]]

There can be no doubt about after breakfast. It is a genuine PP: it consists of a preposition after and its Complement NP breakfast. Together the P+NP make up one phrase, a PP.

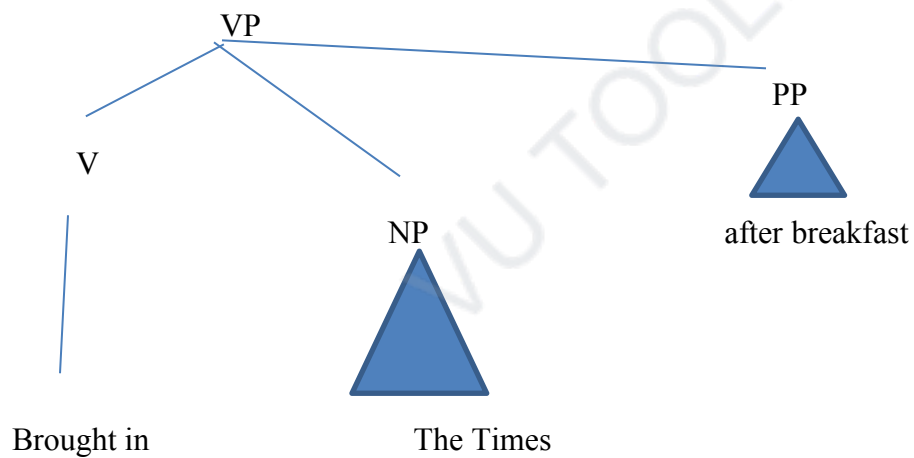
After breakfast is an optional element (an Adverbial Adjunct). According to (7a) in is a P, taking The Times as its Complement. According to (7b) in is somehow part of the verb itself; brought and in seem to make up one unit with word- like properties. In (7c) in is independent of

both brought and of The Times. The three alternatives may be represented by the following tree diagrams

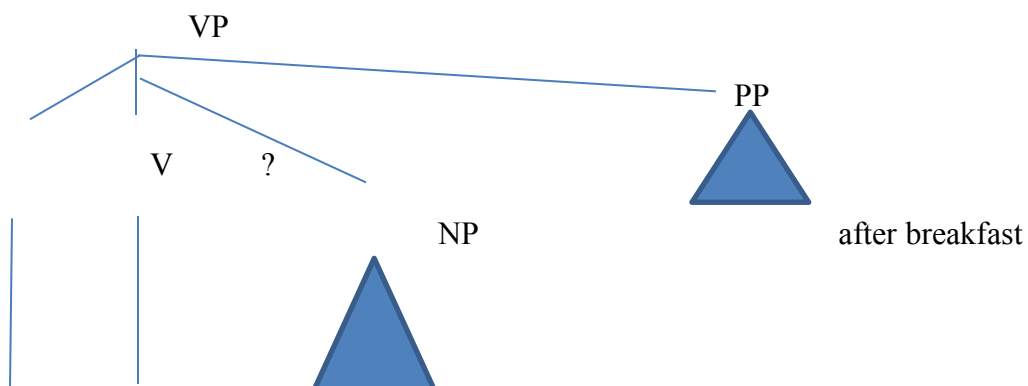
8a



8b



8c



Brought in The Times

The diagrams give description but do not answer our questions.

Topic- 236: Preposition & particles II

Let us consider the behaviour of after breakfast, the optional PP (the brackets around the PP in (8a) -(8c) mark its optionality).

(I) We can front the PP quite easily:

After breakfast Mary Ann brought in The Times.

(II) The PP may serve as the focus element X in a cleft sentence:

It was after breakfast that Mary Ann brought in The Times.

(III) The PP may be replaced by an appropriate Adjunct, here a time Adjunct (then):

Mary Ann brought in The Times then.

(IV) The PP may be questioned by a wh-item like when, where or at what time:

When did Mary Ann bring in The Times?

At what time did Mary Ann bring in The Times?

(V) The PP may be separated from the rest of the sentence by an Adjunct

(Adjuncts can only be inserted between major constituents; they thus mark constituent boundaries):

Mary Ann brought in The Times, as usual, after breakfast.

Tests (I)-(V) show that after and breakfast make up one constituent.

The two elements cannot be separated or inverted, as the following ungrammatical sentences show (tests (VI)-(VIII)):

(VI) *Breakfast Mary brought in The Times after.

(VII) *It was breakfast that Mary brought in The Times after.

(VIII)*May brought in The Times breakfast after.

These three examples show that in this case neither fronting of just the NP, nor clefting with the NP as X, nor changing the word order is possible.

Topic- 237: Preposition & Particles Iii

Let us now turn to the sequence in The Times. Analysis (8a) above suggests that in + The Times make up a PP, just like after breakfast. If the analysis were correct, we would expect some degree of similarity between the two PPs. Notice, however, what happens if we apply the tests (I)-(V) to in The Times:

(I) in+ The Times cannot be fronted together :

*In The Times Mary Ann brought after breakfast.

(II) Neither can in + The Times serve as X in a cleft sentence:

*It was in The Times that Mary Ann brought after breakfast.

(III) In + The Times cannot be replaced by then or there:

*Mary Ann brought {then/there} after breakfast.

Notice, however, what happens if we apply the tests (I)-(V) to in The Times:

(IV) in + The Times cannot be questioned by a wh-word:

{*When/*where} did Mary Ann brought after breakfast.

Nor do we get:

{* In what/* In what newspaper} did Mary Ann bring?

(V) in + The Times cannot be separated from the rest of the sentence by an Adjunct:

*Mary Ann brought, as usual, in The Times after breakfast.

Tests (I)-(V) indicate that *in* and *The Times* are not one constituent. If they are not one constituent, it should be easy to separate the elements or to change the word order. This is indeed confirmed by the following sentences, all of which are grammatical (tests (VI)-(VIII)):

(VI) The Times Mary Ann brought in.

{What newspaper/what} did Mary Ann bring in?

(VII) I was The Times that Mary Ann brought in?

(VIII) Mary Ann brought The Times in after breakfast.

Note, incidentally, that *in* must normally appear in the position on the right of the Object NP if the NP is a personal pronoun:

Mary Ann brought it in.

*Mary Ann brought in it.

Summing up the tests, we see that *after+breakfast* and *in+ The Times* pattern differently in the sentences:

Tests	After+breakfast	in+The Times
(I) Front together	Yes	No
(II) Cleft together	Yes	No
(III) Replace together	Yes	No
(IV) Question together	Yes	No
(V) Adjunct insertion (VI) Front NP	Yes	No
(VII) Cleft NP (VIII) Change order	No	Yes
	No	Yes

The tests show that analysis (8a) above must be wrong: *in The Times* is not a PP.

Topic- 238: Concluding Preposition or Particles I

We are now left with the choice of treating *in* as a particle which goes with *bring* (8b), or as a totally independent constituent (8c). As far as the meaning of the sentence is concerned *bring in* is felt to be a lexical unit, meaning something like 'deliver'. Is there any formal support for this intuition? In other words: do *bring* and *in* behave at all as if they were one constituent? To see this, let us first consider the following sentences:

- (9) John reads *The Times* and Susan reads the *Guardian*.

We have two coordinated clauses here, each with the verb read. In English, there is an operation called gapping, which deletes the second occurrence of a lexical verb in a coordinated pattern:

- (10) John reads The Times and Susan I I I I I the Guardian.

There is a gap at the place of the deletion. Note that PPs are left intact if the verb is gapped. Let's look at the following examples;

- (11) Mary came after breakfast and Bill came after tea.
- (12) Mary came after breakfast and Bill I I I I after tea.

After tea is a PP and is not involved in gapping of V:

- (13) *Mary came after breakfast and Bill I I I I I I I I I I tea.

Consider now:

- (14) Mary Ann brought in The Times and Philip brought in the Guardian.

If we apply verb gapping here, we must also delete in, not just brought:

- (15) Mary Ann brought in The Times and Bill I I I I I I I I I I the Guardian.

The following sentence is ungrammatical:

- (16) *Mary Ann brought in The Times and Philip I I I I I I I I in the Guardian.

This means then that in is not totally independent: it is tightly linked with bring. Together bring and in act as if they were one lexical verb. This suggests that tree diagram (8b) above is the best alternative. We shall say that bring and in make up one lexical unit, consisting of a verb bring and a particle in. Bring in is a phrasal verb, which behaves like an ordinary transitive verb. We propose the following bracketing.

- (17) [s [NP Mary Ann] [VP [V brought in] [NP The Times] [PP after breakfast]]]

Topic- 239: Concluding Preposition or Particles II

In text above, sentence (5b) also contains a phrasal verb: put on. The analysis of that sentence is as follows:

- (18) [s [His wife] [VP [V put on] [NP her bonnet]]]

Now take sentence (5a):

The Vicar settled down to his paper .

Inside the VP we find that V (settle) is followed by down. This string is again a phrasal verb, but this time there is no NP as Complement inside the V P:

- (19) [s[NP The Vicar] [[VP [V settled down] [PP to his paper]]]

Settle down requires the PP to his paper for the meaning intended here (PP/Ac). The Vicar settled down does not mean quite the same. Another instance of a phrasal verb can be found in (5c): She went out.

The phrasal verbs bring in and put up are transitive, and settle down and go out are intransitive. The phrasal verb hand over is:

- (20) She handed over the paper to the Vicar.

is ditransitive.

Trying verb gapping in coordinated structures with put on, settle down, go out, and hand over shows that they constitute one unit. For example:

- (21a) Mrs. Carey put on her bonnet and Mr Carey put on his hat.
- (21b) Mrs. Carey put on her bonnet and Mr Carey I I I I I his hat

Other examples of phrasal verbs are: sit down, take off, switch on, look up, call off, call up, write down, bring up, give back, give in, etc.

Topic- 240: Prepositional verbs I

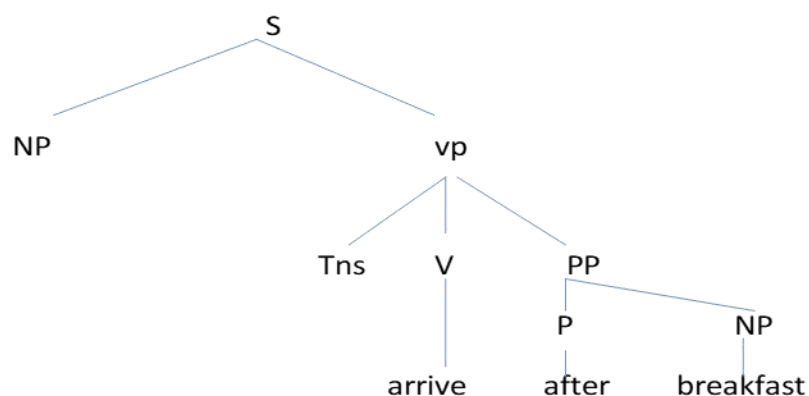
Let's look at the word strings like pray for, look for, hope for and rely on. Let us compare the verbs in the following three examples:

- (1) Mary Ann arrived after breakfast.
- (2) She brought in The Times.
- (3) The Vicar can rely on Mary Ann .

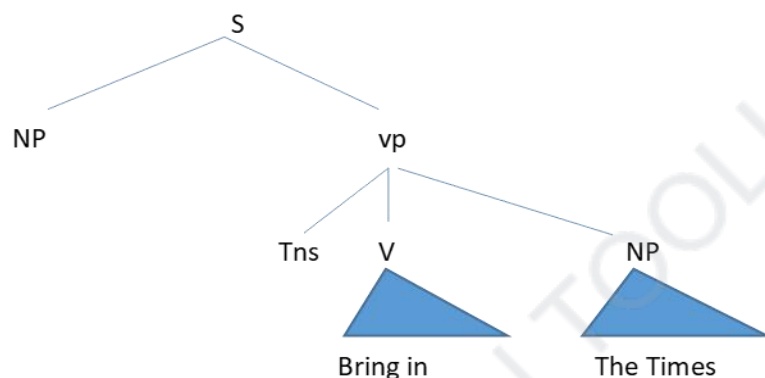
We have seen that after in (1) goes with breakfast rather than with arrived

and we have established that in in (2) structures with brought rather than with The Times.

(1) corresponds to the structure (see (8a) in 6.2):



(2) has the structure (see (8b) in 6.2):



In some respects *rely + on* behaves like *arrive*, whereas in other respects it behaves like *bring in*. To see this, let us apply tests (I)-(VIII) as well as the verb gapping test to (3) (these tests have been applied above to distinguish between V+PP strings and phrasal verbs):

(I) and (VI) Both on Mary Ann (PP) and Mary Ann (NP) can be fronted :

- (4a) On Mary Ann the Vicar can rely.
- (4b) Mary Ann the Vicar can rely on.

(II) and (VII) Both on Mary Ann and Mary Ann may serve as the focus element X in a cleft sentence:

- (5a) It is on Mary Ann that the Vicar can rely.
- (5b) It is Mary Ann that the Vicar can rely on.

(III) On Mary Ann cannot be replaced by the pro-form *then* or *there*

((but Mary Ann can be replaced by *her*):

- (6a) *The Vicar can rely then/ there
- (6b) The Vicar can rely on her.

(VI) On Mary Ann cannot be questioned by a wh-item like when or

Where (but Mary Ann can be questioned by who(m)) :

- (7a) * When/ where can the Vicar rely?
- (7a) Who(m) can the Vicar rely on?
- (7a) Who(m) can the Vicar rely on?

(V) On Mary Ann may be separated from the rest of the sentence by an Adjunct:

- (8a) The Vicar can rely whole-heartedly on Mary Ann.
- (8b) *The Vicar can rely on whole-heartedly Mary Ann.

(VIII) The word order of on Mary Ann cannot be changed into Mary Ann on:

- (9) *The Vicar can rely Mary Ann on.

Verb gapping gives the following result:

- (10a) The Vicar can rely on Mary Ann and Mrs. Carey can rely on Philip.
- (10b) The Vicar can rely on Mary Ann and Mrs. Carey II I I I on

Philip.

- (10c) *The Vicar can rely on Mary Ann and Mrs. Carey I II II I

Philip.

The conclusion is that the structure of the VP with rely on is difficult to determine: according to certain tests the string rely on-NP is like a one-word verb followed by a PP (cf. tests I, II, VIII and gapping. According to other tests it is like a two-word verb followed by an NP (cf. tests III, IV, VI, VII).

Lesson 41

SPLIT PROJECTIONS II

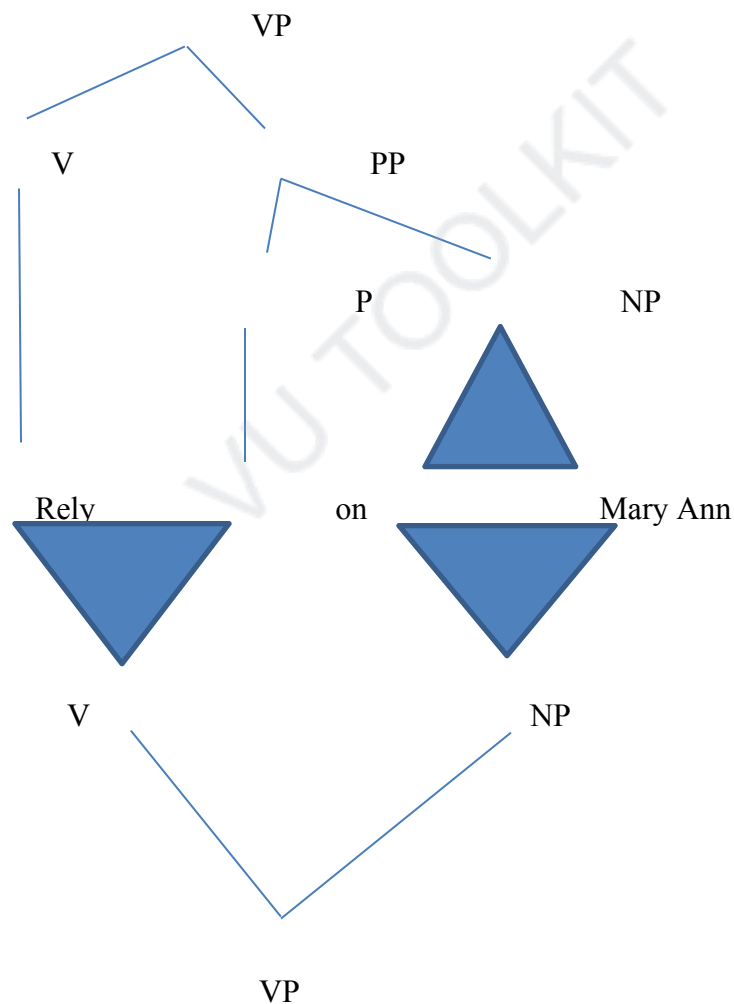
Topic- 241: Prepositional verbs II

In other words, the structure of such V-P-VP strings is indeterminate; it may be either (11a) or (11b) below:

- (11a) [vp (v rely
[PP on NP]]
- (11b) [vp (v rely on] [NP]]

The duality of the structure of rely on ...may be captured by combining two structures into one.

Thus, the structures represented by (11a) and (11b) are combined in tree diagram (11c):



Traditionally, strings such as rely +on are called prepositional verbs. They share grammatical characteristics with V + PP strings as well as with phrasal verbs. Other examples of prepositional

verbs are: look at, look for, look after, call on, call for, deal with, account for, stare at, wait for, etc. In so far as prepositional verbs are two-word verbs they resemble (transitive) phrasal verbs such as bring in, call up, etc. One important characteristic which they share is that both types of verb allow passivisation:

- (12a) Mary Ann brought in The Times.
- (12b) The Times was brought in by Mary Ann.
- (13a) The Vicar can rely on Mary Ann.
- (13b) Mary Ann can be relied on by the Vicar.

Moreover, both types of verb allow wh-question formation with what or who (not when or where,).

For example:

- (12c) What did Mary Ann bring in?
- (13c) Who can the Vicar rely on?

Topic- 242: Prepositional verbs III

We are on the same discussion whether 'rely on' is a particle or a preposition. Let us now look at some differences which are found both in verb plus preposition as a phrasal verb or verb plus prepositional phrase.

Some important differences:

- One point is that the particle of a phrasal verb may be placed before or after the NP, whereas the preposition of the prepositional verb can only occur before the NP (test VIII above).
- Compare (12a) and (13a) with (12d) and (13d):
- (12d) Mary Ann brought The Times in.
- (13d) *The Vicar can rely Mary Ann on.
- One reason for calling *in* a particle in cases like (12d) and *on* a preposition.
- Another difference between prepositional verbs and phrasal verbs is that it is usually fairly easy to insert an Adverbial Adjunct immediately before the preposition, but not immediately before the particle. For example compare:
- (12e) *Mary Ann brought early in The Times.
- (13e) The Vicar can rely whole-heartedly on Mary Ann.
- A third difference is that in relative clauses the particle of the phrasal verb must not precede the relative pronoun.
- Whereas the preposition of the prepositional verb may, but need not, precede the relative pronoun.

Compare:

(12f) The newspaper which Mary Ann brought in ---.

(12g) *The newspaper in which Mary Ann brought ---.

(13f) The girl whom the Vicar can rely on ---.

(13g) The girl on whom the Vicar can rely ---.

So these are some of the differences between a particle and a preposition. A preposition, in certain cases, can move whereas a particle cannot move. In some cases, a particle can move and a preposition cannot move. So, we must keep in our mind all these facts when we are trying to distinguish or determine their place in our syntactic analysis.

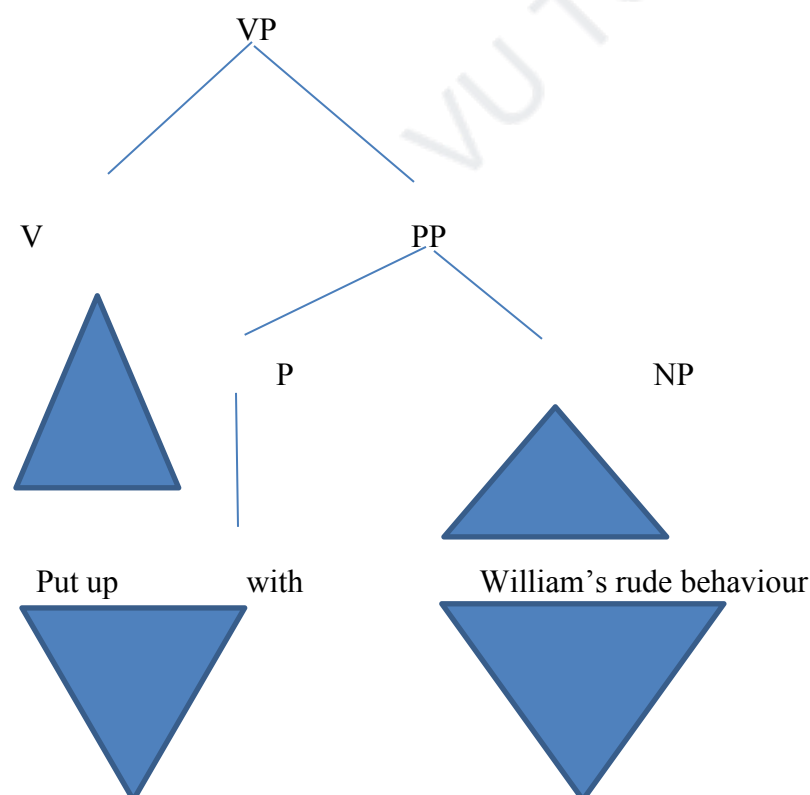
Topic- 243: Phrasal-prepositional verbs and prepositional idioms

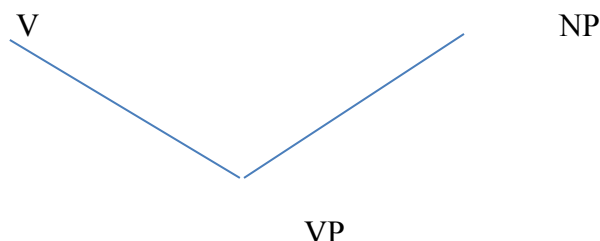
Phrasal-prepositional verbs are combinations of phrasal verbs and prepositional verbs. Examples are: catch up with, come up with, cut down on, give in to, keep up with, look down on, put up with, etc. Like phrasal and prepositional verbs, they may undergo passivisation. For example:

(1a) We have put up with William's rude behaviour too long.

(1b) William's rude behaviour has been put up with too long.

We assume that VP of (1a) has the same dual structure as (3) in 6.3 above (11c)





English also has so called Prepositional idiom. The consist of a verb followed by an NP and a preposition. Examples make mention of make use of, pay attention to, take advantage of, set fire to, etc. These idioms often allow two passives, as a result of their dual structure, as follow:

- (3a) Everyone took advantage of her goodness.
- (3b) Advantage was taken of her goodness.
- (3c)Her goodness was taken advantage of.

What is the structure of the NP take advantage of her goodness? Where, in particular, do the NP advantage and the preposition of belong? The structure of the string might be (4a) (4b)(4c).

- (4a) [VP [V took] [NP advantage] [PP of her goodness]]
- (4b) [VP [v took advantage] [pp of her goodness]]
- (4c) [VP [V took advantage of] [NP her goodness]]

As with prepositional verbs and phrasal-prepositional verbs, the various tests support analysis (4a), or (4b), or (4c). So, again, the structure of the VP is indeterminate. Try some of the tests (fronting, clefting, wh-question formation, verb gapping, etc.) to see whether you agree with our conclusion that there is support for all three analyses.

Topic- 244: Levels of structure

Introduction

In the previous chapters we have looked at constituents at various levels: we have introduced clausal constituents (the sentence/clause. We have discussed phrasal categories such as NP, VP, AdjP, PP, AdvP. We have discussed lexical categories such as N, V, Adj, P Adv. Phrasal categories were directly related to lexical categories in that we argued that each phrasal category has as its Head a lexical category; for example: an NP has an N as its Head. Phrasal categories are sometimes said to be projections of lexical categories. Below we repeat the phrasal categories. We have italicised the lexical category functioning as the Head:

(I) *Noun phrase:*

Det AdjP, AdjP2 N PP, PP2

these nice French students of English from Paris

(2) Verb phrase :

Tns V NP NP

Past buy a car last week

(3) Prepositional phrase:

AdvP P NP

soon after her arrival

(4) Adjective phrase:

AdvP Adj PP

Extremely worried about the future

(5) Adverb Phrase:

AdvP Adv

Very cleverly

We have also shown how the constituents inside the phrase have different GFs: phrases can be described by the following schema:

(6)XP -+ Spec -X- {MOD A}

In the discussion to come we shall look more closely at the internal structure of phrases.

Topic- 245: Flat & layered structures

What are flat structures and layered structures? We have previously described NPs according to the following diagram:

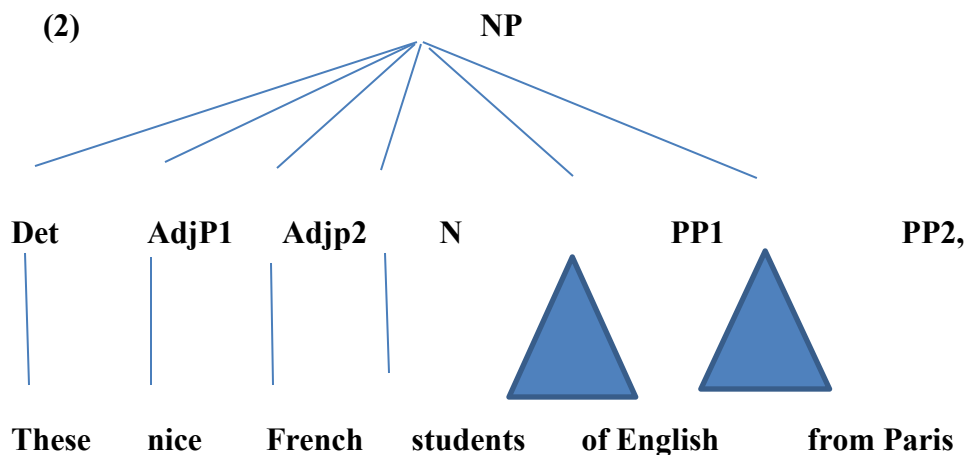
(1) NP :

Category DetAdj P, AdjP 2 N PP1 , PP2

GF Spec Mod (Premodifier) Head Mod (Postmodifier)

thesenice French students of English from Paris

A diagram such as (1) seems to suggest that if we were to use a tree diagram representation, the structure of the NP would look as follows:



In other words, the structure looks 'flat'; NPs are made to look as if they are just strings of elements with various GFs. Tree diagram (2) suggests that the N students takes four Modifiers:

AdjP1 nice

AdjP2 French

PP1 of English

PP2 from Paris

and that all four Modifiers are of equal rank. SO the structure in (2) are arranged to look 'flat'; NPs are made to look as if they are just strings of elements with various GFs. Is the representation such as (2) is semantically adequate?

Topic- 246: Flat & layered structures II

What are flat structures and layered structures? Why is the representation in (2) is semantically adequate or not? Semantically such a representation in (2) is clearly inadequate, since the NP these nice French students of English from Paris does not mean that these students were (a) nice, and (b) French, and (c) of English and (d) from Paris.

Rather, the NP means that these students were (a) students of English,

(b) that 'these students of English' were French,

(c) that 'these French students of English' were from Paris, and that

(d) 'these French students of English from Paris' were nice.

There is thus an internal hierarchy among the Modifiers; of English, for example, is much more closely associated with the N students, than French:

the string students of English is one unit, which is modified by French. In its turn, the string French students of English is modified by from Paris, and finally the string French students of English from Paris is modified by nice. This hierarchy is confirmed by the fact that the order of AdjP1, and AdjP 2 and that of PP1, and PP2 is fairly fixed:

(3) ?These French nice students of English from Paris.

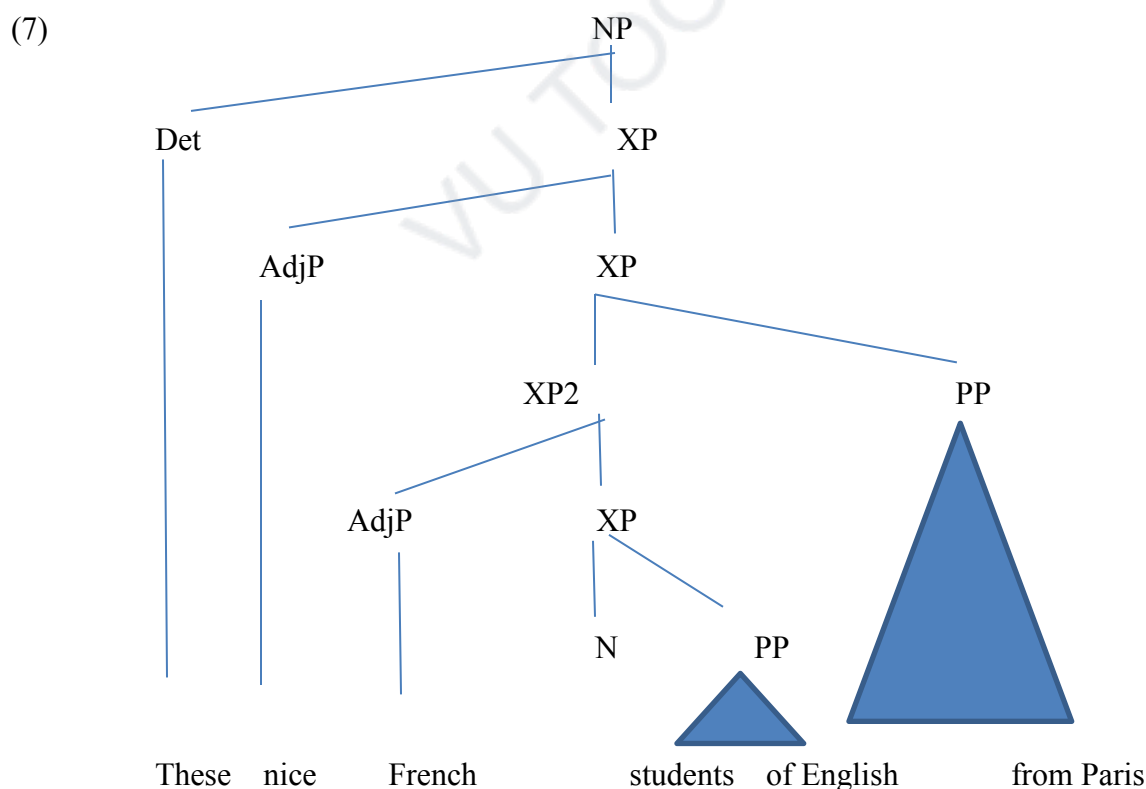
(4) *These nice French students from Paris of English.

(5) *These French nice students from Paris of English.

Using labelled bracketing we arrive at the following build-up:

(6) [NP these [xP1 nice [xp2 [xp3 French [xP4 Students of English]] from Paris]]]

Or in a tree diagram representation:



The diagram in (7) is layered: inside the NP we have levels of structure, indicated provisionally by XP1, etc.

VU TOOLKIT

Lesson 42**A-MOVEMENT (Phases I)****Topic- 247: One-substitution**

Is there any further motivation to support a hierarchical structure such as that in (6) and (7)? One type of argument put forward for constituent structure is that of substitution: pro-forms substitute for constituents. In English there is one type of substitution which typically may affect constituents internal to NPs, called one-substitution. Consider, for example, (8a) and (8b):

(8a) these nice French students and those nice English ones.

(8b) these nice French students and those awful ones.

In (8a) ones stands for students, an N. Obviously, lexical categories are constituents (i.e. units) at some level; they are the minimal constituents of syntax. In (8b), on the other hand, ones stands for French students, i.e. the sequence

[AdjP-N].

Since one-substitution affects this string, it may also be considered a constituent. Provisionally, we may say that one substitutes for either a Head N, or a Head N and one or more Modifiers. As (8b) shows, one-substitution may, but need not, affect AdjPs. Similarly, one-substitution may, but need not, affect PPs (with one major exception):

(9) these nice students from France and those awful ones from Italy

(ones= students).

Let us return once more to the example in (1), renumbered here as (10):

(10) These nice French students of English from Paris.

One-substitution, as we have seen, may affect constituents at more than one level in the representation.

Topic- 248: One-substitution II

- One-substitution, affect constituents at more than one level in the representation. For example, if (10) is conjoined with (10a)-(10d) below, we find that in each case one(s) refers to a constituent at a different level (cf. (7) above):
- (10a) and those ones (ones: nice French students of English from Paris
- = XP1)
- (10b) and those awful ones (ones: French students of English from Paris
- = XP2)

- (10c) and those interesting ones from Boulogne (ones: French students of English = XP3)
- (10d) and those exciting Italian ones (ones: students of English = XP4)

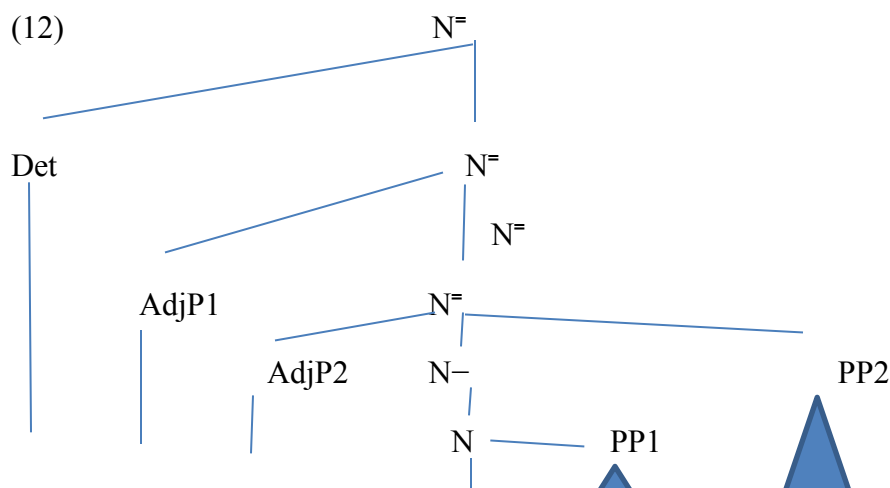
(10a)-(10d) show that the level which one-substitution affects is not fixed. However, consider the ungrammatical (10e):(10e)*and the exciting Greek ones of mathematics from Athens. In the last example ones is meant to replace only the Head N students, leaving the PP1 unaffected. This apparently is impossible: PP1 must be affected by one-substitution. On the basis of the examples above it seems that one-substitution affects the Head N and optionally one or more of any modifiers in the NP, but also that it obligatorily affects constituents of a certain kind such as PP1 in (10).

If we return once more to the example, we see that PP1 of English in fact is not an ordinary Postmodifier. PP1 rather acts like a Complement. The NP students of English corresponds to the VP study English, where English is a complement of V. Like Vs, Ns also may take a Complement, and this Complement (here realised by a PP) is much more closely linked to the N than the Modifiers. Unlike Verb Complements, Noun Complements are syntactically optional.

Topic- 249: Layering

On the basis of one-substitution that there are good grounds for distinguishing two types of PP after the Head N: (1) the ordinary Postmodifiers, (2) the Complement-like elements which are closely linked to the N, like subcategorised categories in VP. One--substitution obligatorily affects the Head N and any Complement-like element. One-substitution may, but need not, involve further Premodifiers and Postmodifiers. The unit consisting of a Head N and its Complements (XP4 above) is oft indicated as N^- (N-bar) or N' (N-prime). N^- is the first projection of N. N^- may further combine with other Modifiers and finally with Det/Spec to form higher levels of structure indicated by $N^=$. The structure provisionally posited in (7), may now be represented as follows:

or in a tree diagram representation:



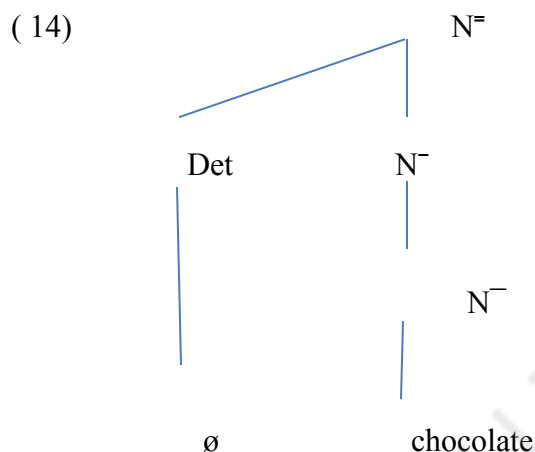
These nice French students of English from Paris

N^- and $N^=$ are projections of N ; the highest N is called the maximal projection; other levels between $N^=$ and the maximal projection are called intermediate projections. Tree diagram (12) can be converted into a labelled bracketing:

(13) [$N^=$ these [$N^=$ nice [$N^=$ [$N^=$ French [N^- students of English]] from Paris]]]

Topic- 250: One-word phrases

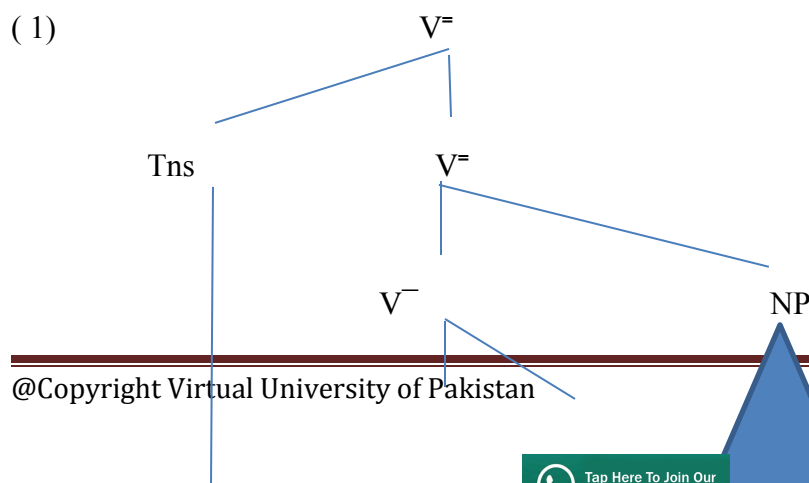
Of course, not all NPs have Premodifiers or Postmodifiers, or N-Complements. However, we have shown that single-word phrases (chocolate, milk, etc.) are also to be treated as NPs. Single-word phrase.

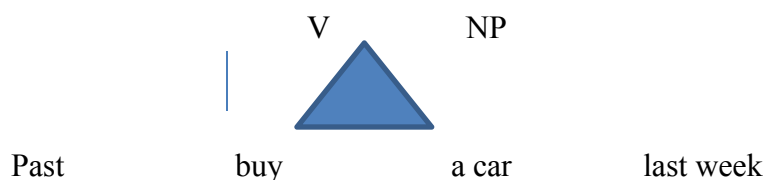


Verb phrases

Layering in VPs

The suggestion put forward for layering NPs can be extended to VPs. Here again we distinguish between the level comprising the Head and its Com-plements (V) and higher projections comprising V^- and any optional Adjunct phrases ($V^=$). For example, the VP bought a car last week has the following structure:





Or in labelled bracketing:

(2)[v⁻ Past [v⁻[v⁻ buy a car] last week]]

Formal evidence

Is there any formal evidence to support the structure above? With respect to NPs we used one-substitution as evidence for internal layering; for VPs we may, by analogy, use do-so substitution to confirm the structure. Again do-so substitution minimally affects the V + Complement(s). Consider:

- (3) John bought a car last week and Mary did so the week before. (did so: bought a car)
- (4) John bought a car last week and Mary did so too. (did so: bought a car last week)
- (5) *John bought a car last week and Mary did so a motor bike the week before. (*did so: bought)

The analogy do-so substitution confirms the structure.

Topic- 251: Prepositional phrases

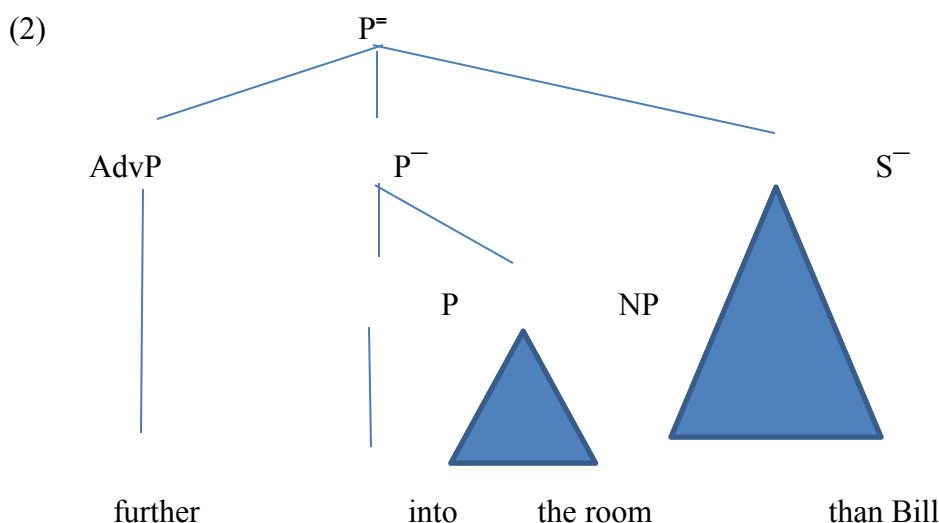
By analogy, the distinction between levels of projection in NP and VP can also be extended to PPs. Again we might distinguish the level of P + Complement (i.e. P⁻) from a higher level of projection in which further Adjuncts or Modifiers are included (P). Take, for example, the PP in (1):

- (1)[P⁻ further [p⁻ into the room] than Bill]

as in

- George ventured further into the room than Bill.

In (1) the room is the Prepc, into is the Head P, and the discontinuous string further ... than Bill may be taken as a reduced form of further ... than Bill ventured, with ellipsis of ventured. If we take further as a Specifier preceding P, then than Bill is directly linked to this Spec, and is clearly less inherently linked to P than the NP the room is. We might present the internal structure of the PP as follows:



The tree diagram is meant to suggest that further and than Bill form a discontinuous unit, and are to be interpreted together.

Topic- 252: Adjective phrases

Complements of Adj_s

We have seen that Adj_s, like V_s, take Complements. Unlike V-Complements, Adj-Complements in English cannot be NP_s, but must be realised by PP_s or S_s. For example:

- (1a)*aware the danger
- (1b) aware of the danger
- (1c) aware that she was in danger

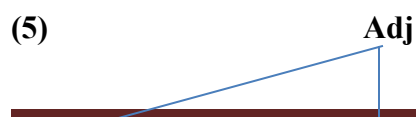
For example:

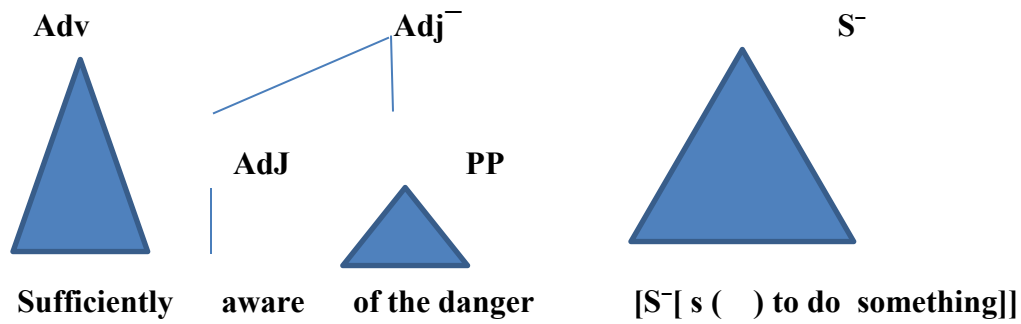
- (2a)*afraid nuclear
- (2b) afraid of nuclear war
- (2c) afraid that they might start a nuclear war

The Adj_s above can be said to subcategorise for a PP introduced by some P, or for an S. Again we might consider Adj_s+ subcategorised PP as Adj_s⁻, and treat further Modifiers or Specifiers as Adj_s⁻-elements. In the AdjPs below the Adj is italicised, and further elements are Adj_s⁻ elements:

- (3)sufficiently aware of the danger to do something
- (4)as afraid of nuclear war as her father

The structure may be represented in a tree diagram (simplified):





Note that the non-finite clause lacks a lexical Subject and also lacks the complementiser *for*: () is a Subject controlled by some other element in S. For example:

(6) Mary was sufficiently aware of the danger () to do something



Adjective phrases also behave in the similar manner.

Lesson 43

AGREEMENT, CASE AND A-MOVEMENT I (Phases II)**Topic- 253: Discontinuous strings**

Above we have introduced discontinuous strings in AdjPs. These are elements which belong together, but which are separated by intervening material. In the following examples such discontinuous strings are italicised:

- (7) He is too nice to do such a thing.
- (8) Susan is more ambitious than her brother.
- (9) This boy is less intelligent than his younger sister.

AdjPs may be split up by the N they modify:

- (10) He is too nice a man to do such a thing.
- (11) You will not find a more ambitious girl than Susan.
- In the examples above the AdjPs are separated by the Head N. In English it is not possible to place an Adj together with its Complement or Post-modifiers before a Head N:
- (12) *He is a too nice to do such a thing man.
- (13) *You will not find a more ambitious than Susan girl.

However, it is possible to find the full AdjP as a Postmodifier:

He is a man too nice to do such a thing.

You will not find a girl more ambitious than Susan.

Adverb phrases

It would be difficult to extend the analysis proposed here to AdvPs since, they lack a Complement.

Topic- 254: Summing up: Layered Structure

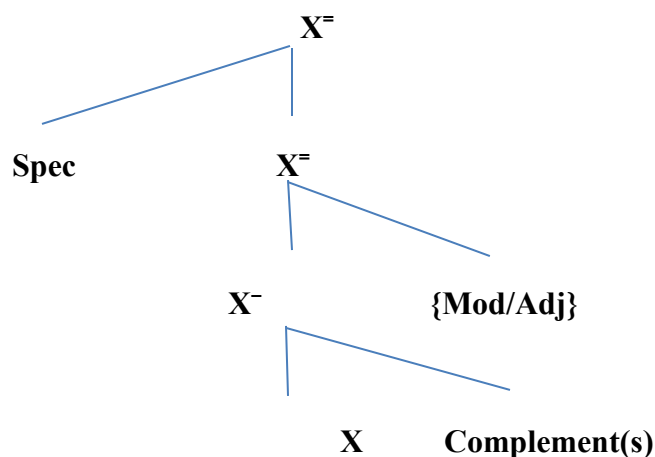
In Chapter 3 we have suggested that one way of talking about phrasal categories is to set up a general schema in which XP stands for any category of the phrasal level, and X for the lexical Head of the phrase:

XP → Spec-X- {Mod ajd } I

We see now that the schema might be expanded and refined by introducing the bar notation for intermediate projections: phrases have different projections:

- (a) the X⁻-level immediately dominates Heads and their Complements;

(b) the $X^=$ -level immediately dominates X^- and Modifiers/ Adjuncts and Specifiers:



Or in PS rules:

(a) $X^= \rightarrow (\text{Spec})-X^- (\{\text{Mod/adj}\})$

(b) $X^- \rightarrow X\text{-Complement(s)}$.

Topic- 255: Theory of X bar

X-bar theory is a theory of syntactic category formation. It embodies two independent claims: One, that phrases may contain intermediate constituents projected from a head X; Two, that this system of projected constituency may be common to more than one category (e.g., N, V, A, P, etc.). The letter X is used to signify an arbitrary lexical category (part of speech); when analyzing a specific utterance, specific categories are assigned. The X may become an N for noun, a V for verb, an A for adjective, or a P for preposition. The term X-bar is derived from the notation representing this structure. Certain structures are represented by X^- (an X with a bar over it). X may be often written as X' , using the prime symbol or with superscript numerals as exponents, e.g., X^1 . In English this is still read as "X bar".

The notation XP stands for X phrase, and is at the equivalent level of X-bar-bar (X with a double over bar), written X'' or X^2 , usually read aloud as X double bar. X-bar theory was first proposed by Noam Chomsky (1970) building on Zellig Harris's 1951 approach to categories, and further developed by Ray Jackendoff (1977). X-bar theory was incorporated into both transformational and nontransformational theories of syntax, including GB, GPSG, LFG, and HPSG. Recent work in the Minimalist Program has largely abandoned X-bar schemata in favor of bare phrase structure approaches.

Topic- 256: A Full Sentence in X-bar

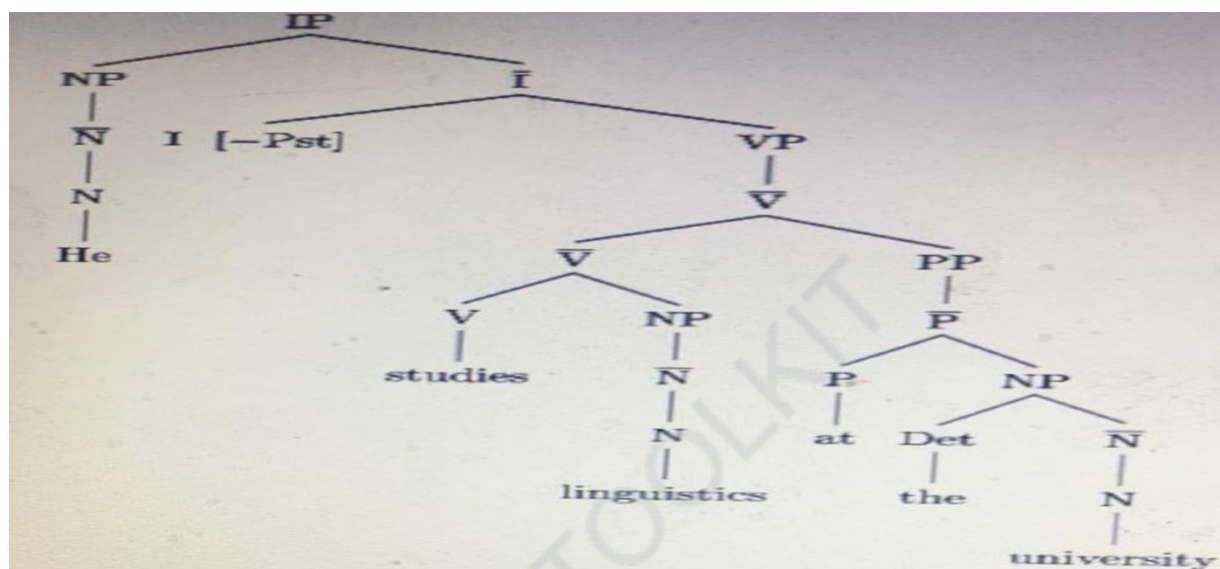
We have discussed XP in phrases. For more complex utterances, different theories of grammar assign X-bar theory elements to phrase types in different ways. Consider the sentence He studies

linguistics at the university. A transformational grammar theory might parse this sentence as the following diagram shows:

X-bar theory graph of the sentence:

"He studies linguistics at the university" (IP = inflectional phrase).

The "IP" is an inflectional phrase.



Its specifier is the noun phrase (NP) which acts as the subject of the sentence. The complement of the IP is the predicate of the sentence, a verb phrase (VP). There is no word in the sentence which explicitly acts as the head of the inflectional phrase, but this slot is usually considered to contain the unspoken "present tense" implied by the tense marker on the verb "studies".

A head-driven phrase structure grammar might parse this sentence differently. In this theory, the sentence is modeled as a verb phrase (VP). The noun phrase (NP) that is the subject of the sentence is located in the specifier of the verb phrase. The predicate parses the same way in both theories.

Topic- 257: Quantity of Sentence Structure

Theories of syntax that build on the X-bar schema tend to posit a large amount of sentence structure. The constituency-based, binary branching structures of the X-bar schema increase the number of nodes in the parse tree to the upper limits of what is possible. The result is highly layered trees (= "tall" trees) that acknowledge as many syntactic constituents as possible. The number of potential discontinuities increases, which increases the role of movement up the tree.

The analysis of phenomena such as inversion and shifting becomes more complex because these phenomena will necessarily involve discontinuities and thus necessitate movement or feature passing. Whether the large amount of sentence structure associated with X-bar schemata is necessary or beneficial is a matter of debate.

Endocentric structures only

When the X-bar schema was introduced and generally adopted into generative grammar in the 1970s, it was replacing a view of syntax that allowed for exocentric structures with one that views all sentence structure as endocentric. In other words, all phrasal units necessarily have a head in the X-bar schema, unlike the traditional binary division of the sentence (S) into a subject noun phrase (NP) and a predicate verb phrase (VP) ($S \rightarrow NP + VP$), which was an exocentric division.

In this regard, the X-bar schema was taking generative grammar one step toward a dependency-based theory of syntax, since dependency-based structures are incapable of acknowledging exocentric divisions. At the same time, the X-bar schema was taking generative grammar two steps away from a dependency-based understanding of syntactic structure insofar as it was allowing for an explosion in the amount of syntactic structure that the theory can posit. Dependency-based structures, in contrast, necessarily restrict the amount of sentence structure to an absolute minimum.

Lesson 44**KEY RESEARCH THEMES IN MORPHOLOGY/SYNTAX****Topic- 258: Morphology in a Wider Setting**

Morphology is at the conceptual centre of linguistics. Not because it is the dominant subdiscipline, but because morphology is the study of word structure, and words are at the interface between phonology, syntax and semantics. Words have phonological properties,

- They articulate together to form phrases and sentences,
- Their form often reflects their syntactic function,
- Their parts are often composed of meaningful smaller pieces.

In addition, words contract relationships with each other by virtue of their form; They form paradigms and lexical groupings. For this reason, morphology is something all linguists have to know about. The centrality of the word brings with it two important challenges. First, there is the question of what governs morphological form: how is allomorphy to be described? The second, what governs the syntactic and semantic function of morphological units, and how these interact with syntax and semantics proper.

There is a less enviable aspect to this centrality. However, the study of morphology in generative linguistics was largely eclipsed by phonology and syntax in the early days (though it is up to historians of linguistics to say exactly why). Ultimately, it came to be that when morphology was considered at all, it was regarded as essentially either a part of phonology or a part of syntax.

Morphology and Syntax

Morphology and lexical semantics, asks such questions as ‘What semanticosyntactic relations can be packaged up inside a single lexeme?’ Morphology in word recognition, presents an overview of recent findings in this field. One of the most powerful tools for investigating the workings of an on-line mechanism is to examine the patterns of errors that mechanism produces. Word production studies, which often involve the careful analysis of large corpora of speech errors, have generated a number of sophisticated models, including connectionist-inspired ones.

Topic- 259: Diachronic Morphology

Diachronic Morphology have painted a fairly complete picture of what morphology is, what constructs are needed in the morphological component of a grammar, and how these constructs interact with one another and with other parts of the grammar. For the most part, the perspective taken on these questions has been purely synchronic; yet, as with all aspects of language (and indeed of human institutions in general), a diachronic perspective is possible as well, focusing on what happens to morphology through time.

Moreover, through the answers given to these questions, especially the first two, several examples of various types of morphological change are presented. 1 What can change? What is stable? 1 What can change? What is stable?

The easy answer here is that just about everything discussed in the previous chapters as constituting morphology is subject to change, especially so once one realizes that regular sound change can alter the shape of morphs without concern for the effect of such a change in pronunciation on the morphological system. Thus, for example, once-distinct case endings can fall together by regular sound change. A change in the realization of number marking alone can be seen in the familiar case of the nominal plural marker /-s/ in English, for it has been spreading at the expense of other plural markers for centuries. For instance, the earlier English form *shoo-n*, as a plural of ‘shoe’, with the plural ending -n still found in *oxen*, has given way to *shoe-s*, with the most frequent, and indeed default, plural ending -s; in this case, the marker has not passed from the language altogether, as *oxen* shows, but the domain of a particular marker has come to be more and more restricted, and that of another has expanded.

Thus it is possible to argue that much – perhaps most – language change has a morphological/morpholexical basis, or at least has some morphological involvement. Such a view would then provide some diachronic justification for the importance of morphology in language in general, and thus for a morphological component in the grammars of particular languages.

Topic- 260: Morphology in Language Acquisition

Children typically begin to say their first words between twelve and twenty months of age. And they produce systematic morphological modulations of those words within their first year of talking. As they move to more complex expression of their meanings, they add grammatical morphemes – prefixes, suffixes, prepositions, postpositions, and clitics. On nouns, for example, they start to add morphemes to mark such distinctions as gender, number, and case; on verbs, they add markers for aspect, tense, gender, number, and person.

In order to acquire noun and verb morphology, children must first analyze the structure of words heard in input, identify stems and affixes, map consistent meanings onto them, and then begin to use those stems and affixes in new combinations. This process of analyzing form and assigning meaning is a prerequisite for the acquisition of inflectional morphology. It is also a prerequisite in the acquisition of word formation. Children begin to use some word-formation processes at around the same time as their first inflections. In particular, they produce novel compounds formed from simple stem combinations (often called root compounds).

Next, during their second year of speech, as some inflectional paradigms become established, they also begin to produce a few derivational affixes in novel word forms. These emerge in

greater numbers between ages three and four, in both derived and compound innovations. Some issues in acquisition The acquisition of morphology in inflection and word formation raises a number of questions about both morphology and the process of acquisition. Languages differ in the extent to which they rely on inflectional morphology to mark grammatical distinctions and grammatical relations.

In essence, languages range from analytic (with virtually no inflectional morphology, as in Chinese) to synthetic (with fairly extensive reliance on inflections, some for grammatical relations and agreement, and many marking several distinctions at once, as in Spanish), to agglutinative (with highly regular and systematic inflections, each marking a separate distinction, as in Turkish). In word formation, there is just as much variety, with some languages relying almost exclusively on compounding to form new words, others relying mainly on derivation, and many others relying on some mix of the two. Derivation may include both affixal and zero-derived forms, and some affixes commonly appear in compounds too. One issue for acquisition, then, is the extent to which language typology affects the Theoretical Linguistics

Topic- 261: Asking Questions about Syntax

The learning of a language is known as linguistics. Linguistics involves many topics of learning a language, including the study of syntax. In Linguistics, the study of syntax involves learning how to structure sentences in a certain language (Smith, 2015). In syntax, you need to construct a basic syntactic description of a language, either a language that you speak well yourself, or one for which you can find a native speaker consultant. You also need the right kind of questions to investigate for a better understanding of that language.

Taking that language as a short case study you need to know how these questions could be answered. The next topic refers to asking questions about syntax in the most literal way. However, I also hope that the discussion we have covered so far has ignited some curiosity about the human language faculty more generally. We may briefly outline some issues and questions surrounding our syntactic abilities that are currently widely debated within linguistics. The study of syntax in linguistics is quite challenging since the learner has to know how to put words in a sentence to make it sensible and avoid ambiguity (Smith, 2015). The choice of the topic on syntax is influenced by the difficulty that is experienced while learning a foreign language (Smith, 2015).

Topic- 262: What Questions to Investigate I

Let's now talk about a framework with which to write a basic syntactic description of a language that you know well, or for which you can access data readily. If appropriate, you can ask one or more native speakers to act as language consultant(s). Make sure you give the source(s) of your data, including attributions to the literature (i.e. cite your sources). Acknowledge any help given by language consultants.

Give the name by which the language is known to its native speakers, plus its English name, if any. State its language family and the principal locations in which it is spoken. Your description should include some or all of the questions outlined in (1) to (13), depending on what features of the language you consider to be most interesting from a syntactic, morphosyntactic and typological point of view. Give enough information on (and illustration of) any feature to make it comprehensible to someone who has no prior knowledge of the language.

All parts of the discussion must be illustrated with appropriate and sufficient data, glossed and translated. Number each example, following the conventions used in this textbook. If your language uses a writing system other than the Roman alphabet, cite data using whatever standard system of transliteration is used for this language. Give a list of abbreviations used in the gloss where these differ from those found in this textbook.

You won't need to discuss syntactic properties that are not manifested in your language. For instance (question (8) below), not all languages mark morphologically the relationship between a head and its dependents. If you were discussing Chinese, question (8) wouldn't be relevant.

In such cases, you can simply state that your language does not, for instance, display head- or dependent-marking. Similarly, you don't need to mention the antipassive construction unless your language has an ergative alignment.

You can collapse questions together where this makes sense for your language. Make sure, when answering each question, that you provide adequate explanations: do not leave the reader to work out for themselves what your data show.

Topic- 263: What Questions to Investigate II

Some basic questions to consider:

- (1) What is the neutral, or unmarked, constituent order (sometimes termed 'word order') in the clause, if there is one? If there is no neutral constituent order, describe the main principles of linearization. You should at least illustrate a transitive clause with two full NP arguments, and an intransitive clause. Are they orders the same in both these clause types?
- (2) What alternative neutral constituent orders are possible, if any? How marked are these?
- (3) What are the main word classes (or syntactic categories) in your language?

Discuss any that have especially interesting properties. Focus on the lexical classes N, V and A. You can expect any language to have a distinct class of nouns and verbs. Most languages will also have a distinct class of adjectives. Most will also have at least one or two (and maybe dozens) of ad positions. Justify all word classes that you posit: in other words, give evidence from its morphosyntactic properties and syntactic distribution to demonstrate that each proposed

class should be regarded as distinct (Chapter 2, Chapter 4). Include some of the main functional categories that your language distinguishes.

(4) Is your language predominantly head-initial or predominantly head-final?

Illustrate with data from more than one word class of heads. Are there any difficulties in establishing a predominant linearization? (Chapter 4). Remember that you are looking at the ordering of heads and their complements here, rather than the position of adjuncts with relation to heads.

(5) How does your language express clausal negation? (Chapter 3).

(6) Describe the main strategies for joining clauses together that are found in your language. What kinds of complementation occur? For instance, does the language have both finite and non-finite complement clauses? Does it have clausal subjects? If so, can they be both finite and non-finite? Does your language rely largely on subordination, as is the case for typical European languages? Does it have nominalized embeddings? Or does it, for instance, use co-ordination or verb serialization? (Chapter 3)

(7) Are the constituent orders occurring in subordinate clauses the same as those in root clauses, or different? If different, describe the differences carefully (Chapter

3). Are there any (other) noteworthy differences between root and embedded clauses?

(8) How, if at all, does your language mark morphosyntactically the relationship between heads and dependents? (Chapter 4). In other words, is your language largely head-marking or largely dependent-marking? Does it display a mix of both strategies? Illustrate at least with reference to the verb and its arguments.

(9) Does your language readily identify distinct constituents? (Chapter 5) If so, give at least two tests for constituency, illustrating with contrasting grammatical and ungrammatical data. Are there distinctive displacement processes for constituents in your language? Perhaps alternatively your language has free word order of the type found in Warlpiri (Chapter 6); if so, illustrate.

(10) Describe the way(s) in which the grammatical functions A, O and S are identified in your language (Chapter 6). Does this rely predominantly on constituent order, on agreement or cross-referencing, or on case-marking? Does your language exemplify an accusative or an ergative alignment? Make sure you give enough data to illustrate this. If your language has ergative alignment, is this purely morphological, or is it also (a much rarer possibility) syntactic? In other words, is there a clear subject relation in your language? Is it possible to identify a syntactic pivot?

(11) Your language almost certainly has some readily identifiable ways to change the grammatical functions or relations, either increasing or decreasing the valency of a verb (Chapter

7). Does it have a passive? If ergative, an antipassive? An impersonal construction? An applicative? A causative?

(12) Describe how wh- (i.e. constituent) questions are formulated (Chapter 8). Does the language have wh-fronting or wh-in-situ, or perhaps both? Discuss the main strategies for forming relative clauses in your language (if appropriate, you could alternatively discuss these under question (6), as part of subordination). Are there other wh-constructions in your language; for instance, is focus movement found, and if so, is it similar to wh-question formation?

(13) Are there any other interesting syntactic constructions that are not covered by these questions? If so, explain and illustrate them.

You may be wondering why it's worthwhile to investigate the grammars of languages. I hope that the preceding chapters have answered this question, but, in case not, you should consider the fact that every week, languages are becoming extinct.

Lesson 45**RESEARCH ISSUES OF THE FIELD OF SYNTAX****Topic- 264: Pedagogical Aspects of Morphology & Syntax**

Both morphology and syntax are highly important for pedagogical purpose of English as a second language. As the knowledge of English as a second language is considered a requirement in many fields, for this reason English is taught as one of the most important subjects, not only in schools but also in universities where the number of hours have increased significantly. That is why; the students of English have to be well prepared. And all we have said is linked with the study of Morphology and Syntax also known as Morphosyntax. Definitely it is not possible to address Morphology without knowing Syntax too as part of it, because they complement each other and share a great link. Learning English means acquiring certain skills such as having good vocabulary awareness, being competent at reading, writing a variety of papers, listening and speaking. All of these abilities are connected, and as teachers what we want to achieve in class is to integrate these skills so that students can produce the language fluently and accurately. Certainly, most of our vocabulary knowledge has been made possible due to our awareness of Morphology; it is known that the more morphological rules and tips you know the better vocabulary you will acquire. Besides, the ability to read involves decoding material, constructing sentences and messages semantically and grammatically correct. The knowledge of Morphosyntax is used for pedagogical purposes in all academic settings. A language cannot develop without a well-established theory of word and sentence formation. We are as teachers very concerned of the acquisition of the rules that govern the language. It is not possible to look into the human brain so as to watch how it processes language. Word formation started to attract the attention of linguists as a consequence of Chomsky's publication in 1970, "Remarks on Nominalization," which cost one of the most important divisions in the linguistic field among the supporters of lexical principles. Linguists succeeded in the identification and analysis of the units of language from the simplest, the phoneme, to the most complex, the sentence, including the intermediate ones such as the morpheme and the word. We want our students to develop a linguistic competence of the language, so that they can be able to produce an infinite number of sentences.

Topic- 265: Enhancing Vocabulary

Decoding and vocabulary development are pivotal to developing strong reading skills. They have been identified as two of the five critical components of reading instruction which include phonemic awareness, decoding, fluency, vocabulary, and comprehension). Other instructional resources have identified decoding and vocabulary development through word recognition/meaning and word study as two of seven dimensions of reading instruction (comprehension, metacognition, language and prior knowledge, word recognition and meaning, word study, fluency, and responding). Morphology is a critical element of successful vocabulary

development and accurate decoding. Awareness of morphology has been shown to be a strong indicator of and positive influence upon reading comprehension (Soifer, 2005). Subsequently, weakness in decoding and vocabulary skills is noted as a potent inhibitor to fully comprehending text.

As teacher we find assessment that indicate a weakness in vocabulary development for most of the students, thus suggesting a need to provide more intense vocabulary instruction with direct instruction in morphology. Morphing conjures mental pictures of children's toys that transform from one form to another through the child's manipulation of the parts. Morphology works in much the same manner, with students manipulating the parts of words to create new meanings or altered, but similar, meanings.

Morphology relates to the segmenting of words into affixes (prefixes and suffixes) and roots or base words, and the origins of words. Understanding that words connected by meaning can be connected by spelling can be critical to expanding a student's vocabulary. Further, parts of words (affixes) can have separate meanings that can transform or morph word meaning.

Topic- 266: Developing Reading Comprehension

Understanding the meaning of prefixes, suffixes, and roots enhances the comprehension of text being read. The manipulation of affixes can impact the part of speech that a word denotes. Having this knowledge enhances text comprehension as well. Direct instruction of morphology is an effective means to help with understanding and applying word structure for decoding, spelling, and vocabulary study (Wilson, 2005). Specifically, students can be taught strategies to segment or manipulate words according to their affixes and roots. As a result, students may be able to recognize an unfamiliar word simply by identifying the affixes and the remaining base word or root (Carreker, 2005).

Textbooks and student writings in the early grades typically use words of Anglo-Saxon origin. Typically, these words are one- to two-syllable, high-frequency words (Berninger & Wolf, 2009). Textbooks and student writings in the upper grades more frequently use words of Latin and Greek origin. In addition, the number of syllables in these words increases and unique spelling patterns emerge. Therefore, the recommended instructional sequence for teaching word origins, affixes, and roots is Anglo-Saxon before Latin and Greek.

Classroom Instruction in Morphology

Prince (2009) suggested four main instructional strategies from Lesaux's work with morphology:

- Morphology should be taught as a distinct component of a vocabulary improvement program throughout the upper elementary years.
- Morphology should be taught as a cognitive strategy to be learned. In order to break a word down into morphemes, students must complete the following four steps:

- o Recognize that they do not know the word.
- o Analyze the word for recognizable morphemes, both in the roots and suffixes.
- o Think of a possible meaning based upon the parts of the word.
- o Check the meaning of the word against the context of the reading.
- Students also need to understand the use of prefixes, suffixes, and roots, and how words get transformed.
- Multisensory Introduction of Affixes

A multisensory-guided discovery approach, as well as the use of an affix card deck, are recommended for teaching affixes. Using this approach, the teacher reads a series of derivatives that have a common trait (e.g., joyful, careful, helpful, graceful, cheerful). The students “discover” the similar sounds and then visually discover the sound-symbol correspondence. The similar sounds and letters are then identified as a prefix or suffix, and the student verbalizes these discoveries to anchor the learning.

Students who understand how words are formed by combining prefixes, suffixes, and roots tend to have larger vocabularies and better reading comprehension than peers without such knowledge and skills (Prince, 2009). Nagy (2007) proposed that the teaching morphological awareness and decoding in school may be the way to narrow the achievement gap for children whose families differ in education and income levels, and ethnic or racial backgrounds. A deep and full knowledge and understanding of vocabulary will improve outcomes for students who struggle.

Topic- 267: Better Understanding of the Language Teaching

Morphology is the study of word formation. As its name suggests, the study of morphology deals important insight of how language works, exposing the different needs of words’ categories and modifying words in various ways (O’Grady and Archibald 2005, 112).

Morphology applied in education helps the teachers particularly EFL teachers to understand the nature of the language system. This is because studying morphology means studying the system of the language works. Unlike Phonology whose main focus simply on the element of the sound, Morphology deals solely with how the words are formed and structured.

Discussing about how the words are formed and structured lead to the morpheme of language. Morpheme is the smallest unit of language that carries the information about meaning or function (O’Grady and Archibald 2005, 113). Moreover, this morpheme is divided into free morpheme and bound morpheme. Free morpheme is a morpheme that can be a word by itself; bound morpheme is a morpheme that must be attached to another element. For instance the word “ boy” is free morpheme because it can be used as a word on its own; in addition, the word ‘plural –s’ is bound morpheme because it must be attached to another word.

From this one advantage the writer mentioned, the students could have formed hundreds or even thousands words. Therefore, one of the ways to enrich students' vocabulary is by introducing them to prefixes and suffixes which are the bound morpheme of English language. Moreover, by knowing the types of morpheme in English language, it will make teacher easier to identify any new word which is a result of derivation. For example in order to understand the word "crucifixion" the teacher needs to find the base of the morpheme which is "crucify (v)". By adding "(x) ion" the meaning changes from "to kill someone by fastening them to a cross" to "the act of killing someone by fastening them to a cross and leaving them to die."

Some strategies for teaching morphology in the classroom based on research by Dr. Nonie Lesaux:

Morphology should be taught as a distinct component of a vocabulary improvement program throughout the upper elementary years.

Morphology should be taught as a cognitive strategy to be learned. In order to break a word down into morphemes, students must complete the following four steps:

- Recognize that they don't know the word.
- Analyze the word for recognizable morphemes, both in the roots and suffixes.
- Think of a possible meaning based upon the parts of the word.
- Check the meaning of the word against the context.
- Students also need to understand the use of prefixes, suffixes, and roots, and how words get transformed.

There is a significant body of research indicating that instruction of prefixes is beneficial, and many vocabulary researchers recommend instruction of the most common prefixes as the first and best component of word analysis (Graves, 2004; Carlisle, 2007).

To conclude, one of the advantages in learning morphology is to understand the nature of words such as how the word is formed and structured. Therefore, learning morphology itself will help students to be more creative in creating new words, for they have learned how to form the words.

Topic- 268: Development of Syntactic Skill

So why do we even study syntax? We can all speak and understand each other; do we need to investigate how? Well, yes! We are humans, it's in our nature to study things and understand how they work. So, whilst studying languages and how they work, we must study syntax and sentence constructions! Studying syntax is relevant to a lot of subject areas in linguistics. We must study syntax to understand how children acquire their language, how they start constructing sentences and what stage do they learn the tacit syntactic rules of the language. It's also good to study syntax so we can understand how bilingual and multilingual speakers are able to construct their sentences despite having different structures for different languages. The sentence structure isn't the same in English as it is in Japanese!

Studying syntax gives us many answers which are necessary for understanding how languages work, as well as being the doorway to future research and theories on all aspects of linguistics! We also study syntax to develop set rules and constraints on the language. We call these parameters. These parameters limit what we can and can't do in a language, helping us establish an effective and working communicative system. Some linguists believe that all languages have the same parameters. This idea is called universal grammar, and was a theory developed by Chomsky in the 1960s. From these ideas of linguistic parameters, we can learn, use and teach the correct way to make sentences. Basically, so we can all understand each other!

Topic- 269: The Knowledge of Morphosyntax

Both morphology and syntax are highly important for pedagogical purpose of English as a second language. As the knowledge of English as a second language is considered a requirement in many fields, for this reason English is taught as one of the most important subjects, not only in schools but also in universities where the number of hours have increased significantly. That is why; the students of English have to be well prepared. And all we have said is linked with the study of Morphology and Syntax also known as Morphosyntax. Definitely it is not possible to address Morphology without knowing Syntax too as part of it, because they complement each other and share a great link. Learning English means acquiring certain skills such as having good vocabulary awareness, being competent at reading, writing a variety of papers, listening and speaking. All of these abilities are connected, and as teachers what we want to achieve in class is to integrate these skills so that students can produce the language fluently and accurately. Certainly, most of our vocabulary knowledge has been made possible due to our awareness of Morphology; it is known that the more morphological rules and tips you know the better vocabulary you will acquire. Besides, the ability to read involves decoding material, constructing sentences and messages semantically and grammatically correct. The knowledge of Morphosyntax is used for pedagogical purposes in all academic settings. A language cannot develop without a well-established theory of word and sentence formation. We are as teachers very concerned of the acquisition of the rules that govern the language. It is not possible to look into the human brain so as to watch how it processes language. Word formation started to attract the attention of linguists as a consequence of Chomsky's publication in 1970, "Remarks on Nominalization," which cost one of the most important divisions in the linguistic field among the supporters of lexical principles. Linguists succeeded in the identification and analysis of the units of language from the simplest, the phoneme, to the most complex, the sentence, including the intermediate ones such as the morpheme and the word. We want our students to develop a linguistic competence of the language, so that they can be able to produce an infinite number of sentences.